

FRBR

object-oriented definition and mapping from FRBR_{ER}, FRAD and FRSAD (version 2.2)

International Working Group on FRBR and CIDOC CRM Harmonisation

Supported by Delos NoE

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Foreword

This document contains a comprehensive description of the object-oriented definition of FRBR, a model in the form of a formal ontology [interpreting FRBR for specific purposes](#), as analysed below. The document comprises the following sections:

- [Section 1](#), The Introduction, describes the [rationale, history and methodology](#) of the development of this model.
- [Section 2](#), The [Description of the Model](#), explains the model in context from a functional perspective with the help of a comprehensive [graphical representation of all constructs](#), describes the format conventions for the formal specifications, and lists the complete class and property definitions that make up the model. Whereas the first serves an overall understanding, the second is the reference for the individual declarations. Here a first reading may stop.
- [Section 3](#) describes the [mapping](#) of the [entity-relationship models](#) of the FRBR family to the object-oriented model FRBR_{OO}. This section defines the transition from one form to the other, and serves as information for further understanding of the intended meaning of the object-oriented definition. It is also a proof that the object-oriented form is an alternative view of the FRBR family, and a proof of completeness of the object-oriented form with respect to the original.
- Since the object-oriented model reuses, wherever appropriate, large parts of [ISO21127](#), the CIDOC Conceptual Reference Model, [section 4](#) provides a comprehensive list of all constructs used from ISO21127, together with their definitions following version 6.0 maintained by CIDOC. Some of these constructs appear only in the mapping in section 3 and not in section 2, because they are generic in nature.
- [Section 5](#) provides a bibliography.
- Section 6 traces [changes](#) that were made in previous versions of the model.

1. Introduction

This document is the definition of the object-oriented version of the **FRBR**¹ family of conceptual models², harmonised with CIDOC CRM, hereafter referred to as **FRBR_{OO}**, a formal ontology that captures and represents the underlying semantics of bibliographic information and therefore facilitates the integration, mediation, and interchange of bibliographic and museum information. Such a common view is necessary for the development of interoperable information systems serving users interested in accessing common or related content. Beyond that, it results in a formalisation which is more suited for the implementation of concepts from the FRBR family of conceptual models with object-oriented tools, and which facilitates the testing and adoption of these concepts in implementations with different functional specifications and beyond the library domain. It applies empirical analysis and ontological structure to the entities and processes associated with the bibliographic universe, to their properties, and to the relationships among them. It thereby reveals a web of interrelationships, which are also applicable to information objects in non-bibliographic arenas³.

The FRBR model was designed as an entity-relationship model by a study group appointed by the International Federation of Library Associations and Institutions (IFLA) during the period 1991-1997, it was approved by the IFLA Cataloguing Section in 1997, and was published in 1998. The original entity-relationship definition of FRBR is referred to hereafter as FRBR_{ER}.

Quite independently, the CIDOC CRM⁴ model was being developed, beginning in 1996, under the auspices of the ICOM-CIDOC (International Council for Museums – International Committee on Documentation) Documentation Standards Working Group. The definition of the CIDOC CRM model was adopted as ISO standard 21127.⁵

The idea that both the library and museum communities might benefit from harmonising their two models was first expressed in 2000, on the occasion of ELAG's (European Library Automation Group) 24th Library Systems Seminar in Paris. This idea led to the formation, in 2003, of the International Working Group on FRBR/CIDOC CRM Harmonisation, that brings together representatives from both communities with the common goals of: a) Expressing the IFLA FRBR model with the concepts, tools, mechanisms, and notation conventions provided by the CIDOC CRM, and: b) Aligning (possibly even merging) the two object-oriented models thus obtained.

The International Working Group on FRBR/CIDOC CRM Harmonisation, formed in 2003 and chaired by Martin Doerr (ICS FORTH, Greece), Patrick Le Bœuf (BnF, France), and Pat Riva (BAnQ, Canada), is affiliated at the same time to the IFLA FRBR Review Group and the CIDOC CRM Special Interest Group (CRM-SIG). The present definition of FRBR_{OO} was developed through email exchange among members of the Working Group, and more importantly during a series of meetings.

Version 1.0 of FRBR_{OO} was finally approved and issued in January 2010; it covered the entities and concepts from FRBR and included an appendix on identifier creation. The focus of later meetings has been to extend the model to fully encompass the published versions of the FRAD and FRSAD models. Version 2.1 is the result of this expansion.

More information on the activities of the Group, minutes of the meetings and all previous versions can be found on http://archive.ifla.org/VII/s13/wgfrbr/FRBR-CRMdialogue_wg.htm and on http://cidoc.ics.forth.gr/frbr_inro.html.

1.1. Purposes

¹ “FRBR” stands for: “Functional Requirements for Bibliographic Records,” after the name of the IFLA Study Group that developed the model. However, current use and understanding of the FRBR model go well beyond that, and the term “FRBR” has now turned into a noun in its own right, used without particular intention to refer to “functionalities,” nor to “requirements,” but rather to the *semantics* of bibliographic records. The *Final Report on Functional Requirements for Bibliographic Records* published in 1998 contained both a study on functional requirements for bibliographic records, and a description of the model known today as “FRBR.”

² In addition to FRBR itself, the FRBR family of conceptual models includes the *Functional Requirements for Authority Data* (FRAD), published in 2009, and the *Functional Requirements for Subject Authority Data* (FRSAD), published in 2011.

³ Coleman, Anita S. 2002. Scientific models as works. *Cataloging & classification quarterly* 33n3/4: 129-59.

⁴ “CIDOC CRM” stands for “Comité international de documentation [= International Committee on Documentation] Conceptual Reference Model,” which, when isolated from any context, is not particularly meaningful (CIDOC is affiliated to ICOM, the International Council of Museums). Just like FRBR, the acronym, rather meaningless in itself, has now turned into a noun in its own right.

⁵ *Information and documentation – a reference ontology for the interchange of cultural heritage information*. ISO 21127:2006. Geneva: ISO, 2006. Revised version: 2014.

This model represents FRBR, FRAD and FRSAD through modelling the conceptualisation of the reality behind library practice, as it is apparent from or implicit in the FRBR family of models. It is important to keep in mind that the aim is not to transform the IFLA models into something conceptually different, but to express the conceptualisation of the FRBR family within the object-oriented methodology instead of the entity-relationship methodology. Furthermore, the intention is to identify the common ground that memory institutions share and to exploit it by pursuing the following objectives.

1.1.1. A Common View of Cultural Heritage Information

The main goal is to reach a common view of cultural heritage information with respect to modelling, standards, recommendations, and practices. Libraries and museums are memory institutions – both strive to preserve cultural heritage objects, and information about such objects, and they often share the same users. Besides, the boundary between them is often blurred: libraries hold a number of museum objects and museums hold a number of library objects; the cultural heritage objects preserved in both types of institutions were created in the same cultural context or period, sometimes by the same agents, and they provide evidence of comparable cultural features. It seems therefore appropriate to build a common conceptualisation of the information gathered by the two types of organisations about cultural heritage.

1.1.2. A Verification of FRBR's Internal Consistency

Expressing the FRBR family in a different formalism than the one in which it was originally developed provides a means to evaluate the models in terms of their internal consistency. It is also a good opportunity to make adjustments to avoid some semantic inconsistencies and imprecisions in the formulation of the FRBR family, which prove to be crucial in the design of an overall model for the integration of cultural heritage related information. Additionally, the clarifications helped in the further development of the FRBR model itself, such as the interpretation of aggregates and aggregating work and understanding the dual nature of Manifestation.

1.1.3. An Enablement of Information Interoperability and Integration

Mediation tools and Semantic Web activities require an integrated, shared ontology for the information accumulated by both libraries and museums for all the collections that they hold, seen as a continuum from highly standardised products such as books, CDs, DVDs, etc., to raw materials such as plants or stones⁶, through “in-between” objects such as draft manuscripts or engraving plates. In addition, such typical “library objects” as books can be about museum objects, and museum objects can represent events or characters found in books (e.g., ‘*Ophelia's death*’) and descriptions of museum objects in museum databases may contain references to bibliographic resources that mention those museum objects: such interrelationships should be either integrated in common information storage, or at least virtually integrated through mediation devices that allow a query to be simultaneously launched on distinct information depositories, which requires common semantic tools such as FRBR_{OO} plugged into CIDOC CRM. Besides, CIDOC CRM is explicitly compatible in formalism with the World Wide Web Consortium's Resource Description Framework (RDF), which is also beneficial for the IFLA models.

1.1.4. An Opportunity for Mutual Enrichment for the FRBR Family and CIDOC CRM

The CIDOC CRM model is influenced by the process of FRBR's re-formulation as well. Modelling bibliographic information highlights some issues that may have been overlooked during the development of CIDOC CRM, and the way such issues are addressed in FRBR_{OO} resulted in some cases in making changes in the CIDOC CRM model. These changes are so significant that a revision of the ISO standard 21127 was required.

1.1.5. An Extension of the FRBR Family and the CIDOC CRM

The harmonisation between the two models is also an opportunity to extend the scope of the CIDOC CRM to bibliographic information, which paves the way for extensions to other domains and formats, such as EAD,

⁶ Natural history museums also are witnesses of “cultural features.” A frog in a museum is not a testimony of “what a frog is,” but of what a human culture, at a given point in time and space, thinks a frog is.

TEI, MPEG7, just to name a few. Consequently, it also extends the scope of the FRBR family of conceptual models to cultural materials, since FRBR_{OO} inherits all concepts of the CIDOC CRM, and opens the way for the IFLA models to benefit from further extensions of the scope of CIDOC CRM, such as the scientific heritage of observations and experiments.

1.1.6. Sources

The main source for the task of translating FRBR into the object-oriented formalism was, quite naturally, the IFLA *Final Report* that contains the complete definition of FRBR_{ER} itself:

IFLA Study Group on the functional requirements for bibliographic records.
Functional requirements for bibliographic records: final report. Munich, Germany:
K. G. Saur, 1998. Also available online from World Wide Web:
<<http://www.ifla.org/en/publications/functional-requirements-for-bibliographic-records>>.

Common awareness of the *Definition of the CIDOC Conceptual Reference Model* provides the required conceptual and technical background:

ICOM/CIDOC Documentation Standards Group; CIDOC CRM Special Interest Group. Definition of the CIDOC Conceptual Reference Model: version 6.0, January 2015. [Heraklion, Greece]: [ICS-FORTH], 2014. Available online at:
<http://www.cidoc-crm.org/docs/cidoc_crm_version_6.0.doc>, or:
<http://www.cidoc-crm.org/docs/cidoc_crm_version_6.0.pdf>.

In the preparation of version 2.1 of FRBR_{OO} the final approved statements of the FRAD and FRSAD models were used.

IFLA Working Group on Functional Requirements and Numbering of Authority Records (FRANAR); Glenn E. Patton, ed., *Functional requirements for authority data: a conceptual model*. München: K.G. Saur, 2009.
<<http://www.ifla.org/publications/functional-requirements-for-authority-data>>.

IFLA Working Group on the Functional Requirements for Subject Authority Records (FRSAR); Marcia Lei Zeng, Maja Žumer and Athena Salaba, ed., *Functional requirements for subject authority data (FRSAD): a conceptual model*. Berlin: De Gruyter Saur, 2011. Also available online from World Wide Web:
<<http://www.ifla.org/node/5849>>.

1.1.7. Understanding the Attributes and Relationships

The methodology consisted in a thorough examination of all attributes and relationships declared in the FRBR family. During its meetings, the International Working Group on FRBR/CIDOC CRM Harmonisation strove to extract the semantics as accurately as possible, to express them as “properties” in the sense of CIDOC CRM, and to relate them to CIDOC CRM properties where possible. Entities, or classes in the terminology adopted by the CIDOC CRM, play a nearly secondary role as the maximal sets of things for which a property is applicable.

1.1.8. Transforming Attributes into Properties

The CIDOC CRM model declares no “attributes” at all, but regards any information element as a “property” (or “relationship”) between two classes. The semantics extracted from FRBR_{ER}, FRAD and FRSAD attributes are therefore rendered in FRBR_{OO} as properties, according to the same principles as the CIDOC CRM model.

1.1.9. By-Product 1: Re-Contextualising Bibliographic Entities

The process of interpreting the precise semantic value of each individual attribute declared in FRBR_{ER} and expressing that semantic value in CRM-like structures resulted also in two by-products.

The first by-product was that it proved necessary to explain and model the general context within which the bibliographic entities isolated in FRBR_{ER} come into being. FRBR_{ER} envisions bibliographic entities as static, ever-

existing things that come from nowhere, and overlooks the complicated path from the initial idea for a new work in a creator's mind to the physical item in a user's hands through the dramatically important decision-making on behalf of publishers, as this complicated path is not explicitly reflected in data actually stored in bibliographic databases and library catalogues, which constituted the domain of reference of the FRBR Study Group. As a matter of fact, bibliographic records *do* contain some implicit information about that complicated path and the relationships it implies between and among bibliographic objects; FRBR_{OO} digs that implicit information out of bibliographic structures, e.g. the precise meaning of "date of publication".

1.1.10. By-Product 2: Adding a Bibliographic Flavour to CIDOC CRM

The second by-product was that the analysis provided for bibliographic processes in FRBR_{OO} and for the processes of naming entities in FRAD and FRSAD, paved the way for the introduction of refinements in the CIDOC CRM. This enabled the museum community's model to give a better account of mass production phenomena (such as the printing of engravings), the relation between creating immaterial content and physical carriers and the practices of identifying or naming things. Further, it introduces a basic model of intellectual conception and derivation applicable to all art forms, which the museum community had been hesitating to formally analyse.

1.2. Differences between FRBR_{ER} and FRBR_{OO}

1.2.1. Introduction of Temporal Entities, Events, and Time Processes

Temporal entities (i.e., phenomena, "perdurants" in philosophy) play a central role in the CIDOC CRM model, as they are the only means to relate objects (either conceptual or physical) to time-spans, locations, and agents. Since FRBR_{OO} borrows structures from the CIDOC CRM to express the concepts declared in FRBR_{ER}, "temporal entities" had inevitably to be introduced into FRBR_{OO}. Besides, some FRBR commentators had already made the point that time issues are insufficiently addressed in FRBR_{ER},⁷ the task of harmonising FRBR with the CIDOC CRM was an opportunity to fix that. Temporal entities were introduced into FRBR_{OO} by declaring some of the classes of FRBR_{OO} as subclasses of the following classes from CIDOC CRM: E65 Creation, E12 Production, E7 Activity, and E13 Attribute Assignment.

Figure 1 shows how the classes F27 Work Conception and F28 Expression Creation serve to link an E39 Actor, an E52 Time-Span and an E53 Place to the F1 Work, F2 Expression and F4 Manifestation Singleton that are created by those processes. In the lower part of the figure the work elaboration process is shown along a time axis. First, the activity F27 Work Conception produces an idea, then the F28 Expression Creation activity produces simultaneously an F2 Expression and its first manifestation (in the form of a F4 Manifestation Singleton), which together realise a work (F1).

⁷ HEANEY, Michael. *Time is of the essence*: some thoughts occasioned by the papers contributed to the International Conference on the Principles and Future Development of AACR [on line]. Oxford: Bodleian Library, 1997 [cited 30 May 2014]. Available from Internet: <<http://ora.ox.ac.uk/objects/uuid%3Aa3f3f1a4-cd70-4bcf-baef-dda55664c430/datastreams/ATTACHMENT01>>.

LAGOZE, Carl. Business unusual: how "event-awareness" may breathe life into the catalog?. In: *Conference on bibliographic control in the new millennium* [on line]. Washington: Library of Congress, October 19, 2000 [cited 19 January 2008]. Available from Internet: <http://lcweb.loc.gov/catdir/bibcontrol/lagoze_paper.html>.

FITCH, Kent. *ALEG Data Model. Inventory* [on line]. [Brisbane]: AustLit Gateway, revised 27 July 2000 [cited 26 March 2004]. Available from World Wide Web: <<http://www.austlit.edu.au:7777/DataModel/inventory.html>>.

DOERR, Martin; HUNTER, Jane; LAGOZE, Carl. Towards a core ontology for information integration. In: *Journal of Digital Information* [on line]. 2003-04-09, Vol. 4, No. 1 [cited 19 January 2008]. Available from Internet: <<http://journals.tdl.org/jodi/article/view/92/91>>.

Work and Time

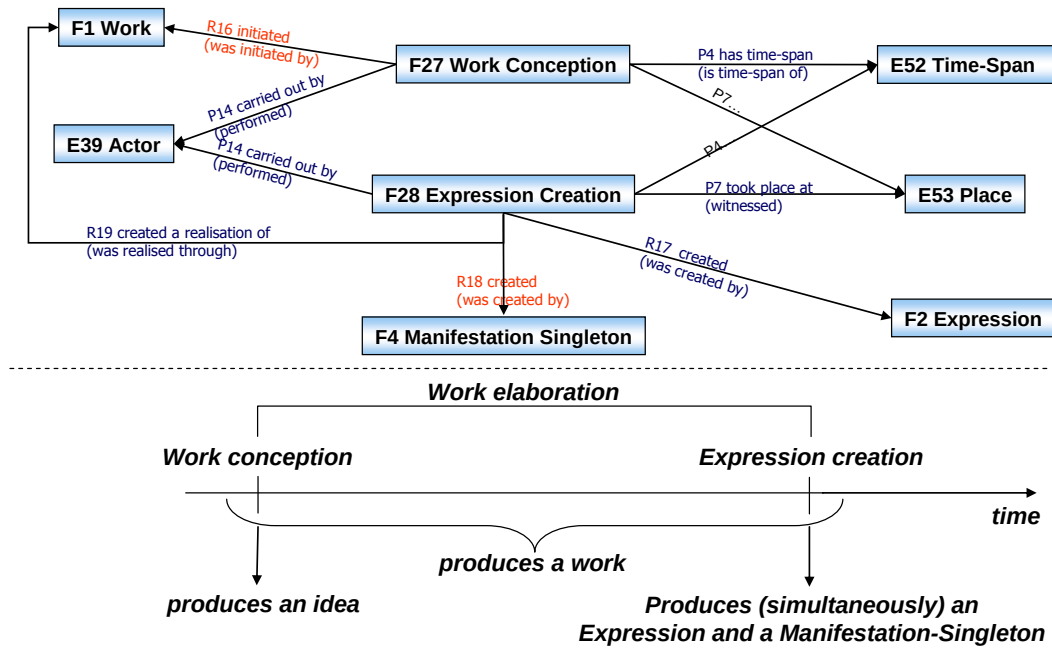


Figure 1

1.2.2. Refinement of Group 1 Entities

The text of FRBR_{ER} in some cases admits of multiple interpretations which introduce some logical inconsistencies, in particular with regard to its “Group 1 entities,” those entities that account for the content of a catalogue record.

The Work entity such as defined in FRBR_{ER} seemed to cover various realities with distinct properties. While the main interpretation intended by the originators of FRBR_{ER} seems to have been that of a set of concepts regarded as commonly shared by a number of individual sets of signs (or “Expressions”), other interpretations were possible as well: that of the set of concepts expressed in one particular set of signs, independently of the materialisation of that set of signs; and that of the overall abstract content of a given publication. FRBR_{OO} retains the vague notion of “Work” as a superclass for the various possible ways of interpreting the FRBR_{ER} definitions: F14 Individual Work corresponds to the concepts associated to one complete set of signs (i.e., one individual instance of F22 Self-Contained Expression); F19 Publication Work comprises publishers’ intellectual contribution to a given publication; and F15 Complex Work is closer to what seems to have been the main interpretation intended in FRBR_{ER}. Additionally, a further subclass is declared for F1 Work: F16 Container Work, which provides a framework for conceptualising works that consist in gathering sets of signs or fragments of sets of signs, of various origins (“aggregates”). Just like any product of the human mind, a Work necessarily begins to exist in the material world at a given point in time; this is the reason why FRBR_{OO} introduces the notion of F27 Work Conception. It makes the meaning of the FRBR_{ER} attribute ‘4.2.3 date of Work’ explicit, and accounts for the relationship between a Work and its creator, which holds even in cases when that creator has no direct participation in the creation of the Expression of that Work which is being catalogued (e.g., in the case of translations).

The Expression entity is relatively clear in FRBR_{ER}, at least from a purely conceptual point of view. However, the need was felt for a distinction between expressions that convey the complete idea of the work they realise, and expressions that convey only a fragment of it: that is, between instances of F22 Self-Contained Expression and instances of F23 Expression Fragment.

The Manifestation entity was defined in FRBR_{ER} in such a way that its definition could be interpreted as covering something physical and conceptual at the same time: it was defined in turn as the “physical embodiment” of an expression of a work and as an entity that represents all the physical objects that bear the same

characteristics. Discussion with members of the original FRBR Study Group⁸ showed that the Manifestation entity was actually meant as an entity all instances of which are *sets*; and sets, in the mathematical sense of the term, can have more than one member, or just one member (in which case they are called singletons). For the sake of clarification, the Working Group felt the need to split the Manifestation entity into two distinct classes, corresponding to the two possible ways of interpreting the ambiguous definition provided for Manifestation in FRBR_{ER}, namely F3 Manifestation Product Type and F4 Manifestation Singleton. Whereas F3 Manifestation Product Type is declared as a subclass of the CIDOC CRM class E55 Type, and therefore as a subclass, too, of the CIDOC CRM class E28 Conceptual Object (a merely abstract notion), F4 Manifestation Singleton is declared as a subclass of the CIDOC CRM class E24 Physical Man-Made Thing, and therefore as a subclass, too, of the CIDOC CRM class E18 Physical Thing.

The Item entity did not pose any particular problem in FRBR_{ER}; but splitting Manifestation into F3 Manifestation Product Type and F4 Manifestation Singleton obliged the Working Group to rethink the articulation between F4 Manifestation Singleton and F5 Item.

All in all, here is a picture of how original FRBR_{ER} entities relate to the classes declared in FRBR_{OO}:

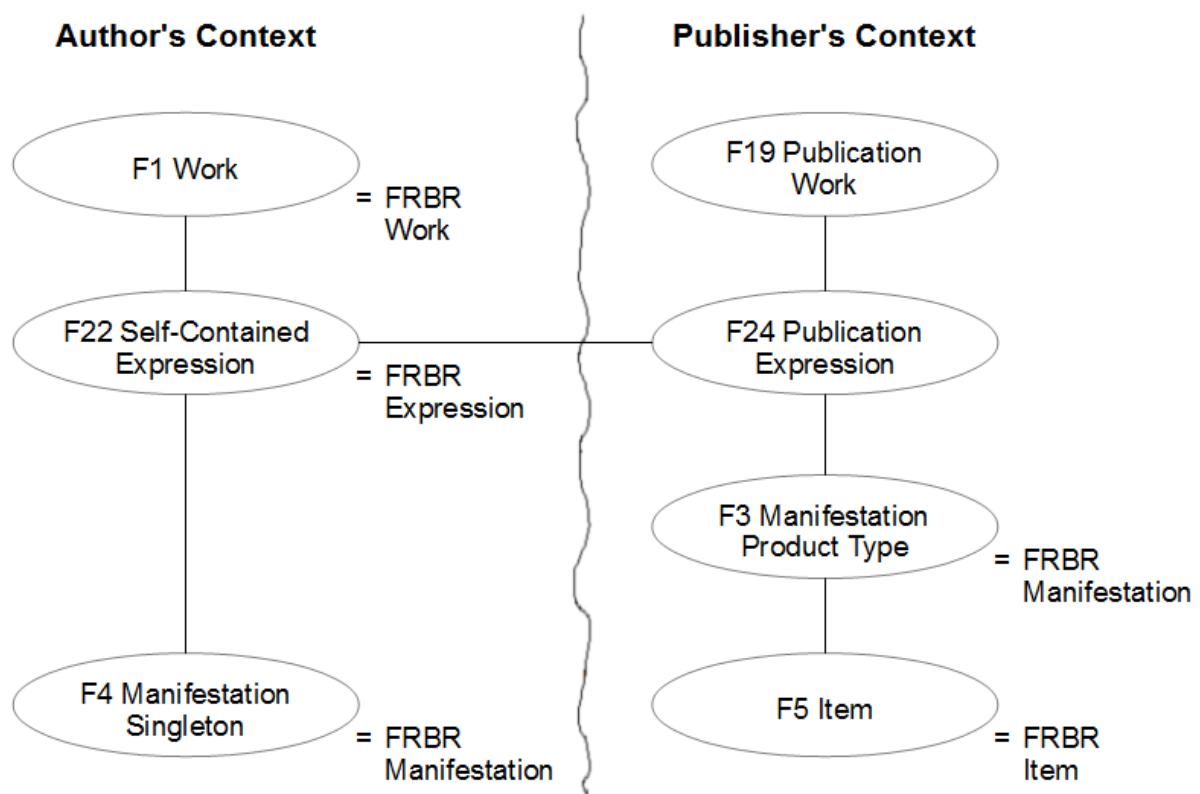


Figure 2

Figure 2 shows how the original FRBR_{ER} entities relate to the classes declared in FRBR_{OO}, particularly the “split” of the FRBR_{ER} Manifestation entity into F3 Manifestation Product Type and F4 Manifestation Singleton. In addition, the figure also shows how FRBR_{OO} makes explicit the publisher’s intellectual contribution, which is not modelled in FRBR_{ER}. Manifestation Product Type embodies a Publication Expression, which in turn comprises both the author’s Expression and the realisation of a Publication Work.

In figure 3 the FRBR_{OO} model of the realisation of a work by an expression is illustrated with a specific example. The overall work is Walt Whitman’s *Leaves of grass* (an instance of F15 Complex Work), which has as a member the “deathbed edition,” itself an instance of F15 Complex Work. The F14 Individual Work which corresponds to the abstract content of the French translation by Léon Bazalgette of that edition is in turn a member of the F15 Complex Work of the “deathbed edition.” The F28 Expression Creation event which produced the translation simultaneously created a realisation of that translation and created the instance of F22 Self-Contained Expression which is the text of that translation.

⁸ Tom Delsey and Beth Dulabahn participated in the Working Group’s first meeting in Paris in 2003.

Work Realisation example

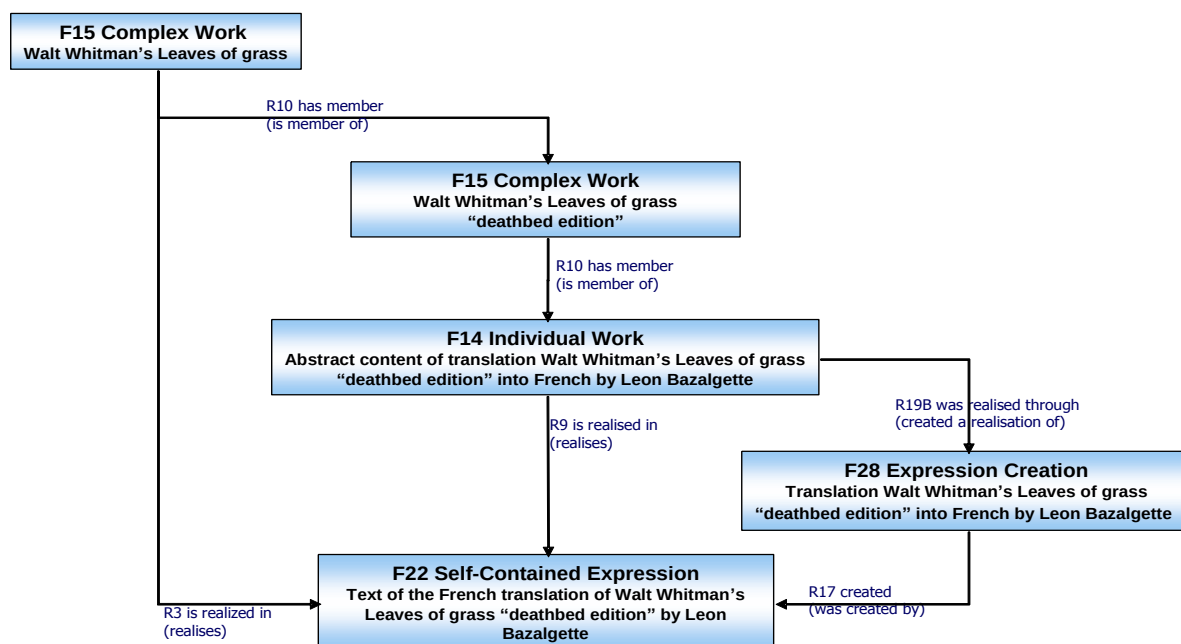


Figure 3

While it can be said that the attributes in $FRBR_{ER}$ still reflect to a certain degree traditional cataloguing policy, this is no longer true in $FRBR_{OO}$. Actually, what a bibliographic record should cover, following the intentions of FRBR, depends on the nature of the thing described, and, to a lesser degree, on the cataloguing policy that was followed when creating it. Interpreting FRBR, $FRBR_{OO}$ strictly associates attributes (or “properties”) with the entity of the bibliographic discourse they actually belong to. Only this form allows for the explanation and reconciliation of the various application dependent simplifications a particular implementation might choose. Some prominent cases are:

When a national bibliographic agency creates a single record for both hardcover and paperback presentations of the same content, that record describes an instance of F24 Publication Expression, and two distinct instances of F3 Manifestation Product Type. But if a library that only holds a copy of, say, the hardcover edition, decides to retain in the record exclusively such information elements which pertain to that edition, then the record can be said to focus on an instance of F3 Manifestation Product Type.

In the case of electronic publishing, since there is no instance of F3 Manifestation Product Type involved (see below, 1.2.3), the bibliographic record can only describe an instance of F24 Publication Expression.

In the case of serials, since the scope note for F18 Serial Work indicates that “there is in general no single Expression or Manifestation representing a complete serial work, unless the serial work has ended,” what the bibliographic record describes is actually an instance of the F18 Serial Work itself. Information elements that, in the $FRBR_{ER}$ conceptualisation, were directly attached to the Expression and Manifestation entities, are in $FRBR_{OO}$ seen as being in reality part of the issuing rule for the serial work (represented as an instance of E29 Design or Procedure). It is at the very core of the definition of F18 Serial Work that it plans that issues are published by a particular publisher and contain texts in a particular form. However, those information elements may change over time while the serial work retains its identity; in that case, the instance of F18 Serial Work has several distinct issuing rules over time, a case not modelled in $FRBR_{ER}$. This is what is meant when a single bibliographic record shows that at a given date, the publisher and/or place of publication have changed.

Any mapping from an existing database to $FRBR_{OO}$ should take all these notions into account.

1.2.3. Analysis of Creation and Production Processes

It proved necessary to analyse creation and production processes, in order to enable a better understanding of interrelations and temporal order.

In particular, the notion of “first externalisation” of a set of signs or expression (and, through the expression, the first externalisation of the individual work realised in the expression) is fully modelled in FRBR₀₀. It is regarded at the same time as a subclass of the creation of something conceptual, and the production of something physical, because the creation of an expression inevitably also affects the physical world, as the recording of the expression causes a physical modification of the object on which it is being recorded. The spatio-temporal circumstances under which the expression is created are necessarily the same spatio-temporal circumstances under which the carrier of the newly created expression is produced. This double phenomenon of conceptual creation/physical production can be represented by the schema presented in figure 4. F28 Expression Creation, which is a subclass of E65 Creation, produces, on the conceptual level, an F14 Individual Work through the property R19 created a realization of, and through R17 created, the F22 Self-Contained Expression which realises that work. Operating simultaneously on the physical level, F28 Expression Creation, a subclass of E12 Production, produces, through R18 created, the F4 Manifestation Singleton which P128 carries the F22 Self-Contained Expression.

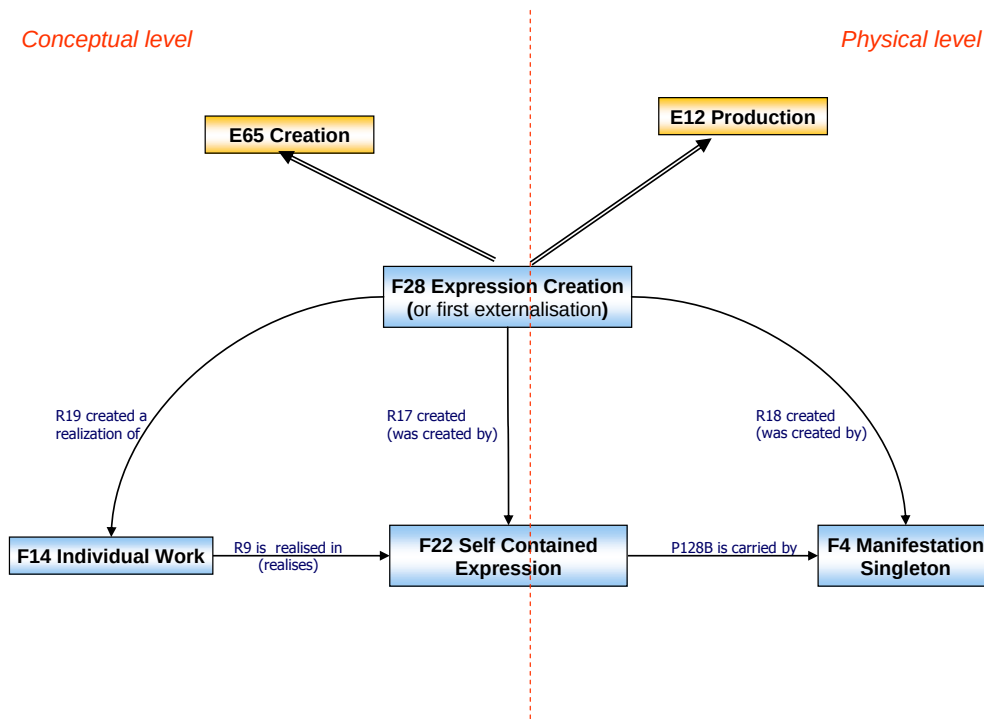


Figure 4

Another topic that is modelled in FRBR₀₀ is the distinction that has to be made between the process of physical publishing and the process of electronic publishing which is illustrated in figure 5. The F5 Items created through physical publishing are the results of an industrial process. As such they are produced by an F32 Carrier Production Event and carry an F24 Publication Expression, yet are also examples of an F3 Manifestation Product Type which *CLR6 should carry* the F24 Publication Expression. In electronic publishing, in contrast, the instances of F53 Material Copy, which are copies on local carriers, still carry the F24 Publication Expression and are produced by an F32 Carrier Production Event without there being any F3 Manifestation Product Type involved in the process. The instances of E29 Design or Procedure involved in the two processes differ: for physical publishing it can be characterised as “how to produce,” while for electronic publishing as “how to download.”

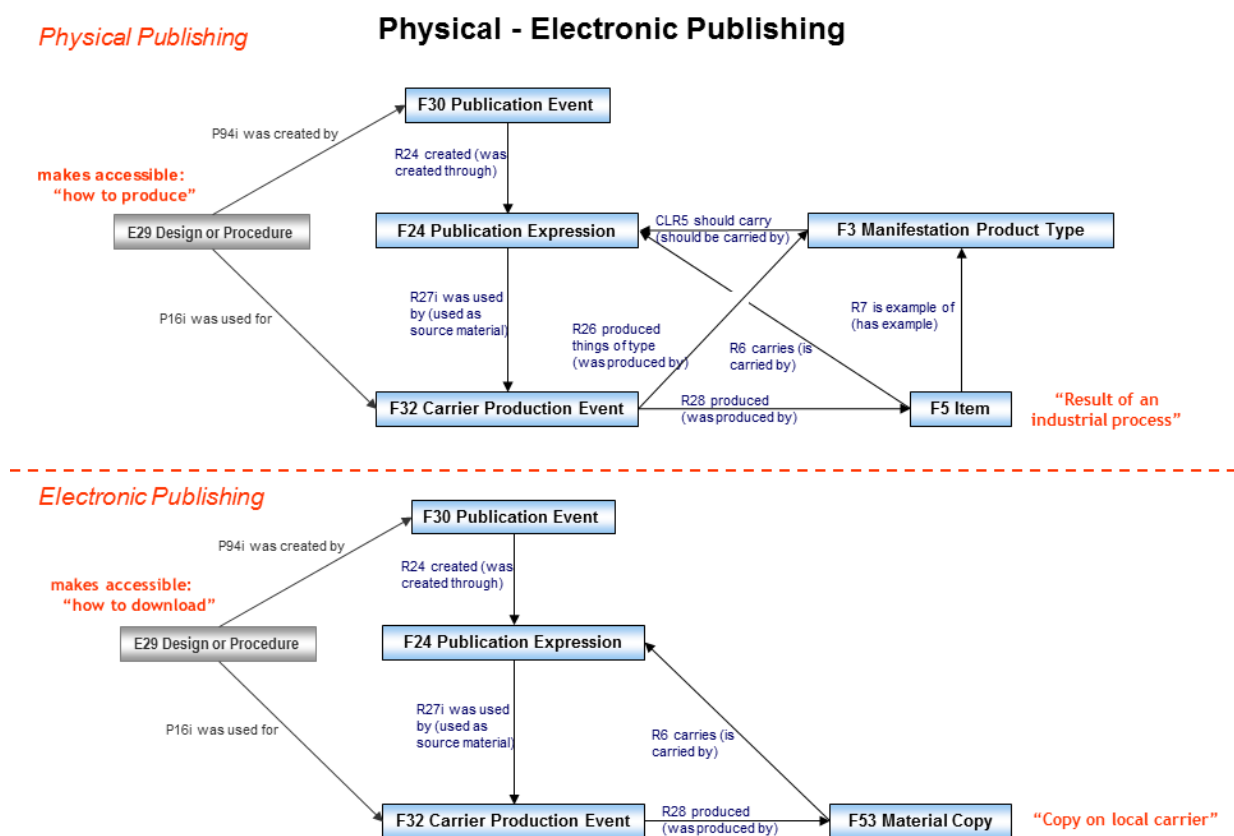


Figure 5

1.3. Differences between FRAD/FRSAD and FRBR₀₀

The FRAD model puts its emphasis on the separation between the bibliographic entities themselves (person, family, corporate body, work), their names as found in the non-bibliographic universe and the controlled access points constructed to represent the instances of the entities in bibliographic contexts through the use of rules as applied by bibliographic agencies.

In FRBR₀₀ the mechanism that underlies these relationships is made explicit by the introduction of classes and properties that capture the link between instances of persons, etc. and the names they use when performing different activities over time. Any activity in which one can see the use of a name in a given context is an F52 Name Use Activity. In the library domain, the F35 Nomen Use Statement records an appellation used within a KOS, including specific identifiers that provide controlled access points, which are generally recorded in authority records. Name authority files, subject headings files, classification systems are all typical library examples that are modelled as F34 KOS.

Taking as an example a Library of Congress name authority record for a corporate body (selected fields shown below), the information recorded in MARC21 field 670 is modelled by means of an F52 Name Use Activity. It indicates that in the book whose title is *Lo Scavo di S. Giovanni di Ruoti ed il periodo tardoantico in Basilicata* published in 1983, the name of the centre is given in Italian on the title page (Centro accademico canadese in Italia) and in English on page 6 (Canadian Academic Centre in Italy). Thus this one book exemplifies two distinct F52 Name Use Activities which associate (R63 named) two different E41 Appellations (R64 used name) with the same instance of F11 Corporate Body by means of two instances of F35 Nomen Use Statement. These statements, created using the appropriate cataloguing rules, record the assignment of the identifiers *Canadian Academic Centre in Italy* (as the preferred form, MARC 21 field 110) and *Centro accademico canadese in Italia* (as a variant form, MARC 21 field 410) as controlled identifiers for this centre. These controlled access points (instances of F35 Nomen Use Statement within the authority record) are R32 warranted by the F52 Name Use Activity which took place in the 1983 book.

```

110 2_ |a Canadian Academic Centre in Italy
410 2_ |a Centro accademico canadese in Italia
670 __ |a Lo Scavo di S. Giovanni di Ruoti ed il periodo tardoantico in
Basilicata, 1983: |b t.p. (Centro accademico canadese in Italia) p. 6
(Canadian Academic Centre in Italy)

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The FRBR_{OO} model clarifies that multiple bibliographic identities (such as pseudonyms coexisting with real names in different publications) are to be understood as instances of multiple name use by a single F10 Person in different contexts. This permits a new interpretation of the FRAD Person (An individual or a persona or identity established or adopted by an individual or group) as different instances of F52 Name Use Activity of the same person taking place in specific contexts (R61). This insight is simpler than defining personas as classes within the model, as well as being considerably more flexible. The same modelling technique subsumes the apparently different situations of multiple pseudonyms used in different contexts, thereby creating multiple bibliographic identities, and variant forms of names used simultaneously. The cardinality of the relationships between instances of F10 Person and the names that persons use can be one-to-one, one-to-many, or many-to-many (allowing for joint pseudonyms).

Similarly, in FRSAD the basic model indicates that thema is distinct from the nomens used to represent it. In FRBR_{OO} this is modelled in the same way: F52 Name Use Activity links the statements found in reference sources that attest to the use of a particular name for a particular concept (each of these statements being an instance of F35 Nomen Use Statement in the context of a particular KOS).

In a typical Library of Congress Subject Headings (LCSH) authority record (selected fields shown below), the Thema, as represented by the record number (sh 8507420 in MARC 21 field 010), is associated with the nomen Lamniformes (an instance of F35 Nomen Use Statement). An instance of F52 Name Use Activity is recorded on page 51 of the book *Fishes of the world* by J.S. Nelson published in 1994 that confirms the use of this term for this thema.

A broader term relationship is stated within LCSH (an instance of a F34 KOS) between this thema and two other themata, this is encoded in the MARC 21 550 fields (code value **g** in subfield **w** indicates broader term). In other terms, the thema (an instance of E55 Type) identified by the nomen “Lamniformes” *P127 has broader term* the thema (a distinct instance of E55 Type) identified by the nomen “Sharks”. This same authority record shows the use of MARC 21 field 053 to encode the assignment of the nomen QL638.94.L36 to this thema, this time within the Library of Congress Classification (LCC).

```

010 __ |a sh 85074230
053 _0 |a QL638.94.L36 |c Zoology
150 __ |a Lamniformes
550 __ |w g |a Chondrichthyes
550 __ |w g |a Sharks
670 __ |a Nelson, J.S. Fishes of the world, 1994: |b p. 51 (Order Lamniformes
(mackerel sharks). Seven families with 10 genera and 16 species)

```

2. Description of the Model

This section explains the model in context from a functional perspective with the help of a comprehensive graphical representation of all constructs, describes the format conventions for the formal specifications, and lists the complete class and property definitions that make up the model. The graphical representation (section 2.1) serves an overall understanding, while the list of definitions (sections 2.6 and 2.7) is the reference for the individual declarations.

2.1. Graphic Overview of the Object-Oriented Definition of FRBR

In this section, FRBR_{OO} is presented in a sequence which follows the intellectual process from Work through Expression to Manifestation. In contrast to FRBR_{ER}, a dynamic view of the respective processes of Expression Creation and of the Publication Work is also presented. Additionally, the dimension of intellectual contributions made by incorporating parts of an Expression in another one is demonstrated using the example of the performing arts. This dimension is only marginally analysed in FRBR_{ER}. This section then contains a presentation of how FRBR_{OO} models the process of identifier creation, which does not belong to the scope of FRBR_{ER} but is an important aspect of FRAD. The section ends with a brief statement about the modelling of photographs and animated images.

2.1.1. Static View of the Work and Expression Classes

Figure 6 shows the relations that exist between works and expressions and the subclasses of both concepts, independently from any dynamic aspects involving the activities of creation and modification. It shows an analysis of the original FRBR_{ER} concepts Work and Expression into the more detailed ones that appear only indirectly in FRBR_{ER} via attributes that are specific to these detailed concepts rather than to Work and Expression in general. The reader may find the actual relation of these concepts to the FRBR_{ER} attributes in section 3.3 below.

Work and Expression, static view

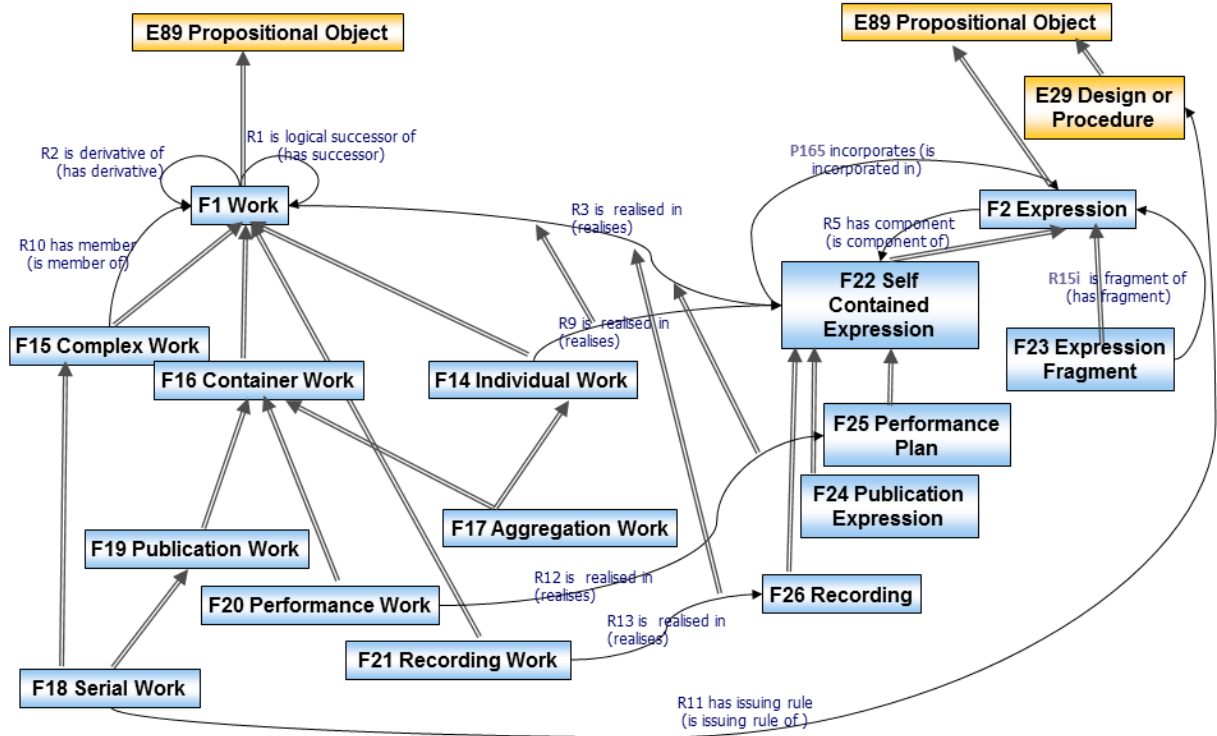


Figure 6

Comments on Figure 6:

- The concepts that make up a work are realised as complete sets of signs. This fact is modelled as: F1 Work *R3 is realised in (realises)* F22 Self-Contained Expression.
- A set of signs may not convey the complete concept of a work; it may just be a fragment of a larger set of signs. This fact is modelled as: F2 Expression *R15 has fragment (is fragment of)* F23 Expression Fragment.
- A complete set of signs may be a structural part of a larger set of signs. This fact is modelled as: F2 Expression *R5 has component (is component of)* F22 Self-Contained Expression.
- A work can present itself as a “continuation” of some other work. This fact is modelled as: F1 Work *R1 is logical successor of (has successor)* F1 Work.
- A work can present itself as “derived” from another work, in many possible ways. This fact is modelled as: F1 Work *R2 is derivative of (has derivative)* F1 Work *R2.1 has type E55 Type [of derivation]*.
- The notion of “work” is actually a vague one, which covers three more specific notions:
 - The sum of concepts conveyed by just one complete set of signs. This is modelled as: F14 Individual Work *is a* F1 Work, and F14 Individual Work *R9 is realised in (realises)* F22 Self-Contained Expression.
 - The concept of re-using some already existing material in order to produce some new creation. This is modelled as: F16 Container Work *is a* F1 Work, F1 Work *R3 is realised in (realises)* F22 Self-Contained Expression, and F22 Self-Contained Expression *P165 incorporates (is incorporated in)* E89 Propositional Object.

- incorporates (is incorporated in)* F2 Expression.
 - The conceptual unity observed across a number of complete sets of signs, which makes it possible to organise publications into “bibliographic families.” This is modelled as: F15 Complex Work *is a* F1 Work, and F15 Complex Work *R10 has member (is member of)* F1 Work.
- g. Additionally, a work can be recognised as being composed of several structural parts. This is also modelled as: F15 Complex Work *is a* F1 Work, and F15 Complex Work *R10 has member (is member of)* F1 Work.⁹
- h. Works that re-use some already existing material or use some event, either natural or involving human activity (i.e., instances of F16 Container Work), are further subdivided into:
 - Works that aggregate already existing expressions of other works. This is modelled as: F17 Aggregation Work *is a* F16 Container Work, F17 Aggregation Work *is a* F14 Individual Work, F14 Individual Work *R9 is realised in (realises)* F22 Self-Contained Expression, and F22 Self-Contained Expression *P165 incorporates (is incorporated in)* F2 Expression.
 - Works that consist in establishing all the features of recordings of sounds and/or images (either natural or involving human activity). This is modelled as: F21 Recording Work *is a* F1 Work, F21 Recording Work *R13 is realised in (realises)* F26 Recording, and F26 Recording *is a* F22 Self-Contained Expression.
 - Works that consist in establishing all the features of a performance. This is modelled as: F20 Performance Work *is a* F16 Container Work, F20 Performance Work *R12 is realised in (realises)* F25 Performance Plan, and F25 Performance Plan *is a* F22 Self-Contained Expression.
 - Works that consist in establishing all the features of a publication. This is modelled as: F19 Publication Work *is a* F16 Container Work, F19 Publication Work *R3 is realised in (realises)* F24 Publication Expression, and F24 Publication Expression *is a* F22 Self-Contained Expression.
 - Works that consist in establishing all the features of serials are a specific case of the latter; but serials have particular constraints as to their frequency of issuance, numbering pattern, etc. This is modelled as: F18 Serial Work *is a* F19 Publication Work, and F18 Serial Work *R11 has issuing rule (is issuing rule of)* E29 Design or Procedure [a CIDOC CRM class].

2.1.2. Dynamic View of the Work and Expression Classes

Figure 7 shows the dynamic process through which products of the mind come into being.

⁹ Alternatively, users who would be eager to avoid the confusion between structural parts and successive members of complex works can use the CIDOC CRM property *P148 has component (is component of)*, instead of *R10 has member*, to record the relationship between an instance of F15 Complex Work and its structural parts.

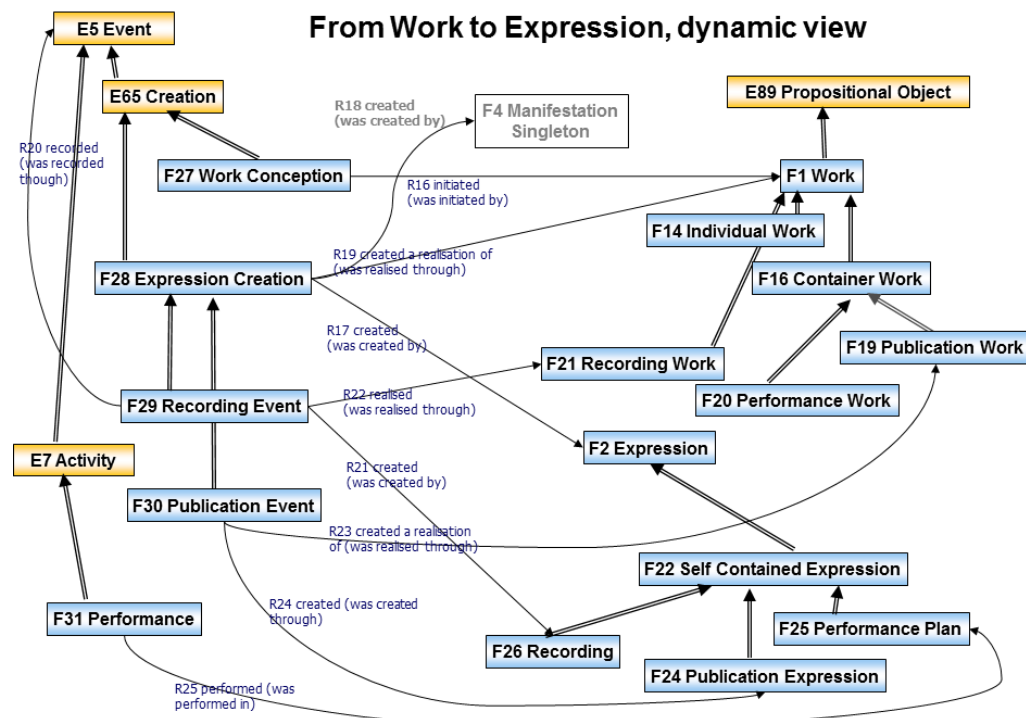


Figure 7

Comments on Figure 7:

- An instance of F1 Work begins to exist from the very moment an individual has the initial idea that triggers a creative process in his or her mind. This is modelled as: F27 Work Conception *R16 initiated (was initiated by)* F1 Work.
- Unless a creator leaves at least one physical sketch for his or her work, the very existence of that instance of F1 Work goes unnoticed, and there is nothing to be catalogued. At least one instance of F2 Expression that *R3i realises* the instance of F1 Work has to be created. This is modelled as: F28 Expression Creation *R19 created a realisation of (was realised through)* F1 Work, and F28 Expression Creation *R17 created (was created by)* F2 Expression. Except for oral tradition and recording in human memory, this very first instance of the respective F2 Expression would be created **simultaneously** on a physical carrier, typically as a unique item or as an electronic file on a specific computer. This is modelled as: F28 Expression Creation *R18 created (was created by)* F4 Manifestation Singleton, as detailed in Figures 8 and 9.
- Sound recordings and moving images are particular cases of expressions, in that they involve both temporal entities (the “things” being recorded, either performances of works, natural events, or states) and decisions made by one or more than one individual (sound engineer, movie director...). This is modelled as: F29 Recording Event *R20 recorded (was recorded through)* E5 Event, F29 Recording Event *R22 realised (was realised through)* F21 Recording Work (i.e., the artistic and technical decisions made about the recording material to be used, the location of microphones and/or cameras, the use of filters, lighting, framing, etc.), and F29 Recording Event *R21 created (was created by)* F26 Recording (i.e., the set of either analogue or digital signs that are inevitably infixed on a carrier at the time they are produced – just like any other kind of expression – but are likely to be conveyed on any other carrier without losing their identity as a distinct expression).
- Publishers make decisions about all the features of a new product, and determine the complete set of signs that will be found on it. This is modelled as: F30 Publication Event *R23 created a realisation of (was realised through)* F19 Publication Work (i.e., a publisher’s concept of a given publication), and F30 Publication Event *R24 created (was created through)* F24 Publication Expression (i.e., the set of *all the signs* present on a given publication, including book cover, title page, page numbers, copyright statement, CD liner notes, text found on a DVD container, etc.).
- Performers make decisions about all the features their performance should display (whether it is an improvisation or it involves some pre-existing work such as a play or a musical composition), and may express these decisions as explicit instructions. This is modelled as: F31 Performance (i.e., the performing activity itself) *R25 performed (was performed in)* F25 Performance Plan (i.e., the set of

instructions for a specific performance, which *P165 incorporates* the text of a play, the content of a musical score, etc.).

2.1.3. Dynamic View of the Manifestation and Item Classes

Figure 8 shows how products of the mind are communicated among human beings through physical carriers that eventually become part of the cultural heritage preserved in memory institutions such as libraries, archives, and museums.

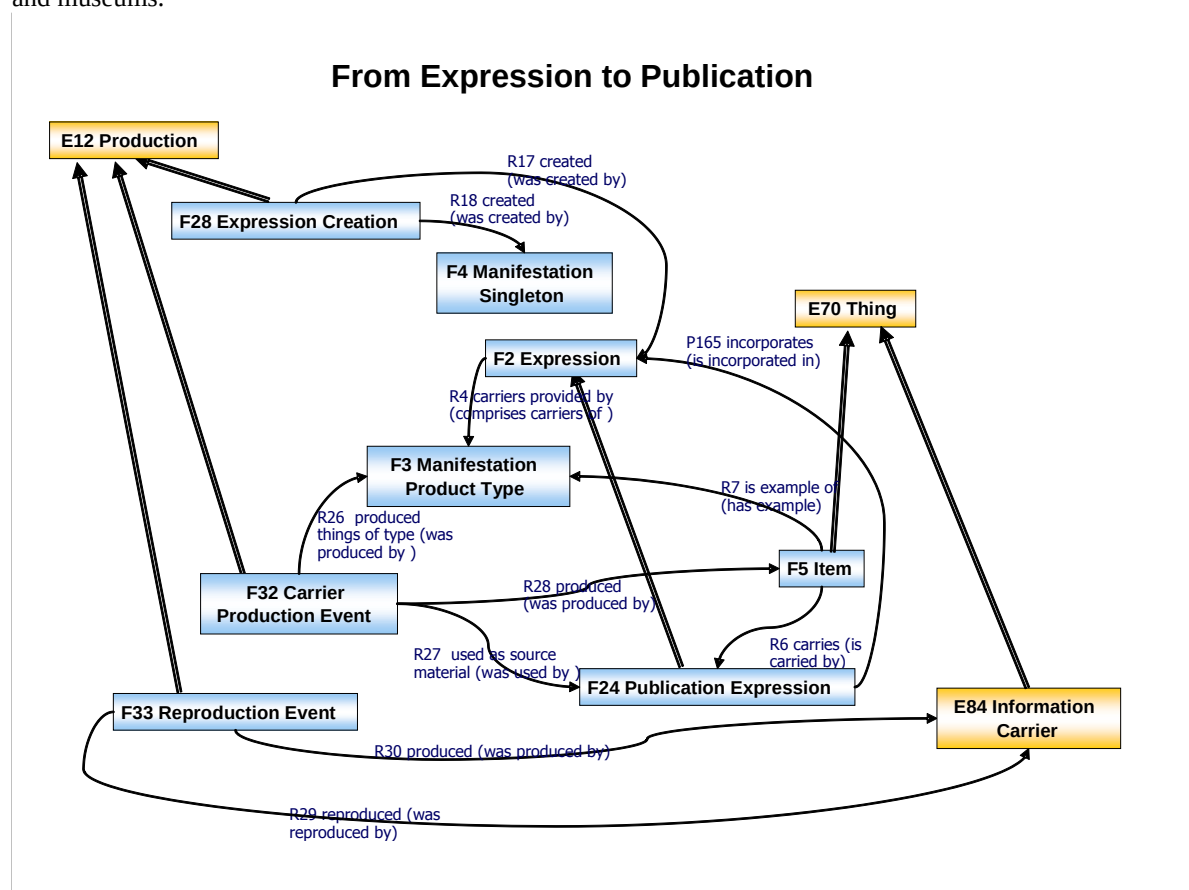


Figure 8

Comments on Figure 8:

Authorial output:

- A creator elaborates an expression (it can be a text, a musical score, a drawing, a map, etc.). This process is modelled as: F28 Expression Creation *R17 created (was created by)* F2 Expression.
- The creator externalises that expression by transforming bits of the physical world into physical carriers of his or her creation. This is modelled as: F28 Expression Creation *R18 created (was created by)* F4 Manifestation Singleton (e.g., a draft manuscript).

Editorial product:

- A publisher elaborates the overall content of a new publication: F30 Publication Event *R24 created (was created through)* F24 Publication Expression (see Figure 2).
- That overall content incorporates the authorial expression such as that found, for instance, on a manuscript provided by the author: F24 Publication Expression *P165 incorporates (is incorporated in)* F2 Expression.

Printing/manufacturing:

- The publisher sends to a manufacturer the overall content of the publication (a mechanical or paste-up, or, most often nowadays, desktop publishing files), along with instructions as to how exemplars of the publication should be manufactured: F32 Carrier Production Event *R27 used as source material (was used by)* F24 Publication Expression.

- As a consequence, all exemplars of the publication are supposed to be similar, i.e., can be identified as belonging to the same type: F32 Carrier Production Event *R26 produced things of type (was produced by)* F3 Manifestation Product Type.
- As a consequence, both the author's expression and the publisher's expression are to be found on all exemplars belonging to that type: F2 Expression *R4 carriers provided by (comprises carriers of)* F3 Manifestation Product Type.
- The manufacturing process results in physical objects, the exemplars themselves: F32 Carrier Production Event *R28 produced (was produced by)* F5 Item.
- Any exemplar is representative for the publication of which it is an exemplar: F5 Item *R7 is example of (has example)* F3 Manifestation Product Type.
- Under normal conditions, any exemplar should display the same overall content defined by the publisher: F5 Item *R6 carries (is carried by)* F24 Publication Expression.

Reproduction:

- Any information carrier can be reproduced by processes that render a similar item to the original used: F33 Reproduction Event *R29 reproduced (was reproduced by)* E84 Information Carrier. This should not be confused with resuming the actual production process itself.
- This process results in a new instance of E84 Information Carrier: F33 Reproduction Event *R30 produced (was produced by)* E84 Information Carrier.

2.1.4. Static View of the Manifestation and Item Classes

Figure 9 shows how FRBR_{OO} renders the meaning of the FRBR_{ER} Manifestation entity and its attributes.

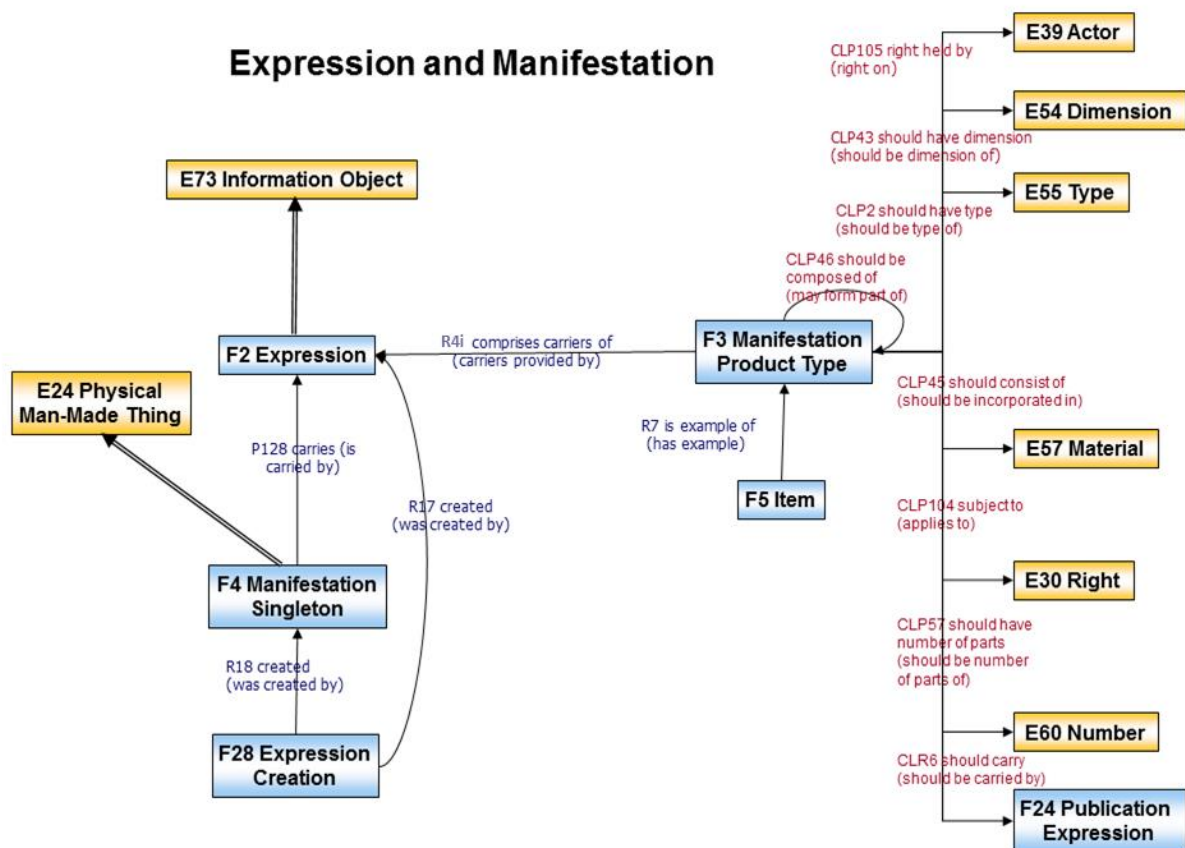


Figure 9

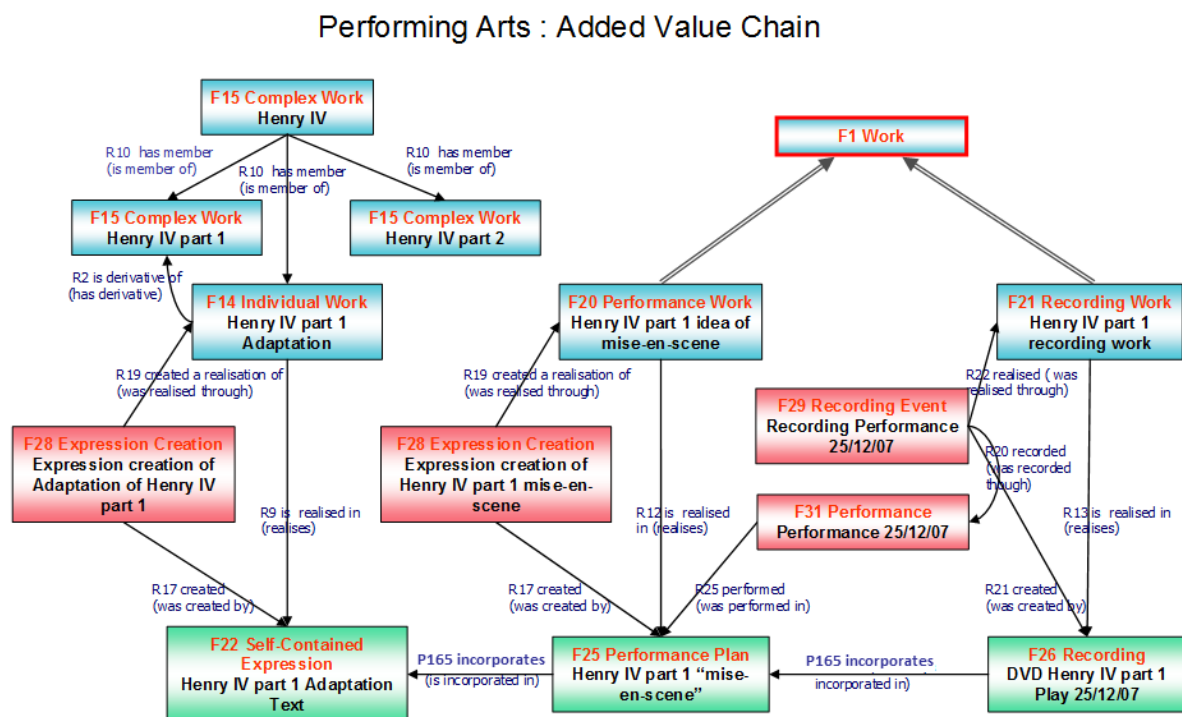
Comments on Figure 9:

Manifestation is split into F4 Manifestation Singleton (a unique, physical object) and F3 Manifestation Product Type (a publication, i.e., an abstract notion only recognisable through its physical exemplars).

Every time a creator drafts a new expression on paper (or on the hard disk of a computer, etc.), that process

results simultaneously in the creation of a new information object and the production of a new physical man-made thing: F28 Expression Creation *R17 created (was created by)* F2 Expression, and F28 Expression Creation *R18 created (was created by)* F4 Manifestation Singleton.

2.1.5. Performing Arts as an Example for the Incorporation of Expressions in Expressions of Other Works



Comments on Figure 10:

of F22 Self-Contained Expression.

The text of this adaptation is incorporated in the stage director's instructions for performance, which realise his or her concept of a mise-en-scène of *Henry IV*.

The play is performed on December 25, 2007.

That performance is filmed.

The resulting recording, which in turn incorporates some aspects of the stage director's performing instructions, in addition to the adapted text of *Henry IV Part 1*, is infixed on a DVD.

Similar considerations hold for the contributions of an editor, illustrator and graphics designer to a manuscript, or the compiling of a collection of poems to the texts selected.

Figure 10 demonstrates three fundamental relationships between instances of F1 Work:

- Continuation or completion of a topic (such as *Henry IV* part 1 and part 2)
- Derivation of an existing Expression in an alternative form (such as original and adaptation)
- Incorporation of an unaltered part of the contents of an Expression (such as play text and performance recording).

2.1.6. Creation and Assignment of Controlled Access Points

During the cataloguing process, one important phase is the creation of what are termed controlled access points (formerly known as "headings"). Controlled access points enable a given instance of a given bibliographic entity to be consistently referred to in a given bibliographic database. Controlled access points are, in general, composed of parts, which consist of signs, and some of which are appellations in their own right. They are created to identify persons, corporate bodies, geographic areas, works, etc.

Specific sections of cataloguing rules specify the steps that have to be followed when creating a controlled access point and how to ensure its uniqueness. The steps include the choice of the preferred form of the name, the choice of qualifiers, their form, punctuation and order. During an F40 Identifier Assignment the cataloguer composes the identifier and makes informed decisions. That complex process is modelled using three distinct classes: F12 Nomen, F13 Identifier, and F50 Controlled Access Point. F13 Identifier corresponds to both standardised strings such as uniform titles (which are instances of F50 Controlled Access Point), and the notion of numeric identifiers such as international standard numbers defined in ISO standards (such as ISNI, ISBN, ISSN, ISRC, etc.). Any qualifier used in cataloguing practice to disambiguate access points is an instance of E90 Symbolic Object (or, in many cases, of its subclass E41 Appellation; e.g., dates are appellations of instances of E52 Time-Span).

Figure 11 is a representation of such an activity.

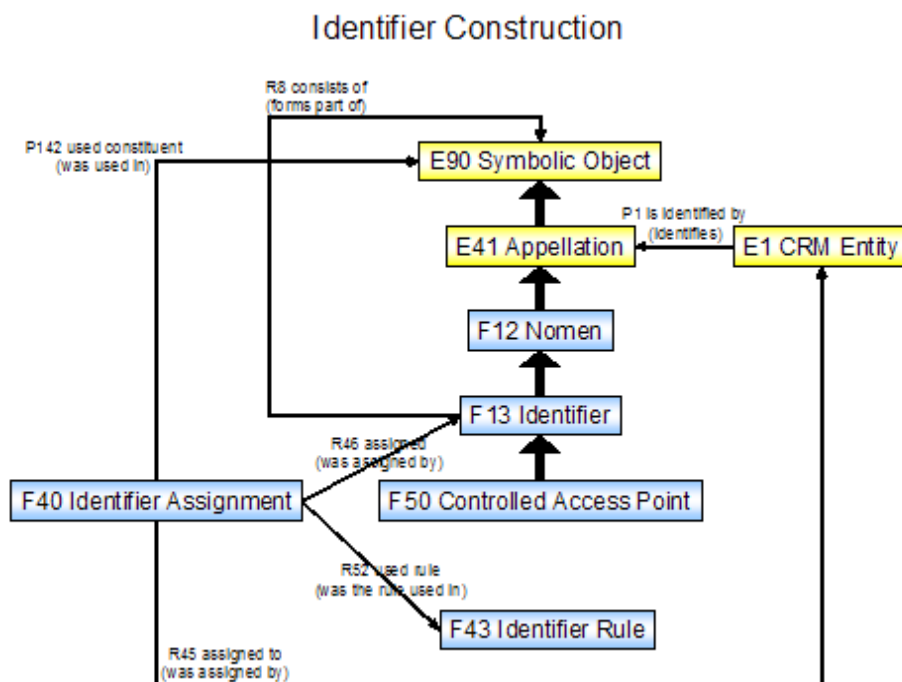


Figure 11

The model described in Figure 11 is relevant beyond library practice. It allows for the implementation of the reasoning processes involved in analysing the information encoded in the parts of an identifier. Therefore the CIDOC CRM has incorporated this model.

The process of creating controlled access points that are as specific, accurate, and as “unique” as possible is particularly interesting when applied to instances of Work and Expression, as they lack a material nature. Figure 12 shows how the instance of an F2 Expression common to some instances of F3 Manifestation Product Type or F4 Manifestation Singleton would be identified with the one manifestation of a particular expression that has been selected to be “representative” for this expression. This model does not correspond to any explicit library information. Rather, it describes an implicit process behind the appellation creation process: The very fact of composing an identifier for an instance of F2 Expression using the identifier of one particular manifestation makes the latter “representative” for the corresponding expression. Note that the “representative” manifestation is not necessarily the historically first or earliest one. It might be the most popular one, or the first one detected. It might be a fragment of a whole detected later. As this introduces a kind of arbitrariness, the authors found this model interesting when discussing the respective practices. Similarly, an instance of F1 Work can only be identified if a corresponding instance of F2 Expression is selected as “representative”.

How to identify Work?

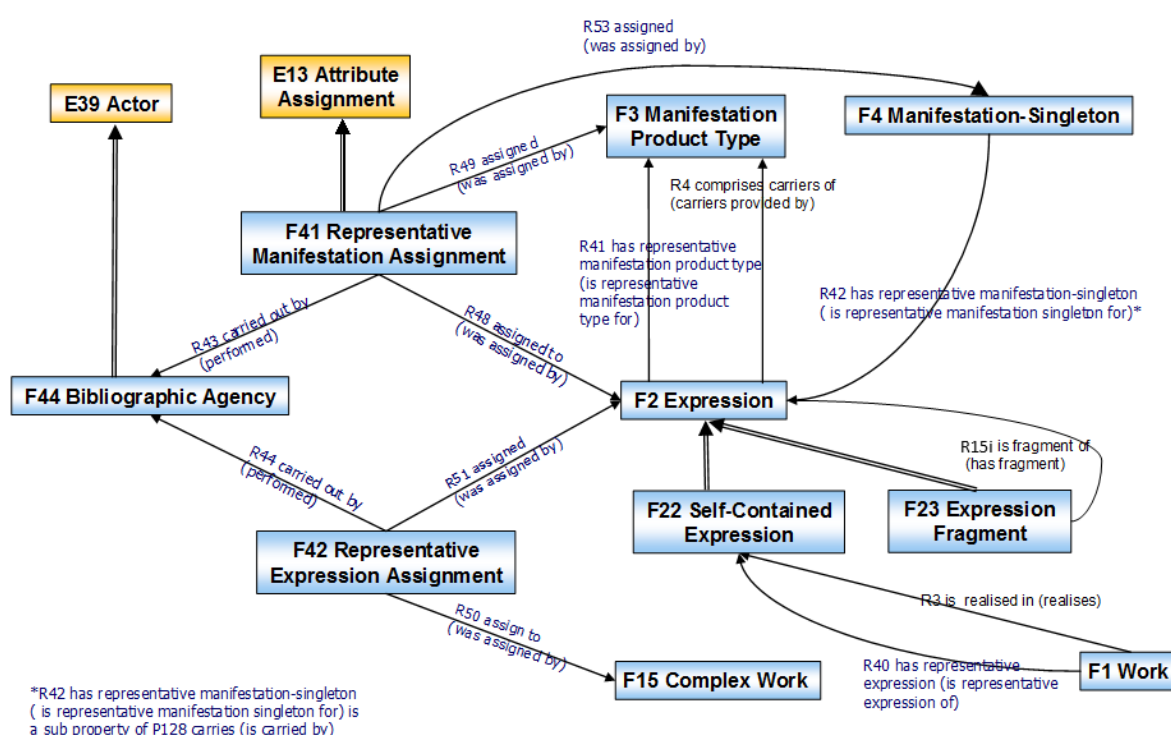


Figure 12

2.1.7. Photographs and animated images

Works realised using the photographic medium are particular, in that their essence mostly resides in the capturing of features of portions of the physical world, using automatic devices (either analogue or digital). That notion may seem close to the notions modelled in FRBR₀₀ as F21 Recording Work, F26 Recording, and F29 Recording Event. However, it is not necessary to consider all kinds of photographs as instances of F21 Recording Work; it depends on their intended functionality. Photographs that were made for the explicit purpose of documenting something are clearly instances of F21 Recording Work, while art photographs can be regarded simply as instances of F1 Work. Besides, double instantiation always makes it possible to deal with any expression of a photographic work as an instance of both F2 Expression and E38 Image, which allows one to use the CIDOC CRM property *P138 represents (has representation)* in order to record the relationship between a

photograph and the sitter who can be recognized on the photograph. Additionally, that relationship is accounted for through the following statement: F28 Expression Creation (*P2 has type: E55 Type {making a photograph}*) *P14 carried out by: E39 Actor (P14.1 in the role of: E55 Type {sitter})*.

In the same way, not all motion pictures are instances of F21 Recording Work. Animated cartoons and computer-generated films cannot be said to “record” anything. Even though fiction movies can be said to record actors’ performances, and even though animated cartoons and computer-generated films, when they include a soundtrack, can be analysed as having a component that is an instance of F26 Recording, these should all be instantiated as F1 Work, rather than F21 Recording Work. The explicit intention to capture features of a given perdurant in order to document that perdurant is what should determine whether an animated image is to be regarded as an instance of F21 Recording Work.

2.2. Naming Conventions

All the classes declared were given both a name and an identifier constructed according to the conventions used in the CIDOC CRM model. For classes, that identifier consists of the letter F followed by a number. Resulting properties were also given a name and an identifier, constructed according to the same conventions. That identifier consists of the letter R followed by a number, which in turn is followed by the letter “i” every time the property is mentioned in its “inverted” form, i.e., from target to domain. “F” and “R” are to be understood as the first two letters of “FRBR” and do not have any other meaning. They correspond respectively to letters “E” and “P” in the CIDOC CRM naming conventions, where “E” originally meant “entity” (although the CIDOC CRM “entities” are now consistently called “classes”), and “P” means “property”. Whenever CIDOC CRM classes are used in FRBR₀₀, they are named by the name they have in the original CIDOC CRM. A number of properties are identified by the letters “CLP” and a number; “CLP” stands for “Class Property” and such properties are taken from Meta-CRM; all of them have F3 Manifestation Product Type as domain, and they indicate that all the exemplars of a given publication “are supposed to” or “should” display the features of the publication they belong to. The publication itself, being an abstract notion, cannot have physical qualities such as, for instance, a given number of pages, but meta-properties are a mechanism borrowed from CIDOC CRM and Meta-CRM that makes it possible to express that a publication is characterised by the number of pages that all of its exemplars, under “ideal” conditions, “should have.”

All classes and properties that were borrowed directly from the CIDOC CRM are named as in CIDOC CRM, i.e., with an identifier beginning with either “E” if it is a class, or “P” if it is a property, and with the original appellation for the class or property in CIDOC CRM.

The choice of the domain of properties, and hence the order of their names, are established in accordance with the following priority list:

- Temporal Entity and its subclasses
- Thing and its subclasses
- Actor and its subclasses
- Other

2.3. Property Quantifiers

Quantifiers for properties are provided for the purpose of semantic clarification only, and should **not** be treated as implementation recommendations. Therefore the term “cardinality constraints” is avoided here, as it typically pertains to implementations.

The following table lists all possible property quantifiers occurring in this document by their notation, together with an explanation in plain words. In order to provide optimal clarity, two widely accepted notations are used redundantly in this document, a verbal and a numeric one. The verbal notation uses phrases such as “one to many”, and the numeric one, expressions such as “(0,n;0,1)”. While the terms “one”, “many” and “necessary” are quite intuitive, the term “dependent” denotes a situation where a range instance cannot exist without an instance of the respective property. In other words, the property is “necessary” for its range.

many to many (0,n;0,n)	Unconstrained: An individual domain instance and range instance of this property can have zero, one or more instances of this property. In other words, this property is optional and repeatable for its domain and range.
one to many (0,n;0,1)	An individual domain instance of this property can have zero, one or more instances of this property, but an individual range instance cannot be referenced by more than one instance of

this property. In other words, this property is optional for its domain and range, but repeatable for its domain only. In some contexts this situation is called a “fan-out”.

many to one (0,1:0,n)	An individual domain instance of this property can have zero or one instance of this property, but an individual range instance can be referenced by zero, one or more instances of this property. In other words, this property is optional for its domain and range, but repeatable for its range only. In some contexts this situation is called a “fan-in”.
many to many, necessary (1,n:0,n)	An individual domain instance of this property can have one or more instances of this property, but an individual range instance can have zero, one or more instances of this property. In other words, this property is necessary and repeatable for its domain, and optional and repeatable for its range.
one to many, necessary (1,n:0,1)	An individual domain instance of this property can have one or more instances of this property, but an individual range instance cannot be referenced by more than one instance of this property. In other words, this property is necessary and repeatable for its domain, and optional but not repeatable for its range. In some contexts this situation is called a “fan-out”.
many to one, necessary (1,1:0,n)	An individual domain instance of this property must have exactly one instance of this property, but an individual range instance can be referenced by zero, one or more instances of this property. In other words, this property is necessary and not repeatable for its domain, and optional and repeatable for its range. In some contexts this situation is called a “fan-in”.
one to many, dependent (0,n:1,1)	An individual domain instance of this property can have zero, one or more instances of this property, but an individual range instance must be referenced by exactly one instance of this property. In other words, this property is optional and repeatable for its domain, but necessary and not repeatable for its range. In some contexts this situation is called a “fan-out”.
one to many, necessary, dependent (1,n:1,1)	An individual domain instance of this property can have one or more instances of this property, but an individual range instance must be referenced by exactly one instance of this property. In other words, this property is necessary and repeatable for its domain, and necessary but not repeatable for its range. In some contexts this situation is called a “fan-out”.
many to one, necessary, dependent (1,1:1,n)	An individual domain instance of this property must have exactly one instance of this property, but an individual range instance can be referenced by one or more instances of this property. In other words, this property is necessary and not repeatable for its domain, and necessary and repeatable for its range. In some contexts this situation is called a “fan-in”.
many to many, necessary, dependent (1,n:1,n)	Both an individual domain instance of this property and an individual range instance can have one or more instances of this property. In other words, this property is necessary and repeatable for both its domain and its range.
one to one (1,1:1,1)	An individual domain instance and range instance of this property must have exactly one instance of this property. In other words, this property is necessary and not repeatable for its domain and for its range.
many to two (2,n:0,n)	An individual domain instance of this property must have at least two instances of this property, but an individual range instance can be referenced by zero, one or more instances of this property.

Some properties are defined as being **necessary** for their **domain** or as being **dependent** from their **range**, following the definitions in the table above. Note that if such a property is not specified for an instance of the respective domain or range, it means that the property exists, but the value on one side of the property is unknown. In the case of optional properties, the methodology proposed by the FRBR_{OO} does not distinguish between a value being unknown or the property not being applicable at all.

2.4. Presentation Conventions

All instances of E41 Appellation are presented within single quotation marks, whether they are used for

themselves or just to refer to the things they name. Any punctuation mark that follows an instance of E41 Appellation is placed outside the single quotation marks, as it does not belong to the appellation itself.

Furthermore, all references to instances of E90 Symbolic Object in the form of a content model are presented within single quotation marks, such as ‘abc’. By content model we mean the symbol sequence the symbolic object consists of.

British spelling is used throughout the original English version of this document, except for occasional quotations and examples.

Double quotation marks are used to indicate an informal name or term. E.g.: *The “lower member” consists of sandstone with minor shale.*

2.5. Class & Property Hierarchies

Although they do not provide comprehensive definitions, compact monohierarchical presentations of the class and property IsA hierarchies have been found to significantly aid in the comprehension and navigation of FRBR_{OO}, and are therefore provided below.

The class hierarchy presented below has the following format:

- Each line begins with a unique class identifier, consisting of a number preceded by the letter “F”.
- A series of em dashes (“—”) follows the unique class identifier, indicating the hierarchical position of the class in the IsA hierarchy.
- The English name of the class appears to the right of the em dashes.
- The index is ordered by hierarchical level, in a “depth first” manner, from the smaller to the larger subhierarchies.
- Classes that appear in more than one position in the class hierarchy as a result of multiple inheritance are shown in an italic typeface.

The property hierarchy presented below has the following format:

- Each line begins with a unique property identifier, consisting of a number preceded by the letter “R”.
- A series of em dashes (“—”) follows the unique property identifier, indicating the hierarchical position of the property in the IsA hierarchy.
- The English name of the property appears to the right of the em dashes, followed by its inverse name in parentheses for reading in the range to domain direction.
- The domain class for which the property is declared.
- The range class that the property references.
- The index is ordered by hierarchical level, in a “depth first” manner, from the smaller to the larger subhierarchies, and by property number between equal siblings.
- Properties that appear in more than one position in the property hierarchy as a result of multiple inheritance are shown in an italic typeface.

2.5.1. FRBR₀₀ Class Hierarchy

F1	Work			
F14	—	Individual Work		
F17	—	—	Aggregation Work	
F15	—	Complex Work		
F18	—	—	Serial Work	
F16	—	Container Work		
F17	—	—	Aggregation Work	
F19	—	—	Publication Work	
F18	—	—	—	Serial Work
F20	—	—	Performance Work	
F21	—	Recording Work		
F2	Expression			
F22	—	Self-Contained Expression		
F24	—	—	Publication Expression	
F25	—	—	Performance Plan	
F26	—	—	Recording	
F23	—	Expression Fragment		
F34	—	KOS		
F35	—	Nomen Use Statement		
F3	Manifestation Product Type			
F4	Manifestation Singleton			
F54	Utilized Information Carrier			
F5	—	Item		
F53	—	Material Copy		
F6	Concept			
F7	Object			
F8	Event			
F9	Place			
F10	Person			
F11	Corporate Body			
F12	Nomen			
F13	—	Identifier		
F50	—	—	Controlled Access Point	
F27	Work Conception			
F28	Expression Creation			
F29	—	Recording Event		
F30	—	Publication Event		
F31	Performance			
F32	Carrier Production Event			
F33	Reproduction Event			
F36	Script Conversion			
F38	Character			
F39	Family			
F40	Identifier Assignment			
F41	Representative Manifestation Assignment			
F42	Representative Expression Assignment			
F43	Identifier Rule			
F44	Bibliographic Agency			
F51	Pursuit			
F52	Name Use Activity			

2.5.2. FRBR₀₀ Class Hierarchy Aligned with (Part of) CIDOC CRM Class Hierarchy

E1	CRM Entity
E2	— Temporal Entity
E3	— — Condition State
E4	— — Period = F8 Event
E5	— — — Event
E7	— — — — Activity
F31	— — — — — Performance
F51	— — — — — Pursuit
E11	— — — — — Modification
E12	— — — — — — Production
F28	— — — — — — — Expression Creation
F29	— — — — — — — Recording Event
F30	— — — — — — — Publication Event
F32	— — — — — — — Carrier Production Event
F33	— — — — — — — Reproduction Event
E13	— — — — — Attribute Assignment
F52	— — — — — — Name Use Activity
E15	— — — — — — Identifier Assignment = F40 Identifier Assignment
F41	— — — — — — Representative Manifestation Assignment
F42	— — — — — — Representative Expression Assignment
E65	— — — — — Creation
E83	— — — — — — Type Creation
F27	— — — — — — Work Conception
F28	— — — — — — <i>Expression Creation</i>
F29	— — — — — — — <i>Recording Event</i>
F30	— — — — — — — <i>Publication Event</i>
E63	— — — — — Beginning of Existence
<i>E12</i>	— — — — — — <i>Production</i>
F28	— — — — — — — <i>Expression Creation</i>
F29	— — — — — — — <i>Recording Event</i>
F30	— — — — — — — <i>Publication Event</i>
F32	— — — — — — — <i>Carrier Production Event</i>
F33	— — — — — — — <i>Reproduction Event</i>
<i>E65</i>	— — — — — — <i>Creation</i>
<i>E83</i>	— — — — — — — <i>Type Creation</i>
F27	— — — — — — — <i>Work Conception</i>
F28	— — — — — — — <i>Expression Creation</i>
F29	— — — — — — — <i>Recording Event</i>
F30	— — — — — — — <i>Publication Event</i>
E77	— Persistent Item
E70	— — Thing
E72	— — — Legal Object
F3	— — — — Manifestation Product Type
E18	— — — — Physical Thing = F7 Object
E19	— — — — — Physical Object

E20	—	—	—	—	—	—	Biological Object
E21	—	—	—	—	—	—	Person =F10 Person
E22	—	—	—	—	—	—	Man-Made Object
E84	—	—	—	—	—	—	Information Carrier
F54	—	—	—	—	—	—	Utilized Information Carrier
F5	—	—	—	—	—	—	Item
F53	—	—	—	—	—	—	Material Copy
E26	—	—	—	—	—	—	Physical Feature
E25	—	—	—	—	—	—	Man-Made Feature
F53	—	—	—	—	—	—	<i>Material Copy</i>
E24	—	—	—	—	—	—	Physical Man-Made Thing
F4	—	—	—	—	—	—	Manifestation Singleton
E22	—	—	—	—	—	—	<i>Man-Made Object</i>
E84	—	—	—	—	—	—	<i>Information Carrier</i>
F54	—	—	—	—	—	—	<i>Utilized Information Carrier</i>
F5	—	—	—	—	—	—	<i>Item</i>
F53	—	—	—	—	—	—	<i>Material Copy</i>
E25	—	—	—	—	—	—	<i>Man-Made Feature</i>
F53	—	—	—	—	—	—	<i>Material Copy</i>
E90	—	—	—	—	—	—	Symbolic Object
E73	—	—	—	—	—	—	Information Object
E31	—	—	—	—	—	—	Document
E32	—	—	—	—	—	—	Authority Document
F34	—	—	—	—	—	—	KOS
F2	—	—	—	—	—	—	Expression
F22	—	—	—	—	—	—	Self-Contained Expression
F24	—	—	—	—	—	—	Publication Expression
F25	—	—	—	—	—	—	Performance Plan
F26	—	—	—	—	—	—	Recording
F23	—	—	—	—	—	—	Expression Fragment
F34	—	—	—	—	—	—	KOS
F35	—	—	—	—	—	—	Nomen Use Statement
F43	—	—	—	—	—	—	Identifier Rule
E29	—	—	—	—	—	—	Design or Procedure
F25	—	—	—	—	—	—	<i>Performance Plan</i>
F36	—	—	—	—	—	—	Script Conversion
F43	—	—	—	—	—	—	<i>Identifier Rule</i>
E41	—	—	—	—	—	—	Appellation
F12	—	—	—	—	—	—	Nomen
F13	—	—	—	—	—	—	<i>Identifier = E42 Identifier</i>
F50	—	—	—	—	—	—	Controlled Access Point
E71	—	—	—	—	—	—	Man-Made Thing
E24	—	—	—	—	—	—	<i>Physical Man-Made Thing</i>
F4	—	—	—	—	—	—	<i>Manifestation Singleton</i>
E22	—	—	—	—	—	—	<i>Man-Made Object</i>
E84	—	—	—	—	—	—	<i>Information Carrier</i>
F54	—	—	—	—	—	—	<i>Utilized Information Carrier</i>
F5	—	—	—	—	—	—	<i>Item</i>
F53	—	—	—	—	—	—	<i>Material Copy</i>

E25	—	—	—	—	—	—	Man-Made Feature
F53	—	—	—	—	—	—	Material Copy
E28	—	—	—	—	—	—	Conceptual Object = F6 Concept
E55	—	—	—	—	—	—	Type
F3	—	—	—	—	—	—	Manifestation Product Type
E89	—	—	—	—	—	—	Propositional Object
F1	—	—	—	—	—	—	Work
F14	—	—	—	—	—	—	Individual Work
F17	—	—	—	—	—	—	Aggregation Work
F15	—	—	—	—	—	—	Complex Work
F18	—	—	—	—	—	—	Serial Work
F16	—	—	—	—	—	—	Container Work
F19	—	—	—	—	—	—	Publication Work
F18	—	—	—	—	—	—	Serial Work
F20	—	—	—	—	—	—	Performance Work
F17	—	—	—	—	—	—	Aggregation Work
F21	—	—	—	—	—	—	Recording Work
E73	—	—	—	—	—	—	Information Object
E31	—	—	—	—	—	—	Document
E32	—	—	—	—	—	—	Authority Document
F34	—	—	—	—	—	—	KOS
F2	—	—	—	—	—	—	Expression
F22	—	—	—	—	—	—	Self-Contained Expression
F24	—	—	—	—	—	—	Publication Expression
F25	—	—	—	—	—	—	Performance Plan
F26	—	—	—	—	—	—	Recording
F23	—	—	—	—	—	—	Expression Fragment
F34	—	—	—	—	—	—	KOS
F35	—	—	—	—	—	—	Nomen Use Statement
F43	—	—	—	—	—	—	Identifier Rule
E29	—	—	—	—	—	—	Design or Procedure
F25	—	—	—	—	—	—	Performance Plan
F36	—	—	—	—	—	—	Script Conversion
F43	—	—	—	—	—	—	Identifier Rule
F13	—	—	—	—	—	—	Identifier = E42 Identifier
E90	—	—	—	—	—	—	Symbolic Object
E73	—	—	—	—	—	—	Information Object
F2	—	—	—	—	—	—	Expression
F22	—	—	—	—	—	—	Self-Contained Expression
F24	—	—	—	—	—	—	Publication Expression
F25	—	—	—	—	—	—	Performance Plan
F26	—	—	—	—	—	—	Recording
F23	—	—	—	—	—	—	Expression Fragment
E29	—	—	—	—	—	—	Design or Procedure
F25	—	—	—	—	—	—	Performance Plan
E41	—	—	—	—	—	—	Appellation = F12 Nomen
F12	—	—	—	—	—	—	Nomen
F13	—	—	—	—	—	—	Identifier = E42 Identifier
F50	—	—	—	—	—	—	Controlled Access Point

F38	—	—	—	—	—	Character
E39	—	—				Actor
E74	—	—	—			Group
F11	—	—	—	—		Corporate Body
F39	—	—	—	—		Family
E40	—	—	—	—		Legal Body
F44	—	—	—	—	—	Bibliographic Agency
<i>E21</i>	—	—	—			<i>Person = F10 Person</i>
E53	—					Place = F9 Place

2.5.3. FRBR₀₀ Property Hierarchy

Property id	Property Name	Entity – Domain	Entity – Range
R1	is logical successor of (has successor)	F1 Work	F1 Work
R2	is derivative of (has derivative)	F1 Work	F1 Work
R3	is realised in (realises)	F1 Work	F22 Self-Contained Expression
—	R9 is realised in (realises)	F14 Individual Work	F22 Self-Contained Expression
—	R12 is realised in (realises)	F20 Performance Work	F25 Performance Plan
—	R13 is realised in (realises)	F21 Recording Work	F26 Recording
—	R40 has representative expression (is representative expression for)	F1 Work	F22 Self-Contained Expression
R4	carriers provided by (comprises carriers of)	F2 Expression	F3 Manifestation Product Type
—	R41 has representative manifestation product type (is representative manifestation product type for)	F2 Expression	F3 Manifestation Product Type
R5	has component (is component of)	F2 Expression	F22 Self-Contained Expression
R6	carries (is carried by)	F5 Item	F24 Publication Expression
R7	is example of (has example)	F5 Item	F3 Manifestation Product Type
R8	consists of (forms part of)	F13 Identifier	E90 Symbolic Object
R10	has member (is member of)	F15 Complex Work	F1 Work
R11	has issuing rule (is issuing rule of)	F18 Serial Work	E29 Design or Procedure
R15	has fragment (is fragment of)	F2 Expression	F23 Expression Fragment
R16	initiated (was initiated by)	F27 Work Conception	F1 Work
R17	created (was created by)	F28 Expression Creation	F2 Expression
—	R21 created (was created through)	F29 Recording Event	F26 Recording
—	R24 created (was created through)	F30 Publication Event	F24 Publication Expression
R18	created (was created by)	F28 Expression Creation	F4 Manifestation Singleton
R19	created a realisation of (was realised through)	F28 Expression Creation	F1 Work
—	R22 created a realisation of (was realised through)	F29 Recording Event	F21 Recording Work
	R23 created a realisation of (was realised through)	F30 Publication Event	F19 Publication Work
R20	recorded (was recorded through)	F29 Recording Event	E5 Event

R25	performed (was performed in)	F31 Performance	F25 Performance Plan
R26	produced things of type (was produced by)	F32 Carrier Production Event	F3 Manifestation Product Type
R27	used as source material (was used by)	F32 Carrier Production Event	F24 Publication Expression
R28	produced (was produced by)	F32 Carrier Production Event	F5 Item
R29	reproduced (was reproduced by)	F33 Reproduction Event	E84 Information Carrier
R30	produced (was produced by)	F33 Reproduction Event	E84 Information Carrier
R31	is reproduction of (has reproduction)	E84 Information Carrier	E84 Information Carrier
R32	is warranted by (warrants)	F35 Nomen Use Statement	F52 Name Use Activity
R33	has content	F12 Nomen	E62 String
R34	has validity period (is validity period of)	F34 KOS	E52 Time-Span
R35	is specified by (specifies)	F35 Nomen Use Statement	F34 KOS
R36	uses script conversion (is script conversion used in)	F35 Nomen Use Statement	F36 Script Conversion
R37	states as nomen (is stated as nomen in)	F35 Nomen Use Statement	F12 Nomen
R38	refers to thema (is thema of)	F35 Nomen Use Statement	E1 CRM Entity
R39	is intended for (is target audience in)	F35 Nomen Use Statement	E74 Group
R42	is representative manifestation singleton for (has representative manifestation singleton)	F4 Manifestation Singleton	F2 Expression
R43	carried out by (performed)	F41 Representative Manifestation Assignment	F44 Bibliographic Agency
R44	carried out by (performed)	F42 Representative Expression Assignment	F44 Bibliographic Agency
R45	assigned to (was assigned by)	F40 Identifier Assignment	E1 CRM Entity
R46	assigned (was assigned by)	F40 Identifier Assignment	F3 Identifier
R48	assigned to (was assigned by)	F41 Representative Manifestation Assignment	F2 Expression
R49	assigned (was assigned by)	F41 Representative Manifestation Assignment	F3 Manifestation Product Type
R50	assigned to (was assigned by)	F42 Representative Expression Assignment	F15 Complex Work
R51	assigned (was assigned by)	F42 Representative Expression Assignment	F2 Expression
R52	used rule (was the rule used in)	F40 Identifier Assignment	F43 Identifier Rule

R53	assigned (was assigned by)	F41 Representative Manifestation Assignment	F4 Manifestation Singleton
R54	has nomen language (is language of nomen in)	F35 Nomen Use Statement	E56 Language
R55	has nomen form (is nomen form in)	F35 Nomen Use Statement	E55 Type
R56	has related use (is related use for)	F35 Nomen Use Statement	F35 Nomen Use Statement
R57	is based on (is basis for)	F38 Character	E39 Actor
R58	has fictional member (is fictional member of)	F38 Character	F38 Character
R59	had typical subject (was typical subject of)	F51 Pursuit	E1 CRM Entity
R60	used to use language (was language used by)	F51 Pursuit	E56 Language
R61	occurred in kind of context (was kind of context for)	F52 Name Use Activity	E55 Type
R62	was used for membership in (was context for)	F52 Name Use Activity	E74 Group
R63	named (was named by)	F52 Name Use Activity	E1 CRM Entity
R64	used name (was name used by)	F52 Name Use Activity	E41 Appellation
R65	recorded aspects of (had aspects recorded through)	F29 Recording Event	E18 Physical Thing
R66	included performed version of (had a performed version through)	F31 Performance	E89 Propositional Object
CLP2	should have type (should be type of)	F3 Manifestation Product Type	E55 Type
CLP43	should have dimension (should be dimension of)	F3 Manifestation Product Type	E54 Dimension
CLP45	should consist of (should be incorporated in)	F3 Manifestation Product Type	E57 Material
CLP46	should be composed of (may form part of)	F3 Manifestation Product Type	F3 Manifestation Product Type
CLP57	should have number of parts	F3 Manifestation Product Type	E60 Number
CLP104	subject to (applies to)	F3 Manifestation Product Type	E30 Right
CLP105	right held by (right on)	F3 Manifestation Product Type	E39 Actor
CLR6	should carry (should be carried by)	F3 Manifestation Product Type	F24 Publication Expression

2.5.4. FRBR_{oo} Property Hierarchy Aligned with (Part of) CIDOC CRM Property Hierarchy

Property id	Property Name	Entity – Domain	Entity – Range
P2	has type (is type of)	E1 CRM Entity	E55 Type
R7	— is example of (has example)	F5 Item	F3 Manifestation Product Type
P3	has note	E1 CRM Entity	E62 String
R33	— R33 has content	F12 Nomen	E62 String
P14	carried out by (performed)	E7 Activity	E39 Actor
R43	— carried out by (performed)	F41 Representative Manifestation Assignment	F44 Bibliographic Agency
R44	— carried out by (performed)	F42 Representative Expression Assignment	F44 Bibliographic Agency
P12	occurred in the presence of (was present at)	E5 Event	E77 Persistent Item
P16	— used specific object (was used for)	E7 Activity	E70 Thing
R19	— — created a realisation of (was realised through)	F28 Expression Creation	F1 Work
R22	— — — created a realisation of (was realised through)	F29 Recording Event	F21 Recording Work
R23	— — — created a realisation of (was realised through)	F30 Publication Event	F19 Publication Work
R27	— — used as source material (was used by)	F32 Carrier Production Event	F24 Publication Expression
R29	— — reproduced (was reproduced by)	F33 Reproduction Event	E84 Information Carrier
P33	— — used specific technique (was used by)	E7 Activity	E29 Design or Procedure
R25	— — — performed (was performed in)	F31 Performance	F25 Performance Plan
R52	— — — used rule (was the rule used in)	F40 Identifier Assignment	F43 Identifier Rule
P142	— — used constituent (was used in)	E15 Identifier Assignment	E90 Symbolic Object
R66	— — included performed version of (had a performed version through)	F31 Performance	E89 Propositional Object
P25	— moved (moved by)	E9 Move	E19 Physical Object
P31	— has modified (was modified by)	E11 Modification	E24 Physical Man-Made Thing
P108	— — has produced (was produced by)	E12 Production	E24 Physical Man-Made Thing
R18	— — — created (was created by)	F28 Expression Creation	F4 manifestation Singleton
R28	— — — produced (was produced by)	F32 Carrier Production Event	F5 Item
R30	— — — produced (was produced by)	F33 Reproduction Event	E84 Information Carrier
P110	— — augmented (was augmented by)	E79 Part Addition	E24 Physical Man-Made Thing
P112	— — diminished (was diminished by)	E80 Part Removal	E24 Physical Man-Made Thing
P92	— brought into existence (was brought into existence by)	E63 Beginning of Existence	E77 Persistent Item
P94	— — has created (was created by)	E65 Creation	E28 Conceptual Object
R16	— — — initiated (was initiated by)	F27 Work Conception	F1 Work
R17	— — — created (was created by)	F28 Expression Creation	F2 Expression
R21	— — — — created (was created by)	F29 Recording Event	F26 Recording
R24	— — — — created (was created through)	F30 Publication Event	F24 Publication Expression
P135	— — — created type (was created by)	E83 Type Creation	E55 Type
P95	— — has formed (was formed by)	E66 Formation	E74 Group
P98	— — brought into life (was born)	E67 Birth	E21 Person
P108	— — has produced (was produced by)	E12 Production	E24 Physical Man-Made Thing
R18	— — — created (was created by)	F28 Expression Creation	F4 manifestation Singleton
R28	— — — produced (was produced by)	F32 Carrier Production Event	F5 Item
R30	— — — produced (was produced by)	F33 Reproduction Event	E84 Information Carrier
P123	— — resulted in (resulted from)	E81 Transformation	E77 Persistent Item

Property id	Property Name	Entity – Domain	Entity – Range
P15	was influenced by (influenced)	E7 Activity	E1 CRM Entity
P16	— <i>used specific object (was used for)</i>	E7 Activity	E70 Thing
R19	— — created a realisation of (was realised through)	F28 Expression Creation	F1 Work
R22	— — — created a realisation of (was realised through)	F29 Recording Event	F21 Recording Work
R23	— — — created a realisation of (was realised through)	F30 Publication Event	F19 Publication work
R27	— — used as source material (was used by)	F32 Carrier Production Event	F24 Publication Expression
R29	— — reproduced (was reproduced by)	F33 Reproduction Event	E84 Information Carrier
P33	— — <i>used specific technique (was used by)</i>	E11 Modification	E29 Design or Procedure
R25	— — — <i>performed (was performed in)</i>	F31 Performance	F25 Performance Plan
R52	— — — <i>used rule (was the rule used in)</i>	F40 Identifier Assignment	F43 Identifier Rule
P142	— — <i>used constituent (was used in)</i>	E15 Identifier Assignment	E90 Symbolic Object
R66	— — <i>included performed version of (had a performed version through)</i>	F31 Performance	E89 Propositional Object
P17	— was motivated by (motivated)	E7 Activity	E1 CRM Entity
P134	— continued (was continued by)	E7 Activity	E7 Activity
P136	— was based on (supported type creation)	E83 Type Creation	E1 CRM Entity
P67	refers to (is referred to by)	E89 Propositional Object	E1 CRM Entity
R32	— is warranted by (warrants)	F35 Nomen Use Statement	F52 Name Use Activity
R37	— states as nomen (is stated as nomen in)	F35 Nomen Use Statement	F12 Nomen
R39	— is intended for (is target audience in)	F35 Nomen Use Statement	E74 Group
R54	— has nomen language (is language of nomen in)	F35 Nomen Use Statement	E56 Language
R55	— has nomen form (is nomen form in)	F35 Nomen Use Statement	E55 Type
P69	has association with (is associated with)	E29 Design or Procedure	E29 Design or Procedure
R56	— has related use (is related use for)	F35 Nomen Use Statement	F35 Nomen Use Statement
P106	is composed of (forms part of)	E90 Symbolic Object	E90 Symbolic Object
R8	— consists of (forms part of)	F13 Identifier	E90 Symbolic Object
R15	— has fragment (is fragment of)	F2 Expression	F23 Expression Fragment
P128	carries (is carried by)	E24 Physical Man-Made Thing	E73 Information Object
R6	— carries (is carried by)	F5 Item	F24 Publication Expression
R42	— is representative manifestation singleton for (has representative manifestation singleton)	F4 Manifestation Singleton	F2 Expression
P65	— shows visual item (is shown by)	E24 Physical Man-Made Thing	E36 Visual Item
P130	shows features of (features are also found on)	E70 Thing	E70 Thing
R1	— is logical successor of (has successor)	F1 Work	F1 Work
R2	— is derivative of (has derivative)	F1 Work	F1 Work
R3	— is realised in (realises)	F1 Work	F22 Self-Contained Expression
R9	— — is realised in (realises)	F14 Individual Work	F22 Self-Contained Expression
R12	— — is realised in (realises)	F20 Performance Work	F25 Performance Plan
R13	— — is realised in (realises)	F21 Recording Work	F26 Recording
R40	— — has representative expression (is representative expression for)	F1 Work	F22 Self-Contained Expression
R31	— is reproduction of (has reproduction)	E84 Information Carrier	E84 Information Carrier
P73	— has translation (is translation of)	E33 Linguistic Object	E33 Linguistic Object
P140	assigned attribute to (was attributed by)	E13 Attribute Assignment	E1 CRM Entity
R45	— assigned to (was assigned by)	F40 Identifier Assignment	E1 CRM Entity
R48	— assigned to (was assigned by)	F41 Representative	F2 Expression

Property id	Property Name	Entity – Domain	Entity – Range
		Manifestation Assignment	
R50	— assigned to (was assigned by)	F42 Representative Expression Assignment	F15 Complex Work
R63	— named (was named by)	F52 Name Use Activity	E1 CRM Activity
P141	assigned (was assigned by)	E13 Attribute Assignment	E1 CRM Entity
P37	— assigned (was assigned by)	E15 Identifier Assignment	E42 Identifier
R46	— — assigned (was assigned by)	F40 Identifier Assignment	F13 Identifier
R49	— assigned (was assigned by)	F41 Representative Manifestation Assignment	F3 Manifestation Product Type
R51	— assigned (was assigned by)	F42 Representative Expression Assignment	F2 Expression
R53	— assigned (was assigned by)	F41 Representative Manifestation Assignment	F4 Manifestation Singleton
R64	— used name (was name used by)	F52 Name Use Activity	E41 Appellation
P148	has component (is component of)	E89 Propositional Object	E89 Propositional Object
R5	— has component (is component of)	F2 Expression	F22 Self-Contained Expression
R10	— has member (is member of)	F15 Complex Work	F1 Work
R35	— is specified by (specifies)	F35 Nomen Use Statement	F34 KOS
R4	carriers provided by (comprises carriers of)	F2 Expression	F3 Manifestation Product Type
R41	— has representative manifestation product type (is representative manifestation product type for)	F2 Expression	F3 Manifestation Product Type
R11	has issuing rule (is issuing rule of)	F18 Serial Work	E29 Design or Procedure
R20	recorded (was recorded through)	F29 Recording Event	E2 Temporal Entity
R26	produced things of type (was produced by)	F32 Carrier Production Event	F3 Manifestation Product Type
R34	has validity period (is validity period of)	F34 KOS	E52 Time-Span
R36	uses script conversion (is script conversion used ni)	F35 Nomen Use Statement	F36 Script Conversion
R38	refers to thema (is thema of)	F35 Nomen Use Statement	E1 CRM Entity
R57	is based on (is basis for)	F38 Character	E39 Actor
R58	has fictional member (is fictional member of)	F38 Character	F38 Character
R59	had typical subject (was typical subject of)	F51 Pursuit	E1 CRM Entity
R60	used to use language (was language used by)	F51 Pursuit	E56 Language
R61	occurred in kind of context (was kind of context for)	F52 Name Use Activity	E55 Type
R62	was used for membership in (was context for)	F52 Name Use Activity	E74 Group
CLP2	should have type (should be type of)	F3 Manifestation Product Type	E55 Type
R65	recorded aspects of (had aspects recorded through)	F29 Recording Event	E18 Physical Thing
CLP43	should have dimension (should be dimension of)	F3 Manifestation Product Type	E54 Dimension
CLP45	should consist of (should be incorporated in)	F3 Manifestation Product Type	E57 Material
CLP46	should be composed of (may form part of)	F3 Manifestation Product Type	F3 Manifestation Product Type
CLP57	should have number of parts	F3 Manifestation Product Type	E60 Number
CLP104	subject to (applies to)	F3 Manifestation Product Type	E30 Right
CLP105	right held by (right on)	F3 Manifestation Product Type	E39 Actor
CLR6	should carry (should be carried by)	F3 Manifestation Product Type	F24 Publication Expression

2.6. *FRBR_{OO}* Class Declaration

The classes of FRBR_{OO} are comprehensively declared in this section using the following format:

- Class names are presented as headings in bold face, preceded by the class's unique identifier;
- The line "Equal to:" declares the CIDOC CRM class that covers the same concept as the FRBR_{OO} class;
- The line "Subclass of:" declares the superclass of the class from which it inherits properties;
- The line "Superclass of:" is a cross-reference to the subclasses of this class;
- The line "Scope note:" contains the textual definition of the concept the class represents;
- The line "Examples:" contains a bulleted list of examples of instances of this class. If the example is also instance of a subclass of this class, the unique identifier of the subclass is added in parenthesis. If the example instantiates two classes, the unique identifiers of both classes is added in parenthesis. Non-fictitious examples may be followed by an explanation in brackets.
- The line "Properties:" declares the list of the class's properties;
- Each property is represented by its unique identifier, its forward and reverse names, and the range class that it links to, separated by colons;
- Inherited properties are not represented;
- Properties of properties are provided indented and in parentheses beneath their respective domain property.

F1 Work

Subclass of: [E89](#) Propositional Object

Superclass of: [F14](#) Individual Work
[F15](#) Complex Work
[F16](#) Container Work
[F21](#) Recording Work

Scope note: This class comprises distinct concepts or combinations of concepts identified in artistic and intellectual expressions, such as poems, stories or musical compositions. Such concepts may appear in the course of the coherent evolution of an original idea into one or more expressions that are dominated by the original idea. A Work may be elaborated by one or more Actors simultaneously or over time. The substance of Work is ideas. A Work may have members that are works in their own right.

A Work can be either *individual* or *complex*. If it is individual its concept is completely realised in a single F22 Self-Contained Expression. If it is complex its concept is embedded in an F15 Complex Work. An F15 Complex Work consists of alternative members that are either F15 Complex Works themselves or F14 Individual Works.

A Work is the product of an intellectual process of one or more persons, yet only indirect evidence about it is at our hands. This can be contextual information such as the existence of an order for a work, reflections of the creators themselves that are documented somewhere, and finally the expressions of the work created. As ideas normally take shape during discussion, elaboration and implementation, it is not reasonable to assume that a work starts with a complete concept. In some cases, it can be very difficult or impossible to define the whole of the concept of a work at a particular time. The objective evidence for such a notion can only be based on a stage of expressions at a given time. In this sense, the sets of ideas that constitute particular self-contained expressions may be regarded as a kind of “snap-shot” of a work.

A Work may include the concept of aggregating expressions of other works into a new expression. For instance, an anthology of poems is regarded as a work in its own right that makes use of expressions of the individual poems that have been selected and ordered as part of an intellectual process. This does not make the contents of the aggregated expressions part of this work, but only parts of the resulting expression.

Examples: Abstract content of Giovanni Battista Piranesi’s ‘Carcere XVI: the pier with chains: 1st state’ (F14)

‘La Porte de l’Enfer’ by Auguste Rodin conceived between 1880 and 1917 (F15)

‘Hamlet’ by William Shakespeare (F15)

Properties: [R1](#) is logical successor of (has successor): [F1](#) Work
[R2](#) is derivative of (has derivative): [F1](#) Work
(R2.1 has type: [E55](#) Type)
[R3](#) is realised in (realises): [F22](#) Self-Contained Expression
[R40](#) has representative expression (is representative expression for): [F22](#) Self-Contained Expression

F2 Expression

Subclass of: [E73](#) Information Object

Superclass of: [F22](#) Self-Contained Expression
[F23](#) Expression Fragment
[F34](#) KOS
[F35](#) Nomen Use Statement

[F43](#) Identifier Rule

Scope note: This class comprises the intellectual or artistic realisations of *works* in the form of identifiable immaterial objects, such as texts, poems, jokes, musical or choreographic notations, movement pattern, sound pattern, images, multimedia objects, or any combination of such forms that have objectively recognisable structures. The substance of F2 Expression is signs.

Expressions cannot exist without a physical carrier, but do not depend on a specific physical carrier and can exist on one or more carriers simultaneously. Carriers may include human memory.

Inasmuch as the form of F2 Expression is an inherent characteristic of the F2 Expression, any change in form (e.g., from alpha-numeric notation to spoken word, a poem created in capitals and rendered in lower case) is a new F2 Expression. Similarly, changes in the intellectual conventions or instruments that are employed to express a *work* (e.g., translation from one language to another) result in the creation of a new F2 Expression. Thus, if a text is revised or modified, the resulting F2 Expression is considered to be a new F2 Expression. Minor changes, such as corrections of spelling and punctuation, etc., are normally considered variations within the same F2 Expression. On a practical level, the degree to which distinctions are made between variant *expressions* of a *work* will depend to some extent on the nature of the F1 Work itself, and on the anticipated needs of users.

The genre of the work may provide an indication of which features are essential to the expression. In some cases, aspects of physical form, such as typeface and page layout, are not integral to the intellectual or artistic realisation of the *work* as such, and therefore are not distinctive criteria for the respective expressions. For another work, features such as layout may be essential. For instance, the author or a graphic designer may wrap a poem around an image.

An expression of a work may include expressions of other works within it. For instance, an anthology of poems is regarded as a work in its own right that makes use of expressions of the individual poems that have been selected and ordered as part of an intellectual process. This does not make the contents of the aggregated expressions part of this work, but only parts of the resulting expression.

If an instance of F2 Expression is of a specific form, such as text, image, etc., it may be simultaneously instantiated in the specific classes representing these forms in CIDOC CRM. Thereby one can make use of the more specific properties of these classes, such as language (which is applicable to instances of E33 Linguistic Object only).

Examples: The Italian text of Dante's 'Divina Commedia' as found in the authoritative critical edition '*La Commedia secondo l'antica vulgata a cura di Giorgio Petrocchi*', Milano: Mondadori, 1966-67 (= Le Opere di Dante Alighieri, Edizione Nazionale a cura della Società Dantesca Italiana, VII, 1-4) (F22 and E33)

The Italian text of Dante's 'Inferno' as found in the same edition (F22 and E33)

'Nel mezzo del cammin di nostra vita

mi ritrovai per una selva oscura

ché la diritta via era smarrita' [the Italian text of the first stanza of Dante's 'Inferno' and 'Divina Commedia'] (F23 and E33)

The signs which make up Christian Morgenstern's 'Fisches Nachtgesang' [a poem consisting simply of '—' and '¨' signs, arranged in a determined combination] (F22)

Properties: [R4](#) carriers provided by (comprises carriers of): [F3](#) Manifestation Product Type

[R5](#) has component (is component of): [F22](#) Self-Contained Expression

[R15](#) has fragment (is fragment of): [F23](#) Expression Fragment

[R41](#) has representative manifestation product type (is representative manifestation product type for): [F3](#) Manifestation Product Type

F3 Manifestation Product Type

Subclass of: [E55](#) Type

[E72](#) Legal Object

Scope note: This class comprises the definitions of publication products.

An instance of F3 Manifestation Product Type is the “species”, and all copies of a given object are “specimens” of it. An instance of F3 Manifestation Product Type defines all of the features or traits that instances of F5 Item normally display in order that they may be recognised as copies of a particular publication. However, due to production problems or subsequent events, one or more instances of F5 Item may not exhibit all these features or traits; yet such instances still retain their relationship to the same instance of F3 Manifestation Product Type.

The features that characterise a given instance of F3 Manifestation Product Type include: one instance of F24 Publication Expression, containing one or more than one instance of F2 Expression, reflecting the authors’ content of the manifestation and all additional input by the publisher; and the appropriate types of physical features for that form of the object. For example, hardcover and paperback are two distinct publications (i.e. two distinct instances of F3 Manifestation Product Type) even though authorial and editorial content are otherwise identical in both publications. The activity of cataloguing aims at the most accurate listing of features or traits of an instance of F3 Manifestation Product Type that are sufficient to distinguish it from another instance of F3 Manifestation Product Type.

Examples: The publication product containing the text entitled ‘Harmonie universelle’ (authored by the person named ‘Marin Mersenne’), issued in 1636 in Paris by the publisher named ‘Sébastien Cramoisy’

The publication product containing a modern reprint of Marin Mersenne’s ‘Harmonie universelle’, issued in 1986 in Paris by the publisher named ‘Les éditions du CNRS’, and identified by ISBN ‘2-222-00835-2’

The publication product containing the third edition of the combination of texts and graphics titled ‘Codex Manesse: die Miniaturen der großen Heidelberger Liederhandschrift, herausgegeben und erläutert von Ingo F. Walther unter Mitarbeit von Gisela Siebert’, issued by the publisher named ‘Insel-Verlag’ in 1988

The publication product containing the cartographic resource titled ‘Ordnance Survey Explorer Map 213, Aberystwyth & Cwm Rheidol’, issued in May 2005 by the publisher named ‘Ordnance Survey’ and identified by ISBN ‘0-319-23640-4’ (folded), 1:25,000 scale

The publication product containing the recordings of musical works performed by the person named ‘Florence Foster Jenkins’ gathered under the title ‘The Glory (????) of the human voice’, identified by label and label number ‘RCA Victor Gold Seal GD61175’ (*Note: the four question marks within parentheses belong to the title itself*)

Properties: [CLP2](#) should have type (should be type of): [E55](#) Type
[CLP43](#) should have dimension (should be dimension of): [E54](#) Dimension
[CLP45](#) should consist of (should be incorporated in): [E57](#) Material
[CLP46](#) should be composed of (may form part of): [F3](#) Manifestation Product Type
[CLP57](#) should have number of parts: [E60](#) Number
[CLP104](#) subject to (applies to): [E30](#) Right
[CLP105](#) right held by (right on): [E39](#) Actor
[CLR6](#) should carry (should be carried by): [F24](#) Publication Expression

F4 Manifestation Singleton

Subclass of: [E24](#) Physical Man-Made Thing

Scope note: This class comprises physical objects that each carry an instance of F2 Expression, and that were produced as unique objects, with no siblings intended in the course of their production. It should be noted that if all but one copy of a given publication are destroyed, then that copy does not become an instance of F4 Manifestation Singleton, because it was produced together with sibling copies, even though it now happens to be unique. Examples of instances of F4 Manifestation Singleton include manuscripts, preparatory sketches and the final clean draft sent by an author or a composer to a publisher.

Examples:	The manuscript known as ‘The Book of Kells’ The manuscript score of Charles Racquet’s ‘Organ fantasy’, included in Marin Mersenne’s personal copy of his own ‘Harmonie universelle’ [Marin Mersenne planned a second edition of his ‘Harmonie universelle’ after it had been first published in 1636, and he asked the composer Charles Racquet to compose his organ fantasy especially for that planned second edition; but Mersenne died before he could finish and publish the second edition and Racquet’s score remained until the 20th century as a manuscript addition to Mersenne’s copy, held in Paris by the Library of the Conservatoire national des arts et métiers] Marin Mersenne’s personal copy, held in Paris by the Library of the Conservatoire national des arts et métiers, of his own ‘Harmonie universelle’, containing all of his manuscript additions for a planned second edition that never took place before his death, but that served as a basis for the modern reprint published in 1986
Properties:	R42 is representative manifestation singleton for (has representative manifestation singleton): F2 Expression

F5 Item

Subclass of:	F54 Utilized Information Carrier
Scope note:	This class comprises physical objects (printed books, scores, CDs, DVDs, CD-ROMS, etc.) that carry a F24 Publication Expression and were produced by an industrial process involving an F3 Manifestation Product Type.
Examples:	Marin Mersenne’s personal copy of his own ‘Harmonie universelle’ without any manuscript addition and without Charles Racquet’s manuscript score, as a mere witness of the 1st edition of ‘Harmonie universelle’, Paris, 1636 [the same physical object can be regarded at the same time as an instance of F5 Item inasmuch as it is a witness of a publication, and as an instance of F4 Manifestation Singleton inasmuch as it contains manuscript annotations and additions and as it served as the basis for a subsequent production process] Any other copy of the original edition of Marin Mersenne’s ‘Harmonie universelle’, Paris, 1636 Any copy of the modern reprint publication of Marin Mersenne’s ‘Harmonie universelle’, Paris, 1986, ISBN ‘2-222-00835-2’
Properties:	R7 is example of (has example): F3 Manifestation Product Type

F6 Concept

Equal to:	E28 Conceptual Object
Scope note:	An abstract notion or idea. [FRBR_{ER}] Includes fields of knowledge, disciplines, schools of thought (philosophies, religions, political ideologies, etc.), etc. Includes theories, processes, techniques, practices, etc. <i>[Definition from the FRAD model, unchanged]</i> This class comprises non-material products of our minds and other human produced data that have become objects of a discourse about their identity, circumstances of creation or historical implication. The production of such information may have been supported by the use of technical devices such as cameras or computers. Characteristically, instances of this class are created, invented or thought by someone, and then may be documented or communicated between persons. Instances of E28 Conceptual Object have the ability to exist on more than one particular carrier at the same time, such as paper, electronic signals, marks, audio media, paintings, photos, human memories, etc. They cannot be destroyed. They exist as long as they can be found on at least one carrier or in at least one human memory. Their existence ends when the last carrier and the last memory are lost. <i>[Scope note for E28 Conceptual Object in CIDOC CRM version 5.0.1]</i>
Examples:	Natural history of whales

Cultural history of Wales

The appreciation of Victor Hugo's works in Germany between 1870 and 1914

F7 Object

Equal to: [E18](#) Physical Thing

Scope Note: This class comprises all persistent physical items with a relatively stable form, man-made or natural.

[This is the beginning of scope note for E18 Physical Object in CIDOC CRM version 5.0.1]

Examples: Buckingham Palace

The *Lusitania*

Apollo 11

The Eiffel Tower

F8 Event

Equal to: [E4](#) Period

Scope note: This class comprises sets of coherent phenomena or cultural manifestations bounded in time and space.

It is the social or physical coherence of these phenomena that identify an E4 Period and not the associated spatio-temporal bounds. These bounds are a mere approximation of the actual process of growth, spread and retreat. Consequently, different periods can overlap and coexist in time and space, such as when a nomadic culture exists in the same area as a sedentary culture.

Typically this class is used to describe prehistoric or historic periods such as the 'Neolithic Period', the 'Ming Dynasty' or the 'McCarthy Era'. There are however no assumptions about the scale of the associated phenomena. In particular all events are seen as synthetic processes consisting of coherent phenomena. Therefore E4 Period is a superclass of E5 Event. For example, a modern clinical E67 Birth can be seen as both an atomic E5 Event and as an E4 Period that consists of multiple activities performed by multiple instances of E39 Actor.

There are two different conceptualisations of "artistic style", defined either by physical features or by historical context. For example, Impressionism can be viewed as a period lasting from approximately 1870 to 1905 during which paintings with particular characteristics were produced by a group of artists that included (among others) Monet, Renoir, Pissarro, Sisley and Degas. Alternatively, it can be regarded as a style applicable to all paintings sharing the characteristics of the works produced by the Impressionist painters, regardless of historical context. The first interpretation is an E4 Period, and the second defines morphological object types that fall under E55 Type.

Another specific case of an E4 Period is the set of activities and phenomena associated with a settlement, such as the populated period of Nineveh.

[This is the Scope note for E4 Period in CIDOC CRM version 5.0.1]

[Note that in CIDOC CRM, E12 Production, E13 Attribute Assignment, and E65 Creation are indirect subclasses of E4 Period = F8 Event; as a consequence, F8 Event is an indirect superclass of: F27 Work Conception, F28 Expression Creation, F40 Identifier Assignment, F41 Representative Manifestation Assignment, F42 Representative Expression Assignment, F32 Carrier Production Event, F33 Reproduction Event, and F30 Publication Event]

Examples: The battle of Trafalgar

Printing for the publisher named 'Doubleday' in 2003 all the copies of the first print run of the novel entitled 'Da Vinci Code' (F32)

Having the initial idea that eventually resulted in the existence of the opera entitled ‘Der fliegende Holländer’ (F27)

Creating for Mozart’s 41st symphony the uniform title that was thereafter consistently used to refer unambiguously to that symphony everywhere in the Library of Congress’s catalogue (F40)

F9 Place

Equal to: [E53](#) Place

Scope note: This class comprises extents in space, in particular on the surface of the earth, in the pure sense of physics: independent from temporal phenomena and matter. The instances of E53 Place are usually determined by reference to the position of immobile objects such as buildings, cities, mountains, rivers, or dedicated geodetic marks. A Place can be determined by combining a frame of reference and a location with respect to this frame. It may be identified by one or more instances of E44 Place Appellation.

It is sometimes argued that instances of E53 Place are best identified by global coordinates or absolute reference systems. However, relative references are often more relevant in the context of cultural documentation and tend to be more precise. In particular, we are often interested in position in relation to large, mobile objects, such as ships. For example, the Place at which Nelson died is known with reference to a large mobile object – H.M.S Victory. A resolution of this Place in terms of absolute coordinates would require knowledge of the movements of the vessel and the precise time of death, either of which may be revised, and the result would lack historical and cultural relevance.

Any object can serve as a frame of reference for E53 Place determination. The model foresees the notion of a section of an E19 Physical Object as a valid E53 Place determination. [*Scope Note for E53 Place in CIDOC CRM version 5.0.1*]

Note that Places may be determined by the location of historical or contemporary objects, geographic features, events or geo-political units.

Examples: The area referred to as ‘Lutèce’

The area referred to as ‘verso of the title page of the Library of Congress’s copy of the 1st edition of the novel entitled ‘Da Vinci Code’

F10 Person

Equal to: [E21](#) Person

Scope note: This class comprises real persons who live or are assumed to have lived. Bibliographic identities or personae assumed by an individual or a group should be modelled as F12 Nomen and connected to the relevant person or group with an instance of F35 Nomen Use Statement, even if nothing more can be said about this person or group. In a bibliographic context, a name presented following the conventions usually employed for personal names will be assumed to correspond to an actual real person (F10 Person), unless evidence is available to indicate that this is not the case. The fact that a persona may erroneously be classified as an instance of F10 Person does not imply that the concept comprises personae.

Examples: Margaret Atwood

Hans Christian Andersen

Queen Victoria

F11 Corporate Body

Subclass of: [E74](#) Group

Superclass of: [E40](#) Legal Body

Scope note: This class comprises organisations and groups of two or more people and/or organisations acting as a unit.

To be considered an F11 Corporate Body a gathering of people needs to bear a name and exhibit organisational characteristics sufficient to allow the body as a whole to participate in the creation, modification or production of an E73 Information Object. Groups such as conferences, congresses, expeditions, exhibitions, festivals, fairs, etc. are modelled as F11 Corporate Bodies when they are named and can take collective action, such as approving a report or publishing their proceedings.

Examples: The International Machaut Society
The British Library
The Jackson Five
The Regional Municipality of Ottawa-Carleton
Symposium on Glaucoma

F12 Nomen

Subclass of: [E41](#) Appellation

Superclass of: [F13](#) Identifier

Scope note: This class comprises any sign or arrangements of signs following a specific syntax (sequences of alphanumeric characters, chemical structure symbols, sound symbols, ideograms etc.) that are used or can be used to refer to and identify a specific instance of some class or category within a certain context. The scripts or type sets for the types of symbols used to compose an instance of F12 Nomen have to be explicitly specified. The identity of an instance of F12 Nomen is given by the order of its symbols and their individual role with respect to their scripts, regardless of the semantics of the larger structural components it may be built from. Structural tags occurring in the nomen string are regarded as symbols constituting the nomen. Spelling variants are regarded as different nomina, whereas the use of different fonts (visual representation variants) or different digital encodings do not change the identity.

Examples: ‘杜甫’ [the name of a Chinese poet of the 8th century, in simplified Chinese characters]
‘Du Fu’ [Pinyin romanised form of the name of a Chinese poet of the 8th century]
‘Tu Fu’ [another romanised form of the name of a Chinese poet of the 8th century]
‘Thơ Đô Phủ’ [Vietnamese form of the name of a Chinese poet of the 8th century]
‘جامعة صفاقس’ [Arabic name of the Sfax University (Tunisia), in Arabic script]
‘Ġāmi‘at Ṣafāqīs’ [Arabic name of the Sfax University (Tunisia), transliterated]
‘Université de Sfax’ [French name of the Sfax University (Tunisia)]
‘3-[(2S)-1-methylpyrrolidin-2-yl]pyridine’ [the IUPAC systematic name for nicotine]
‘Murders in the rue Morgue’ [English title of a textual work] (to put the image of the formula)
‘Poe, Edgar Allan, 1809-1849. Murders in the rue Morgue’ (F50) [controlled author/title access point for a textual work]
‘modelling’ [not the activity, just the written signs that represent its English name in British spelling]
‘modeling’ [not the activity, just the written signs that represent its English name in American spelling]

Properties: [R33](#) has content: [E62](#) String
(R33.1 has encoding: [E55](#) Type)

F13 Identifier

Subclass of: [F12](#) Nomen

Superclass of: [F50](#) Controlled Access Point

Equal to: [E42](#) Identifier

Scope note: This class comprises strings or codes assigned to instances of E1 CRM Entity in order to identify them uniquely and permanently within the context of one or more organisations. Such codes are often known as inventory numbers, registration codes, etc. and are typically composed of alphanumeric sequences. The class E42 Identifier is not normally used for machine-generated identifiers used for automated processing unless these are also used by human agents. *[Adapted from the Scope Note of CIDOC CRM E42 Identifier ver. 5.0.1]*

Examples: ISSN ‘0041-5278’
ISRC ‘FIFIN8900116’
Shelf mark ‘Res 8 P 10’
‘Guillaume de Machaut (1300?-1377)’ (F50) [a controlled personal name access point that follows the French rules]
‘Guillaume, de Machaut, ca. 1300-1377’ (F50) [a controlled personal name access point that follows the AACR rules]
‘Rite of spring (Choreographic work: Bausch)’ (F50)

Properties: [R8](#) consists of (forms part of): [E90](#) Symbolic Object

F14 Individual Work

Subclass of: [F1](#) Work

Superclass of: [F17](#) Aggregation Work

Scope note: This class comprises works that are realised by one and only one self-contained expression, i.e., works representing the concept as expressed by precisely this expression.

Inherent to the notion of work is the completion of recognisable outcomes of the work. These outcomes, i.e. the Self-Contained Expressions, are regarded as the symbolic equivalents of Individual Works, which form the atoms of a complex work. Normally creators would characterise an outcome of a work as finished. In other cases, one could recognise an outcome of a work as complete from the elaboration or logical coherence of its content, or if there is any historical knowledge about the creator deliberately or accidentally never finishing (completing) that particular expression. In all those cases, one would regard the corresponding expression as equivalent to one Individual Work.

Examples: Abstract content of Giovanni Battista Piranesi’s ‘Carcere XVI: the pier with chains: 1st state’
Abstract content of Giovanni Battista Piranesi’s ‘Carcere XVI: the pier with chains: 2nd state’ [explanation: these are two states of the same etching, but with so many and so significant differences between them that they can scarcely be recognised as conveying the same work; more generally speaking, each individual state of an etching, as a Self-Contained Expression, conveys its own F14 Individual Work (even if the differences are not so blatant as in the case of ‘Carcere XVI’), and is regarded as part of the larger, abstract F15 Complex Work that encompasses all distinct states of the same etching]

Abstract content of the recording made of performances of Johann Sebastian Bach’s ‘Toccatina in C minor BWV 911’ by Glenn Gould on May 15 & 16, 1979, in Toronto, Eaton’s Auditorium

Properties: [R9](#) is realised in (realises): [F22](#) Self-Contained Expression

F15 Complex Work

Subclass of: [F1](#) Work

Superclass of: [F18](#) Serial Work

Scope note: This class comprises works that have other works as members. The members of a Complex Work may constitute alternatives to, derivatives of, or self-contained components of other members of the same Complex Work.

In practice, no clear line can be drawn between parallel and subsequent processes in the evolution of a work. One part may not be finished when another is already revised. An initially monolithic work may be taken up and evolve in pieces. The member relationship of Work is based on the conceptual relationship, and should not be confused with the internal structural parts of an individual expression. The fact that an expression may contain parts from other work(s) does not make the expressed work complex. For instance, an anthology for which only one version exists is not a complex work.

The boundaries of a Complex Work have nothing to do with the value of the intellectual achievement but only with the dominance of a concept. Thus, derivations such as translations are regarded as belonging to the same Complex Work, even though in addition they constitute an Individual Work themselves. In contrast, a Work that significantly takes up and merges concepts of other works so that it is no longer dominated by the initial concept is regarded as a new work. In cataloguing practice, detailed rules are established prescribing which kinds of derivation should be regarded as crossing the boundaries of a complex work. Adaptation and derivation graphs allow the recognition of distinct sub-units, i.e. a complex work contained in a larger complex work.

As a Complex Work can be taken up by any creator who acquires the spirit of its concept, it is never finished in an absolute sense.

Examples: Work entitled ‘La Porte de l’Enfer’ by Auguste Rodin
Work entitled ‘Hamlet’ by William Shakespeare
Work entitled ‘Der Ring der Nibelungen’ by Richard Wagner
Work entitled ‘Carceri d’invenzione’ by Giovanni Battista Piranesi
Work entitled ‘Mass in B minor BWV 232’ by Johann Sebastian Bach

Properties: [R10](#) has member (is member of): [F1](#) Work

F16 Container Work

Subclass of: [F1](#) Work

Superclass of: [F17](#) Aggregation Work
[F19](#) Publication Work
[F20](#) Performance Work

Scope note: This class comprises works whose essence is to enhance or add value to expressions from one or more other works without altering them, by the selection, arrangement and/or addition of features of different form, such as layout to words, recitation and movement to texts, instrumentation to musical scores etc. This does not make the contents of the incorporated expressions part of the Container Work, but only part of the resulting expression. Container Work may include the addition of new, original parts to the incorporated expressions, such as introductions, graphics, etc.

A new version of a container work does not make the resulting complex work a Container Work as well. The inclusion of expressions from a complex work in a Container Work does not make the Container Work itself complex.

Examples: The aggregation and arrangement concept of the anthology entitled ‘American Women Poets of the Nineteenth Century: An Anthology’, edited by Cheryl Walker and published by Rutgers

University Press in July 1992 (F17)

The concept for the layout created by printer Guido Morris for the text of Michael Hamburger's English translation of 12 poems by Georg Trakl for publication in 1952 (F19)

The concept by the publisher named 'Dell' of issuing together, in 2002, three novels entitled 'The Partner', 'The Street Lawyer', and 'A time to kill', by the author named 'John Grisham', with just the statement 'Three #1 bestsellers by John Grisham' as a collective title (F19)

The concept of Sergei Radlov's mise-en-scène of a Yiddish translation of the textual work entitled 'King Lear' in Moscow in 1935 (F20)

The concept of putting together the English text of 'King Lear' and a Spanish translation thereof in a bilingual edition of 'King Lear' (F17)

F17 Aggregation Work

Subclass of: [F14](#) Individual Work
[F16](#) Container Work

Scope note: This class comprises works whose essence is the selection and/or arrangement of expressions of one or more other works. This does not make the contents of the aggregated expressions part of this work, but only part of the resulting expression. F17 Aggregation Work may include additional original parts.

An expression of a work may include expressions of other works within it. For instance, an anthology of poems is regarded as a work in its own right that makes use of expressions of the individual poems that have been selected and ordered as part of an intellectual process.

A new version of an aggregate work does not make the resulting complex work an aggregate work as well. The inclusion of expressions from a complex work in an aggregation work does not make the aggregation work itself complex.

Examples: The aggregation and arrangement concept of the anthology entitled 'American Women Poets of the Nineteenth Century: An Anthology', edited by Cheryl Walker and published by Rutgers University Press in July 1992

The aggregation and arrangement concept of the Web site named 'IFLANET'

The aggregation and arrangement concept of the collection of articles entitled 'Marij Kogoj (1892-1992): zbornik referatov s kolokvija ob stoletnici skladateljevega rojstva 7.10.1992 v Ljubljani = Marij Kogoj (1892-1992): proceedings from the colloquium held in Ljubljana at the centenary of the composer's birth on October 7th, 1992' and edited by a person named 'Ivan Klemenčič'

F18 Serial Work

Subclass of: [F15](#) Complex Work
[F19](#) Publication Work

Scope note: This class comprises works that are, or have been, planned to result in sequences of Expressions or Manifestations with common features. Whereas a work can acquire new members during the time it evolves, Expressions and Manifestations are identified with a certain state achieved at a particular point in time. Therefore there is in general no single Expression or Manifestation representing a complete serial work, unless the serial work has ended.

Serial Works may or may not have a plan for an overall expression.

The retrospective reprinting of all issues of a Serial Work at once, in the form of a monograph, is regarded to be another member of a Complex Work, which contains the Serial Work and the Individual Work realised in the monograph. This does not make the monograph part of the Serial Work.

- Examples: The periodical entitled ‘The UNESCO Courier’, ISSN ‘0041-5278’
- The periodical entitled ‘Courrier de l’UNESCO’, ISSN ‘0304-3118’ [French edition of the periodical titled ‘The UNESCO Courier’, ISSN ‘0041-5278’]
- The series entitled ‘L’évolution de l’humanité’, ISSN ‘0755-1843’ [a monograph series comprising volumes that were published from 1920 on, and some of which were reprinted, with different physical features and rearranged in a different order, from 1968 on, in a distinct series also entitled ‘L’évolution de l’humanité’, ISSN ‘0755-1770’]
- Properties: [R11](#) has issuing rule (is issuing rule of): [E29](#) Design or Procedure

F19 Publication Work

- Subclass of: [F16](#) Container Work
- Superclass of: [F18](#) Serial Work
- Scope note: This class comprises works that have been planned to result in a manifestation product type or an electronic publishing service and that pertain to the rendering of expressions from other works.
- Examples: The concept of publishing Stephen Crane’s complete poems (as edited by Joseph Katz), which includes the idea that every time a stanza jumps over a page change, the statement ‘[NO STANZA BREAK]’ should be printed as a warning for readers that the new page continues the same stanza
- The concept, on behalf of publisher named ‘Verlag Neue Kunsthandlung’, of issuing together, around 1925, three formerly independent publications (‘Emil Orlik’ by Max Osborn – vol. 2 within the series named ‘Graphiker der Gegenwart’, published in 1920; ‘Anders Zorn’ by Paul Friedrich – vol. 10 within the series named ‘Graphiker der Gegenwart’, published in 1924; and ‘Max Slevogt’ by Julius Elias – vol. 11 within the series named ‘Graphiker der Gegenwart’, published in 1923) as one, new publication, entitled ‘102 Bilder aus der Sammlung *Graphiker der Gegenwart*’
- The concept, on behalf of publisher named ‘Dell’, of issuing together in 2002 three novels, titled ‘The partner’, ‘The street lawyer’, and ‘A time to kill’, by author named ‘John Grisham’, with just the statement ‘Three #1 bestsellers by John Grisham’ as a collective title

F20 Performance Work

- Subclass of: [F16](#) Container Work
- Scope note: This class comprises the sets of concepts for rendering a particular or a series of like performances.
- F20 Performance Work is declared as a subclass of F16 Container Work. This implies that the incorporated expressions (such as the text of the staged play, the text of the argument for the ballet, the recorded music to be used for the ballet, or the content of the musical score to be used for a concert, etc.) are not by themselves a part of the expression of this F1 Work. Rather, an expression (F25 Performance Plan) of the instructions the stage production, choreography or musical performance consists of *incorporates* (P165) that textual or musical content. In other words, the text of ‘Hamlet’ is not a component of the concepts that underlie a given mise-en-scène of ‘Hamlet’, but any staging directions (F25 Performance Plan) that convey a given director’s vision of ‘Hamlet’ must necessarily incorporate the text of ‘Hamlet’.
- Examples: The conceptual content of Sergei Radlov’s mise-en-scène of a Yiddish translation of the textual work entitled ‘King Lear’ in Moscow in 1935
- The conceptual content of Pina Bausch’s choreography of the ballet entitled ‘Rite of spring’ in Wuppertal in 1975
- The conceptual content of Bruno Walter’s performance of Gustav Mahler’s 9th symphony in

1961

The conceptual content of the “performance handbook” for Luigi Nono’s musical work entitled ‘À Pierre’

Properties: [F12](#) is realised in (realises): [F25](#) Performance Plan

F21 Recording Work

Subclass of: [F1](#) Work

Scope note: This class comprises works that conceptualise the capturing of features of perdurants. The characteristics of the manifestation of a recording work are those of the product of the capture process. The characteristics of any other works recorded are distinct from those of the recording work itself. In the case where the recorded perdurant expresses some Work, the respective instance of F21 is also an F16 Container Work.

The concept of recording is not necessarily linked to the use of modern devices that allow for mechanical recording, such as tape recorders or cameras. However, in practice, library catalogues tend to regard as recordings only the products of such mechanical devices.

But the concept of recording is very much linked to the notion that there is something that is recorded. In general, photographs or animated images are not to be regarded as instances of F21 Recording Work just because of the use of the medium, but simply as instances of F1 Work (or F15 Complex Work). Only such photographs and animated images that can be used as documentation are to be regarded as recordings.

Examples: The concept of recording the Swedish 17th century warship Vasa in August 1959 to April 1961

The concept of documenting the Live Aid concerts July 13, 1985, London, Philadelphia, Sydney and Moscow

The concept of making a photograph of the three Allied leaders at Yalta in February 1945

Oceania Project’s concept of making a large digital acoustic data archive dedicated to East Australian humpback whale songs

The concept of recording Louise Bourgeois’s activity in the documentary movie entitled ‘Louise Bourgeois: The Spider, the Mistress, and the Tangerine’

Properties: [F13](#) is realised in (realises): [F26](#) Recording

F22 Self-Contained Expression

Subclass of: [F2](#) Expression

Superclass of: [F24](#) Publication Expression
[F25](#) Performance Plan
[F26](#) Recording

Scope note: This class comprises the immaterial realisations of individual works at a particular time that are regarded as a complete whole. The quality of wholeness reflects the intention of its creator that this expression should convey the concept of the work. Such a whole can in turn be part of a larger whole.

Inherent to the notion of work is the completion of recognisable outcomes of the work. These outcomes, i.e. the Self-Contained Expressions, are regarded as the symbolic equivalents of Individual Works, which form the atoms of a complex work. A Self-Contained Expression may contain expressions or parts of expressions from other work, such as citations or items collected in anthologies. Even though they are incorporated in the Self-Contained Expression, they are not regarded as becoming members of the expressed container work by their inclusion in the expression, but are rather regarded as foreign or referred to elements.

F22 Self-Contained Expression can be distinguished from F23 Expression Fragment in that an F23 Expression Fragment was not intended by its creator to make sense by itself. Normally creators would characterise an outcome of a work as finished. In other cases, one could recognise an outcome of a work as complete from the elaboration or logical coherence of its content, or if there is any historical knowledge about the creator deliberately or accidentally never finishing (completing) that particular expression. In all those cases, one would regard an expression as self-contained.

- Examples:
- The Italian text of Dante's 'Inferno' as found in the authoritative critical edition *La Commedia secondo l'antica vulgata a cura di Giorgio Petrocchi*, Milano: Mondadori, 1966-67 (= Le Opere di Dante Alighieri, Edizione Nazionale a cura della Società Dantesca Italiana, VII, 1-4)
 - The musical notation of Franz Schubert's lied known as 'Ave Maria'
 - The musical notation of Franz Schubert's lieder cycle entitled 'Seven Songs after Walter Scott's The Lady of the Lake', of which 'Ave Maria' is a distinct part
 - The musical notation of Franz Liszt's piano transcription of Franz Schubert's lied known as 'Ave Maria'

F23 Expression Fragment

Subclass of: [F2](#) Expression

Scope note: This class comprises parts of Expressions and these parts are not Self-Contained Expressions themselves.

The existence of an instance of F23 Expression Fragment can be due to accident, such as loss of material over time, e.g. the only remaining manuscript of an antique text being partially eaten by worms, or due to deliberate isolation, such as excerpts taken from a text by the compiler of a collection of excerpts.

An F23 Expression Fragment is only identified with respect to its occurrence in a known or assumed whole. The size of an instance of F23 Expression Fragment ranges from more than 99% of an instance of F22 Self-Contained Expression to tiny bits (a few words from a text, one bar from a musical composition, one detail from a still image, a two-second clip from a movie, etc.).

- Examples:
- The only remnants of Sappho's poems
 - The words 'Beati pauperes spiritu' (excerpted from Matthew's Gospel 5,3 in Latin translation)

F24 Publication Expression

Subclass of: [F22](#) Self-Contained Expression

Scope note: This class comprises complete sets of signs present in publications, reflecting publishers' final decisions as to both selection of content and layout of the publications. Frequently the creation of a Publication Expression includes both adding graphical form and fonts to Expressions consisting of words alone and selecting illustrations and other content. As such, an instance of Publication Expression incorporates all Expressions combined for the resulting final form of rendering, whether visual, audio or tactile. An instance of Publication Expression is one entity regardless of the number of independent Expressions published within it, as long as it represents one unit of release. The published third party content can be associated via the property *P165 incorporates (is incorporated in)*.

- Examples:
- The text, its layout and the textual and graphic (Saur's logo on p. [i]) content of front and back cover, spine (spine title), and p. [i-iv] of the publication entitled 'Functional Requirements for Bibliographic Records: final report', published by K. G. Saur in 1998, identified by ISBN '3-598-11382-X'
 - The overall content of the book identified by ISBN '0-8014-9130-4': the text of Stephen Crane's

complete poems as edited by Joseph Katz, the numbering system introduced by Joseph Katz in order to identify each individual poem by Stephen Crane, page numbers, the text of Joseph Katz's dedication, preface, acknowledgements, and introduction, the table of contents, the index of first lines, the statements found on title page, back of title page (including CIP bibliographic record), cover front, back front, and spine, and the layout of the publication; for one of Stephen Crane's longer poems, printed on p. 142-143, a statement reads at bottom of p. 142: '[NO STANZA BREAK]': obviously, this statement does not belong to the Self-Contained Expression intended by Stephen Crane, and presumably not to the one intended by editor Joseph Katz either, but was more probably added by the publishing team, due to characteristics of the layout of the publication: a cautious reader can easily interpret '[NO STANZA BREAK]' as non-belonging to the poem itself, but an OCR process would not make the distinction between the text of the poem and the statement made by the publisher; '[NO STANZA BREAK]' belongs to the Publication Expression, although it does not belong to the Self-Contained Expression intended by Stephen Crane and Joseph Katz

The overall content of the LP sound recording identified by label and label number 'CBS 34-61237': a recorded performance of Terry Riley's musical work 'In C', the text of liner notes by Paul Williams translated into French by Bernard Weinberg, technical statements such as 'Stereo,' publisher's logo, series logo, title and statement of responsibility on front, back, and spine of the cover and on the recording itself, duration statement, cover art by G. Joly, overall layout, etc.; a special, shunting sound was added at the end of side one and beginning of side two, as Terry Riley's work is in the form of a continuous musical flow without any interruption and the technical possibilities of vinyl LPs did not allow the complete performance to be contained on just one side: that special, shunting sound was not intended in Riley's score nor in the performance but was added by the publisher (with or without Riley's consent, this detail is not documented), and as such it is part of the Publication Expression although it is not part of the composer's and the performers' Self-Contained Expression (this shunting sound was no longer needed in subsequent releases on CD)

The overall content of the DVD entitled 'The Aviator (2-Disc Full Screen Edition)', released in 2004: Martin Scorsese's movie itself; layout of the box and the two DVDs contained in the box; pictures on the DVDs themselves; English, Spanish, and French subtitles; English and French audio tracks; and bonuses: commentaries by director Martin Scorsese, editor Thelma Schoonmaker, and producer Michael Mann; a deleted scene ('Howard Tells Ava About His Car Accident'); and featurettes 'A Life Without Limits: The Making of The Aviator'; 'The Role of Howard Hughes in Aviation History'; 'Modern Marvels: Howard Hughes, A Documentary by the History Channel'; 'The Visual Effects of The Aviator'; 'The Affliction of Howard Hughes: Obsessive Compulsive Disorder'; 'The Age of Glamour: The Hair And Makeup of The Aviator'; 'Costuming The Aviator: The Work of Sandy Powell'; 'Constructing The Aviator: The Work of Dante Ferretti'; 'An evening with Leonardo DiCaprio and Alan Alda'; 'OCD Panel Discussion With Leonardo DiCaprio, Martin Scorsese, and Howard Hughes' Widow Terry Moore'; 'Still Gallery'; 'Scoring The Aviator: The Work Of Howard Shore'; and 'The Wainwright Family – Loudon, Rufus and Martha'

F25 Performance Plan

Subclass of: [F22](#) Self-Contained Expression
[E29](#) Design or Procedure

Scope note: This class comprises sets of directions to which individual performances of theatrical, choreographic, or musical works and their combinations should conform.

In the case of theatrical performances, such directions incorporate, but are not limited nor reducible to, the text of a given version of the play performed (e.g., a translated text, some passages of which are deliberately omitted, with some rephrased lines, etc.).

In the case of choreographic performances, such directions may incorporate, but are neither limited nor reducible to, the notation of choreographic movements in systems such as labanotation.

In the case of musical performances, such directions may incorporate, but are neither limited nor reducible to, the musical score. In case of electronic music, they may incorporate software instructions.

These directions may or may not completely determine the form of the intended performance. Depending on the nature of the directions, the form of the intended performance, such as the sets of movements or the sound characteristics, may or may not be predictable from the directions.

Note that a performance plan may be more or less elaborate, and may even foresee just improvisation.

- Examples:
- The set of instructions for the production of a Yiddish translation of the textual work entitled ‘King Lear’, as directed by Sergei Radlov in Moscow in 1935
 - The set of instructions for the production of the ballet entitled ‘Rite of spring’, as choreographed by Pina Bausch in Wuppertal in 1975
 - The set of instructions by Bruno Walter for performing Gustav Mahler’s 9th symphony, delivered by him to the Columbia Symphony Orchestra during rehearsals in Hollywood in 1961 (as partially documented in the CD entitled ‘Bruno Walter conducts and talks about Mahler symphony No. 9: rehearsal & performance’)
 - The set of instructions contained in the “performance handbook” for Luigi Nono’s musical work entitled ‘À Pierre’

F26 Recording

- Subclass of: [F22](#) Self-Contained Expression
- Scope note: This class comprises expressions which are created in instances of F29 Recording Event. A recording is intended to convey (and preserve) the features of one or more perdurants.
- Examples:
- The set of signs that make up the third alternate take of the musical work entitled ‘Blue Hawaii’ as performed by Elvis Presley in Hollywood, Calif., Radio Recorders, on March 22nd, 1961
 - The set of signs that make up the famous photograph of the three Allied leaders at Yalta in February 1945
 - The set of signs that make up the sound recording of an East Australian humpback whale song in 1994
 - The set of signs that make up the sequences from the documentary movie entitled ‘Louise Bourgeois: The Spider, the Mistress, and the Tangerine’ in which Louise Bourgeois is seen at work

F27 Work Conception

- Subclass of: [E65](#) Creation
- Scope note: This class comprises beginnings of evolutions of works.
- An instance of F27 Work Conception marks the initiation of the creation of a work. The work, as an intellectual construction, evolves from this point on, until the last known expression of it. The instance of E39 Actor with which a work is associated through the chain of properties F1 Work *R16i was initiated by* F27 Work Conception *P14 carried out by* E39 Actor corresponds to the notion of the “creator” of the work. In the case of commissioned works, it is not the commissioning that is regarded as the work conception, but the acceptance of the commission.
- This event does not always correlate with the date assigned in common library practice to the work, which is usually a later event (such as the date of completion of the first clean draft).
- In addition, F27 Work Conception can serve to document the circumstances that surrounded the appearance of the original idea for a work, when these are known.

- Examples: Ludwig van Beethoven's having the first ideas for his fifth symphony
- Pablo Picasso's acceptance, in 1930, of Ambroise Vollard's commission for a set of 100 etchings, now known as the 'Vollard Suite'
- René Goscinny's and Albert Uderzo's first collaborative ideas for the comic book entitled 'Asterix in Britain' [*comment: Goscinny wrote the script and Uderzo made the drawings; both are regarded as co-creators of that collaborative, at the same level of creative input, and no attempt is made to ascertain whether the ideas for the script preceded the ideas for the drawings, or vice-versa*]
- The combination of activities, carried out, among others, by Alfred Hitchcock, that began the process which eventually resulted in the movie entitled 'Psycho' coming into being
- Oscar Wilde's having by May 1897 the initial idea of writing his poem entitled 'The ballad of the Reading gaol', inspired by his stay in the Reading prison from November 20, 1895 to May 18, 1897, and the execution of Charles Thomas Woolridge on July 7, 1896
- Properties: [R16](#) initiated (was initiated by): [F1](#) Work

F28 Expression Creation

- Subclass of: [E12](#) Production
[E65](#) Creation
- Superclass of: [F29](#) Recording Event
[F30](#) Publication Event
- Scope note: This class comprises activities that result in instances of F2 Expression coming into existence. This class characterises the externalisation of an Individual Work.
- Although F2 Expression is an abstract entity, a conceptual object, the creation of an expression inevitably also affects the physical world: when you scribble the first draft of a poem on a sheet of paper, you produce an instance of F4 Manifestation Singleton; F28 Expression Creation is a subclass of E12 Production because the recording of the expression causes a physical modification of the carrying E18 Physical Thing. The work becomes manifest by being expressed on a physical carrier different from the creator's brain. The spatio-temporal circumstances under which the expression is created are necessarily the same spatio-temporal circumstances under which the first instance of F4 Manifestation Singleton is produced. The mechanisms through which *oral tradition* (of myths, tales, music, etc.) operates are not further investigated in this model. As far as bibliographic practice is concerned, only those instances of F2 Expression that are externalised on physical carriers other than both the creator's brain and the auditor's brain are taken into account (for a discussion of the modelling of oral traditions, see: Nicolas, Yann. 'Folklore Requirements for Bibliographic Records: oral traditions and FRBR.' In: *Cataloging & Classification Quarterly* (2005). Vol. 39, No. 3-4. P. 179-195).
- It is possible to use the *P2 has type (is type of)* property in order to specify that the creation of a given expression of a given work played a particular role with regard to the overall bibliographic history of that work (e.g., that it was the creation of the progenitor expression on which all other expressions of the same work are based; or that it was the creation of the critical edition that served as the basis for canonical references to the work).
- Examples: The creation of the original manuscript score of 'Uwertura tragiczna' by Andrzej Panufnik in 1942 in Warsaw
- The reconstruction from memory of the manuscript score of 'Uwertura tragiczna' by Andrzej Panufnik in 1945 after the original score was destroyed during the war
- The creation, by Lord Byron, of the English text of his work entitled 'Manfred' (*P2 has type* E55 Type {major original contribution})
- The creation, by Woldemar Starke, of his German translation of Lord Byron's text entitled 'Manfred' (*P2 has type* E55 Type {translation})
- The recording of the third alternate take of 'Blue Hawaii' performed by Elvis Presley in

Hollywood, Calif., Radio Recorders, on March 22nd, 1961 [each individual take is a distinct instance of F2 Expression]

Properties: [R17](#) created (was created by): [F2](#) Expression
[R18](#) created (was created by): [F4](#) Manifestation Singleton
[R19](#) created a realisation of (was realised through): [F1](#) Work

F29 Recording Event

Subclass of: [F28](#) Expression Creation

Scope note: This class comprises activities that intend to convey (and preserve) the features of perdurants in a recording, such as a live recording of a performance, a documentary, or other capture of a perdurant. Such activities may follow the directions of a recording plan. They may include postproduction.

Examples: The making of the recording of the third alternate take of the musical work titled ‘Blue Hawaii’ as performed by Elvis Presley in Hollywood, Calif., Radio Recorders, on March 22nd, 1961

The making of the photograph of the three Allied leaders at Yalta in February 1945

The making of the recording of an East Australian humpback whale song in 1994 in the framework of the Oceania Project

Filming Louise Bourgeois at work in the context of the shooting of the documentary movie entitled ‘Louise Bourgeois: The Spider, the Mistress, and the Tangerine’

Properties: [R20](#) recorded (was recorded through): [E2](#) Temporal Entity
[R21](#) created (was created through): [F26](#) Recording
[R22](#) created a realisation of (was realised through): [F21](#) Recording Work
[R65](#) recorded aspects of (had aspects recorded through): [E18](#) Physical Thing

F30 Publication Event

Subclass of: [F28](#) Expression Creation

Scope note: This class comprises the activities of publishing. Such an event includes the creation of an F24 Publication Expression and setting up the means of production. The end of this event is regarded as the date of publication, regardless of whether the carrier production is started. Publishing can be either physical or electronic. Electronic publishing is regarded as making an instance of F24 Publication Expression available in electronic form on a public network. Electronic Publishing does not mean producing a physical instance of F5 Item by partially electronic means. Making an electronic file available on a physical carrier can be regarded as equivalent to setting up the means of production; downloading the file is regarded as the electronic equivalent of F32 Carrier Production Event.

Examples: Publishing Amerigo Vespucci’s ‘Mundus novus’ in Paris ca. 1503-1504

Establishing in 1972 the layout, features, and prototype for the publication of ‘The complete poems of Stephen Crane, edited with an introduction by Joseph Katz’ (ISBN ‘0-8014-9130-4’), which served for a second print run in 1978

Making available online the article by Allen Renear, Christopher Phillippe, Pat Lawton, and David Dubin, entitled ‘An XML document corresponds to which FRBR Group 1 entity?’
<<http://conferences.idealliance.org/extreme/html/2003/Lawton01/EML2003Lawton01.html>>

Properties: [R23](#) created a realisation of (was realised through): [F19](#) Publication Work
[R24](#) created (was created through): [F24](#) Publication Expression

F31 Performance

Subclass of: [E7](#) Activity

Scope note: This class comprises activities that follow the directions of a performance plan, such as a theatrical play, an expression of a choreographic work or a musical work; i.e., they are intended to communicate directly or indirectly to an audience.

Such activities can be identified at various levels of granularity, and can be contiguous or not. Any individual performance (with or without intermissions) is a single instance of F31 Performance. In addition, a complete run of performances can also be seen as an instance of F31 Performance, with individual performances as parts. A complete run of performances may comprise an original run plus any of its extensions and tours.

Note that a performance plan may be more or less elaborate, and may even foresee just improvisation.

Examples: Performing the first performance of a Yiddish translation of the textual work entitled ‘King Lear’, as directed by Sergei Radlov, in Moscow, at the Moscow State Jewish Theatre, on February 10, 1935 [individual performance]

Performing the ballet entitled ‘Rite of spring’, as choreographed by Pina Bausch, in Avignon, at the Popes’ Palace, on July 7, 1995 [individual performance]

Performing the operatic work entitled ‘Dido and Aeneas’, as directed by Edward Gordon Craig and conducted by Martin Shaw, in London, Hampstead Conservatoire, on May 17, 18, and 19, 1900 [run of performances]

Properties: [R25](#) performed (was performed in): [F25](#) Performance Plan

[R66](#) included performed version of (had a performed version through): [E89](#) Propositional Object

F32 Carrier Production Event

Subclass of: [E12](#) Production

Scope note: This class comprises activities that result in instances of F54 Utilized Information Carrier coming into existence. Both the production of a series of physical objects (printed books, scores, CDs, DVDs, CD-ROMS, etc.) and the creation of a new copy of a file on an electronic carrier are regarded as instances of F32 Carrier Production Event.

Typically, the production of copies of a publication (no matter whether it is a book, a sound recording, a DVD, a cartographic resource, etc.) strives to produce items all as similar as possible to a prototype that displays all the features that all the copies of the publication should also display, which is reflected in property *R27 used as source material* F24 Publication Expression.

Examples: The printing of copies of the 3rd edition of ‘Codex Manesse: die Miniaturen der großen Heidelberger Liederhandschrift, herausgegeben und erläutert von Ingo F. Walther unter Mitarbeit von Gisela Siebert’, Insel-Verlag, 1988 [a fac-simile edition of an illuminated mediaeval manuscript]

The printing of copies of the ‘Ordnance Survey Explorer Map 213, Aberystwyth & Cwm Rheidol’, ISBN 0-319-23640-4 (folded), 1:25,000 scale, released in May 2005 [a cartographic resource]

The production of copies of the sound recording titled ‘The Glory (???) of the human voice’, RCA Victor Gold Seal GD61175, containing recordings of musical works performed by Florence Foster Jenkins [a sound recording; the question marks in parentheses belong to the original title]

My clicking now on the link http://cidoc.ics.forth.gr/docs/cidoc_crm_version_4.0.pdf, and thus downloading on my PC a reproduction of the electronic file titled ‘Definition of the CIDOC Conceptual Reference Model... version 4.0’ that is stored on the ICS FORTH’s servers in Heraklion, Crete

The second print run, in 1978, of ‘The complete poems of Stephen Crane, edited with an introduction by Joseph Katz’ (ISBN ‘0-8014-9130-4’), a publication dated 1972 [publication of a printed text]

Properties: [R26](#) produced things of type (was produced by): [F3](#) Manifestation Product Type
[R27](#) used as source material (was used by): [F24](#) Publication Expression
[R28](#) produced (was produced by): [F54](#) Utilized Information Carrier

F33 Reproduction Event

Subclass of: [E12](#) Production

Scope note: This class comprises activities that consist in making copies, more or less mechanically, of an instance of E84 Information Carrier (such as an F5 Item or an F4 Manifestation Singleton which is also instance of E84 Information Carrier), preserving the expression carried by it. A Reproduction Event results in new instances of E84 Information Carrier coming into existence. In general, the copy will have different attributes from the original and they are therefore not regarded as siblings.

This class makes it possible to account for the legal distinction between private copying for the purpose of “fair use,” and mass production for the purpose of dissemination.

It can prove difficult to determine where to draw the line between F33 Reproduction Event and F32 Carrier Production Event in cases where multiple copies are produced. In this case, the copies, but not the original, may be regarded as instances of F5 Item. It is the existence of an explicit production plan that makes the difference. As a consequence, F33 Reproduction Event and F32 Carrier Production Event are not declared as *disjoint*, which makes it possible to account for such situations that could be regarded as instances of both Production Event and Reproduction Event.

Examples: My photocopying now for my own private use an exemplar of the article entitled ‘Federal Court’s Ruling Against Photocopying Chain Will Not Destroy “Fair Use”’ by Kenneth D. Crews, issued in ‘Chronicle of higher education’, 17 April 1991, A48

The BnF’s producing in 1997 the microfilm identified by call number ‘Microfilm M-12169’ of the exemplar identified by shelf mark ‘Res 8 P 10’ of Amerigo Vespucci’s ‘Mundus novus’ published in Paris ca. 1503-1504

The BnF’s reproducing in 2001 the exemplar identified by call number ‘NC His Master’s Voice HC 20’ of a 78 rpm phonogram released by Gramophone in 1932, as part of the CD identified by call number ‘SDCR 2120’

The BnF’s making in 2003 a digitisation, identified by call number ‘IFN 7701015’, of the collection of drawings (held by the BnF) that were made by Étienne-Louis Boullée in 1784 for his project of a ‘Newton Cenotaph’

Properties: [R29](#) reproduced (was reproduced by): [E84](#) Information Carrier
[R30](#) produced (was produced by): [E84](#) Information Carrier

F34 KOS

Subclass of: [E32](#) Authority Document
[E29](#) Design or Procedure
[F2](#) Expression

Scope note: This class comprises documents that establish controlled terminology (nomina) for consistent use. They may also describe relationships between entities and controlled terminology and relationships between entities. Note that any meaningful change in a Knowledge Organisation System (KOS) that affects the validity status of its elements defines a new release (Expression) of the KOS. Note that identifiers created following a rule in a KOS are to be regarded as being taken from this KOS, even though not explicitly spelled out. This definition of KOS reflects

current library practice and not the use of the term in general.

- Examples: LCSH February 20 to March 19 2012
DDC 19 [19th English edition, published only in print by Forest Press in 1979]
- Properties: [R34](#) has validity period (is validity period of): [E52](#) Time-Span

F35 Nomen Use Statement

- Subclass of: [F2](#) Expression
[E29](#) Design or Procedure
- Scope note: This class comprises statements relating a Thema with a particular Nomen and its usage in the context of a common Complex Work realized by one or more KOS.
- Examples: '010 __|a sh 85082387'...'150 __|a Maxwell equations' [MARC21 encoding of the preferred subject access point from LCSH, <http://lccn.loc.gov/sh85082387>, as of 19 November 2012]
'010 __|a sh 85082387'...'450 __|a Equations, Maxwell' [MARC21 encoding of a variant subject access point, from the same source]
'001 FRBNF119547493'...'100 w.0..barus.\$aGončarova\$mNatal'â Sergeevna\$d1881-1962'
[INTERMARC encoding of the preferred access point for a personal name, from the authority file of the National Library of France, <http://catalogue.bnf.fr/ark:/12148/cb119547494/INTERMARC>, as of 15 June 2012]
'001 FRBNF119547493'...'100 w.0..c.rus.\$aГончарова\$mНаталья Сергеевна\$d1881-1962'
[INTERMARC encoding of a parallel access point from the same source]
'001 FRBNF119547493'...'400 \$w....b.eng.\$aGoncharova\$mNatalia\$d1881-1962'
[INTERMARC encoding of a variant access point from the same source]
- Properties: [R32](#) is warranted by (warrants): [F52](#) Name Use Activity
[R35](#) is specified by (specifies): [F34](#) KOS
(R35.1 has status: [E55](#) Type)
[R36](#) uses script conversion (is script conversion used in): [F36](#) Script Conversion
[R37](#) states as nomen (is stated as nomen in): [F12](#) Nomen
[R38](#) refers to thema (is thema of): [E1](#) CRM Entity
[R39](#) is intended for (is target audience in): [E74](#) Group
[R54](#) has nomen language (is language of nomen in): [E56](#) Language
[R55](#) has nomen form (is nomen form in): [E55](#) Type
[R56](#) has related use (is related use for): [F35](#) Nomen Use Statement
(R56.1 has type: [E55](#) Type)

F36 Script Conversion

- Subclass of: [E29](#) Design or Procedure
- Scope note: This class comprises rule sets for converting signs or arrangements of signs from one script or type set to another.
- Examples: ISO 9:1995

F38 Character

- Subclass of: [E28](#) Conceptual Object
- Scope note: This class comprises fictional or iconographic individuals or groups of individual appearing in works in a way relevant as subjects. Characters may be purely fictitious or based on real persons

or groups, but as characters they may exhibit properties that would be inconsistent with a real person or group. Rather than merging characters with real persons, they should be described as disjoint, but related entities.

Examples: Harry Potter [in J.K. Rowling's series of novels and the films based on them]
Sinuhe the Egyptian [in Mika Waltari's novel]
The Knights of the Round Table [in fiction]

Properties: [R57](#) is based on (is basis for): [E39](#) Actor
[R58](#) has fictional member (is fictional member of): [F38](#) Character

F39 Family

Subclass of: [E74](#) Group

Scope note: This class comprises groups of two or more persons presented as a family justified by relationships of birth, marriage, adoption, civil union, or similar social or legal status and an assumed common tradition, including examples such as royal families, dynasties, houses of nobility, etc.

Examples: House of Tudor
The Grimm brothers

F40 Identifier Assignment

Equal to: [E15](#) Identifier Assignment

Scope note: This class comprises activities that result in the allocation of an identifier to an instance of any subclass of E1 CRM Entity. An F40 Identifier Assignment may include the creation of the identifier from multiple constituents. Explicit reference to the used constituents can be made using the property *P142 used constituent (was used in)*. The syntax of the identifier and the kinds of constituents to be used in constructing it may be declared in a rule. The construction of controlled access points for the names of persons, families and corporate bodies following specific cataloguing rules is a typical library application of identifier assignment. F40 Identifier Assignment also includes the assignment of uniform titles as controlled access points for works or expressions.

Examples: Assigning the name heading 'William, Prince, Duke of Cambridge, 1982-' as a controlled access point for a personal name using the *Anglo-American Cataloguing Rules*, 2nd edition
Assigning the name heading 'Library and Archives Canada' as an authorised controlled access point for a corporate body name using the *Anglo-American Cataloguing Rules*, 2nd edition
Assigning the name heading 'Bibliothèque et Archives Canada' as an authorised controlled access point for a corporate body name using the *Règles de catalogage anglo-américaines*, 2e édition
Assigning the author-title heading 'Goethe, Johann Wolfgang von, 1749-1832. Faust. 1. Theil.' as a uniform title for a work
Assigning the title heading 'Bible. English. American Standard' as a uniform title for an expression

Properties: [R45](#) assigned to (was assigned by): [E1](#) CRM Entity
[R46](#) assigned (was assigned by): [F13](#) Identifier
[R52](#) used rule (was the rule used in): [F43](#) Identifier Rule

F41 Representative Manifestation Assignment

Subclass of: [E13](#) Attribute Assignment

Scope note: This class comprises activities through which an Agency declares (implicitly or explicitly) that a given instance of F3 Manifestation Product Type or F4 Manifestation Singleton is representative for a given F2 Expression, i.e., that some features found on that instance of F3 Manifestation Product Type or F4 Manifestation Singleton (most prominently, information about the title) can be inferred to also apply to that instance of F2 Expression, no matter within which manifestation it is embodied.

The reasoning behind is that the Work title is known through the title of an Expression that is deemed representative of the Work, and the title of the representative Expression is known through the title proper of a Manifestation that is deemed representative of the Expression representative of the Work.

Examples: Selecting the 1775 edition of Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' as the representative manifestation for the text of Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' and citing that publication as the source for the authority record created by the Library of Congress for that work

Selecting the 2007 edition of John Tavener's musical work entitled 'The Eternal Sun' as the representative manifestation for the notation of John Tavener's musical work entitled 'The Eternal Sun' and citing that publication as the source for the authority record created by the National Library of France for that work

Selecting the 1983 edition of Stanley Karnow's textual work entitled 'Vietnam, the war nobody won' as the representative manifestation for a partial expression of the series entitled 'Headline series' and citing that publication as the source for the authority record created by the Library of Congress for that series

Selecting the issue dated October 2002 of the periodical entitled 'The New Courier' as the representative manifestation for a partial expression of the periodical entitled 'The New Courier' and citing that publication as the source for the bibliographic record created by the National Library of France for that periodical

Selecting the manuscript identified by shelfmark 'MS-8282' within the collections of the National Library of France, Department for Music, as representative for the notation of Stanislas Champein's musical work entitled 'Vichnou' and citing that manuscript as the source for the authority record created by the National Library of France for that work

Properties: [R43](#) carried out by (performed): [F44](#) Bibliographic Agency
[R48](#) assigned to (was assigned by): [F2](#) Expression
[R49](#) assigned (was assigned by): [F3](#) Manifestation Product Type
[R53](#) assigned (was assigned by): [F4](#) Manifestation Singleton

F42 Representative Expression Assignment

Subclass of: [E13](#) Attribute Assignment

Scope note: This class comprises activities through which an Agency declares (implicitly or explicitly) that a given instance of F2 Expression is representative for a given F15 Complex Work, i.e., that some attributes of that instance of F2 Expression (most prominently, information about the title) can be inferred to also apply to that instance of F15 Complex Work, no matter in which particular expression it is realised.

The reasoning behind this is that the Work title is known through the title of an Expression that is deemed representative of the Work, and the title of the representative Expression is known through the title of a Manifestation that is deemed representative of the Expression that is representative of the Work.

For instance, by using the qualified uniform title 'Poe, Edgar Allan, 1809-1849. Murders in the rue Morgue (French)' for the French rendition of Poe's 'Murders in the rue Morgue' by

Baudelaire, an Agency implicitly states that the French text does not constitute a representative F2 Expression for Poe's F1 Work, however the original English text does constitute a representative F2 Expression for Poe's F1 Work.

Examples: Selecting the text embodied in the 1775 edition of Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' as the representative expression for Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' and citing that publication as the source for the authority record created by the Library of Congress for that work

Selecting the musical notation embodied in the 2007 edition of John Tavener's musical work entitled 'The Eternal Sun' as the representative expression for John Tavener's musical work entitled 'The Eternal Sun' and citing that publication as the source for the authority record created by the National Library of France for that work

Selecting the publication expression of the 1983 edition of Stanley Karnow's textual work entitled 'Vietnam, the war nobody won' as the representative expression for the series entitled 'Headline series' and citing that publication as the source for the authority record created by the Library of Congress for that series

Selecting the publication expression of the issue dated October 2002 of the periodical entitled 'The New Courier' as the representative expression of the periodical entitled 'The New Courier' and citing that publication as the source for the bibliographic record created by the National Library of France for that periodical

Selecting the content of the manuscript identified by shelfmark 'MS-8282' within the collections of the National Library of France, Department for Music, as the representative expression of Stanislas Champein's musical work entitled 'Vichnou' and citing that manuscript as the source for the authority record created by the National Library of France for that work

Properties: [R44](#) carried out by (performed): [F44](#) Bibliographic Agency
[R50](#) assigned to (was assigned by): [F15](#) Complex Work
[R51](#) assigned (was assigned by): [F2](#) Expression

F43 Identifier Rule

Subclass of: [E29](#) Design or Procedure
[F2](#) Expression

Scope note: This class comprises sets of instructions relating to the formulation of a unique identifier.

Examples: AACR2R 25.25-25.35F1

RAK-Musik (Revidierte Ausgabe 2003), Chapter 6

AFNOR Z 44-079

F44 Bibliographic Agency

Subclass of: [E40](#) Legal Body

Scope note: This class comprises agents who create the bibliographic description of publications and perform the authority control associated with such descriptions, for the description of copies of such publications actually held by libraries, and for the description of unique documents (manuscripts, objects...) held by libraries.

The activity of creating such descriptions implies that one has to make decisions (as to the uniform title for a work, as to whether an arrangement still belongs to the same work or is definitely a new work, etc.). Since such decisions always are debatable and different agencies can make different decisions about the same real-world entities, it is important to document which agency made which decision.

Examples: The National Library of France, identified in bibliographic and authority records by the code 'FRBNF' at the beginning of INTERMARC field 001

F50 Controlled Access Point

Subclass of: [E13](#) Identifier

Scope note: This class comprises identifiers that are not only designed to be unique for the thing they identify, but also to ensure, by following adequate rules based on widely known and accepted properties for their generation, that an independent agency using the same rule would create the same identifier for the same thing.

F50 Controlled Access Point covers the notion of both “preferred” and “variant” forms. It does not cover the notion of “cross references”. A cross reference may not uniquely identify one entity, but can be shared by two or more entities, regardless of whether it displays the same structural characteristics as preferred controlled access points.

Examples: ‘Maxwell equations’ [preferred subject access point from LCSH, <http://lccn.loc.gov/sh85082387>, as of 19 November 2012]

‘Equations, Maxwell’ [variant subject access point, from the same source]

‘Gončarova, Natal’â Sergeevna (1881-1962)’ [preferred access point for a personal name, from the authority file of the National Library of France, <http://catalogue.bnf.fr/ark:/12148/cb119547494/PUBLIC>, as of 15 June 2012]

‘Гончарова, Наталья Сергеевна (1881-1962)’ [parallel access point from the same source]

‘Goncharova, Natalia (1881-1962)’ [variant access point from the same source]

F51 Pursuit

Subclass of: [E7](#) Activity

Scope note: This class comprises periods of continuous activity of an Actor in a specific professional or creative domain or field.

Examples: Natalya Goncharova working as a set and costume designer, painter, illustrator and poet in Russia and France in the first half of the 20th century

Satyajit Ray working as a filmmaker, writer, composer and graphic designer in India in the second half of the 20th century

Folger Shakespeare Library in Washington studying the works of William Shakespeare

M. & N. Hanhart working in lithographic publishing (1839-1882)

Properties: [R59](#) had typical subject (was typical subject of): [E1](#) CRM Entity
[R60](#) used to use language (was language used by): [E56](#) Language
(R60.1 has type of use: [E55](#) Type)

F52 Name Use Activity

Subclass of: [E13](#) Attribute Assignment

Scope note: This class comprises periods of continuous use of a specific instance of E41 Appellation for a particular instance of E1 CRM Entity by an E39 Actor. It includes in particular the use of the name by its carrier. Characteristically, actors performing an activity may choose a particular appellation for themselves in the context of this activity. Such cases should be modelled by additionally classifying these activities as instances of F52 Name Use Activity.

It is possible to specify the type of name use, through the *P2 has type* property, e.g.: use of a pseudonym, use of a married name, use of a birth name, use of a blended name, use of a religious name, etc.

Examples: Using the pseudonym ‘Prince’ until 1993, and again from 2000 on

Using the pseudonym ‘Love Symbol’ from 1993 to 2000

Using the pseudonym ‘Lewis Carroll’ when authoring works of fiction (*P2 has type* E55 Type {use of a pseudonym})

Using the name ‘Charles Dodgson’ when authoring works of mathematics and logics (*P2 has type* E55 Type {use of a birth name})

Using the name ‘Mother Teresa’ instead of ‘Agnes Gonxha Bojaxhiu’ when becoming head of the Missionaries of Charity (*P2 has type* E55 Type {use of a religious name})

Using the name ‘Elizabeth Barrett Browning’ instead of ‘Elizabeth Barrett Barrett’ after marrying Robert Browning (*P2 has type* E55 Type {use of a married name})

Using the name ‘Antonio Villaraigosa’ instead of ‘Antonio Villar’ after marrying Corina Raigosa (*P2 has type* E55 Type {use of a blended name}) [*comment*: when former mayor of Los Angeles Antonio Villar and Corina Raigosa got married in 1987, the two spouses decided they would merge their two last names into one]

Properties: [R61](#) occurred in kind of context (was kind of context for): [E55](#) Type
[R62](#) was used for membership in (was context for): [E74](#) Group
[R63](#) named (was named by): [E1](#) CRM Entity
[R64](#) used name (was name used by): [E41](#) Appellation

F53 Material Copy

Subclass of: [F54](#) Utilized Information Carrier
[E25](#) Man-Made Feature

Scope note: This class comprises the features created on an instance of E84 Information Carrier when an F24 Publication Expression is copied to it by an F32 Carrier Production Event. This is the typical result of an electronic publishing process.

Examples: The physical features created on my PC’s hard drive when I clicked on the link <http://cidoc.ics.forth.gr/docs/cidoc_crm_version_4.0.pdf>, and thus downloaded a reproduction of the electronic file titled ‘Definition of the CIDOC Conceptual Reference Model... version 4.0’ that is stored on the ICS FORTH’s servers in Heraklion, Crete

F54 Utilised Information Carrier

Subclass of: [E84](#) Information Carrier
Superclass of: [F53](#) Material Copy
[F5](#) Item

Scope note: This class comprises physical objects that carry one or more instances of F24 Publication Expression.

Examples: The physical features created on my PC’s hard drive when I clicked on the link <http://cidoc.ics.forth.gr/docs/cidoc_crm_version_4.0.pdf>, and thus downloaded a reproduction of the electronic file titled ‘Definition of the CIDOC Conceptual Reference Model... version 4.0’ that is stored on the ICS FORTH’s servers in Heraklion, Crete (F53)

Any copy of the modern reprint publication of Marin Mersenne’s ‘Harmonie universelle’, Paris, 1986, ISBN ‘2-222-00835-2’ (F5)

Properties: [R6](#) carries (is carried by): F24 Publication Expression

2.7. FRBR Property Declaration

The properties of FRBR_{OO} are comprehensively declared in this section using the following format:

- Property names are presented as headings in bold face, preceded by unique property identifiers;
- The line “Domain:” declares the class for which the property is defined;
- The line “Range:” declares the class to which the property points, or that provides the values for the property;
- The line “Equal to:” declares the CIDOC CRM property that covers the same concept as the FRBR_{OO} property;
- The line “Is covered by shortcut:” declares the CIDOC CRM property that constitutes a shortcut for a more detailed path of which the FRBR_{OO} property is a part;
- The line “Superproperty of:” is a cross-reference to any subproperties the property may have;
- The line “Subproperty of:” is a cross-reference to any superproperties the property may have, in either CIDOC CRM or FRBR_{OO}. All FRBR_{OO} properties that fall under the scope of the CIDOC CRM are, either directly or indirectly, subproperties of at least one CIDOC CRM property. However, this line remains empty for FRBR_{OO} properties that are shortcuts of more developed paths that involve CIDOC CRM properties and/or their FRBR_{OO} subproperties;
- The line “Quantification:” declares the possible number of occurrences for domain and range class instances for the property. Possible values are enumerated in section 2.3;
- The line “Scope note:” contains the textual definition of the concept the property represents;
- The line “Examples:” contains a bulleted list of examples of instances of this property. If the example is also instance of a subproperty of this property, the unique identifier of the subclass is added in parenthesis. If the example instantiates two properties, the unique identifiers of both properties is added in parenthesis;
- The line “Properties:” introduces any properties the property may have.
- In some cases the superproperty of a property is listed as *Out of scope*. This indicates that the property that should be the superproperty is outside of the coverage of CIDOC CRM.

R1 is logical successor of (has successor)

Domain: [F1](#) Work

Range: [F1](#) Work

Subproperty of: [E70](#) Thing. [P130](#) shows features of (features are also found on): [E70](#) Thing

Quantification: (0,n;0,n)

Scope note: This property associates an instance of F1 Work which logically continues the content of another instance of F1 Work with the latter.

Examples: Albrecht Dürer's woodcut from 'The Large Woodcut Passion' entitled 'The Agony in the Garden' (F1, conceived ca 1496-98) *R1 is logical successor of* Albrecht Dürer's woodcut from 'The Large Woodcut Passion' entitled 'The Last Supper' (F1, dated 1510)

The first 'Star wars' trilogy (F15, 1977-1983) *R1 is logical successor of* The second 'Star wars' trilogy (F15, 1999-2005) [Note that the logical order does not follow, in either of these two examples, the chronological order]

R2 is derivative of (has derivative)

Domain: [F1](#) Work

Range: [F1](#) Work

Subproperty of: [E70](#) Thing. [P130](#) shows features of (features are also found on): [E70](#) Thing

Quantification: (0,n;0,n)

Scope note: This property associates an instance of F1 Work which modifies the content of another instance of F1 Work with the latter. The property *R2.1 has type* of this property allows for specifying the kind of derivation, such as adaptation, summarisation etc.

Examples: William Schuman's orchestration of Charles Ives's 'Variations on America' (F15) *R2 is derivative of* Charles Ives's 'Variations on America' (F15) *R2.1 has type* orchestration (E55)

Charles Ives's musical work entitled 'Variations on America' (F15) *R2 is derivative of* the musical work titled 'America' (F15) *R2.1 has type* variations (E55)

The musical work entitled 'America' (F15) *R2 is derivative of* the musical work entitled 'God save the King' (F15) *R2.1 has type* same tune with different lyrics (E55)

Properties: *R2.1 has type*: [E55](#) Type

R3 is realised in (realises)

Domain: [F1](#) Work

Range: [F22](#) Self-contained Expression

Superproperty of: [F14](#) Individual Work. [R9](#) is realised in (realises): [F22](#) Self-Contained Expression
[F20](#) Performance Work. [R12](#) is realised in (realises): [F25](#) Performance Plan
[F21](#) Recording Work. [R13](#) is realised in (realises): [F26](#) Recording
[F1](#) Work. [R40](#) has representative expression (is representative expression for): [F22](#) Self-Contained Expression

Subproperty of: [E70](#) Thing. [P130](#) shows features of (features are also found on): [E70](#) Thing

Quantification: (0,n;1,1)

Scope note:	This property associates an instance of F22 Self-Contained Expression with an instance of F1 Work. This property expresses the association that exists between an expression (F22) and the work that this expression conveys. The semantics of the association will be different depending on what specific subtype of F1 Work the work is an instance of. If the work is an instance of F14 Individual Work, the F22 Self-Contained Expression completely conveys the individual work. If the work is an instance of F15 Complex Work, the F22 Self-Contained Expression conveys an alternative member of the complex work. Our factual knowledge of how a given work is realised into an expression is often limited and this property makes it possible to express the association between instances of F22 Self-Contained Expression and the work it conveys without using the more developed paths. The property <i>R3.1 has type</i>: E55 Type allows for specifying the role played by the referred to expression in the overall bibliographic history of the work (e.g., ‘progenitor expression’, on which all other expressions of the same work are based; ‘reference for canonical citations’, in the sense of the HuCit ontology developed by Matteo Romanello and Michele Pasin; ‘earliest draft’, ‘intermediate draft’, ‘final clean draft’, ‘princeps edition’, etc.).
Examples:	Dante’s work entitled ‘Inferno’ (F15) <i>R3 is realised in</i> the Italian text of Dante’s ‘Inferno’ as found in the authoritative critical edition <i>La Commedia secondo l’antica vulgata a cura di Giorgio Petrocchi</i>, Milano: Mondadori, 1966-67 (= <i>Le Opere di Dante Alighieri</i>, Edizione Nazionale a cura della Società Dantesca Italiana, VII, 1-4) (F22) <i>R3.1 has type</i> authoritative critical edition (E55) Mozart’s work entitled ‘Il dissoluto punito ossia il Don Giovanni’ (F15) <i>R3 is realised in</i> the notated music of the Prague version, as found on manuscript Ms 1548 of the National Library of France (F22) <i>R3.1 has type</i> autograph version (E55)
Properties:	R3.1 has type: E55 Type

R4 carriers provided by (comprises carriers of)

Domain:	F2 Expression
Range:	F3 Manifestation Product Type
Superproperty of:	F2 Expression. R41 has representative manifestation product type (is representative manifestation product type for): F3 Manifestation Product Type
Subproperty of:	E73 Information Object. P128 is carried by: E24 Physical Man-Made Thing. P2 has type: E55 Type
Quantification:	(1,n;0,n)
Scope note:	This property associates a publication, i.e. an instance of F3 Manifestation Product Type, with an instance of F2 Expression, which all exemplars of that publication should carry, as long as they are recognised as complete exemplars of that publication. Typically, this property is observed on one exemplar of a publication, and extrapolated to all other exemplars of the same publication. This property is a shortcut of: F2 Expression <i>P165i is incorporated in</i> F24 Publication Expression <i>CLR6i should be carried by</i> F3 Manifestation Product Type.
Examples:	The text of Marin Mersenne’s ‘Harmonie universelle’ (F22) <i>R4 carriers provided by</i> publication identified by ISBN ‘2-222-00835-2’ (F3) A recording of the Atrium Musicae Ensemble’s performance of a fragment of Euripides’ textual and musical work entitled ‘Orestes’ (F26) <i>R4 carriers provided by</i> the CD entitled ‘Musique de la Grèce antique = Ancient Greek music = Griechische Musik der Antike’, released in 2000 and identified by UPC/EAN ‘794881601622’ (F3)

R5 has component (is component of)

Domain:	F2 Expression
Range:	F22 Self-Contained Expression
Subproperty of:	E89 Propositional Object. P148 has component (is component of): E89 Propositional Object
Quantification:	(0,n;0,n)
Scope note:	This property associates an F2 Expression X with a structural component Y that conveys in itself the complete concept of a work that is member of (R10) the overall work realized by X. It does not cover the relationship that exists between pre-existing expressions that are re-used in a new, larger expression and that new, larger expression. Such a relationship is modelled by <i>P165 incorporates</i>.
Examples:	The Italian text of Dante's textual work entitled 'Divina Commedia' (F22) <i>R5 has component</i> the Italian text of Dante's textual work entitled 'Inferno' (F22) The musical notation of Mozart's Singspiel entitled 'Die Zauberflöte' (F22) <i>R5 has component</i> the musical notation of Mozart's aria entitled 'Der Hölle Rache', also known as 'The Queen of the Night's Aria' (F22) The visual content of the map entitled 'Wales – The Midlands – South West England', scale 1:400,000, issued by Michelin in 2005 (F22) <i>R5 has component</i> the visual content of the inset entitled 'Liverpool', scale 1:200,000, set within the compass of the map titled 'Wales – The Midlands – South West England', scale 1:400,000, issued by Michelin in 2005 (F22)

R6 carries (is carried by)

Domain:	F54 Utilised Information Carrier
Range:	F24 Publication Expression
Subproperty of:	E24 Physical Man-Made Thing. P128 carries (is carried by): E73 Information Object
Quantification:	(1,1;0,n)
Scope note:	This property associates an instance of F54 Utilized Information Carrier with the unique instance of F24 Publication Expression it carries.
Examples:	The British Library's holding identified by shelfmark 'DSC 9078.177 vol 19' (F5) <i>R6 carries</i> the entire content (text, layout, publisher logo, etc.) of the publication entitled 'Functional Requirements for Bibliographic Records: final report', issued by publisher named 'K. G. Saur' in 1998 (F24)

R7 is example of (has example)

Domain:	F5 Item
Range:	F3 Manifestation Product Type
Subproperty of:	E1 CRM Entity. P2 has type (is type of): E55 Type
Quantification:	(1,1;0,n)
Scope note:	This property associates a publication with one of its exemplars. It is a shortcut of the more developed path: F5 Item <i>R28i was produced by</i> F32 Carrier Production <i>R26 produced things of type (was produced by)</i>: F3 Manifestation Product Type.
Examples:	The item held by the National Library of France and identified by shelf mark 'Res 8 P 10' (F5) <i>R7 is example of</i> the edition of Amerigo Vespucci's textual and cartographic work entitled 'Mundus novus' issued in Paris ca. 1503-1504 (F3)

R8 consists of (forms part of)

Domain:	F13 Identifier
Range:	E90 Symbolic Object
Subproperty of:	E90 Symbolic Object. P106 is composed of (forms part of): E90 Symbolic Object
Quantification:	(0,n;0,n)
Scope note:	This property associates an instance of F13 Identifier with one of the non-syntactic instances of E90 Symbolic Object which form part of it.
Examples:	Uniform title ‘The Adoration of the Shepherds (Coventry)’ (F50) <i>R8 consists of</i> ‘The Adoration of the Shepherds’ (E35), and <i>R8 consists of</i> ‘Coventry’ (E48) Uniform title ‘Rite of spring (Choreographic Work : Bausch)’ (F50) <i>R8 consists of</i> ‘Rite of spring’ (E35), <i>R8 consists of</i> ‘Choreographic Work’ (F12), and <i>R8 consists of</i> ‘Bausch’ (F12) Uniform title ‘King Kong (1933)’ (F50) <i>R8 consists of</i> ‘King Kong’ (E35), and <i>R8 consists of</i> ‘1933’ (E50) Controlled access point ‘Guillaume, de Machaut, ca. 1300-1377’ (F50) <i>R8 consists of</i> ‘Guillaume, de Machaut’ (F12), and <i>R8 consists of</i> ‘ca. 1300-1377’ (E90) Controlled access point ‘Univerza v Ljubljani. Oddelek za bibliotekarstvo’ (F50) <i>R8 consists of</i> ‘Univerza v Ljubljani’ (F12), and <i>R8 consists of</i> ‘Oddelek za bibliotekarstvo’ (F12) ISBN ‘978-002-002-0’ (F13) <i>R8 consists of</i> ‘978’ (E90) indicating the Nigerian ISBN Agency, <i>R8 consists of</i> ‘002’ (E90) indicating the Nigerian Institute of International Affairs, <i>R8 consists of</i> ‘002’ (E90) used for the publication entitled ‘Nigeria’s international economic relations’, and <i>R8 consists of</i> ‘0’ (E90)

R9 is realised in (realises)

Domain:	F14 Individual Work
Range:	F22 Self-Contained Expression
Subproperty of:	F1 Work. R3 is realised in (realises): F22 Self-Contained Expression
Quantification:	(1,1;1,1)
Scope note:	This property associates an F14 Individual Work with the unique F22 Self-Contained Expression that completely conveys it. It is a short cut for the more developed path: F14 Individual Work <i>R19i was realised through</i> F28 Expression Creation <i>R17 created</i> F22 Self-Contained Expression.
Examples:	Abstract content of Giovanni Battista Piranesi’s graphic work entitled ‘Carcere XVI: the pier with chains: 2nd state’ (F14) <i>R9 is realised in</i> Giovanni Battista Piranesi’s graphic work entitled ‘Carcere XVI: the pier with chains: 2nd state’ (F22) Abstract content of the English text of the 1855 edition of Walt Whitman’s textual work entitled ‘Leaves of Grass’ (F14) <i>R9 is realised in</i> the English text of the 1855 edition of Walt Whitman’s textual work entitled ‘Leaves of Grass’ (F22)

R10 has member (is member of)

Domain:	F15 Complex Work
Range:	F1 Work
Subproperty of:	E89 Propositional Object. P148 has component (is component of): E89 Propositional Object

Quantification:	(2,n:0,n)
Scope note:	This property associates an instance of F15 Complex Work with an instance of F1 Work that forms part of it.
Examples:	Dante's textual work entitled 'Divina Commedia' (F15) <i>R10 has member</i> Dante's textual work entitled 'Inferno' (F15) Dante's textual work entitled 'Inferno' (F15) <i>R10 has member</i> the abstract content of the pseudo-old French text of Émile Littré's translation entitled 'L'Enfer mis en vieux langage françois et en vers' [a 19th century translation of Dante's 'Inferno' into old French] published in Paris in 1879 (F14) Giovanni Battista Piranesi's graphic work entitled 'Carceri' (F15) <i>R10 has member</i> Giovanni Battista Piranesi's graphic work entitled 'Carcere XVI: the pier with chains' (F15) Giovanni Battista Piranesi's graphic work entitled 'Carcere XVI: the pier with chains' (F15) <i>R10 has member</i> the abstract content of Giovanni Battista Piranesi's graphic work entitled 'Carcere XVI: the pier with chains: 2nd state' (F14)

R11 has issuing rule (is issuing rule of)

Domain:	F18 Serial Work
Range:	E29 Design or Procedure
Subproperty of:	
Quantification:	(0,n:0,n)
Scope note:	This property associates an instance of F18 Serial Work with the instance of E29 Design or Procedure that specifies the issuing policy planned by this Work, such as sequencing pattern, expected frequency and expected regularity. This property is a shortcut of the full path: F18 Serial Work <i>R23i was realised through</i> F30 Publication Event <i>P16 used specific object</i> E29 Design or Procedure.
Examples:	The serial entitled 'Quarterly journal of pure and applied mathematics', identified by ISSN '1549-6724' (F18) <i>R11 has issuing rule</i> to be issued every three months, on a regular basis, with each issue being numbered according to the pattern 'Vol. 1, no. 1 (2005)' that was observed by the Library of Congress's cataloguers on an exemplar of the first issue (E29)

R12 is realised in (realises)

Domain:	F20 Performance Work
Range:	F25 Performance Plan
Subproperty of:	F1 Work. R3 is realised in (realises): F22 Self-Contained Expression
Quantification:	(0,n:1,1)
Scope note:	This property associates an instance of F20 Performance Work with an instance of F25 Performance Plan that consists of signs (words, figures, etc.) which express the directions the instance of F20 Performance Work consists of.
Examples:	The concept of Sergei Radlov's mise-en-scène of a Yiddish translation of the textual work entitled 'King Lear' in Moscow in 1935 (F20) <i>R12 is realised in</i> the set of instructions for the production of a Yiddish translation of the textual work entitled 'King Lear', as directed by Sergei Radlov in Moscow in 1935 (F25) The concept of Pina Bausch's choreography of the ballet entitled 'Rite of spring' in Wuppertal in 1975 (F20) <i>R12 is realised in</i> the set of instructions for the production of the ballet entitled 'Rite of spring', as choreographed by Pina Bausch in Wuppertal in 1975 (F25)

The concept of Bruno Walter's performance of Gustav Mahler's 9th symphony in 1961 (F20) *R12 is realised in* the set of instructions by Bruno Walter for performing Gustav Mahler's 9th symphony, delivered by him to the Columbia Symphony Orchestra during rehearsals in Hollywood in 1961 (as partially documented in the CD entitled 'Bruno Walter conducts and talks about Mahler symphony No. 9: rehearsal & performance') (F25)

The concept of the "performance handbook" for Luigi Nono's musical work entitled 'À Pierre' (F20) *R12 is realised in* the set of instructions contained in the performance handbook for Luigi Nono's musical work entitled 'À Pierre' (F25)

R13 is realised in (realises)

Domain: [F21](#) Recording Work

Range: [F26](#) Recording

Subproperty of: [F1](#) Work. [R3](#) is realised in (realises): [F22](#) Self-Contained Expression

Quantification: (0,n;0,1)

Scope note: This property associates an instance of F21 Recording Work with an instance of F26 Recording realising the instance of F21 Recording Work.

This is a shortcut of the more elaborated path through R22 *was realised through*, F29 Recording Event and R21 *created*, which should be used when information about the recording event is available.

Examples: The concept of the third alternate take of the musical work entitled 'Blue Hawaii' as performed by Elvis Presley in Hollywood, Calif., Radio Recorders, on March 22nd, 1961 (F21) *R13 is realised in* the set of signs that make up the third alternate take of the musical work entitled 'Blue Hawaii' as performed by Elvis Presley in Hollywood, Calif., Radio Recorders, on March 22nd, 1961 (F26)

The concept of making a photograph of the three Allied leaders at Yalta in February 1945 (F21) *R13 is realised in* the visual content of the famous photograph of the three Allied leaders at Yalta in February 1945 (F26)

Oceania Project's concept of making a large digital acoustic data archive dedicated to East Australian humpback whale songs (F21) *R13 is realised in* the audio content of the album entitled 'Songlines – Songs of the East Australian Humpback Whales' released in 2011 (F26)

The concept of recording Louise Bourgeois's artistic activity in the documentary movie entitled 'Louise Bourgeois: The Spider, the Mistress, and the Tangerine' (F21) *R13 is realised in* the audiovisual content of the documentary movie entitled 'Louise Bourgeois: The Spider, the Mistress, and the Tangerine' (F26)

R15 has fragment (is fragment of)

Domain: [F2](#) Expression

Range: [F23](#) Expression Fragment

Subproperty of: [E90](#) Symbolic Object. [P106](#) is composed of (forms part of): [E90](#) Symbolic Object

Quantification: (0,n;0,n)

Scope note: This property associates the fragment of an expression and the expression of which it is a fragment.

Examples: The ancient Greek text of the four stanzas from an ode by Sappho that were quoted by Pseudo-Longinus in his textual work entitled 'On the sublime' (F23) *R15 is fragment of* the complete ancient Greek text, now irremediably lost, of Sappho's ode currently identified as Sappho's poem #2 (F22)

The statement ‘fasc. 111’ (abridgement for ‘fascicle no. 111’) indicating the sequential position of the publication identified by ISBN ‘2-7018-0037-4’ within the series entitled ‘Bibliothèque des Écoles françaises d’Athènes et de Rome’ and identified by ISSN ‘0257-4101’ (F23) *R15 is fragment of* the overall content of the publication identified by ISBN ‘2-7018-0037-4’ (F24)

R16 initiated (was initiated by)

Domain: [F27](#) Work Conception

Range: [F1](#) Work

Subproperty of: [E65](#) Creation. [P94](#) has created (was created by): [E28](#) Conceptual Object

Quantification: (1,n;1,1)

Scope note: This property associates the first conception of a work and the work itself that ensued from a given initial idea.

It marks the origin of the causality chain that results in a work’s coming into existence.

Examples: Ludwig van Beethoven’s decision to compose a fifth symphony (F27) *R16 initiated* Ludwig van Beethoven’s Fifth Symphony (F15)

Pablo Picasso’s acceptance, in 1930, of Ambroise Vollard’s commission for a set of 100 etchings, now known as the ‘Vollard Suite’ (F27) *R16 initiated* the ‘Vollard Suite’ (F15)

René Goscinny’s and Albert Uderzo’s decision to collaborate on the comic book entitled ‘Asterix in Britain’ (F27) *R16 initiated* the comic book entitled ‘Asterix in Britain’ (F15)

The creative spark that motivated Oscar Wilde, by May 1897, to write a poem inspired by his stay in the Reading prison in 1895-1897 (F27) *R16 initiated* Oscar Wilde’s poem entitled ‘The ballad of the Reading gaol’ (F15)

R17 created (was created by)

Domain: [F28](#) Expression Creation

Range: [F2](#) Expression

Superproperty of: [F29](#) Recording Event. [R21](#) created (was created by): [F26](#) Recording
[F30](#) Publication Event. [R24](#) created (was created through): [F24](#) Publication Expression

Subproperty of: [E65](#) Creation. [P94](#) has created (was created by): [E28](#) Conceptual Object

Quantification: (1,1;1,n)

Scope note: This property associates the expression that was first externalised during a particular creation event with that particular creation event.

Examples: Richard Wagner’s writing the original manuscript of his opera entitled ‘Der fliegende Holländer’ (F28) *R17 created* the notational content of the original manuscript of Richard Wagner’s opera entitled ‘Der fliegende Holländer’ (F22)

Oscar Wilde’s writing the original manuscript of his poem entitled ‘The ballad of the Reading gaol’ (F28) *R17 created* the English text of Oscar Wilde’s poem entitled ‘The ballad of the Reading gaol’ (F22)

R18 created (was created by)

Domain: [F28](#) Expression Creation

Range: [F4](#) Manifestation Singleton

Subproperty of: [E12](#) Production. [P108](#) has produced (was produced by): [E24](#) Physical Man-Made Thing

Quantification: (1,n;0,1)

Scope note: This property associates an instance of F28 Expression Creation with the first physical objects in which the resulting instance of F2 Expression was embodied.

Examples: Emily Dickinson's creating the text of one of the several extant versions of her poem known as 'Safe in their alabaster chambers' (F28) *R18 created* the manuscript now identified as 'Massachusetts Cambridge Harvard University Houghton Library bMS Am 1118.3 (203c, 203d)' (F4)

Emily Dickinson's creating the text of another one of the several extant versions of her poem known as 'Safe in their alabaster chambers' (F28) *R18 created* the manuscript now identified as 'Massachusetts Cambridge Harvard University Houghton Library bMS Am 1118.5 (74c)' (F4)

The recording of the third alternate take of the musical work entitled 'Blue Hawaii' performed by Elvis Presley in Hollywood, Calif., Radio Recorders, on March 22nd, 1961 (F28) *R18 created* the master tape of the 3rd alternate take of the musical work entitled 'Blue Hawaii' performed by Elvis Presley in Hollywood, Calif., Radio Recorders, on March 22nd, 1961 (F4) (each individual take is a distinct expression)

The resource (a drawing) held by the New York Public Library and identified by call number '*MGZGB Far P Cop 1' (F4) *R18i was created by* the creation, by the artist named 'Peter Farmer', of a costume design for the character named 'War' in the Act III Masque of the seasons, in the Festival Ballet of London production of the choreographic work entitled 'Coppélia', with choreography by Jack Carter after Petipa (F28)

R19 created a realisation of (was realised through)

Domain: [F28](#) Expression Creation

Range: [F1](#) Work

Superproperty of: [F29](#) Recording Event. [R22](#) created a realisation of (was realised through): [F21](#) Recording Work

[F30](#) Publication Event. [R23](#) created a realisation of (was realised through): [F19](#) Publication Work

Subproperty of: [E7](#) Activity. [P16](#) used specific object (was used for): [E70](#) Thing

Quantification: (1,n;1,1)

Scope note: This property associates an instance of F28 Expression Creation with the corresponding instance of F14 Individual Work or an instance of F15 Complex Work of which the corresponding instance of F14 Individual Work is a member.

Examples: Giovanni Battista Piranesi's creating the image identified as 'Carcere XVI: the pier with chains: 2nd state' (F28) *R19 created a realisation of* the concept of Giovanni Battista Piranesi's graphic work entitled 'Carcere XVI: the pier with chains: 2nd state' (F14)

Recording Glenn Gould's performance of Johann Sebastian Bach's musical work entitled 'Toccata in C minor BWV 911' on May 15 & 16, 1979, in Toronto, Eaton's Auditorium (F29) *R19 created a realisation of* the concept of the recorded performance of Johann Sebastian Bach's musical work entitled 'Toccata in C minor BWV 911' by Glenn Gould on May 15 & 16, 1979, in Toronto, Eaton's Auditorium (F21)

R20 recorded (was recorded through)

Domain: [F29](#) Recording Event

Range: [E2](#) Temporal Entity

Subproperty of: [E7](#) Activity. [P15](#) was influenced by (influenced): [E5](#) Event. [P9B](#) forms part of: [E5](#) Event. [P9](#) consists of: [E5](#) Event

Quantification: (1,n:0,n)

Scope note: This property associates an instance of F29 Recording Event with the instance of E2 Temporal Entity which was captured.

Examples: The making of the recording of the third alternate take of the musical work entitled ‘Blue Hawaii’ as performed by Elvis Presley in Hollywood, Calif., Radio Recorders, on March 22nd, 1961 (F29) *R20 recorded* Elvis Presley’s performance of the musical work entitled ‘Blue Hawaii’ in Hollywood, Calif., Radio Recorders, on March 22nd, 1961 (F31)

R21 created (was created through)

Domain: [F29](#) Recording Event

Range: [F26](#) Recording

Subproperty of: [F28](#) Expression Creation. [R17](#) created (was created by): [F2](#) Expression

Quantification: (1,n:1,n)

Scope note: This property associates an instance of F29 Recording Event with the instance of F26 Recording that was created.

Examples: The making of the recording of the third alternate take of the musical work entitled ‘Blue Hawaii’ as performed by Elvis Presley in Hollywood, Calif., Radio Recorders, on March 22nd, 1961 (F29) *R21 created* the set of signs that make up the third alternate take of the musical work entitled ‘Blue Hawaii’ as performed by Elvis Presley in Hollywood, Calif., Radio Recorders, on March 22nd, 1961 (F26)

The making of the photograph of the three Allied leaders at Yalta in February 1945 (F29) *R21 created* the set of signs that make up the famous photograph of the three Allied leaders at Yalta in February 1945 (F26)

The making of the recording of an East Australian humpback whale song in 1994 in the framework of the Oceania Project (F29) *R21 created* the set of signs that make up the sound recording of an East Australian humpback whale song in 1994 (F26)

Filming Louise Bourgeois at work in the context of the shooting of the documentary movie entitled ‘Louise Bourgeois: The Spider, the Mistress, and the Tangerine’ (F29) *R21 created* the set of signs that make up the sequences from the documentary movie entitled ‘Louise Bourgeois: The Spider, the Mistress, and the Tangerine’ in which Louise Bourgeois is seen at work (F26)

R22 created a realisation of (was realised through)

Domain: [F29](#) Recording Event

Range: [F21](#) Recording Work

Subproperty of: [F28](#) Expression Creation. [R19](#) created a realisation of (was realised through): [F1](#) Work

Quantification: (0,1:0,n)

Scope note: This property associates an instance of F29 Recording Event with the instance of F21 Recording Work it realised.

Examples: The making of the recording of the third alternate take of the musical work entitled ‘Blue Hawaii’ as performed by Elvis Presley in Hollywood, Calif., Radio Recorders, on March 22nd, 1961 (F29) *R22 created a realisation of* the concept of the third alternate take of the musical work entitled ‘Blue Hawaii’ as performed by Elvis Presley in Hollywood, Calif.

The making of the photograph of the three Allied leaders at Yalta in February 1945 (F29) *R22*

created a realisation of the concept of making a photograph of the three Allied leaders at Yalta in February 1945 (F21)

The making of the recording of an East Australian humpback whale song in 1994 in the framework of the Oceania Project (F29) *R22 created a realisation of* Oceania Project's concept of making a large digital acoustic data archive dedicated to East Australian humpback whale songs (F21)

Filming Louise Bourgeois at work in the context of the shooting of the documentary movie entitled 'Louise Bourgeois: The Spider, the Mistress, and the Tangerine' (F29) *R22 created a realisation of* the concept of recording Louise Bourgeois's artistic activity in the documentary movie entitled 'Louise Bourgeois: The Spider, the Mistress, and the Tangerine' (F21)

R23 created a realisation of (was realised through)

Domain: [F30](#) Publication Event
Range: [F19](#) Publication Work
Subproperty of: [F28](#) Expression Creation. [R19](#) created a realisation of (was realised through): [F1](#) Work
Quantification: (0,1:0,n)
Scope note: This property associates an instance of F30 Publication Event with the instance of F19 Publication Work it realised.
Examples: Establishing in 1972 the layout, features, and prototype for the publication of Stephen Crane's complete poems (F30) *R23 created a realisation of* Cornell University Press's concepts for an edition of Stephen Crane's complete poems (F19)

R24 created (was created through)

Domain: [F30](#) Publication Event
Range: [F24](#) Publication Expression
Subproperty of: [F28](#) Expression Creation. [R17](#) created (was created by): [F2](#) Expression
Quantification: (1,n:1,n)
Scope note: This property associates the instance of F24 Publication Expression that was created during a particular F30 Publication Event with that F30 Publication Event.
Examples: Establishing in 1972 the layout, features, and prototype for the publication of Stephen Crane's complete poems (F30) *R24 created* the set of signs and instructions as to manufacturing established by Cornell University Press for a publication of Stephen Crane's complete poems (F24)

R25 performed (was performed in)

Domain: [F31](#) Performance
Range: [F25](#) Performance Plan
Subproperty of: [E7](#) Activity. [P33](#) used specific technique (was used by): [E29](#) Design or Procedure
Quantification: (0,n:0,n)
Scope note: This property associates an instance of F31 Performance with the instance of F25 Performance Plan to which all those participating in the performance were supposed to conform.
Examples: Performing the first performance of a Yiddish translation of 'King Lear', as directed by Sergei Radlov, in Moscow, at the Moscow State Jewish Theatre, on February 10, 1935 (F31) *R25*

performed the set of instructions for the production of a Yiddish translation of ‘King Lear’, directed by Sergei Radlov in Moscow in 1935 (F25)

Performing the ballet ‘Rite of spring’, as choreographed by Pina Bausch, in Avignon, at the Popes’ Palace, on July 7, 1995 (F31) *R25 performed* the set of instructions for the production of the ballet ‘Rite of spring’, as choreographed by Pina Bausch (F25)

R26 produced things of type (was produced by)

Domain: [F32](#) Carrier Production Event

Range: [F3](#) Manifestation Product Type

Subproperty of: [E12](#) Production. [P108](#) has produced: [E24](#) Physical Man-MadeThing. [P2](#) has type: [E55](#) Type

Quantification: (1,n;0,n)

Scope note: This property associates an instance of F32 Carrier Production Event with the instance of F3 Manifestation Product Type it produced items of.

Examples: The production of copies of the publication entitled ‘Codex Manesse: die Miniaturen der großen Heidelberger Liederhandschrift, herausgegeben und erläutert von Ingo F. Walther unter Mitarbeit von Gisela Siebert’, 3rd edition, Insel-Verlag, 1988 (F32) *R26 produced things of type* the publication identified as ‘Codex Manesse: die Miniaturen der großen Heidelberger Liederhandschrift, herausgegeben und erläutert von Ingo F. Walther unter Mitarbeit von Gisela Siebert’, 3rd edition, Insel-Verlag, 1988 (F3)

The production of copies of the publication entitled ‘Ordnance Survey Explorer Map 213, Aberystwyth & Cwm Rheidol’, ISBN ‘0-319-23640-4’ (folded), 1:25,000 scale, released in May 2005 (F32) *R26 produced things of type* the publication identified by ISBN ‘0-319-23640-4’ (F3)

The production of copies of the sound recording entitled ‘The Glory (????) of the human voice’, RCA Victor Gold Seal GD61175, containing recordings of musical works performed by Florence Foster Jenkins (F32) *R26 produced things of type* the publication entitled ‘The Glory (????) of the human voice’ and identified by the label and label number ‘RCA Victor Gold Seal GD61175’ (F3)

The production of a second print run, in 1978, of the publication titled ‘The complete poems of Stephen Crane, edited with an introduction by Joseph Katz’ (identified by ISBN ‘0-8014-9130-4’) (F32) *R26 produced things of type* the publication, dated 1972, entitled ‘The complete poems of Stephen Crane, edited with an introduction by Joseph Katz’ (identified by ISBN ‘0-8014-9130-4’) (F3)

R27 used as source material (was used by)

Domain: [F32](#) Carrier Production Event

Range: [F24](#) Publication Expression

Subproperty of: [E7](#) Activity. [P16](#) used specific object (was used for): [E70](#) Thing

Quantification: (0,n;0,n)

Scope note: This property associates an instance of F32 Carrier Production Event with the set of signs provided by the publisher to be carried by all of the produced items.

Examples: The production of copies of the publication identified by ISBN ‘1-86197-612-7’ (F32) *R27 used as source material* the final set of signs sent by the publisher named ‘Profile Books’ to their printer for the production of copies of the publication identified by ISBN ‘1-86197-612-7’ (F24)

R28 produced (was produced by)

Domain:	F32 Carrier Production Event
Range:	F54 Utilized Information Carrier
Subproperty of:	E12 Production. P108 has produced (was produced by): E24 Physical Man-Made Thing
Quantification:	(0,n:1,1)
Scope note:	This property associates an instance of F32 Carrier Production Event with any one of the produced items (i.e., the instances of F5 Item or F53 Material Copy).
Examples:	The production of copies of the publication entitled ‘Codex Manesse: die Miniaturen der großen Heidelberger Liederhandschrift, herausgegeben und erläutert von Ingo F. Walther unter Mitarbeit von Gisela Siebert’, 3rd edition, Insel-Verlag, 1988 (F32) <i>R28 produced</i> the National Library of France’s holding identified by shelf mark ‘C-1604(2)’ (F5) The production of copies of the publication entitled ‘Ordnance Survey Explorer Map 213, Aberystwyth & Cwm Rheidol’, ISBN 0-319-23640-4 (folded), 1:25,000 scale, released in May 2005 (F32) <i>R28 produced</i> the National Library of Wales’ holding identified by holding information ‘MAP, STORFA/STACK ; FLAT MAP, C16 (20/1), Sheet 213, c.135/5/2’ (F5) The production of copies of the sound recording entitled ‘The Glory (????) of the human voice’, RCA Victor Gold Seal GD61175 (F32) <i>R28 produced</i> the London Public Library’s holding identified by call number ‘R J416.G1’ (F5) The second print run, occurring in 1978, of the publication dated of 1972 and entitled ‘The complete poems of Stephen Crane, edited with an introduction by Joseph Katz’ (identified by ISBN ‘0-8014-9130-4’) (F32) <i>R28 produced</i> Universitätsbibliothek Passau’s holding identified by call number ‘00/HT 4801.978 K2’ (F5)

R29 reproduced (was reproduced by)

Domain:	F33 Reproduction Event
Range:	E84 Information Carrier
Subproperty of:	E7 Activity. P16 used specific object (was used for): E70 Thing
Quantification:	(1,n:0,n)
Scope note:	This property associates an instance of F33 Reproduction Event with an instance of E84 Information Carrier it reproduces.
Examples:	Making a photocopy of an exemplar of Eran Guter’s dissertation entitled ‘Where languages end: Ludwig Wittgenstein at the crossroads of music, language, and the world’ (F33) <i>R29 reproduced</i> one of the original exemplars of Eran Guter’s dissertation (E84)

R30 produced (was produced by)

Domain:	F33 Reproduction Event
Range:	E84 Information Carrier
Subproperty of:	E12 Production. P108 produced (was produced by): E24 Physical Man-Made Thing
Quantification:	(1,n:0,1)
Scope note:	This property associates an instance of F33 Reproduction Event with an instance of E84 Information Carrier it produces.
Examples:	Making a photocopy of an exemplar of Eran Guter’s dissertation entitled ‘Where languages end: Ludwig Wittgenstein at the crossroads of music, language, and the world’ (F33) <i>R30</i>

produced the New York Public Library holding identified by call number ‘JMD 04-1060’ (E84)

R31 is reproduction of (has reproduction)

Domain: [E84](#) Information Carrier

Range: [E84](#) Information Carrier

Subproperty of: [E70](#) Thing. [P130](#) shows features of (features are also found on): [E70](#) Thing

Quantification: (0,1:0,n)

Scope note: This property associates an instance of E84 Information Carrier which is a reproduction of another instance of E84 Information Carrier with the latter. It is considered that a reproduction of multiple originals resulting in a single product requires a merging of those objects prior to the reproduction. Therefore an Information Carrier is regarded to be a reproduction of one and only one original. This property is a shortcut of the more fully developed path from E84 Information Carrier through *R30 produced (was produced by)*, F33 Reproduction Event *R29 reproduced (was reproduced by)* to E84 Information Carrier.

Examples: The New York Public Library holding identified by call number ‘JMD 04-1060’ (E84) *R31 is reproduction of* one of the original exemplars of Eran Guter’s dissertation (E84)

R32 is warranted by (warrants)

Domain: [F35](#) Nomen Use Statement

Range: [F52](#) Name Use Activity

Subproperty of: [E89](#) Propositional Object. [P67](#) refers to (is referred to by): [E1](#) CRM Entity

Quantification: (0,n:0,n)

Scope note: This property associates an instance of F35 Nomen Use Statement with an instance of F52 Name Use Activity which provides evidence for the use of the particular nomen in the stated sense. The association between this name use activity and a source for it may be described by the property *P70 documents (is documented in)*.

Examples: The variant controlled access point in record n 79021736, tagged in MARC21 format, ‘400 1_ |a Пруст, Марсель, |d 1871-1922’ found in the Library of Congress authorities as of 15 June 2012 (F35) *R32 is warranted by* the use of the name ‘Марсель Пруст’ in Cyrillic script (F52), for which evidence can be found in the publication referred to as ‘Andreev, L.G. Marsel’ Prust, 1968’ in the authority record established by the Library of Congress for Marcel Proust

The preferred access point in record sh 85109469, tagged in MARC21 format, ‘150 __ |a Quantum theory’ found in the Library of Congress authorities as of 15 June 2012 (F35) *R32 is warranted by* the use of the phrase ‘quantum theory’ in the English language (F52), as attested to by references to the NASA and INSPEC thesauri in the authority record established by the Library of Congress

The preferred access point in record sh 85074230 in the LCSH authority file tagged in MARC 21 in the field ‘150__ |a Lamniformes’ (F35) *R32 is warranted by* the use of the term ‘Lamniformes’ for mackerel sharks in the entry on page 51 of *Fishes of the world* by J.S. Nelson published in 1994 (F52)

The preferred access point in record n 85118480 in the Library of Congress name authority file as of 19 November 2012 tagged in MARC 21 in the field ‘110 2_ |a Canadian Academic Centre in Italy’ (F35) *R32 is warranted by* the use of the name ‘Canadian Academic Centre in Italy’ on page 6 of the book ‘Lo Scavo di S. Giovanni di Ruoti ed il periodo tardoantico in Basilicata’ published in 1983

R33 has content

Domain:	F12 Nomen
Range:	E62 String
Subproperty of:	E1 CRM Entity. P3 has note: E62 String
Quantification:	(1,n;0,n)
Scope note:	This property associates an instance of F12 Nomen with one or more equivalent serialized content models for it. In digital form the symbol arrangement constituting an instance of F12 Nomen can only be represented through a particular encoding, for example ASCII or Latin1 for the Latin script. We call such a representation a content model. The property <i>R33.1 has encoding</i> : E55 Type allows for specifying the encoding of a particular associated content model. Together with this specification, a content model allows for unambiguously defining a nomen independently from the encoding used for representing the content.
Examples:	The term ‘earth’ encoded as ASCII (F12) <i>R33 has content</i> ‘0x65 0x61 0x72 0x74 0x68’ (E62) <i>R33.1 has encoding</i> ASCII (E55) The term ‘earth’ encoded as UNICODE UTF16 (F12) <i>R33 has content</i> ‘0x0065 0x0061 0x0072 0x0074 0x0068’ <i>R33.1 has encoding</i> UNICODE UTF16 (E55) The term ‘earth’ in Latin Arial font (F12) <i>R33 has content</i> ‘earth’ (E62) <i>R33.1 has encoding</i> printed Latin Arial (E55)
Properties:	R33.1 has encoding: E55 Type

R34 has validity period (is validity period of)

Domain:	F34 KOS
Range:	E52 Time-Span
Subproperty of:	Out of CRM Scope.
Quantification:	(1,1;0,n)
Scope note:	This property associates an instance of F34 KOS with the instance of E52 Time-Span describing the period for which the particular KOS Expression was regarded as valid by its maintainers. Each change of validity status of a nomen use statement within a KOS should be associated with a release change of the KOS. The individual time-span of a validity state of a nomen would be the union of the time-spans of the KOS in which the Nomen was declared to have the particular validity status (provisional, accepted, obsolete etc...).
Examples:	LCSH February 20 to March 19 2012 (F34) <i>R34 has validity period</i> February 20 to March 19 2012 (E52) DDC 19 (F34) <i>R34 has validity period</i> 1979 to 1989 (E52)

R35 is specified by (specifies)

Domain:	F35 Nomen Use Statement
Range:	F34 KOS
Subproperty of:	E89 Propositional Object. P148 has component (is component of): E89 Propositional Object
Quantification:	(1,1;1,n)
Scope note:	This property associates an instance of F35 Nomen Use Statement with an instance of F34 KOS in which the Nomen Use Statement has a given status. The property R35.1 allows for specifying the particular status of the nomen use statement within the KOS. An instance of R35

is specified by should have only one status.

- Examples: ‘acoustic surface wave device’ (F35) *R35 is specified by* INSPEC Thesaurus version January 1973 (F34) *R35.1 has status* valid (E55)
- ‘acoustic surface wave device’ (F35) *R35 is specified by* INSPEC Thesaurus version June 1978 (F34) *R35.1 has status* obsolete (E55)
- Properties: R35.1 has status: [E55](#) Type

R36 uses script conversion (is script conversion used in)

- Domain: [F35](#) Nomen Use Statement
- Range: [F36](#) Script Conversion
- Quantification: (0,1:0,n)
- Scope note: This property associates an instance of F35 Nomen Use Statement with the instance of F36 Script Conversion that was used to create the Nomen referred to in the Nomen Use Statement. The source of this conversion may or may not be explicitly mentioned. Is shortcut of: F35 Nomen Use Statement *R17 was created by (created)* F28 Expression Creation *P33 used specific technique (was used by)* E29 Design or Procedure.
- Examples: ‘001 FRBNF120864715’...‘100 \$w.0..ba....\$aDu\$mFu\$d0712-0770’ (F35) *R36 uses script conversion* Pinyin (F36)
- ‘001 FRBNF119547493’...‘100 w.0..barus.\$aGončarova\$mNatal’â Sergeevna\$d1881-1962’ (F35) *R36 uses script conversion* ISO 9:1995 (F36)

R37 states as nomen (is stated as nomen in)

- Domain: [F35](#) Nomen Use Statement
- Range: [F12](#) Nomen
- Subproperty of: [E89](#) Propositional Object. [P67](#) refers to (is referred to by): [E1](#) CRM Entity
- Quantification: (1,1:0,n)
- Scope note: This property associates an instance of F35 Nomen Use Statement with the instance of F12 Nomen for which it declares usage.
- Examples: ‘001 FRBNF120864715’...‘100 \$w.0..ba....\$aDu\$mFu\$d0712-0770’ (F35) *R37 states as nomen* ‘Du, Fu (0712-0770)’ (F50)
- ‘001 FRBNF119547493’...‘100 w.0..barus.\$aGončarova\$mNatal’â Sergeevna\$d1881-1962’ (F35) *R37 states as nomen* ‘Gončarova, Natal’â Sergeevna (1881-1962)’ (F50)
- ‘010 __ |a n 79021736’...‘400 1_ |a Пруст, Марсель, |d 1871-1922’ (F35) *R37 states as nomen* ‘Пруст, Марсель, 1871-1922’ (F50)
- ‘010 __ |a sh 85074230’...‘150 __ |a Lamniformes’ (F35) *R37 states as nomen* ‘Lamniformes’ (F12)
- ‘010 __ |a sh 85074230’...‘053 _0 |a QL638.94.L36 |c Zoology’ (F35) *R37 states as nomen* the Library of Congress classification number QL638.94.L36 (F12)
- ‘ID: 300024668’...‘navaja (C,U,Spanish,UF,U,SN)’ *R37 states as nomen* ‘navaja’(F12) (“used for” term, Getty Art & Architecture Thesaurus Online, retrieved 19/11/2012)
- ‘ID: 7010879’...‘Candia (H,V,Greek (transliterated),U) ...Venetian corruption of Arabic name, used from 13th cen.’ *R37 states as nomen* ‘Candia’ (F12) (“historical term”, Getty Thesaurus of Geographic Names Online, retrieved 19/11/2012)

‘ID: 7010879’...‘Ērakleion (NA,V,Greek (transliterated),U)’ *R37 states as nomen* ‘Ērakleion’ (F12) (“not-applicable term”, Getty Thesaurus of Geographic Names Online, retrieved 19/11/2012)

R38 refers to thema (is thema of)

Domain: [F35](#) Nomen Use Statement

Range: [E1](#) CRM Entity

Is covered by shortcut: [E32](#) Authority Document. [P71](#) lists: [E1](#) CRM Entity is shortcut of: [F34](#) KOS. [R35B](#) specifies: [F35](#) Nomen Use Statement. [R38](#) refers to thema: [E1](#) CRM Entity

Quantification: (1,1:0,n)

Scope note: This property associates an instance of F35 Nomen Use Statement with the instance of E1 CRM Entity for which it declares the usage of a nomen.

Examples: ‘010 __ |a sh 85109469’...‘150 __ |a Quantum theory’ [preferred subject access point from LCSH, <http://lccn.loc.gov/sh85109469>, as of 15 June 2012] (F35) *R38 refers to thema* the branch of physics known as quantum theory (F6 = E28)

R39 is intended for (is target audience in)

Domain: [F35](#) Nomen Use Statement

Range: [E74](#) Group

Subproperty of: [E89](#) Propositional Object. [P67](#) refers to (is referred to by): [E1](#) CRM Entity

Quantification: (0,n:0,n)

Scope note: This property associates an instance of F35 Nomen Use Statement with an instance of E74 Group which is the target audience for which the associated nomen use is intended. To indicate that the associated nomen is declared as preferred in all contexts, the instance of E74 Group is ‘mankind’.

Examples: ‘010 __ |a sj 96004896’...‘150 __ |a Belly button’ (F35) *R39 is intended for* Children (E74)

R40 has representative expression (is representative expression for)

Domain: [F1](#) Work

Range: [F22](#) Self-Contained Expression

Subproperty of: [F1](#) Work. [R3](#) is realised in (realises): [F22](#) Self-Contained Expression

Quantification: (0,n:0,n)

Scope note: This property identifies an instance of F22 Self-Contained Expression that has been chosen as the most characteristic expression of the instance of F1 Work of which it is an expression. There is no other semantic implication to this notion of being characteristic than to be an adequate candidate to uniquely identify the Work realized by it. Prototypically, this is the instance of F22 Self-Contained Expression that is deemed characteristic of an instance of F15 Complex Work.

Typically, any expression that is not regarded as “representative” for the work it expresses, would require a uniform title, with qualifiers specifying the differences between that expression and a representative expression, although this may not always be done in practice. The title of a Work may not be one taken from a representative expression.

A given work can have more than one representative expression, provided the differences

between these expressions are not deemed “substantial.” If the anticipated needs of users are not considered to call for bibliographic distinctions between variant expressions of a work, then even expressions that differ significantly from each other can be regarded as equally representative for the work. (See *FRBR: Final Report*, p. 19-20).

A given expression can be deemed representative for a work with regard to some of its aspects (e.g., the text contained in an edition the title proper of which reads ‘The tragicall historie of HAMLET Prince of Denmarke’, and the language of that text), and not representative for it with regard to some other aspects (e.g., the title proper ‘The tragicall historie of HAMLET Prince of Denmarke’ itself, which, being different from the title that is regarded as “representative” for Shakespeare’s work, will require the use of a uniform title).

R40 has representative expression is a shortcut of the more developed path F1 Work *R50i was assigned by* F42 Representative Expression Assignment *R51 assigned* F2 Expression.

Examples: Richard Brinsley Sheridan’s textual work entitled ‘St. Patrick’s Day’ (F15) *R40 has representative expression* the linguistic content of the 1775 edition of Richard Brinsley Sheridan’s textual work entitled ‘St. Patrick’s Day’, mentioned in the ‘Encyclopaedia Britannica’, 15th edition, cited as the source for the authority record created for that work by the Library of Congress (F22)

John Tavener’s musical work entitled ‘The Eternal Sun’ (F15) *R40 has representative expression* the notational content embodied in the 2007 edition of John Tavener’s ‘The Eternal Sun’ cited as the source for the authority record created for that work by the National Library of France (F22)

The series entitled ‘Headline series’ (F18) *R40 has representative expression* the overall content of the publication entitled ‘Vietnam, the war nobody won’ by S. Karnow, which belongs to the series entitled ‘Headline series’ and was used by the Library of Congress as the basis for creating the authority record for that series (F24)

The periodical entitled ‘The New Courier’, released by UNESCO, and described by the National Library of France in a bibliographic record that contains the following statement: ‘Notice réd. d’après le n° d’octobre 2002’ (i.e., ‘description based on the issue dated October 2002’) (F18) *R40 has representative expression* the overall content of the October 2002 issue of UNESCO’s periodical entitled ‘The New Courier’ (F24)

R41 has representative manifestation product type (is representative manifestation product type for)

Domain: [F2](#) Expression

Range: [F3](#) Manifestation Product Type

Subproperty of: [F2](#) Expression. [R4](#) carriers provided by (comprises carriers of): [F3](#) Manifestation Product Type

Quantification: (0,n;0,n)

Scope note: This property identifies an instance of F3 Manifestation Product Type that has been chosen as the most characteristic Manifestation Product Type of the instance of F2 Expression of which it is a manifestation.

Identifying an instance of F3 Manifestation Product Type that is representative for an instance of F2 Expression makes it possible in turn to identify an instance of F2 Expression that is representative for an instance of F1 Work, and to decide what should be regarded as the title of the work.

The title of an Expression may not be one taken from a representative Manifestation Product Type or Manifestation Singleton.

A given expression can have more than one Representative Manifestation Product Type.

R41 has representative manifestation product type is a shortcut of the more developed path F2 Expression *R48i was assigned by* F41 Representative Manifestation Assignment *R49 assigned* F3 Manifestation Product Type.

- Examples:
- The original text of Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' (F22) *R41 has representative manifestation product type* the 1775 edition of Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day', mentioned in the 'Encyclopaedia Britannica', 15th edition, cited as the source for the authority record created for that work by the Library of Congress (F3)
- The original notation of John Tavener's musical work entitled 'The Eternal Sun' (F22) *R41 has representative manifestation* the 2007 edition of John Tavener's 'The Eternal Sun' cited as the source for the authority record created for that work by the National Library of France (F3)
- The textual content of the series entitled 'Headline series' (F2) *R41 has representative manifestation* the publication entitled 'Vietnam, the war nobody won' by S. Karnow, which belongs to the series entitled 'Headline series' and was used by the Library of Congress as the basis for creating the authority record for that series (F3)
- The textual content of the periodical entitled 'The New Courier' (F2) *R41 has representative manifestation* the October 2002 issue of the periodical entitled 'The New Courier' (F3), which was used as the source for the bibliographic record created by the National Library of France

R42 is representative manifestation singleton for (has representative manifestation singleton)

- Domain: [F4](#) Manifestation Singleton
- Range: [F2](#) Expression
- Subproperty of: [E24](#) Physical Man-Made Thing. [P128](#) carries (is carried by): [E73](#) Information Object
- Quantification: (0,n;0,n)
- Scope note: This property identifies an instance of Manifestation Singleton that has been declared as the unique representative for an instance of F2 Expression by some bibliographic agency.
- This property identifies an instance of F4 Manifestation Singleton that has been chosen as the most characteristic Manifestation Singleton of the instance of F2 Expression of which it is a manifestation.
- Identifying an instance of F4 Manifestation Singleton that is representative for an instance of F2 Expression makes it possible in turn to identify an instance of F2 Expression that is representative for an instance of F1 Work, and to decide what should be regarded as the title of the work.
- The title of an Expression may not be one taken from a representative Manifestation Product Type or Manifestation Singleton.
- A given expression can have more than one representative Manifestation Singleton.
- It is a shortcut for the more developed path: F2 Expression *R48i was assigned by* F41 Representative Manifestation Assignment *R53 assigned* F4 Manifestation Singleton.
- Examples: The musical text of Stanislas Champein's opera 'Vichnou' (F22) *R42 has representative manifestation singleton* the manuscript identified by shelfmark 'MS-8282' within the collections of the National Library of France, Department for Music (F4) [explanation: the BnF's Department for Music holds 3 manuscript scores (identified by shelfmarks 'MS-8282', 'MS-13778', and 'MS-17321') for this opera; the title inscribed on MS-8282 is 'Vichnou', while MS-13778 and MS-17321 are titled 'Vistnou'; the authorised form chosen by cataloguers and reference tools such as the Grove Dictionary for Opera is 'Vichnou', while 'Vistnou' is recorded in the BnF's authority file as a variant form only]

R43 carried out by (performed)

- Domain: [F41](#) Representative Manifestation Assignment

Range: [F44](#) Bibliographic Agency

Subproperty of: [E7](#) Activity. [P14](#) carried out by (performed): [E39](#) Actor

Quantification: (1,1:0,n)

Scope note: This property associates a bibliographic agency (represented by one or more of its cataloguers) and the assigning of which Manifestation (i.e., which instance of F3 Manifestation Product Type or F4 Manifestation Singleton) is representative for a given expression.

In cataloguing practice, such a relationship is usually just implicit. However, it can become explicit, for example when a bibliographic agency creates an authority record for a given work and fills the “Source” field with information about the publication that contains the expression that was used by the bibliographic agency to establish the uniform title for the work realised in that expression.

Examples: Assigning the manuscript identified by shelfmark ‘MS-8282’ within the collections of the National Library of France, Department for Music, as representative for the musical text of Stanislas Champein’s opera ‘Vichnou’ (F41) *R43 carried out by* the National Library of France (F44), identified by code ‘FRBNF’ at the beginning of field 001 in the INTERMARC authority record created for Stanislas Champein’s opera ‘Vichnou’

Assigning the 1775 edition of Richard Brinsley Sheridan’s textual work entitled ‘St. Patrick’s Day’ (F41), mentioned in the ‘Encyclopaedia Britannica’, 15th edition, *R43 carried out by* the Library of Congress (F44), identified by code ‘DLC’ in field 040 of the MARC21 authority record created for Richard Brinsley Sheridan’s textual work entitled ‘St. Patrick’s Day’

R44 carried out by (performed)

Domain: [F42](#) Representative Expression Assignment

Range: [F44](#) Bibliographic Agency

Subproperty of: [E7](#) Activity. [P14](#) carried out by (performed): [E39](#) Actor

Quantification: (1,1:0,n)

Scope note: This property associates a bibliographic agency (represented by one or more of its cataloguers) and the assigning of which expression is representative for a given Work.

In cataloguing practice, such a relationship is usually just implicit. However, it can become explicit, for example when a bibliographic agency creates an authority record for a given work and fills the “Source” field with information about the publication that contains the expression that was used by the bibliographic agency to establish the uniform title for the work realised in that expression.

Examples: Assigning the musical text contained in the manuscript identified by shelfmark ‘MS-8282’ within the collections of the National Library of France, Department for Music, as representative for Stanislas Champein’s opera ‘Vichnou’ (F41) *R44 carried out by* the National Library of France, identified by code ‘FRBNF’ at the beginning of field 001 in the INTERMARC authority record for Stanislas Champein’s opera ‘Vichnou’ (F44)

Assigning the text embodied in the 1775 edition of Richard Brinsley Sheridan’s textual work entitled ‘St. Patrick’s Day’ as the representative expression for Richard Brinsley Sheridan’s textual work entitled ‘St. Patrick’s Day’ (F42) *R44 carried out by* the bibliographic agency identified, in field 040 of a MARC21 authority record for Richard Brinsley Sheridan’s textual work entitled ‘St. Patrick’s Day’, by the code ‘DLC’ (i.e., the Library of Congress) (F44)

R45 assigned to (was assigned by)

Domain: [F40](#) Identifier Assignment

Range: [E1](#) CRM Entity

Subproperty of: [E13](#) Attribute Assignment. [P140](#) assigned attribute to (was attributed by): [E1](#) CRM Entity

Quantification: (1,1:0,n)

Scope note: This property identifies the entity to which an actor, such as a bibliographic agency, assigned an instance of F13 Identifier.

Examples: Assigning the uniform title ‘The Adoration of the Shepherds (Coventry)’ (F40) *R45 assigned to the anonymous textual work otherwise simply known as ‘The Adoration of the Shepherds’* (F15) [assignment of an Identifier to a Work]

Assigning the uniform title ‘Rite of spring (Choreographic Work: Bausch)’ (F40) *R45 assigned to Pina Bausch’s choreographic work entitled ‘Rite of spring’* (F15) [assignment of an Identifier to a Work]

Assigning the uniform title ‘King Kong (1933)’ (F40) *R45 assigned to the motion picture directed in 1933 by Merian C. Cooper and Ernest B. Schoedsack, entitled ‘King Kong’* (F15) [assignment of an Identifier to a Work]

Assigning the controlled access point ‘Guillaume, de Machaut, ca. 1300-1377’ (F40) *R45 assigned to Guillaume de Machaut* (F10) [assignment of an Identifier to a Person]

Assigning the controlled access point ‘Univerza v Ljubljani. Oddelek za bibliotekarstvo’ (F40) *R45 assigned to the Department for library science of the University of Ljubljana* (F11) [assignment of an Identifier to a Corporate Body]

R46 assigned (was assigned by)

Domain: [F40](#) Identifier Assignment

Range: [F13](#) Identifier

Equal to: [E15](#) Identifier Assignment: [P37](#) assigned (was assigned by): [E42](#) Identifier

Quantification: (1,1:0,n)

Scope note: This property associates the instance of F13 Identifier assigned to an instance of E1 CRM Entity and the event of assigning it.

Examples: Assigning a uniform title to the anonymous textual work known as ‘The Adoration of the Shepherds’, a title shared by another, distinct anonymous textual work (F40) *R46 assigned the uniform title ‘The Adoration of the Shepherds (Coventry)’* (F50)

Assigning a uniform title to Pina Bausch’s choreographic work entitled ‘Rite of spring’ (F40) *R46 assigned the uniform title ‘Rite of spring (Choreographic Work: Bausch)’* (F50)

Assigning a uniform title to the motion picture directed in 1933 by Merian C. Cooper and Ernest B. Schoedsack and entitled ‘King Kong’ (F40) *R46 assigned the uniform title ‘King Kong (1933)’* (F50)

Assigning a controlled access point to Guillaume de Machaut (F40) *R46 assigned ‘Guillaume, de Machaut, ca. 1300-1377’* (F50)

Assigning a controlled access point to the Department for library science of the University of Ljubljana (F40) *R46 assigned ‘Univerza v Ljubljani. Oddelek za bibliotekarstvo’* (F50)

Assigning a subject heading (in an authority record) to the concept of knowledge representation (F40) *R46 assigned ‘Conceptual structures (Information theory)’* (F50)

Assigning a subject heading (in a bibliographic record) to the concept of the appreciation of Victor Hugo’s works in Germany between 1870 and 1914 (F40) *R46 assigned ‘Hugo, Victor, 1802-1885 – Appreciation – Germany – 1870-1914’* (F13)

R48 assigned to (was assigned by)

Domain:	F41 Representative Manifestation Assignment
Range:	F2 Expression
Subproperty of:	E13 Attribute Assignment. P140 assigned attribute to (was attributed by): E1 CRM Entity
Quantification:	(1,1:0,n)
Scope note:	This property associates the event of assigning a representative instance of F3 Manifestation Product Type or F4 Manifestation Singleton with the expression to which it was assigned.
Examples:	Selecting the 1775 edition of Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' as the representative manifestation for the text of Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' (F41) <i>R48 assigned to</i> the text of Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' (F22) Selecting the 2007 edition of John Tavener's musical work entitled 'The Eternal Sun' as the representative manifestation for the notation of John Tavener's musical work entitled 'The Eternal Sun' (F41) <i>R48 assigned to</i> the notation of John Tavener's musical work entitled 'The Eternal Sun' (F22) Selecting the 1983 edition of Stanley Karnow's textual work entitled 'Vietnam, the war nobody won' as the representative manifestation for a partial expression of the series entitled 'Headline series' (F41) <i>R48 assigned to</i> the content of the series entitled 'Headline series' (F22) Selecting the issue dated October 2002 of the periodical entitled 'The New Courier' as the representative manifestation for a partial expression of the periodical entitled 'The New Courier' (F41) <i>R48 assigned to</i> the content of the periodical entitled 'The New Courier' (F22) Selecting the manuscript held by the National Library of France and identified by shelf mark 'MS-8282' as the representative Manifestation Singleton for the notation of Stanislas Champein's opera 'Vichnou' (F41) <i>R48 assigned to</i> the notation of Stanislas Champein's opera 'Vichnou' (F22)

R49 assigned (was assigned by)

Domain:	F41 Representative Manifestation Assignment
Range:	F3 Manifestation Product Type
Subproperty of:	E13 Attribute Assignment. P141 assigned (was assigned by): E1 CRM Entity
Quantification:	(0,n:0,n)
Scope note:	This property associates the event of assigning a representative instance of F3 Manifestation Product Type with the F3 Manifestation Product Type which has been assigned.
Examples:	Selecting the representative manifestation for the text of Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' (F41) <i>R49 assigned</i> the 1775 edition of Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' (F3) Selecting the representative manifestation for the notation of John Tavener's musical work entitled 'The Eternal Sun' (F41) <i>R49 assigned</i> the 2007 edition of John Tavener's musical work entitled 'The Eternal Sun' (F3) Selecting the representative manifestation for a partial expression of the series entitled 'Headline series' (F41) <i>R49 assigned</i> the 1983 edition of Stanley Karnow's textual work entitled 'Vietnam, the war nobody won' (F3) Selecting the representative manifestation for a partial expression of the periodical entitled 'The New Courier' (F41) <i>R49 assigned</i> the issue dated October 2002 of the periodical entitled 'The New Courier' (F22)

R50 assigned to (was assigned by)

Domain:	F42 Representative Expression Assignment
Range:	F15 Complex Work
Subproperty of:	E13 Attribute Assignment. P140 assigned attribute to (was attributed by): E1 CRM Entity
Quantification:	(1,1:0,n)
Scope note:	This property associates the event of assigning a representative instance of F2 Expression with the instance of F15 Complex Work to which it was assigned.
Examples:	Selecting the text embodied in the 1775 edition of Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' as the representative expression for Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' (F42) <i>R50 assigned to</i> Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' (F15) Selecting the musical notation embodied in the 2007 edition of John Tavener's musical work entitled 'The Eternal Sun' as the representative expression for John Tavener's musical work entitled 'The Eternal Sun' (F42) <i>R50 assigned to</i> John Tavener's musical work entitled 'The Eternal Sun' (F15) Selecting the publication expression of the 1983 edition of Stanley Karnow's textual work entitled 'Vietnam, the war nobody won' as the representative expression for the series entitled 'Headline series' (F42) <i>R50 assigned to</i> the series entitled 'Headline series' (F18) Selecting the publication expression of the issue dated October 2002 of the periodical entitled 'The New Courier' as the representative expression of the periodical entitled 'The New Courier' (F42) <i>R50 assigned to</i> the periodical entitled 'The New Courier' (F18) Selecting the content of the manuscript identified by shelfmark 'MS-8282' within the collections of the National Library of France, Department for Music, as the representative expression of Stanislas Champein's musical work entitled 'Vichnou' (F42) <i>R50 assigned to</i> Stanislas Champein's musical work entitled 'Vichnou' (F15)

R51 assigned (was assigned by)

Domain:	F42 Representative Expression Assignment
Range:	F2 Expression
Subproperty of:	E13 Attribute Assignment. P141 assigned (was assigned by): E1 CRM Entity
Quantification:	(1,n:0,n)
Scope note:	This property associates the event of assigning a representative instance of F2 Expression with the F2 Expression which has been assigned.
Examples:	Selecting the representative expression for Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' (F42) <i>R51 assigned</i> the text embodied in the 1775 edition of Richard Brinsley Sheridan's textual work entitled 'St. Patrick's Day' (F22) Selecting the representative expression for John Tavener's musical work entitled 'The Eternal Sun' (F42) <i>R51 assigned</i> the musical notation embodied in the 2007 edition of John Tavener's musical work entitled 'The Eternal Sun' (F15) Selecting the representative expression for the series entitled 'Headline series' (F42) <i>R51 assigned</i> the publication expression of the volume of the series entitled 'Headline series' that consists of the 1983 edition of Stanley Karnow's textual work entitled 'Vietnam, the war nobody won' (F24) Selecting the representative expression of the periodical entitled 'The New Courier' (F42) <i>R51 assigned</i> the publication expression of the issue dated October 2002 of the periodical entitled 'The New Courier' (F24)

Selecting the representative expression of Stanislas Champein's musical work entitled 'Vichnou' (F42) *R51 assigned* the content of the manuscript identified by shelfmark 'MS-8282' within the collections of the National Library of France, Department for Music (F22)

R52 used rule (was the rule used in)

Domain: [F40](#) Identifier Assignment

Range: [F43](#) Identifier Rule

Subproperty of: [E7](#) Activity. [P33](#) used specific technique: [E29](#) Design or Procedure

Quantification: (0,n;0,n)

Scope note: This property associates the event of assigning an instance of F13 Identifier with the instructions followed by an actor, such as a Bibliographic Agency, in creating that identifier.

Examples: Assigning the uniform title 'Bach, Johann Sebastian, 1685-1750. Concertos, violins (2), string orchestra, BWV 1043, D minor' to Johann Sebastian Bach's Double Concerto in D minor, BWV 1043 (F40) *R52 used rule* AACR2R 25.25-25.35F1 (F43)

Assigning the uniform title 'Bach, Johann Sebastian [Konzerte, VI 1 2 Orch BWV 1043]' to Johann Sebastian Bach's Double Concerto in D minor, BWV 1043 (F40) *R52 used rule* RAK-Musik (Revidierte Ausgabe 2003), Chapter 6 (F43)

Assigning the uniform title 'Bach, Johann Sebastian (1685-1750). – [Concertos. Violons (2), orchestre à cordes. BWV 1043. Ré mineur]' to Johann Sebastian Bach's Double Concerto in D minor, BWV 1043 (F40) *R52 used rule* AFNOR Z 44-079 (F43)

Assigning the controlled access point 'Guillaume de Machaut (1300?-1377)' (F40) *R52 used rule* AFNOR Z 44-061 (F43)

Assigning the controlled access point 'Guillaume, de Machaut, ca. 1300-1377' (F40) *R52 used rule* AACR2R 22 (F43)

R53 assigned (was assigned by)

Domain: [F41](#) Representative Manifestation Assignment

Range: [F4](#) Manifestation Singleton

Subproperty of: [E13](#) Attribute Assignment. [P141](#) assigned (was assigned by): [E1](#) CRM Entity

Quantification: (0,n;0,n)

Scope note: This property associates the event of assigning a representative instance of F4 Manifestation Singleton with the F4 Manifestation Singleton which has been assigned.

Examples: Selecting the representative Manifestation Singleton for the notation of Stanislas Champein's opera 'Vichnou' (F41) *R53 assigned* the manuscript held by the National Library of France and identified by shelf mark 'MS-8282' (F4)

R54 has nomen language (is language of nomen in)

Domain: [F35](#) Nomen Use Statement

Range: [E56](#) Language

Subproperty of: [E89](#) Propositional Object. [P67](#) refers to (is referred to by): [E1](#) CRM Entity

Quantification: (0,n;0,n)

Scope note: This property associates an instance of F35 Nomen Use Statement with an instance of E56

Language which is a target language for the associated nomen use.

- Examples: '001 FRBNF119304566'...'100 \$w.0..b.spa.\$aColón\$mCristóbal\$d1450?-1506' (F35) *R54 has nomen language* the language referred to in subfield \$w by the ISO code 'spa,' i.e., Spanish (E56)
- '001 FRBNF119304566'...'400 \$w....b.eng.\$aColumbus\$mChristopher\$d1450?-1506' (F35) *R54 has nomen language* the language referred to in subfield \$w by the ISO code 'eng,' i.e., English (E56)
- '001 FRBNF118726828'...'110 \$w20..b.fre.\$aConseil international des musées' (F35) *R54 has nomen language* the language referred to in subfield \$w by the ISO code 'fre,' i.e., French (E56)
- '001 FRBNF118726828'...'110 \$w20..b.ger.\$aInternationaler Museumsrat' (F35) *R54 has nomen language* the language referred to in subfield \$w by the ISO code 'ger,' i.e., German (E56)
- 'ID: 300024668'...'navaja (C,U,Spanish,UF,U,SN)' *R54 has nomen language* Spanish (E56) ("used for" term, Getty Art & Architecture Thesaurus Online, retrieved 19/11/2012)

R55 has nomen form (is nomen form in)

- Domain: [F35](#) Nomen Use Statement
- Range: [E55](#) Type
- Subproperty of: [E89](#) Propositional Object. [P67](#) refers to (is referred to by): [E1](#) CRM Entity
- Quantification: (0,1;0,n)
- Scope note: This property associates an instance of F35 Nomen Use Statement with the instance of E55 Type that characterizes the Nomen referred to in the Nomen Use Statement, such as abbreviation, full name etc. In the case of abbreviations, the source of this form may or may not be explicitly mentioned.
- Examples: '010 __ |a n 78004438'...'410 2_ |a IFLA' to refer to the International Federation of Library Associations and Institutions (F35) *R55 has nomen form* acronym (E55)

R56 has related use (is related use for)

- Domain: [F35](#) Nomen Use Statement
- Range: [F35](#) Nomen Use Statement
- Subproperty of: [E29](#) Design or Procedure. [P69](#) has association with (is associated with): [E29](#) Design or Procedure
- Quantification: (0,n;0,n)
- Scope note: This property associates an instance of F35 Nomen Use Statement with another instance of F35 Nomen Use Statement which has a related use in some context, such as alternative, lexical variant, replacing former use etc. The property R56.1 allows for specifying the particular kind of the nomen use statement relationship.
- Examples: '001 FRBNF122597517'...'14506\$w.1..b.fre.\$aDu sublime' (F35) *R56 has related use*
- '001 FRBNF122597517'...'14506\$w.0..g.grp \$aΠερί Ὑπσους' (F35) *R56.1 has type* parallel form (E55)
- '001 FRBNF126866954'...'100 \$w.0..b.....\$aTyrrell\$mGeorge\$d1861-1909' (F35) *R56 has related use*
- '001 FRBNF126866954'...'466 \$w....b \$aTyrell\$oSaffaire\$g1907' (F35) *R56.1 has type* has variant access point for a personal name, intended to be displayed in a subject index only, and not in a personal names index (E55) [Explanation: in the InterMarc format used at the National Library of France, tag 466 in an authority record for a person serves to

introduce a topical term which is displayed as a variant form for the personal name in the subject index, but is not displayed in the name index]

‘010 __ |a n 78004438’...‘410 2_ |a IFLA’ (F35) *R56 has related use* ‘010 __ |a n 78004438’...‘110 2_ |a International Federation of Library Associations and Institutions’ *R56.1 has type* has full form (in MARC 21 field 410 serves as a reference to the accepted form which appears in field 110)

Properties: R56.1 has type: [E55](#) Type

R57 is based on (is basis for)

Domain: [F38](#) Character

Range: [E39](#) Actor

Shortcut of: F38 Character. P94i was created by (has created): E65 Creation. P17 was motivated by (motivated): E39 Actor

Quantification: (0,n;0,n)

Scope note: This property associates an instance of F38 Character with an instance of E39 Actor that the character is motivated by or is intended to represent. An instance of F38 Character may be based on a combination of features taken from several actors. This property is a shortcut of the more fully developed path from E28 Conceptual Object, restricted to F38 Character, through the inverse of P94 has created (was created by), E65 Creation P17 was motivated by (motivated) to E1 CRM Entity restricted to E39 Actor.

Examples: The Character ‘Sinuhe’ (F38) in Mika Waltari’s ‘Sinuhe the Egyptian: A Novel’ *R57 is based on* Sinuhe (E21) (documented in the autobiographic narrative in fragments carried by The Ramesside Papyrus, Pap. Berlin 10499, Pap. Berlin 3022, The Amherst fragments (m-q) and other Egyptian sources)

The Character ‘Alexander’ (F38) in Mary Renault’s ‘Fire from Heaven’ *R57 is based on* Alexander the Great of Macedon (356-323) (E21)

R58 has fictional member (is fictional member of)

Domain: [F38](#) Character

Range: [F38](#) Character

Subproperty of: out of CRM Scope

Quantification: (0,n;0,n)

Scope note: This property associates an instance of F38 Character representing a group with another instance of F38 Character that is presented in relevant fiction as a member of the fictional group.

Examples: Argonauts (F38) *R58 has fictional member* Jason (F38)

R59 had typical subject (was typical subject of)

Domain: [F51](#) Pursuit

Range: [E1](#) CRM Entity

Subproperty of: E65 Creation. P94 has created (was created by): E89 Propositional Object. P129 is about (is subject of): E1 CRM Entity

Quantification: (0,n;0,n)

Scope note: This property associates an instance of F51 Pursuit with the instance of E1 CRM Entity that is the typical subject of the associated activity, such as an area of expertise in which the actor is engaged or was engaged.

Examples: John Dover Wilson's activity as a Shakespeare scholar (F51) *R59 had typical subject* William Shakespeare (F10)

R60 used to use language (was language used by)

Domain: [F51](#) Pursuit

Range: [E56](#) Language

Shortcut of: E65 Creation. P94 has created (was created by): E33 Linguistic Object. P72 has language (is language of): E56 Language

Quantification: (0,n;0,n)

Scope note: This property associates an instance of F51 Pursuit with the instance of E56 Language that was characteristically used for the products of the associated activity.

The property *R60.1 has type of use* allows for specifying a particular form of use.

Examples: Samuel Beckett's activity as author of English texts (F51) *R60 used to use language* English (E56) *R60.1 has type of use* Authorship (E55)

Samuel Beckett's activity as author of French texts (F51) *R60 used to use language* French (E56) *R60.1 has type of use* Authorship (E55)

Samuel Beckett's activity as translator of English texts into French (F51) *R60 used to use language* French (E56) *R60.1 has type of use* Translation – target language (E55)

Samuel Beckett's activity as translator of English texts (F51) *R60 used to use language* English (E56) *R60.1 has type of use* Translation – source language (E55)

Properties: R60.1 has type of use: [E55](#) Type

R61 occurred in kind of context (was kind of context for)

Domain: [F52](#) Name Use Activity

Range: [E55](#) Type

Shortcut of: F52 Name Use Activity. P9i forms part of: F51 Pursuit. P2 has type: E55 Type

Quantification: (0,n;0,n)

Scope note: This property associates an instance of F52 Name Use Activity with the instance of E55 Type that characterises the kind of role or context within which the associated name was used.

Examples: Charles Lutwidge Dodgson using the name 'Lewis Carroll' (F52) *R61 occurred in kind of context* writing for children (E55)

Charles Lutwidge Dodgson using the name 'Charles Dodgson' (F52) *R61 occurred in kind of context* writing in mathematics (E55)

R62 was used for membership in (was context for)

Domain: [F52](#) Name Use Activity

Range: [E74](#) Group

Shortcut of: [E7](#) Activity. P17 was motivated by (motivated): E85 Joining. P144 joined with: E74 Group

Quantification: (0,n;0,n)

Scope note: This property associates an instance of F52 Name Use Activity with the instance of E74 Group that characterises the context within which the associated name was used for membership in that group.

Examples: Using the name 'John Paul I' (F52) *R62 was used for membership in the corporate body identified in the Library of Congress's authority file as 'Catholic Church. Pope' (F11)*

R63 named (was named by)

Domain: [F52](#) Name Use Activity

Range: [E1](#) CRM Entity

Subproperty of: [E13](#) Attribute Assignment. [P140](#) assigned attribute to (was attributed by): [E1](#) CRM Entity

Quantification: (1,1;0,n)

Scope note: This property associates an instance of F52 Name Use Activity with the instance of E1 CRM Entity that the associated name was used for.

Examples: The recording of the MARC 21 field '110 2_ |a Canadian Academic Centre in Italy' (F52) *R63 named* the instance of F11 Corporate Body represented by the Library of Congress authority record number n 85118480

The appearance of the name 'Centro accademico canadese in Italia' on the title page of the book 'Lo Scavo di S. Giovanni di Ruoti ed il periodo tardoantico in Basilicata' published in 1983 (F52) *R63 named* the instance of F11 Corporate Body represented by the Library of Congress authority record number n 85118480

The statement 'The International Federation of Library Associations and Institutions (IFLA) is the leading international body representing the interests of library and information services and their users' cited from the 'About IFLA' page of the website <http://www.ifla.org/> (F35) *R63 named* the instance of F11 Corporate Body represented by the Library of Congress authority record number n 78004438

R64 used name (was name used by)

Domain: [F52](#) Name Use Activity

Range: [E41](#) Appellation

Subproperty of: [E13](#) Attribute Assignment. [P141](#) assigned (was assigned by): [E1](#) CRM Entity

Quantification: (1,1;0,n)

Scope note: This property associates an instance of F52 Name Use Activity with the instance of E41 Appellation that was used for the associated entity.

Examples: The appearance of the name 'Lewis Carroll' on the title page of 'Le avventure d'Alice nel paese delle meraviglie', published in 1872 in London by Macmillan and Co. (F52) *R64 used name* 'Lewis Carroll' (E41)

The appearance of the name 'Centro accademico canadese in Italia' on the title page of the book 'Lo Scavo di S. Giovanni di Ruoti ed il periodo tardoantico in Basilicata' published in 1983 (F52) *R64 used name* 'Centro accademico canadese in Italia' (E41)

The appearance of the name 'Canadian Academic Centre in Italy' on page 6 of the book 'Lo Scavo di S. Giovanni di Ruoti ed il periodo tardoantico in Basilicata' published in 1983 (F52) *R64 used name* 'Canadian Academic Centre in Italy' (E41)

The appearance of the name 'IFLA' as an acronym for 'The International Federation of Library Associations and Institutions' in the sentence (F2) 'The International Federation of Library Associations and Institutions (IFLA) is the leading international body representing the

interests of library and information services and their users' cited from the 'About IFLA' page of the website <http://www.ifla.org/> (F52) *R64 used name 'IFLA'* (E41)

R65 recorded aspects of (had aspects recorded through)

Domain: [F29](#) Recording Event

Range: [E18](#) Physical Thing

Shortcut of: [F29](#) Recording Event. [R20](#) recorded: [E3](#) Condition State. [P44i](#) is condition of: [E18](#) Physical Thing

Subproperty of shortcut of: [F29](#) Recording Event. [R20](#) recorded: [E5](#) Event. [P12](#) occurred in the presence of: [E18](#) Physical Thing

Quantification: (0,n;0,n)

Scope note: This property associates an instance of F29 Recording Event with an instance of E18 Physical Thing some of whose features, at the time of recording, were recorded in that process.

Examples: The making of the photograph of the three Allied leaders at Yalta in February 1945 (F29) *R65 recorded aspects of Stalin* (E21)

Filming Louise Bourgeois at work in the context of the shooting of the documentary movie entitled 'Louise Bourgeois: The Spider, the Mistress, and the Tangerine' (F29) *R65 recorded aspects of Louise Bourgeois* (E21)

R66 included performed version of (had a performed version through)

Domain: [F31](#) Performance

Range: [E89](#) Propositional Object

Subproperty of: [E7](#) Activity. [P16](#) used specific object (was used for): [E70](#) Thing

Quantification: (0,n;0,n)

Scope note: This property associates an instance of F31 Performance with a product of the mind that was performed in the course of that instance of F31 Performance.

According to the level of knowledge available about the performance, the range of this property can actually be specialised as either an instance of F1 Work (if nothing is known as to which specific expression of the work was performed), or of F2 Expression (if there is a reasonable amount of certainty as to which specific expression—e.g., a well identified translation of a play—of the work was performed).

In addition to being a subproperty of P16 used specific object (was used for), this property also is a shortcut of the fully developed path that goes from F31 Performance to F1 Work through: R25 performed F25 Performance Plan P165 incorporates F22 Self-Contained Expression R3i realises. In this fully developed path, the specific instance of F22 Self-Contained Expression can be precisely identified and described for its own sake, or it can just be known to have necessarily existed.

Examples: The performance of 'Hamlet' on 17 June 1909 in Berlin, Deutsches Theater, by Alexander Moissi, directed by Max Reinhardt (F31) *R66 included performed version of William Shakespeare's work 'Hamlet'* (F15) [Note: the specific German translation that was performed is not mentioned in the documentation available from <http://www.glopap.org/pi/en/record/production/1001207>]

The performance of 'Hamlet' on 6 June 1964 in Zurich, Schauspielhaus, by Compagnia Proclemer-Albertazzi, directed by Franco Zeffirelli (F31) *R66 included performed version of Gerardo Guerrieri's Italian translation* (F22) of William Shakespeare's work 'Hamlet'

CLP2 should have type (should be type of)

Domain: [F3](#) Manifestation Product Type

Range: [E55](#) Type

Subproperty of:

Quantification: (0,n;0,n)

Scope note: This property associates a publication, i.e. an instance of F3 Manifestation Product Type, with an instance of E55 Type, which all exemplars of that publication should belong to, as long as they are recognised as exemplars of that publication. Typically, this property is observed on one exemplar of a publication, and extrapolated to all other exemplars of the same publication. This logical inference is an induction along the path that can be modelled as: F3 Manifestation Product Type *R7i has example* F5 Item *P41i was classified by* E17 Type Assignment *P42 assigned* E55 Type.

It can happen that a given exemplar, or subset of exemplars, originally produced, or intended to be produced, with that characteristic, accidentally lacks it. This fact should be recorded as a property of F5 Item, and not of F3 Manifestation Product Type.

Examples: The sound recording entitled ‘The Glory (????) of the human voice’, identified by label and label number ‘RCA Victor Gold Seal GD61175’, containing recordings of musical works performed by Florence Foster Jenkins (F3) *CLP2 should have type* sound recording (E55)

The sound recording entitled ‘The Glory (????) of the human voice’, identified by label and label number ‘RCA Victor Gold Seal GD61175’, containing recordings of musical works performed by Florence Foster Jenkins (F3) *CLP2 should have type* kind of sound: monaural (E55)

CLP43 should have dimension (should be dimension of)

Domain: [F3](#) Manifestation Product Type

Range: [E54](#) Dimension

Subproperty of:

Quantification: (1,n;1,1)

Scope note: This property associates a publication, i.e. an instance of F3 Manifestation Product Type, with an instance of E54 Dimension, which all exemplars of that publication should have, as long as they are recognised as exemplars of that publication. Typically, this property is observed on one exemplar of a publication, and extrapolated to all other exemplars of the same publication. This logical inference is an induction along the path that can be modelled as: F3 Manifestation Product Type *R7i has example* F5 Item *P39i was measured by* E16 Measurement *P40 observed dimension* E54 Dimension.

It can happen that a given exemplar, or subset of exemplars, originally produced, or intended to be produced, with that characteristic, accidentally lacks it. This fact should be recorded as a property of F5 Item, and not of F3 Manifestation Product Type.

Examples: The publication entitled ‘Functional Requirements for Bibliographic Records: final report’, published by K. G. Saur in 1998, identified by ISBN ‘3-598-11382-X’ (F3) *CLP43 should have dimension* height of the individual copy of ‘Functional Requirements for Bibliographic Records: final report’ that I have at hand and that I observed while describing it (E54) *P3 has note* ‘24 cm’ (E62) [or, alternatively: *P90 has value* ‘24’ (E60) and *P91 has unit* ‘cm’ (E58)]

The jigsaw puzzle entitled ‘Map of the New York city subway system’, designed by Stephen J. Voorhies and released around 1954 by the Union Dimes Savings Bank (F3) *CLP43 should have dimension* length and height of the exemplar held and catalogued by the Library of Congress (E54) *P3 has note* ‘46 x 29 cm’ (E62)

CLP45 should consist of (should be incorporated in)

Domain: [F3](#) Manifestation Product Type

Range: [E57](#) Material

Subproperty of:

Quantification: (0,n:0,n)

Scope note: This property associates a publication, i.e. an instance of F3 Manifestation Product Type, with an instance of E57 Material, which all exemplars of that publication should consist of, as long as they are recognised as exemplars of that publication. Typically, this property is observed on one exemplar of a publication, and extrapolated to all other exemplars of the same publication. This logical inference is an induction along the path that can be modelled as: F3 Manifestation Product Type *R7i* has example F5 Item *P41i* was classified by E17 Type Assignment *P42* assigned E57 Material.

It can happen that a given exemplar, or subset of exemplars, originally produced, or intended to be produced, with that characteristic, accidentally lacks it. This fact should be recorded as a property of F5 Item, and not of F3 Manifestation Product Type.

Examples: The jigsaw puzzle entitled ‘Map of the New York city subway system’, designed by Stephen J. Voorhies and released around 1954 by the Union Dimes Savings Bank (F3) *CLP45 should consist of* cardboard (E57)

CLP46 should be composed of (may form part of)

Domain: [F3](#) Manifestation Product Type

Range: [F3](#) Manifestation Product Type

Subproperty of:

Quantification: (0,n:0,n)

Scope note: This property associates an instance of F3 Manifestation Product Type which prescribes that all its Items will contain as parts an Item of another instance of F3 Manifestation Product Type with that instance of F3 Manifestation Product Type.

Examples: The publication product identified by ISBN ‘0618260587’ and consisting of a 3-volume edition of J.R.R. Tolkien’s ‘The Lord of the rings’ (F3) *CLP46 should be composed of* the publication product identified by ISBN ‘0618260595’ and consisting of an edition of J.R.R. Tolkien’s ‘The two towers’ (F3)

The publication product issued by Deutsche Grammophon in 1998 and consisting of a recording of Richard Wagner’s ‘Der fliegende Holländer’ as performed in 1991 by Plácido Domingo, Cheryl Studer et al., and conducted by Giuseppe Sinopoli (F3) *CLP46 should be composed of* the publication product consisting of printed programme notes and libretto with French and English translations (F3)

CLP57 should have number of parts

Domain: [F3](#) Manifestation Product Type

Range: [E60](#) Number

Subproperty of:

Quantification: (1,1:0,n)

Scope note: This property associates a publication, i.e. an instance of F3 Manifestation Product Type, with an instance of E60 Number, which denotes the number of physical units all exemplars of that

publication should consist of, as long as they are recognised as complete exemplars of that publication. Typically, this property is observed on one exemplar of a publication, and extrapolated to all other exemplars of the same publication. This logical inference is an induction along the path that can be modelled as: F3 Manifestation Product Type *R7i* has example F5 Item *P57* has number of parts E60 Number.

It can happen that a given exemplar, or subset of exemplars, originally produced, or intended to be produced, with that characteristic, accidentally lacks it. This fact should be recorded as a property of F5 Item, and not of F3 Manifestation Product Type.

Examples: The jigsaw puzzle entitled ‘Map of the New York city subway system’, designed by Stephen J. Voorhies and released around 1954 by the Union Dimes Savings Bank (F3) *CLP57 should have number of parts* 76 (E60) [Number of physical units of the exemplar held by the Library of Congress, as observed by a cataloguer from the Library of Congress when he/she catalogued that particular exemplar and recorded the statement: ‘1 jigsaw puzzle (ca. 76 pieces)’]

The publication entitled ‘History of costume: in slides, notes, and commentaries’ by Jeanne Button, Patricia Quinn Stuart, and Stephen Sbarge, released by Slide Presentations (New York) ca. 1975 (F3) *CLP57 should have number of parts* 1,491 (E60) [Number of physical units of the exemplar held by the Gelman Library of the George Washington University, as observed by a cataloguer from the Gelman Library of the George Washington University when he/she catalogued that particular exemplar and recorded the statement: ‘1,491 slides in 14 slide trays + 6 ring binders in cases (30 x 29 cm.)’]

CLP104 subject to (applies to)

Domain: [F3](#) Manifestation Product Type

Range: [E30](#) Right

Subproperty of:

Quantification: (0,n:1,1)

Scope note: This property associates a publication, i.e. an instance of F3 Manifestation Product Type, with an instance of E30 Right, which applies to all exemplars of that publication, as long as they are recognised as exemplars of that publication.

The rights covered by this property may include: acquisition or access authorisation; terms of availability; access restrictions on the Manifestation Product Type; etc.

Examples: The publication entitled ‘Recent poems’ by the author named ‘Stephen Spender’, released by the publisher named ‘Anvil Press Poetry’ in 1978 and identified by ISBN ‘0856460516’ (F3) *CLP104 subject to* availability restricted to Anvil Press Poetry subscribers (E30) [*P3 has note* ‘This edition [...] is available only to Anvil Press Poetry subscribers’ (E62)]

CLP105 right held by (right on)

Domain: [F3](#) Manifestation Product Type

Range: [E39](#) Actor

Subproperty of:

Quantification: (0,n:0,n)

Scope note: This property associates a publication, i.e. an instance of F3 Manifestation Product Type, with an instance of E39 Actor, who holds an instance of E30 Right on all exemplars of that publication, as long as they are recognised as exemplars of that publication.

Examples: The publication entitled ‘Recent poems’ by the author named ‘Stephen Spender’, released by the publisher named ‘Anvil Press Poetry’ in 1978 and identified by ISBN ‘0856460516’ (F3) *CLP105 right held by* Anvil Press Poetry (F11)

CLR6 should carry (should be carried by)

Domain: [F3](#) Manifestation Product Type

Range: [F24](#) Publication Expression

Subproperty of:

Quantification: (1,1:0,n)

Scope note: This property associates a publication, i.e. an instance of F3 Manifestation Product Type, with an instance of F24 Publication Expression, which all exemplars of that publication should carry, as long as they are recognised as complete exemplars of that publication. Typically, this property is observed on one exemplar of a publication, and extrapolated to all other exemplars of the same publication. This logical inference is an induction along the path that can be modelled as: F3 Manifestation Product Type *R7i has example* F5 Item *R6 carries* F24 Publication Expression.

It can happen that a given exemplar, or a subset of exemplars, originally produced, or intended to be produced with that characteristic, accidentally lacks part of the publication expression. This fact should be recorded as a property of F5 Item, and not of F3 Manifestation Product Type.

Examples: The publication, dated 1972, entitled ‘The complete poems of Stephen Crane, edited with an introduction by Joseph Katz’ (ISBN ‘0-8014-9130-4’) (F3) *CLR6 should carry* the overall content of the book identified by ISBN ‘0-8014-9130-4’, i.e.: the text of Stephen Crane’s complete poems as edited by Joseph Katz, the numbering system introduced by Joseph Katz in order to identify each individual poem by Stephen Crane, page numbers, the text of Joseph Katz’s dedication, preface, acknowledgements, and introduction, the table of contents, the index of first lines, the statements found on title page, back of title page (including CIP bibliographic record), cover front, back front, and spine, and the layout of the publication, and the occasional statement ‘[NO STANZA BREAK]’ (F24)

3. FRBR_{ER} Family to FRBR_{OO} Mappings

3.1. Introduction

This chapter defines the mapping between the FRBR_{ER} model (**Functional Requirements for Bibliographic Records** http://www.ifla.org/files/cataloguing/frbr/frbr_2008.pdf) and FRBR_{OO} (FRBR object-oriented definition and mapping of the FRBR_{ER}). The mapping includes a listing of the entities, relationships and attributes defined in FRBR_{ER} and shows how the same information can be expressed using FRBR_{OO}. These mappings can be seen on one hand as an intellectual definition of the relationship between both models. On the other hand, they are in a format that could be turned more or less mechanically into an algorithm to automatically transform data structured following the one form into data in the other form, i.e. they can be used to implement an automatic data translation.

The FRBR_{OO} model includes a more elaborated set of classes and for this reason there are several FRBR_{OO} classes listed for some of the FRBR_{ER} entities.

FRBR_{OO} is defined as an extension to the CIDOC CRM model which regards any information element as a property (or relationship) between two classes. For this reason each FRBR attribute or relationship is defined using a path of subsequent properties of FRBR_{OO} that includes the domain, the property name and the range of each property in the path.

The attributes in FRBR_{ER} have been defined at a logical level and express the characteristics of an entity as they might be viewed and composed by a user when creating an information system. In contrast to that, FRBR_{OO} tries to model the things and processes of the reality that librarians deal with and that FRBR_{ER} entities refer to explicitly or implicitly. Therefore corresponding paths expressed using FRBR_{OO} are in many cases complex paths that include the intermediate classes and properties that are needed to make explicit the implicit meaning and structure of the FRBR attributes. Using the same method of mapping, FRBR_{OO} can also be used to describe the meaning of data structures and models other than FRBR and so give an account of the degree to which such data structures or models represent library concepts.

Some FRBR_{ER} attributes and relationships will correspond to more than one path in FRBR_{OO} depending on the interpretation and use of an attribute or a relationship. This is particularly evident when inspecting manifestation attributes where attributes can be mapped to one path if they are used for uncontrolled text entries and a different path if they are used for codes or terms from a controlled set of values.

3.2. Explanation of Types Used in the Mapping

FRBR_{OO} is a core ontology in the sense of the CIDOC CRM, i.e., it specifies only the concepts necessary to describe the basic relationships between things in the selected domain of discourse. The same holds for the FRBR_{ER} model. Therefore, FRBR_{OO} adopts the CRM model's use of external types, i.e. terms that appear as data, declared as instances of the class E55 Type, to declare specialisations of concepts or relationships that are considered not to contribute to basic structure of the core ontology.

Some of the FRBR_{ER} attributes and relationships are considered to be too specific in this sense and for this reason are expressed in the mapping using paths that include instantiation of E55 Type in different ways. The following summary lists the different ways external types are used in the mapping. The listing of types is not exhaustive but is included as examples for the types (terms) that are needed here and there to express all the semantics of bibliographic data elements in FRBR_{OO}. Quotation marks indicate type values (instances of E55 Type) whereas type names without quotation marks identify whole type vocabularies.

FRBR_{ER} attributes mapped to FRBR_{OO} using *P3 has note* in combination with *P3.1 has type*.

This solution is used to map data elements that are uncontrolled text entries. They are all mapped to the property ***P3 has note*** for the respective class. In the sequence, *P3.1 has type* is used to differentiate between different special meanings of the *P3 has note* property and to express the particular meaning of the note, such as:

- *P3.1 has type*: E55 Type = {"Capture mode", "Collation", "Colour", "Extent of the carrier", "File characteristics", "Foliation", "Generation", "Groove width", "Kind of cutting", "Kind of sound", "Physical medium", "Playing speed", "Presentation format", "Reduction ratio", "Reproduction characteristics",

“Scheduled treatment”, “Series statement”, “System requirements”, “Tape configuration”, “Technique”, “Type size”, “Typeface”,...}

FRBR_{ER} attributes mapped to FRBR_{OO} using *P2 has type*.

This solution is used for data elements that are coded values or terms from a controlled vocabulary and express categorical characteristics of an entity. In the sequence, the use of the respective terminology to instantiate E55 Type should be restricted to specific vocabularies as indicated in the following list:

- E55 Type = *superclass of* (Capture mode, Colour, File characteristics, Foliation, Form of carrier, Generation, Groove width, Kind of cutting, Kind of sound, Playing speed, Presentation format, Reduction Ratio, Reproduction characteristics, System requirements, Tape configuration, Type size, Typeface,...)

FRBR_{ER} attributes mapped to FRBR_{OO} using *P148 has component (is component of)* of *E33 Linguistic Object P2 has type*.

This solution is used for data elements that are transcribing information appearing in the content of a bibliographic object, typically the title page. The property *P2 has type E55 Type* is used to define the category for this information, such as:

- E55 Type = {“Edition/Issue designation”, “Series statement”, “Statement of responsibility”}

FRBR_{ER} relationships mapped to FRBR_{OO} using *R2 is derivative of in combination with R2.1 has type*.

This solution is used to parametrise the specialisations of the more generic FRBR_{OO} property *R2 is derivative of* by the corresponding FRBR_{ER} subproperties:

- R2.1 has type E55 Type = {“Abridgement”, “Adaptation”, “Arrangement”, “Imitation”, “Revision”, “Summary”, “Transformation”, “Translation”}

3.3. List of FRBR_{ER} Mappings

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
3.2.1	Work		F1 Work
3.2.1	Work		F15 Complex Work
3.2.1	Work		F14 Individual Work
3.2.1	Work		F17 Aggregation Work
3.2.1	Work		F19 Publication Work
3.2.1	Work		F18 Serial Work
3.2.1	Work		F16 Container Work
3.2.1	Work		F20 Performance Work
3.2.1	Work		F21 Recording work
4.2.1	Work: Title of the work		F1 Work P102 has title E35 Title
4.2.2	Work: Form of work		F1 Work P2 has type E55 Type E55 Type {Form}
4.2.3	Work: Date of the work		F1 Work R16i was initiated by F27 Work Conception P4 has time-span E52 Time-Span P78 is identified by E50 Date
4.2.3	Work: Date of the work		F1 Work R19i was realised through F28 Expression Creation P2 has type E55 Type {“Earliest known externalisation”} and F28 Expression Creation P4 has time-span E5 Time- Span P78 is identified by E50 Date
4.2.4	Work: Other distinguishing characteristics		F1 Work P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
4.2.5	Work: Intended termination	if no intended termination it is an instance of F18 Serial Work	F18 Serial Work
4.2.5	Work: Intended termination	if it has an intended termination it is an instance of F14 Individual Work	F14 Individual Work
4.2.6	Work: Intended audience		F1 Work P103 was intended for E55 Type
4.2.6	Work: Intended audience		F1 Work P67 refers to {P67.1 has type E55 Type = “intended audience”} E74 Group
4.2.7	Work: Context for the work		F1 Work R16i was initiated by F27 Work Conception P15 was influenced by E1 CRM Entity
4.2.8	Work: Medium of performance (Musical work)		F1 Work P2 has type E55 Type {Medium}
4.2.8	Work: Medium of performance (Musical work)	used as part of an identifier	F1 Work P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object
4.2.8	Work: Medium of performance (Musical work)		F1 Work R40 has representative expression F22 Self-Contained Expression P103 was intended for E55 Type {Medium of performance}
4.2.9	Work: Numeric designation (Musical work)		F1 Work P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object
4.2.10	Work: Key (Musical work)	used as part of an identifier	F1 Work P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object
4.2.10	Work: Key (Musical work)		F1 Work R40 has representative expression F22 Self-Contained Expression P2 has type E55 Type {Key}
4.2.11	Work: Coordinates (Cartographic work)		F1 Work P129 is about E27 Site P59i is located in or within E53 Place P87 is identified by E47 Spatial Coordinates
4.2.12	Work: Equinox (Cartographic work)		F1 Work P129 is about E27 Site P59i is located in or within E53 Place P87 is identified by E47 Spatial Coordinates
5.2.1	Work: is realized through (Expression)		F1Work R3 is realised in F22 Self-Contained Expression
5.2.2	Work: is created by (Person, Corporate body)		F1 Work R16i was initiated by F27 Work Conception P14 carried out by {P14.1 in the role of: E55 Type = “Creator”} E39 Actor
5.2.3	Work: has as subject (all other entities)		F1 Work P129 is about E1 CRM Entity
5.2.3	Work: is subject of (Work)		F1 Work P129i is subject of F1 Work
5.3.1	Work: has a successor (Work)		F1 Work R1i has successor F1 Work
5.3.1	Work: is a successor to (Work)		F1 Work R1 is logical successor of F1 Work
5.3.1	Work: has a supplement (Work)		F1 Work R16i was initiated by F27 Work Conception P16 used specific object {P16.1 mode of use: E55 Type = “supplemented work”} F1 Work
5.3.1	Work: supplements (Work)		F1 Work P16i was used for {P16.1 mode of use: E55 Type = “supplemented work”} F27 Work Conception R16 initiated F1 Work

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
5.3.1	Work: has a complement (Work)		F15 Complex Work R10 has member F1 Work R3 is realised in F22 Self-Contained Expression P165 incorporates F22 Self-Contained Expression R3i realises F1 Work
5.3.1	Work: complements (Work)		F1 Work R3 is realised in F22 Self-Contained Expression P165i is incorporated in F22 Self-Contained Expression R3i realises F1 Work R10i is member of F15 Complex Work
5.3.1	Work: is a summary of (Work)		F1 Work R2 is derivative of {R2.1 has type E55 Type = "Summary"} F1 Work
5.3.1	Work: has a summary (Work)		F1 Work R2i has derivative {R2.1 has type E55 Type = "Summary"} F1 Work
5.3.1	Work: is an adaptation of (Work, Expression)		F1 Work R2 is derivative of {R2.1 has type E55 Type = "Adaptation"} F1 Work
5.3.1	Work: has adaptation (Work)		F1 Work R2i has derivative {R2.1 has type E55 Type = "Adaptation"} F1 Work
5.3.1	Work: is a transformation of (Work, Expression)		F1 Work R2 is derivative of {R2.1 has type E55 Type = "Transformation"} F1 Work
5.3.1	Work: has a transformation (Work)		F1 Work R2i has derivative {R2.1 has type E55 Type = "Transformation"} F1 Work
5.3.1	Work: is an imitation of (Work, Expression)		F1 Work R2 is derivative of {R2.1 has type E55 Type = "Imitation"} F1 Work
5.3.1	Work: has an imitation (Work)		F1 Work R2i has derivative {R2.1 has type E55 Type = "Imitation"} F1 Work
5.3.1.1	Work: has part (Work)		F15 Complex Work R10 has member F1 Work
5.3.1.1	Work: is part of (Work)		F1 Work R10i is member of F15 Complex Work
3.2.2	Expression		F2 Expression
3.2.2	Expression		F22 Self-Contained Expression
3.2.2	Expression		F24 Publication Expression
3.2.2	Expression		F23 Expression Fragment
3.2.2	Expression		F26 Recording
3.2.2	Expression		F25 Performance Plan
4.3.1	Expression: Title of the expression		F2 Expression P102 has title E35 Title
4.3.2	Expression: Form of the expression		F2 Expression P2 has type E55 Type {Form}
4.3.3	Expression: Date of the expression		F2 Expression R17i was created by F28 Expression Creation P4 has time-span E52 Time-Span P78 is identified by E50 Date
4.3.4	Expression: Language of the expression		F2 Expression (instantiated as E33 Linguistic Object) P72 has language E56 Language
4.3.5	Expression: Other distinguishing characteristics		F2 Expression P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object
4.3.6	Expression: Extensibility of expression		F22 Self-Contained Expression R3i realises F1 Work R10i is member of F18 Serial Work P3 has note {P3.1 has type E55 Type = "Extensibility"} E62 String
4.3.7	Expression: Revisability of expression		F22 Self-Contained Expression R3i realises F1 Work R10i is member of F18 Serial Work P3 has note {P3.1 has type E55 Type = "Revisability"} E62 String
4.3.8	Expression: Extent of the expression		F2 Expression P43 has dimension E54 Dimension

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
4.3.9	Expression: Summarization of content		F24 Expression R5 has component F2 Expression {P2 has type E55 Type = "Summary"}
4.3.10	Expression: Context for the expression		F2 Expression R17i was created by F28 Expression Creation P15 was influenced by E1 CRM Entity
4.3.11	Expression: Critical response to the expression		F2 Expression P129i is subject of F1 Work R3 is realised in F22 Self-Contained Expression
4.3.12	Expression: Use restrictions on the expression		F2 Expression P104 is subject to E30 Right
4.3.13	Expression: Sequencing pattern (Serial)		F22 Self-Contained Expression R3i realises F18 Serial work P3 has note {P3.1 has type E55 Type = "Sequencing pattern"} E62 String
4.3.13	Expression: Sequencing pattern (Serial)		F22 Self-Contained Expression R3i realises F18 Serial work R11 has issuing rules E29 Design or Procedure
4.3.14	Expression: Expected regularity of issue (Serial)		F22 Self-Contained Expression R3i realises F18 Serial work P3 has note {P3.1 has type E55 Type = "Expected regularity"} E62 String
4.3.14	Expression: Expected regularity of issue (Serial)		F22 Self-Contained Expression R3i realises F18 Serial Work R11 has issuing rules E29 Design or Procedure
4.3.15	Expression: Expected frequency of issue (Serial)		F22 Self-Contained Expression R3i realises F18 Serial work P3 has note {P3.1 has type E55 Type = "Expected frequency"} E62 String
4.3.15	Expression: Expected frequency of issue (Serial)		F22 Self-Contained Expression R3i realises F18 Serial Work R11 has issuing rules E29 Design or Procedure
4.3.16	Expression: Type of score (Musical notation)		F2 Expression P2 has type E55 Type {Type of score}
4.3.17	Expression: Medium of performance (Musical notation or recorded sound)		F2 Expression P2 has type E55 Type {Medium of performance}
4.3.17	Expression: Medium of performance (Musical notation)		F2 Expression P103 was intended for E55 Type {being performed on medium of performance N...}
4.3.17	Expression: Medium of performance (Recorded sound)		F26 Recording R21i was created through F29 Recording Event R20 recorded F31 Performance P125 used object of type E55 Type {Medium of performance}
4.3.18	Expression: Scale (Cartographic image/object)		F2 Expression (instantiated as E36 Visual Item P2 has type E55 Type {Cartographic image}) P138 represents {P138.1 has type E55 Type = "Scale"} E1 CRM Entity
4.3.19	Expression: Projection (Cartographic image/object)		F2 Expression (instantiated as E36 Visual Item P2 has type E55 Type {Cartographic image}) P138 represents {P138.1 has type E55 Type = "Projection"} E1 CRM Entity
4.3.19	Expression: Projection (Cartographic image/object)		F2 Expression R17i was created by F28 Expression Creation P32 used general technique E55 Type {Projection}
4.3.20	Expression: Presentation technique (Cartographic image/Object)		F2 Expression P2 has type E55 Type {Technique}
4.3.20	Expression: Presentation technique (Cartographic image/object)		F2 Expression R17i was created by F28 Expression Creation P32 used general technique E55 Type {Presentation technique}
4.3.21	Expression: Representation of relief (Cartographic image/object)		F2 Expression P2 has type E55Type {Technique}

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
4.3.21	Expression: Representation of relief (Cartographic image/object)		F2 Expression R17i was created by F28 Expression Creation P32 used general technique E55 Type {Representation of relief}
4.3.22	Expression: Geodetic, grid, and vertical measurement (Cartographic image/object)		F2 Expression P2 has type E55 Type {Different types of Geodetic, grid, and vertical measurement}
4.3.22	Expression: Geodetic, grid, and vertical measurement (Cartographic image/object)		F2 Expression R17i was created by F28 Expression Creation P32 used general technique E55 Type {Geodetic, grid, and vertical measurement}
4.3.23	Expression: Recording technique (Remote sensing image)		F2 Expression P2 has type E55 Type {Recording technique}
4.3.23	Expression: Recording technique (Remote sensing image)		F2 Expression R17i was created by F28 Expression Creation P32 used general technique E55 Type {Recording technique}
4.3.24	Expression: Special characteristics (Remote sensing image)		F2 Expression P3 has note {P3.1 has type E55 Type = “Special characteristics”} E62 String
4.3.25	Expression: Technique (Graphic of projected image)		F2 Expression P2 has type E55 Type {Technique}
4.3.25	Expression: Technique (Graphic of projected image)		F2 Expression R17i was created by F28 Expression Creation P32 used general technique E55 Type {Technique}
5.2.1	Expression: is a realization of		F22 Self-Contained Expression R3i realises F1 Work
5.2.1	Expression: is embodied in		F2 Expression R4 carriers provided by F3 Manifestation Product Type
5.2.1	Expression: is embodied in		F2 Expression P165i is incorporated in F24 Publication Expression CLR6i should be carried by F3 Manifestation Product Type
5.2.1	Expression: is embodied in		F2 Expression P128R is carried by F4 Manifestation Singleton
5.2.1	Expression: is embodied in		F2 Expression R17i was created by F28 Expression Creation R18 created F4 Manifestation Singleton
5.2.1	Expression: is realized by		F2 Expression R17i was created by F28 Expression Creation P14 carried out by {P14.1 in the role of E55 Type = e.g. “Translator”} E39 Actor
5.2.3	Expression: is subject of		F2 Expression P129i is subject of F1 Work
5.3.2	Expression: has an abridgement		F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = “Abridgement”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work
5.3.2	Expression: is an abridgement of		F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type E55 Type = “Abridgement”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
5.3.2	Expression: has a revision		F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = “Revision”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work
5.3.2	Expression: is a revision of		F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type E55 Type = “Revision”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work
5.3.2	Expression: has a translation		F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = “Translation”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work
5.3.2	Expression: has a translation		F22 Self-Contained Expression (instantiated as E33 Linguistic Object) P73 has translation E33 Linguistic Object
5.3.2	Expression: is a translation of		F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type E55 Type = “Translation”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work
5.3.2	Expression: is a translation of		F22 Self-Contained Expression (instantiated as E33 Linguistic Object) P73i is translation of E33 Linguistic Object
5.3.2	Expression: has an arrangement		F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = “Arrangement”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work
5.3.2	Expression: is an arrangement of		F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type E55 Type = “Arrangement”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work
5.3.2	Expression: has a successor		F22 Self-Contained Expression R9i realises F14 Individual Work R1i has successor F1 Work
5.3.2	Expression: is a successor to		F22 Self-Contained Expression R9i realises F14 Individual Work R1 is logical successor of F1 Work
5.3.2	Expression: has a supplement		F22 Self-Contained Expression R17i was created by F28 Expression Creation P16 used specific object {P16.1 mode of use E55 Type = “supplemented expression”} F2 Expression
5.3.2	Expression: supplements		F2 Expression P16 was used in {P16.1 mode of use E55 Type = “supplemented expression”} F28 Expression Creation R17 created F22 Self-Contained Expression
5.3.2	Expression: has a complement		F22 Self-Contained Expression P165 incorporates F2 Expression

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
5.3.2	Expression: complements		F2 Expression P165 is incorporated in F22 Self-Contained Expression
5.3.2	Expression: has a summary		F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = "Summary"} F1 Work R3 is realised in F22 Self-Contained Expression
5.3.2	Expression: is a summary of		F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type E55 Type = "Summary"} F1 Work R3 is realised in F22 Self-Contained Expression
5.3.2	Expression: has an adaptation		F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = "Adaptation"} F1 Work R3 is realised in F22 Self-Contained Expression
5.3.2	Expression: has an adaptation		F2 Expression P16 was used in {P16.1 mode of use E55 Type = "adapted source"} F28 Expression Creation R17 created F22 Self-Contained Expression
5.3.2	Expression: is an adaptation of		F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type E55 Type = "Adaptation"} F1 Work R3 is realised in F22 Self-Contained Expression
5.3.2	Expression: is an adaptation of		F22 Self-Contained Expression R17i was created by F28 Expression Creation P16 used specific object {P16.1 mode of use E55 Type = "adapted source"} F2 Expression
5.3.2	Expression: has a transformation		F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = "Transformation"} F1 Work R3 is realised in F22 Self-Contained Expression
5.3.2	Expression: has a transformation		F2 Expression P16 was used in {P16.1 mode of use E55 Type = "transformed source"} F28 Expression Creation R17 created F22 Self-Contained Expression
5.3.2	Expression: is a transformation of		F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type E55 Type = "Transformation"} F1 Work R3 is realised in F22 Self-Contained Expression
5.3.2	Expression: is a transformation of		F22 Self-Contained Expression R17i was created by F28 Expression Creation P16 used specific object {P16.1 mode of use E55 Type = "transformed source"} F2 Expression
5.3.2	Expression: has an imitation		F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = "Imitation"} F1 Work R3 is realised in F22 Self-Contained Expression
5.3.2	Expression: has an imitation		F2 Expression P16 was used in {P16.1 mode of use E55 Type = "imitated source"} F28 Expression Creation R17 created F22 Self-Contained Expression
5.3.2	Expression: is an imitation of		F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type: E55 Type = "Imitation"} F1 Work R3 is realised in F22 Self-Contained Expression
5.3.2	Expression: is an imitation of		F22 Self-Contained Expression R17i was created by F28 Expression Creation P16 used specific object {P16.1 mode of use E55 Type = "imitated source"} F2 Expression
5.3.2.1	Expression: has part		F2 Expression R15 has fragment F23 Expression Fragment
5.3.2.1	Expression: has part		F2 Expression R5 has component F22 Self-Contained Expression

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
5.3.2.1	Expression: is part of		F23 Expression Fragment R15i is fragment of F2 Expression
5.3.2.1	Expression: is part of		F22 Self-Contained Expression R5i is component of F2 Expression
3.2.3	Manifestation	If it is a published item, or something that is produced as multiple copies	F3 Manifestation Product Type
3.2.3	Manifestation	If it is a unique manifestation (most particularly a manuscript)	F4 Manifestation Singleton
4.4.1	Manifestation: Title of the manifestation		F3 Manifestation Product Type P102 has title E35 Title
4.4.1	Manifestation: Title of the manifestation		F4 Manifestation Singleton P102 has title E35 Title
4.4.2	Manifestation: Statement of responsibility		F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P148 has component E33 Linguistic Object P2 has type E55 Type {"Statement of responsibility"}
4.4.2	Manifestation: Statement of responsibility		F4 Manifestation Singleton P128 carries F2 Expression P148 has component E33 Linguistic Object P2 has type E55 Type {"Statement of responsibility"}
4.4.3	Manifestation: Edition/Issue designation		F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P148 has component E33 Linguistic Object P2 has type E55 Type {"Edition/Issue designation"}
4.4.3	Manifestation: Edition/Issue designation		F4 Manifestation Singleton P128 carries F2 Expression P148 has component E33 Linguistic Object P2 has type E55 Type {"Edition/Issue designation"}
4.4.4	Manifestation: Place of publication/distribution	Publishing:	F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P94 was created by F30 Publication Event P14 carried out by E39 Actor P74 has current or former residence E53 Place P87 is identified by E44 Place Appellation
4.4.4	Manifestation: Place of publication/distribution	Publishing:	F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P148 has component E33 Linguistic Object P2 has type E55 Type {Place of publication/distribution}
4.4.4	Manifestation: Place of publication/distribution	[Understood as the place where a manuscript was made]	F4 Manifestation Singleton R18i was created by [F28 Expression Creation or E12 Production, according to whether the manuscript is an original or a copy] P14 carried out by E39 Actor P74 has current or former residence E53 Place P87 is identified by E44 Place Appellation
4.4.4	Manifestation: Place of publication/distribution	Distribution:	F3 Manifestation Product Type P104 is subject to E30 Right {P2 has type E55 Type = "Distribution right"} P75i is possessed by E39 Actor P74 has current or former residence E53 Place P87 is identified by E44 Place Appellation

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
4.4.5	Manifestation: Publisher/distributor	Publishing:	F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P94 was created by F30 Publication Event P14 carried out by {P14.1 in the role of E55 Type = “Publisher”} E39 Actor P131 is identified by E82 Actor Appellation
4.4.5	Manifestation: Publisher/distributor	Publishing:	F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P148 has component E33 Linguistic Object P2 has type E55 Type {Publisher/distributor}
4.4.5	Manifestation: Publisher/distributor	[Understood as the person or institution who commissioned the manuscript]	F4 Manifestation Singleton R18i was created by [F28 Expression Creation or E12 Production, according to whether the manuscript is an original or a copy] P14 carried out by {P14.1 in the role of E55 Type = “Commissioner”} E39 Actor P131 is identified by E82 Actor Appellation
4.4.6	Manifestation: Date of publication/distribution		F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P148 has component E33 Linguistic Object P2 has type E55 Type {Date of publication/distribution}
4.4.6	Manifestation: Date of publication/distribution		F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P94 was created by F30 Publication Event P4 has time-span E52 Time-Span P78 is identified by E49 Time Appellation
4.4.6	Manifestation: Date of publication/distribution		F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P94 was created by E65 Creation Event P4 has time-span E52 Time-Span P82 at some time within E61 Time Primitive
4.4.6	Manifestation: Date of publication/distribution		F4 Manifestation-Singleton P128 carries E73 Information Object P148 has component E33 Linguistic Object P2 has type E55 Type {Date of publication/distribution}
4.4.6	Manifestation: Date of publication/distribution		F4 Manifestation-Singleton R18 was created by [F28 Expression Creation or E12 Production, according to whether the manuscript is an original or a copy] P4 has time-span E52 Time-Span P82 at some time within E61 Time Primitive
4.4.7	Manifestation: Fabricator/manufacturer		F3 Manifestation Product Type R26i was produced by F32 Carrier Production Event P14 carried out by E39 Actor P131 is identified by E82 Actor Appellation
4.4.7	Manifestation: Fabricator/manufacturer		F4 Manifestation-Singleton R18 was created [F28 Expression Creation or E12 Production, according to whether the manuscript is an original or a copy] P14 carried out by E39 Actor P131 is identified by E82 Actor Appellation
4.4.8	Manifestation: Series statement		F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P148 has component E33 Linguistic Object P2 has type E55 Type {“Series statement”}
4.4.9	Manifestation: Form of carrier		F3 Manifestation Product Type CLP2 should have type E55 Type {Form of carrier}
4.4.9	Manifestation: Form of carrier		F4 Manifestation Singleton P2 has type E55 Type {Form of carrier}
4.4.10	Manifestation: Extent of the carrier		F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = “Extent of the carrier”} E62 String

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
4.4.10	Manifestation: Extent of the carrier		F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Extent of the carrier"} E62 String
4.4.10	Manifestation: Extent of the carrier		F3 Manifestation Product Type CLP57 should have number of parts E60 Number
4.4.10	Manifestation: Extent of the carrier		F4 Manifestation Singleton P57 has number of parts E60 Number
4.4.11	Manifestation: Physical medium		F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Physical medium"} E62 String
4.4.11	Manifestation: Physical medium		F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Physical medium"} E62 String
4.4.11	Manifestation: Physical medium		F3 Manifestation Product Type CLP45 should consist of E57 Material
4.4.11	Manifestation: Physical medium		F4 Manifestation Singleton P45 consists of E57 Material
4.4.12	Manifestation: Capture mode		F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Capture mode"} E62 String
4.4.12	Manifestation: Capture mode		F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Capture mode"} E62 String
4.4.12	Manifestation: Capture mode		F3 Manifestation Product Type CLP2 should have type E55 Type {Capture mode}
4.4.12	Manifestation: Capture mode		F4 Manifestation Singleton P2 has type E55 Type {Capture mode}
4.4.13	Manifestation: Dimensions of the carrier		F3 Manifestation Product Type CLP43 should have dimension E54 Dimension
4.4.13	Manifestation: Dimensions of the carrier		F4 Manifestation Singleton P43 has dimension E54 Dimension
4.4.14	Manifestation: Manifestation identifier		F3 Manifestation Product Type P1 is identified by F13 Identifier
4.4.14	Manifestation: Manifestation identifier		F4 Manifestation Singleton P1 is identified by E42 Identifier
4.4.15	Manifestation: Source for acquisition/access authorization		F3 Manifestation Product Type CLP104 is subject to E30 Right
4.4.15	Manifestation: Source for acquisition/access authorization		F3 Manifestation Product Type CLP105 right held by E39 Actor P131 is identified by E82 Actor Appellation
4.4.15	Manifestation: Source for acquisition/access authorization		F4 Manifestation Singleton P104 is subject to E30 Right
4.4.15	Manifestation: Source for acquisition/access authorization		F4 Manifestation Singleton P105 right held by E39 Actor P131 is identified by E82 Actor Appellation
4.4.15	Manifestation: Source for acquisition/access authorization		F4 Manifestation Singleton P49 has former or current keeper E39 Actor P131 is identified by E82 Actor Appellation
4.4.15	Manifestation: Source for acquisition/access authorization		F4 Manifestation Singleton P51 has former or current owner E39 Actor P131 is identified by E82 Actor Appellation
	Manifestation: Terms of availability		F3 Manifestation Product Type CLP104 is subject to E30 Right
4.4.16	Manifestation: Terms of availability		F4 Manifestation Singleton P104 is subject to E30 Right
4.4.17	Manifestation: Access restrictions on the manifestation		F3 Manifestation Product Type CLP104 is subject to E30 Right
4.4.17	Manifestation: Access restrictions on the manifestation		F4 Manifestation Singleton P104 is subject to E30 Right

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4.4.18	Manifestation: Typeface (Printed book)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Typeface"} E62 String
4.4.18	Manifestation: Typeface (Printed book)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {Typeface}
4.4.18	Manifestation: Typeface (Printed book)	descriptive form	F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Typeface"} E62 String
4.4.18	Manifestation: Typeface (Printed book)	coded form	F4 Manifestation Singleton P2 has type E55 Type {Typeface}
4.4.18	Manifestation: Type size (Printed book)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Type size"} E62 String
4.4.18	Manifestation: Type size (Printed book)	coded form	F3 Manifestation Product Type CLR6 should carry F24 Publication Expression R24i was created through F30 Publication Event P94 created E29 Design or Procedure P67 refers to {P67.1 has type: E55 Type = Type size} E54 Dimension
4.4.18	Manifestation: Type size (Printed book)	descriptive form	F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Type size"} E62 String
4.4.18	Manifestation: Type size (Printed book) [for typescripts and digital typescripts]	coded form	F4 Manifestation Singleton P128 carries E73 Information Object P43 has dimension E54 Dimension P2 has type E55 Type {Type size}
4.4.20	Manifestation: Foliation (Hand-printed book)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Foliation"} E62 String
4.4.20	Manifestation: Foliation (Hand-printed book)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {Foliation}
4.4.20	Manifestation: Foliation (Hand-printed book)	descriptive form	F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Foliation"} E62 String
4.4.20	Manifestation: Foliation (Hand-printed book)	coded form	F4 Manifestation Singleton P2 has type E55 Type {Foliation}
4.4.21	Manifestation: Collation (Hand-printed book)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Collation"} E62 String
4.4.21	Manifestation: Collation (Hand-printed book)	descriptive form	F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Collation"} E62 String
4.4.22	Manifestation: Publication status (Serial)	coded form	F18 Serial Work P2 has type E55 Type {Publication status}
4.4.22	Manifestation: Publication status (Serial)	descriptive form	F18 Serial Work P2 has note {P3.1 has type E55 Type = "Publication status"} E62 String
4.4.23	Manifestation: Numbering (Serial)		F3 Manifestation Product Type P1 is identified by F13 Identifier R8 consists of F12 Nomen
4.4.24	Manifestation: Playing speed (Sound recording)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Playing speed"} E62 String
4.4.24	Manifestation: Playing speed (Sound recording)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {Playing speed}
4.4.24	Manifestation: Playing speed (Sound recording)	descriptive form	F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Playing speed"} E62 String
4.4.24	Manifestation: Playing speed (Sound recording)	coded form	F4 Manifestation Singleton P2 has type E55 Type {Playing speed}
4.4.25	Manifestation: Groove width (Sound recording)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Groove width"} E62 String
4.4.25	Manifestation: Groove width (Sound recording)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {Groove width}
4.4.25	Manifestation: Groove width (Sound recording)	descriptive form	F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Groove width"} E62 String

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4.4.25	Manifestation: Groove width (Sound recording)	coded form	F4 Manifestation Singleton P2 has type E55 Type {Groove width}
4.4.26	Manifestation: Kind of cutting (Sound recording)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Kind of cutting"} E62 String
4.4.26	Manifestation: Kind of cutting (Sound recording)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {Kind of cutting}
4.4.26	Manifestation: Kind of cutting (Sound recording)	descriptive form	F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Kind of cutting"} E62 String
4.4.26	Manifestation: Kind of cutting (Sound recording)	coded form	F4 Manifestation Singleton P2 has type E55 Type {Kind of cutting}
4.4.27	Manifestation: Tape configuration (Sound recording)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Tape configuration"} E62 String
4.4.27	Manifestation: Tape configuration (Sound recording)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {Tape configuration}
4.4.27	Manifestation: Tape configuration (Sound recording)	descriptive form	F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Tape configuration"} E62 String
4.4.27	Manifestation: Tape configuration (Sound recording)	coded form	F4 Manifestation Singleton P2 has type E55 Type {Tape configuration}
4.4.28	Manifestation: Kind of sound (Sound recording)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Kind of sound"} E62 String
4.4.28	Manifestation: Kind of sound (Sound recording)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {Kind of sound}
4.4.28	Manifestation: Kind of sound (Sound recording)	descriptive form	F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Kind of sound"} E62 String
4.4.28	Manifestation: Kind of sound (Sound recording)	coded form	F4 Manifestation Singleton P2 has type E55 Type {Kind of sound}
4.4.29	Manifestation: Special reproduction characteristics (Sound recording)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Reproduction characteristics"} E62 String
4.4.29	Manifestation: Special reproduction characteristics (Sound recording)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {Reproduction characteristics}
4.4.29	Manifestation: Special reproduction characteristics (Sound recording)	descriptive form	F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Reproduction characteristics"} E62 String
4.4.29	Manifestation: Special reproduction characteristics (Sound recording)	coded form	F4 Manifestation Singleton P2 has type E55 Type {Reproduction characteristics}
4.4.30	Manifestation: Colour (Image)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Colour"} E62 String
4.4.30	Manifestation: Colour (Image)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {Colour}
4.4.30	Manifestation: Colour (Image)	descriptive form	F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Colour"} E62 String
4.4.30	Manifestation: Colour (Image)	coded form	F4 Manifestation Singleton P2 has type E55 Type {Colour}
4.4.31	Manifestation: Reduction ratio (Microform)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Reduction ratio"} E62 String

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4.4.31	Manifestation: Reduction ratio (Microform)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {Reduction Ratio}
4.4.32	Manifestation: Polarity (Microform or visual projection)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Tape configuration"} E62 String
4.4.32	Manifestation: Polarity (Microform or visual projection)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {Tape configuration}
4.4.33	Manifestation: Generation (Microform or visual projection)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = Generation} E62 String
4.4.33	Manifestation: Generation (Microform or visual projection)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {Generation}
4.4.34	Manifestation: Presentation format (Visual projection)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "Presentation format"} E62 String
4.4.34	Manifestation: Presentation format (Visual projection)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {Presentation format}
4.4.34	Manifestation: Presentation format (Visual projection)	coded form	F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P103 was intended for E55 Type {being projected in format N...}
4.4.35	Manifestation: System requirements (Electronic resource)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "System requirements"} E62 String
4.4.35	Manifestation: System requirements (Electronic resource)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {System requirements}
4.4.35	Manifestation: System requirements (Electronic resource)	coded form	F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P103 was intended for E55 Type {being used in environment N...}
4.4.36	Manifestation: File characteristics (Electronic resource)	descriptive form	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = "File characteristics"} E62 String
4.4.36	Manifestation: File characteristics (Electronic resource)	coded form	F3 Manifestation Product Type CLP2 should have type E55 Type {File characteristics}
4.4.37	Manifestation: Mode of access (Remote access electronic resource)		F3 Manifestation Product P3 has note {P3.1 has type E55 Type = "Mode of access"} E62 String
4.4.38	Manifestation: Access address (Remote access electronic resource)		F3 Manifestation Product P3 has note {P3.1 has type E55 Type = "Access address"} E62 String
5.2.1	Manifestation: is the embodiment of		F3 Manifestation Product Type R4 comprises carriers of F2 Expression
5.2.1	Manifestation: is the embodiment of		F3 Manifestation Product Type CLR6 should carry F24 Publication Expression
5.2.1	Manifestation: is the embodiment of	(in the case of an original manuscript only)	F4 Manifestation Singleton R18i was created by F28 Expression Creation R17 created F2 Expression
5.2.1	Manifestation: is the embodiment of	(in all cases)	F4 Manifestation Singleton P128 carries F2 Expression
5.2.1	Manifestation: is exemplified by		F3 Manifestation Product Type R7i has example F5 Item
5.3.5	Manifestation: is produced by		F3 Manifestation Product Type R26i was produced by F32 Carrier Production Event P14 carried out by E39 Actor P131 is identified by E82 Actor Appellation

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
5.2.2	Manifestation: is produced by		F4 Manifestation Singleton R18 was created by [F28 Expression Creation or E12 Production, according to whether the manuscript is an original or a copy] P14 carried out by E39 Actor P131 is identified by E82 Actor Appellation
5.2.3	Manifestation: is subject of		F3 Manifestation Product Type P129i is subject of F1 Work
5.2.3	Manifestation: is subject of		F4 Manifestation Singleton P129i is subject of F1 Work
5.3.4.1	Manifestation: has part		F3 Manifestation Product Type CLP46 should be composed of F3 Manifestation Product Type
5.3.4.1	Manifestation: has part		F4 Manifestation Singleton P46 is composed of F4 Manifestation Singleton
5.3.4.1	Manifestation: is part of		F3 Manifestation Product Type CLP46i may form part of F3 Manifestation Product Type
5.3.4.1	Manifestation: is part of		F4 Manifestation Singleton P46i forms part of F4 Manifestation Singleton
5.3.4	Manifestation: has a reproduction (a Manifestation)	[generic case]	[F3 Manifestation Product Type or F4 Manifestation Singleton] P130i features are also found on {P130.1 kind of similarity: E55 Type = "Reproduction"} [F3 Manifestation Product Type or F4 Manifestation Singleton]
5.3.4	Manifestation: has a reproduction (a Manifestation)	[from F3 to F3]	F3 Manifestation Product Type P125 was type of object used in F33 Reproduction Event R30 produced E84 Information Carrier P128 carries E90 Symbolic Object P165i is incorporated in F24 Publication Expression CLR6i should be carried by F3 Manifestation Product Type
5.3.4	Manifestation: has a reproduction (a Manifestation)	[from F3 to F4]	F3 Manifestation Product Type P125 was type of object used in F33 Reproduction Event R30 produced E84 Information Carrier (also instantiated as F4 Manifestation Singleton)
5.3.4	Manifestation: has a reproduction (a Manifestation)	[from F4 to F3]	F4 Manifestation Singleton P16i was used for {P16.1 mode of use: E55 Type = "reproduced source"} F33 Reproduction Event R30 produced E84 Information Carrier P128 carries E90 Symbolic Object P165i is incorporated in F24 Publication Expression CLR6i should be carried by F3 Manifestation Product Type
5.3.4	Manifestation: has a reproduction (a Manifestation)	[from F4 to F4]	F4 Manifestation Singleton (also instantiated as E84 Information Carrier) R29i was reproduced by F33 Reproduction Event R30 produced E84 Information Carrier (also instantiated as F4 Manifestation Singleton)
5.3.4	Manifestation: is a reproduction of (a Manifestation)	[generic case]	[F3 Manifestation Product Type or F4 Manifestation Singleton] P130 shows features of {P130.1 kind of similarity: E55 Type = "Reproduction"} [F3 Manifestation Product Type or F4 Manifestation Singleton]
5.3.4	Manifestation: is a reproduction of (a Manifestation)	[from F3 to F3]	F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P165 incorporates E90 Symbolic Object P128i is carried by E84 Information Carrier R30i was produced by F33 Reproduction Event P125 used object of type F3 Manifestation Product Type
5.3.4	Manifestation: is a reproduction of (a Manifestation)	[from F3 to F4]	F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P165 incorporates E90 Symbolic Object P128i is carried by E84 Information Carrier R30i was produced by F33 Reproduction Event R29 reproduced E84 Information Carrier (also instantiated as F4 Manifestation Singleton)

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
5.3.4	Manifestation: is a reproduction of (a Manifestation)	[from F4 to F3]	F4 Manifestation Singleton (also instantiated as E84 Information Carrier) R30i was produced by F33 Reproduction Event P125 used object of type F3 Manifestation Product Type
5.3.4	Manifestation: is a reproduction of (a Manifestation)	[from F4 to F4]	F4 Manifestation Singleton R30i was produced by F33 Reproduction Event R29 reproduced E84 Information Carrier (also instantiated as F4 Manifestation Singleton)
5.3.4	Manifestation: has an alternate	(if the relationship is expressed as a note)	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = “has an alternate”} E62 String
5.3.4	Manifestation: has an alternate	(if the relationship is expressed as a link)	F3 Manifestation Product Type P130i features are also found on {P130.1 kind of similarity: E55 Type = “[type of alternate format]”} F3 Manifestation Product Type
5.3.4	Manifestation: is an alternate to	(if the relationship is expressed as a note)	F3 Manifestation Product Type P3 has note {P3.1 has type E55 Type = “is an alternate to”} E62 String
5.3.4	Manifestation: is an alternate to	(if the relationship is expressed as a link)	F3 Manifestation Product Type P130 shows features of {P130.1 kind of similarity: E55 Type = “[type of alternate format]”} F3 Manifestation Product Type
5.3.5	Manifestation: is a reproduction of (an Item)	[generic case]	[F3 Manifestation Product Type or F4 Manifestation Singleton] P130 shows features of {P130.1 kind of similarity: E55 Type = “Reproduction”} F5 Item
5.3.5	Manifestation: is a reproduction of (an Item)	[from F3 to F5]	F3 Manifestation Product Type CLR6 should carry F24 Publication Expression P165 incorporates E90 Symbolic Object P128i is carried by E84 Information Carrier R30i was produced by F33 Reproduction Event R29 reproduced F5 Item
5.3.5	Manifestation: is a reproduction of (an Item)	[from F4 to F5]	F4 Manifestation Singleton R30i was produced by F33 Reproduction Event R29 reproduced F5 Item
3.2.4	Item		F4 Manifestation Singleton
3.2.4	Item		F5 Item
4.5.1	Item: Item identifier		F4 Manifestation Singleton P1 is identified by E42 Identifier
4.5.1	Item: Item identifier		F5 Item P1 is identified by E42 Identifier
4.5.2	Item: Fingerprint		F4 Manifestation Singleton P1 is identified by E41 Appellation
4.5.2	Item: Fingerprint		F5 Item P1 is identified by E41 Appellation
4.5.3	Item: Provenance of the item		F4 Manifestation Singleton P49 has former or current keeper E39 Actor
4.5.3	Item: Provenance of the item		F4 Manifestation Singleton P51 has former or current owner E39 Actor
4.5.3	Item: Provenance of the item		F5 Item P49 has former or current keeper E39 Actor
4.5.3	Item: Provenance of the item		F5 Item P51 has former or current owner E39 Actor
4.5.4	Item: Marks/inscriptions		F4 Manifestation Singleton P65 shows visual item E37 Mark
4.5.4	Item: Marks/inscriptions		F5 Item P65 shows visual item E37 Mark
4.5.5	Item: Exhibition history		F4 Manifestation Singleton P12i was present at E7 Activity

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
4.5.5	Item: Exhibition history		F5 Item P12i was present at E7 Activity
4.5.6	Item: Condition of the item		F4 Manifestation Singleton P44 has condition state E3 Condition State
4.5.6	Item: Condition of the item		F5 Item P44 has condition state E3 Condition State
4.5.7	Item: Treatment history		F4 Manifestation Singleton P31i was modified by E11 Modification Event
4.5.7	Item: Treatment history		F5 Item P31i was modified by E11 Modification Event
4.5.8	Item: Scheduled treatment		F4 Manifestation Singleton P3 has note {P3.1 has type E55 Type = "Scheduled treatment"} E62 String
4.5.8	Item: Scheduled treatment		F5 Item P3 has note {P3.1 has type E55 Type = "Scheduled treatment"} E62 String
4.5.9	Item: Access restrictions on the item		F4 Manifestation Singleton P104 is subject to E30 Right
4.5.9	Item: Access restrictions on the item		F5 Item P104 is subject to E30 Right
5.2.1	Item: exemplifies		F5 Item R7 is example of F3 Manifestation Product Type
5.2.3	Item: is subject of		F5 Item P129i is subject of F1 Work
5.2.2	Item: is owned by		P5 Item P51 has former or current owner E39 Actor
5.2.2	Item: is owned by		P5 Item P50 has current keeper E39 Actor
5.3.5	Item: has a reproduction (a Manifestation)	[generic case]	F5 Item P130i features are also found on {P130.1 kind of similarity: E55 Type = "Reproduction"} [F3 Manifestation Product Type or F4 Manifestation Singleton]
5.3.5	Item: has a reproduction (a Manifestation)	[from F5 to F3]	F5 Item R29i was reproduced by F33 Reproduction Event R30 produced E84 Information Carrier P128 carries E90 Symbolic Object P165i is incorporated in F24 Publication Expression CLR6i should be carried by F3 Manifestation Product Type
5.3.5	Item: has a reproduction (a Manifestation)	[from F5 to F4]	F5 Item R29i was reproduced by F33 Reproduction Event R30 produced F4 Manifestation Singleton
5.3.6.1	Item: has part		F5 Item P46 is composed of F5 Item
5.3.6.1	Item: has part		F4 Manifestation Singleton P46 is composed of F4 Manifestation Singleton
5.3.6.1	Item: is part of		F5 Item P46i forms part of F5 Item
5.3.6.1	Item: is part of		F4 Manifestation Singleton P46i forms part of F4 Manifestation Singleton
5.3.6	Item: has reconfiguration	("bound with")	F5 Item P16 was used for {P16.1 mode of use: E55 Type = "Bound item"} E7 Activity P16 used specific object {P16.1 mode of use: E55 Type = "Bound item"} F5 Item
5.3.6	Item: has reconfiguration	("split into", "extracted from")	F5 Item P112i was diminished by E80 Part Removal P113 removed E84 Information Carrier [portion of an instance of F5 Item]
5.3.6	Item: is a reconfiguration of	("bound with")	F5 Item P16 was used for {P16.1 mode of use: E55 Type = "Bound item"} E7 Activity P16 used specific object {P16.1 mode of use: E55 Type = "Bound item"} F5 Item
5.3.6	Item: is a reconfiguration of	("split into", "extracted from")	E84 Information Carrier [portion of an instance of F5 Item] P113i was removed by E80 Part Removal P112 diminished F5 Item

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
5.3.6	Item: has a reproduction (an Item)		F5 Item P130i features are also found on {P130.1 kind of similarity: E55 Type = “Reproduction”} F5 Item
5.3.6	Item: has a reproduction (an Item)		F5 Item R29 was reproduced by F33 Reproduction Event R30 produced F5 Item
5.3.6	Item: is a reproduction of (an Item)		F5 Item P130 shows features of {P130.1 kind of similarity: E55 Type = “Reproduction”} F5 Item
5.3.6	Item: is a reproduction of (an Item)		F5 Item R30i was produced by F33 Reproduction Event R29 reproduced F5 Item
3.2.5	Person		E21 Person
4.6.1	Person: Name of person		E21 Person P131 is identified by E82 Actor Appellation
4.6.2	Person: Dates of person	Date of birth	E21 Person P98 was born E67 Birth P4 has time-span E52 Time-Span P78 is identified by E50 Date
4.6.2	Person: Dates of person	Date of death	E21 Person P100 died in E69 Death P4 has time-span E52 Time-Span P78 is identified by E50 Date
4.6.2	Person: Dates of person	was active in period	E21 Person P14i performed F51 Pursuit P4 has time-span E52 Time-Span P78 is identified by E50 Date
4.6.2	Person: Dates of person	part of identifier	F21 Person P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object
4.6.3	Person: Title of person		E21 Person P2 has type E55 Type {Title}
4.6.3	Person: Title of person		E21 Person P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object
4.6.4	Person: Other designation associated with the person		E21 Person P1 is identified by F50 Controlled Access Point R8 consists E90 Symbolic Object
5.2.2	Person: has created		E21 Person P14 performed F27 Work Conception R16 initiated F1 Work
5.2.2	Person: has realized		E21 Person P14i performed F28 Expression Creation R17 created F2 Expression
5.2.2	Person: has produced	(in the sense of: “publishing”)	E21 Person P14i performed F30 Publication Event R24 created F24 Publication Expression CLR6i should be carried by F3 Manifestation Product Type
5.2.2	Person: has produced	(in the sense of: “distributing”)	E21 Person P75 possesses E30 Right {P2 has type E55 Type = “Distribution right”} P104i applies to F3 Manifestation Product Type
5.2.2	Person: has produced	(in the sense of: “fabricating or manufacturing”)	E21 Person P14i performed F32 Carrier Production Event R26 produced things of type F3 Manifestation Product Type
5.2.2	Person: has produced	(in the sense of: “fabricating or manufacturing”; for an original unique document)	E21 Person P14i performed F28 Expression Creation R18 created F4 Manifestation Singleton
5.2.2	Person: has produced	(in the sense of: “fabricating or manufacturing”; for a unique document that is a copy of another one)	E21 Person P14i performed E12 Production P108 produced F4 Manifestation Singleton
5.2.2	Person: is owner of		E21 Person P51i is former or current owner of F5 Item
5.2.2	Person: is owner of		E21 Person P51i is former or current owner of F4 Manifestation Singleton

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
4.6.1	Person: is subject of		E21 Person P129i is subject of F1 Work
3.2.6	Corporate Body		F11 Corporate Body
4.7.1	Corporate Body: Name of the corporate body		F11 Corporate Body P131 is identified by E82 Actor Appellation
4.7.2	Corporate Body: Number associated with the corporate body	(element of identifier)	F11 Corporate Body P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object
4.7.3	Corporate Body: Place associated with the corporate body	Place associated with an activity / event	F11 Corporate Body P14i performed E7 Activity P7 took place at E53 Place P87 is identified by E44 Place Appellation
4.7.3	Corporate Body: Place associated with the corporate body	Location with which the corporate body is otherwise associated	F11 Corporate Body P74 has current or former residence E53 Place P87 is identified by E44 Place Appellation
4.7.3	Corporate Body: Place associated with the corporate body	(element of identifier)	F11 Corporate Body P1 is identified by F50 Controlled Access Point R8 consists of E44 Place Appellation
4.7.4	Corporate Body: Date associated with the corporate body	Formation of a group	F11 Corporate Body P95i was formed by E66 Formation P4 has time-span E52 Time-Span P78 is identified by E50 Date
4.7.4	Corporate Body: Date associated with the corporate body	Time-span of the event (conference)	F11 Corporate Body P14i performed E7 Activity P4 has time-span E52 Time-Span P78 is identified by E50 Date
4.7.4	Corporate Body: Date associated with the corporate body	(element of identifier)	F11 Corporate Body P1 is identified by F50 Controlled Access Point R8 consists of E50 Date
4.7.5	Corporate Body: Other designation associated with the corporate body	(element of identifier)	F11 Corporate Body P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object
5.2.2	Corporate Body: has created		F11 Corporate Body P14i performed F27 Work Conception R16 initiated F1 Work
5.2.2	Corporate Body: has realized		F11 Corporate Body P14i performed F28 Expression Creation R17 created F2 Expression
5.2.2	Corporate Body: has produced	(in the sense of: “publishing”)	F11 Corporate Body P14i performed F30 Publication Event R24 created F24 Publication Expression CLR6i should be carried by F3 Manifestation Product Type
5.2.2	Corporate Body: has produced	(in the sense of: “distributing”)	F11 Corporate Body P75 possesses E30 Right {P2 has type E55 Type = “Distribution right”} P104i applies to F3 Manifestation Product Type
5.2.2	Corporate Body: has produced	(in the sense of: “fabricating or manufacturing”)	F11 Corporate Body P14i performed F32 Carrier Production Event R26 produced things of type F3 Manifestation Product Type
5.2.2	Corporate Body: has produced	(in the sense of: “fabricating or manufacturing”; for an original unique document)	F11 Corporate Body P14i performed F28 Expression Creation R18 created F4 Manifestation Singleton
5.2.2	Corporate Body: has produced	(in the sense of: “fabricating or manufacturing”; for a unique document that is a copy of another one)	F11 Corporate Body P14i performed E12 Production P108 produced F4 Manifestation Singleton
5.2.2	Corporate Body: is owner of		F11 Corporate Body P51i is former or current owner of F5 Item
5.2.2	Corporate Body: is owner of		F11 Corporate Body P51i is former or current owner of F4 Manifestation Singleton
5.2.3	Corporate Body: is subject of		F11 Corporate Body P129i is subject of F1 Work

FRBR _{ER} Section numbers	Unit of Information	Condition	Mapping
3.2.7	Concept		F6 Concept
4.8.1	Concept: Term for the concept		F6 Concept P1 is identified by E41 Appellation
5.2.3	Concept: is subject of		F6 Concept P129i is subject of F1 Work
3.2.8	Object		E18 Physical Thing
4.9.1	Object: Term for the object		E18 Physical Thing P1 is identified by E41 Appellation
5.2.3	Object: is subject of		E18 Physical Thing P129i is subject of F1 Work
3.2.9	Event		E4 Period
4.10.1	Event: Term for the event		E4 Period P1 is identified by E41 Appellation
5.2.3	Event: is subject of		E4 Period P129i is subject of F1 Work
3.2.10	Place		E53 Place
4.11.1	Place: Term for the place		E53 Place P87 is identified by E44 Place Appellation
5.2.3	Place: is subject of		E53 Place P129i is subject of F1 Work

3.4. FRSAD to FRBR_{oo} Mappings

FRSAD section number	Unit of Information	Condition	Mapping	Comment
[3.2]	Work		F1 Work	
5.1	Work: has subject		F1 Work P129 is about E1 CRM Entity	
3.4	Thema		E1 CRM Entity	Appears as the reference to any entity without specifying a specific way in which the entity appears in the work. MD: relational expression to anything => class = E1
4.1.1	Thema: Type of thema		E1 CRM Entity. P2 has type E55 Type	
4.1.2	Thema: Scope note		E1 CRM Entity. P3 has note E62 String	
5.1	Thema: is subject of		E1 CRM Entity P1029i is subject of F1 Work	
5.2	Thema: has appellation		E1 CRM Entity P1 is identified by F12 Nomen	

FRSAD section number	Unit of Information	Condition	Mapping	Comment
5.3.1.1	Thema: Generic relationship		IsA OR E55 Type P127 has broader term (has narrower term) E55 Type	
5.3.1.2	Thema: Whole-part relationship		a) for E55 Type: SKOS broader partitive b) a series of CRM properties relating to inclusion to be listed	
5.3.1.3	Thema: Instance relationship		E55 Type P2i is type of E1 CRM Entity	
5.3.1.4	Thema: Polyhierarchical relationship			This is a statement about the cardinality of the relationships enumerated in section 5.3 of FRSAD, rather than a distinct relationship in its own right
5.3.1.5	Thema: Other hierarchical relationship			
5.3.2	Thema: Associative relationships		Any other “related to” relationship.	No model necessary
3.5	Nomen	F12 Nomen		
4.2.1	Nomen: Type of nomen		F12 Nomen P2 has type E55 Type	
4.2.2	Nomen: Scheme		F35 Nomen Use Statement R35 is specified by F34 KOS	FRSAD:Scheme = KOS, FRAD “authority file” □ ⊆ E32 Authority Document. The scheme in which the nomen is established, including value encoding schemes (subject heading lists, thesauri, classification systems, name authority lists, etc.) and syntax encoding schemes (standards for encoding dates, etc.).
4.2.3	Nomen: Reference source of nomen	Warning! Only if Nomen is uniquely defined by one symbolic form	F35 Nomen Use Statement R32 is warranted by F52 Name Use Activity	
4.2.4	Nomen: Representation		F12 Nomen. P2 has type E55 Type	

FRSAD section number	Unit of Information	Condition	Mapping	Comment
4.2.5	Nomen: Language		F35 Nomen Use Statement (instantiated as E33 Linguistic Object) R54 has nomen language E56 Language	
4.2.6	Nomen: Script	The script type in the sense of ISO15924	F12 Nomen P2 has type E55 Type	
4.2.7	Nomen: Script conversion		F35 Nomen Use Statement R36 uses script conversion F36 Script Conversion	
4.2.8	Nomen: Form		F35 Nomen Use Statement R55 has nomen form E55 Type	
4.2.9	Nomen: Time of validity of nomen		F35 Nomen Use Statement R35 is specified by F34 KOS R34 has validity period E52 Time-Span	Only validity status of participation in KOS release
4.2.10	Nomen: Audience		F35 Nomen Use Statement R39 is intended for E74 Group	Audience is not usage status
4.2.11	Nomen: Status		F35 Nomen Use Statement R35 is specified by {R35.1 has status E55 Type} F34 KOS	
5.2	Nomen: is appellation of		F12 Nomen P1i identifies E1 CRM Entity	
5.4.1	Nomen: Equivalence relationship		F35 Nomen Use Statement R56 has related use {R56.1 has type E55 Type = "equivalence"} F35 Nomen Use Statement	
5.4.2	Nomen: Whole-part relationship		F35 Nomen Use Statement P106 is composed of E90 Symbolic Object	

3.5. FRAD to FRBR_{oo} Mappings

FRAD section number	Unit of Information	Condition	Mapping	Comment
3.4	Person		union of F10 = E21 Person and F38 Character	

FRAD section number	Unit of Information	Condition	Mapping	Comment
4.1	Person: Dates associated with the person		a) F10 Person P98i was born E67 Birth P4 has time-span E52 Time-Span P78 is identified by E50 Date b) F10 Person P100i died in E69 Death P4 has time-span E52 Time-Span P78 is identified by E50 Date c) F10 Person P14i performed F51 Pursuit. P4 has time-span: E52 Time-Span P78 is identified by E50 Date d) F10 Person P12i was present at E5 Event (was living) e) F10 Person P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object	
4.1	Person: Title of person		F10 Person P2 has type E55 Type F10 Person P107i is current or former member of E74 Group (the office justifying the title) or F10 Person P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object	“title” seen as category of people that are allowed to carry a name of this category as additive to their names, such as “Dr.” “Mrs.” etc., or the name of the office the person holds or held that justified the title or an element in the identifier for the person
4.1	Person: Gender		F10 Person P2 has type E55 Type	
4.1	Person: Place of Birth		F10 Person P98i was born E67 Birth P7 took place at E53 Place	
4.1	Person: Place of Death		F10 Person P100i died in E69 Death P7 took place at E53 Place	
4.1	Person: Country		It may be one of the following: a) place of citizenship b) place of Pursuit F51 Pursuit P7 took place at E53 Place c) place of birth d) typical subject of work: F51 Pursuit R59.had typical subject E1 CRM Entity e) typical place of publishing: P3 has note	There are different local interpretations of which relationship is relevant for identification
4.1	Person: Place of residence		F10 Person P74 has current or former	

FRAD section number	Unit of Information	Condition	Mapping	Comment
			residence E53 Place	
4.1	Person: Affiliation		F10 Person P107i is current or former member of E74 Group	“cultural identity” maps to: P2 has type E55 Type (such as El Greco “Spanish Painter”)
4.1	Person: Address		a) F10 Person P76 has contact point E51 Contact Point (E45 Address is subclass of E51) b) F10 Person P107i is current or former member of E74 Group P76 has contact point E51 Contact Point c) F10 Person P74 has current or former residence E53 Place P87 is identified by E45 Address	A place name, includes E51 Contact Points, E45 Address. These are specializations of E44 Placename
4.1	Person: Language		F10 Person.P14i performed F51 Pursuit R60 used to use language E56 Language	
4.1	Person: Field of activity		a) F10 Person P14i performed F51 Pursuit P2 has type E55 Type b) F10 Person P14i performed F51 Pursuit R59 had typical subject E1 CRM Entity	
4.1	Person: Profession/occupation		a) F10 Person P2 has type E55 Type b) F10 Person P14i performed {P14.1 in the role of E55 Type} F51 Pursuit.P2 has type E55 Type	Either a classification or the Pursuit type.
4.1	Person: Biography/history		F10 Person P3 has note E62 String (each part of it is P12 was present at E5 Event) In case of a reference: F10 Person P70i is documented in E31 Document	
4.1	Person: Other information associated with the person		F10 Person P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object	These are typically constituents of Nomens
5.3.1	Person: Pseudonymous relationship		Implicit by the connection of two instances of F52 Name Use Activity via R63 named: the same instance of F10 Person	

FRAD section number	Unit of Information	Condition	Mapping	Comment
5.3.1	Person: Secular relationship / Religious relationship		as above and one or both instances of F52 referring to a particular context: F52 Name Use Activity R61 occurred in kind of context E55 Type	May be confused with a name change.
5.3.1	Person: Official relationship		as above and one or both instances of F52 referring to a particular membership context: F52 Name Use Activity R62 was used for membership in E74 Group	
5.3.1	Person: Attributive relationship		E39 Actor. P14i performed E65 Creation P94 has created F1 Work P94i was created by E65 Creation P14 was carried out by E39 Actor	One of the two work assignments is regarded to be false.
5.3.1	Person: Collaborative relationship		Implicit: one instance of F52 Name Use Activity carried out by several instances of E21 Person using one name (via R64 used name) in the same context	The FRAD text does not reflect the intention. It is actually about collaborating under a single persona
5.3.1	Person: Sibling relationship		Implicit through following connection: F10 Person {instance A} P152 has parent F10 Person {instance B} and F10 Person {instance C} P152 has parent F10 Person {instance B} (A and C are siblings through their common parent B)	
5.3.1	Person: Parent/child relationship		F10 Person P152 has parent F10 Person	
5.3.2	Person: Membership relationship		F10 Person P107i is current or former member of F39 Family	
5.3.3	Person: Membership relationship		F10 Person P107i is current or former member of F11 Corporate Body	
3.4	Family		F39 Family	
4.2	Family: Type		F39 Family P2 has type E55 Type	
4.2	Family: Dates		a) F39 Family P95i was formed by E66 Formation P4 has time-span E52 Time-Span P78 is	

FRAD section number	Unit of Information	Condition	Mapping	Comment
			identified by E50 Date b) F39 Family P99i was dissolved by E68 Dissolution P4 has time-span E52 Time-Span P78 is identified by E50 Date c) F39 Family P14i performed F51 Pursuit P4 has time-span E52 Time-Span P78 is identified by E50 Date d) F39 Family P12i was present at E5 Event P4 has time-span E52 Time-Span P78 is identified by E50 Date f) F39 Family P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object	
4.2	Family: Places associated with family		a) F39 Family P74 has current or former residence E53 Place b) F39 Family P51i is former or current owner of E27 Site P53 has former or current location E53 Place c) E39 Family P14i performed F51 Pursuit P7 took place at E53 Place	
4.2	Family: Language		F39 Family P14i performed F51 Pursuit R60 used to use language E56 Language	added in errata, published Nov 2011.
4.2	Family: Field of activity		a) F39 Family P14i performed F51 Pursuit P2 has type E55 Type b) F39 Family P14i performed F51 Pursuit R59 had typical subject E1 CRM Entity	
	Family: History		F39 Family P3 has note E62 String (each part of it is P12 was present at E5 Event) In case of a reference: F39 Family P70i is documented in E31 Document	
5.3.2	Family: Membership relationship		F39 Family P107 has current or former member F10 Person	
5.3.4	Family: Genealogical relationship		F39 Family P95i was formed by E66 Formation P151 was formed from	

FRAD section number	Unit of Information	Condition	Mapping	Comment
			F39 Family	
5.3.5	Family: Founding relationship		F39 Family P14i performed E66 Formation P95 formed F11 Corporate Body	
5.3.5	Family: Ownership relationship		F39 Family P107i is current or former member of {P107.1 kind of member E55 Type = "owner"} F11 Corporate Body	Regarded as a kind of membership. Other interpretations of ownership should be represented by a note
3.4	Corporate Body		F11 Corporate Body $\cup$ F38 Character representing a group	
4.3	Corporate Body: Place associated with the corporate body		a) F11 Corporate Body P74 has current or former residence E53 Place b) F11 Corporate Body P51i is former or current owner of E27 Site P53 has former or current location E53 Place c) F11 Corporate Body P14i performed F51 Pursuit P7 took place at E53 Place d) F11 Corporate Body P1 is identified by F50 Controlled Access Point R8 consists of E44 Place Appellation	
4.3	Corporate Body: Dates associated with the corporate body		a) F11 Corporate Body P95i was formed by E66 Formation P4 has time-span E52 Time-Span P78 is identified by E50 Date b) F11 Corporate Body P99i was dissolved by E68 Dissolution P4 has time-span E52 Time-Span P78 is identified by E50 Date c) F11 Corporate Body P14i performed F51 Pursuit P4 has time-span E52 Time-Span P78 is identified by E50 Date d) F11 Corporate Body P12i was present at E5 Event P4 has time-span E52 Time-Span P78 is identified by E50 Date f) F11 Corporate Body P1 is identified by F50 Controlled Access Point R8	

FRAD section number	Unit of Information	Condition	Mapping	Comment
			consists of E90 Symbolic Object	
4.3	Corporate Body: Language		F11 Corporate Body P14i performed F51 Pursuit R60 used to use language E56 Language	
4.3	Corporate Body: Address		a) F11 Corporate Body P76 has contact point E51 Contact Point (E45 Address is subclass of E51) b) F11 Corporate Body P107i is current or former member of E74 Group P76 has contact point E51 Contact Point c) F11 Corporate Body P74 has current or former residence E53 Place P87 is identified by E45 Address	
4.3	Corporate Body: Field of activity		a) F11 Corporate Body P14i performed F51 Pursuit P2 has type E55 Type b) F11 Corporate Body P14i performed F51 Pursuit R59 had typical subject E1 CRM Entity	
4.3	Corporate Body: History		F11 Corporate Body P3 has note E62 String (each part of it is P12 was present at E5 Event) In case of a reference: F11 Corporate Body P70i is documented in E31 Document	
4.3	Corporate Body: Other information associated with the corporate body		F11 Corporate Body P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object	
5.3.3	Corporate Body: Membership relationship		F11 Corporate Body P107 has current or former member F10 Person	
5.3.5	Corporate Body: Founding relationship		F11 Corporate Body P95i was formed by E66 Formation P14 carried out by F39 Family	
5.3.5	Corporate Body: Ownership relationship		F11 Corporate Body P107 has current or former member {P107.1 kind of member E55 Type = "owner"} F39 Family	Regarded as a kind of membership. Other interpretations of ownership should be represented by a note

FRAD section number	Unit of Information	Condition	Mapping	Comment
5.3.6	Corporate Body: Hierarchical relationship		F11 Corporate Body P107i is current or former member of F11 Corporate Body	
5.3.6	Corporate Body: Sequential relationship		a) conference series: each “conference” is a member of the series; the formation of one is “P120 occurs before” the formation of the next b) change of name: see F52 Name Use Activity c) group merging or splitting: instance of E81 Transformation	
3.4	Work		F1 Work	Attributes already mapped in 3.3 are not repeated here
4.4	Work: Form of work		F1 Work P2 has type E55 Type	See also attribute Form of work in FRBR
4.4	Work: Subject of the work			FRSAD supersedes FRAD as far as the subject relationship is concerned
4.4	Work: Place of origin of the work		F1 Work R16i was initiated by F27 Work Conception P7 took place at E53 Place F2 Expression R17i was created by F28 Expression Creation P7 took place at E53 Place	This is related either to work conception or to the first expression of the work. Place of origin of a cinematographic work is tied to the place of citizenship of the producer (and not the director)
4.4	Work: Other distinguishing characteristic		two mappings: full path from work to fragment or component of any expression/manifestation realizing. b) “representative expression/manifestation” in FRBRoo c) F1 Work P1 is identified by F50 Controlled Access Point R8 consists of E90 Symbolic Object	“Any characteristic that serves to differentiate the work from another work with the same title. [FRBR] Includes parts of intellectual or artistic content. Includes musical incipits.”
3.4	Expression		F2 Expression	Attributes already mapped in 3.3 are not repeated here
3.4	Manifestation		F3 Manifestation Product Type $\cup$ F4 Manifestation	Attributes already mapped in 3.3 are not repeated here

FRAD section number	Unit of Information	Condition	Mapping	Comment
			Singleton	
3.4	Item		F5 Item	
4.7	Item: Location of item		a) F5 Item P55 has current location E53 Place b) F5 Item P46i forms part of E78 Collection c) F5 Item P46i forms part of E78 Collection P55 has current location E53 Place d) F5 Item P50 has current keeper E39 Actor e) F5 Item P50 has current keeper E39 Actor P74 has current or former residence E53 Place	The collection and/or institution in which the item is held, stored, or made available for access. Probably meant how to access it – via keeper
4.7	Item: Custodial history of item		P30i custody transferred through: E10 Transfer of Custody	Although FRAD uses the term ‘ownership’ the intended meaning is custody. This justifies this mapping.
4.7	Item: Immediate source of acquisition of item		F5 Item P24i changed ownership through E8 Acquisition P23 transferred title from E39 Actor	The source from which an item was directly acquired and the circumstances under which it was acquired
3.4	Concept		F6 Concept	
3.4	Object		F7 Object	
3.4	Event		F8 Event	
3.4	Place		F9 Place	Place name normalization is not in scope of FRAD, not even in the context of describing publishers. Fictitious places are treated as concepts (Themata etc).
3.4	Name		F12 Nomen	E41 Appellation ⊃ FRSAD:Nomen = F12 Nomen ⊃ FRBR Name ⊃ FRAD: Name E41 may include names in the sense of linguistics with a history of evolution. Attribute FRBR Name restricted to characters, FRSAD adds signs.

FRAD section number	Unit of Information	Condition	Mapping	Comment
				FRAD name excludes ad-hoc constructed identifiers
4.12	Name: Type of name		F52 Name Use Activity R63 named E1 CRM Entity P2 has type E55 Type	Actually the type of the thema
4.12	Name: Name string		F12 Nomen R33 has content{R33.1has encoding E55 Type} E62 String	A sequence of numeric and/or alphabetic characters or symbols that represents the name of an entity. Decision: We regard that any instance/subclass of Nomen should foresee a content string that completely represents the identity of a Nomen instance regardless of the semantics of the structural components it is built from. A nomen identity may not extend to the interpretation of equivalence of structural components. Occurrences of structural tags in the nomen string are regarded as part of the content symbols.
4.12	Name: Scope of usage		F52 Name Use Activity R61 occurred in kind of context E55 Type	
4.12	Name: Dates of usage		F52 Name Use Activity P4 had time-span E52 Time-Span	
4.12	Name: Language of name			Covered by FRSAD model
5.2	Name: is appellation of		F12 Nomen P1i identifies E1 CRM Entity	
5.4.1	Earlier/Later name relationship		Connect two F52 Name Use Activities by P120 occurs after	
5.4.1, 2, 3, 4	Alternative linguistic form relationship		E1 CRM Entity P38i is thema of F35 Nomen Use Statement (R37 states as nomen F12 Nomen, P14 carried out by E39 Actor),	
5.4.1, 3, 4	Other variant name relationships		E1 CRM Entity P38i is thema of F35 Nomen Use Statement R37 states as nomen F12 Nomen	
5.4.3	Expanded name relationship (Acronym / initials / abbreviations relationship)		E1 CRM Entity P38i is thema of F35 Nomen Use Statement R37 states as nomen F12 Nomen	

FRAD section number	Unit of Information	Condition	Mapping	Comment
3.4	Identifier		E42 Identifier = F13 Identifier = (FRAD Identifier $\cup$ FRAD Controlled Access Point).	FRAD Identifier $\cap$ FRAD Controlled Access Point = $\emptyset$ If no attribute specific to FRAD Identifier only can be found, no reason to model disjointness, and use F13 Identifier, but model Controlled Access Point
4.13	Identifier: Type of an identifier		F13 Identifier P2 has Type E55 Type	We include types of CAPs in the sense of structural differentiation of the Identifier composition which may come with different Entity types they are designed for
5.2	Identifier: is assigned to		F13 Identifier P1i identifies E1 CRM Entity	
3.4	Controlled Access Point		F50 Controlled Access Point	
4.14	CAP: Status of controlled access point			covered by FRSAD and is described in the scope notes of F34 KOS and R34 <i>has validity period (is validity period of)</i>
4.14	CAP: Designated usage of controlled access point		F50 Controlled Access Point R37i is stated as nomen in F35 Nomen Use Statement P2 has type E55 Type	
4.14	CAP: Undifferentiated access point		F50 Controlled Access Point R37i is stated as nomen in F35 Nomen Use Statement P2 has type E55 Type [=‘Possibly non-unique’]	
4.14	CAP: Language of base access point		F50 Controlled Access Point R37i is stated as nomen in F35 Nomen Use Statement R54 has nomen language E56 Language	
4.14	CAP: Transliteration scheme of base access point		F50 Controlled Access Point R37i is stated as nomen in F35 Nomen Use Statement R36 used script conversion F36 Script Conversion	describe how constituents of CAPs appear in their own F35s...
4.14	CAP: Transliteration scheme of cataloguing		F50 Controlled Access Point R37i is stated as nomen in F35 Nomen Use Statement R36 used script conversion F36 Script Conversion	
4.14	CAP: Source of controlled access point		F50 Controlled Access Point R37i is stated as nomen in F35 Nomen Use Statement R32 is warranted by F52 Name Use Activity P70i is	The publication or reference source used in establishing the form of name or title on which the controlled access point is based. Includes

FRAD section number	Unit of Information	Condition	Mapping	Comment
			documented in E31 Document	Adopt FRSAD model, adequately to nomen constituents (via R8 or P106).
4.14	CAP: Base access point		F50 Controlled Access Point R8 consists of E90 Symbolic Object P2 has type E55 Type {"Base access point"}	
4.14	CAP: Addition		F50 Controlled Access Point R8 consists of E90 Symbolic Object P2 has type E55 Type {"Addition"}	
5.2	CAP: is based on		F50 Controlled Access Point R46i was assigned by F40 Identifier Assignment P142 used constituent E90 Symbolic Object	
5.2	CAP: is governed by		F50 Controlled Access Point R46i was assigned by F40 Identifier Assignment R52 used rule F43 Identifier Rule	
5.2	CAP: is created/modified by		F50 Controlled Access Point R46i was assigned by F40 Identifier Assignment P14 carried out by F44 Bibliographic Agency	An identifier is never modified. A new instance of F35 Nomen Use Statement may be created, and the "previous one" declared obsolete or deleted.
5.5	CAP: Parallel language		F35 Nomen Use Statement R56 has related use {R56.1 has type E55 Type = "parallel language"} F35 Nomen Use Statement	
5.5	CAP: Alternate script		F35 Nomen Use Statement R56 has related use {R56.1 has type E55 Type = "alternate script"} F35 Nomen Use Statement	
5.5	CAP: Different rules		F35 Nomen Use Statement R56 has related use {R56.1 has type E55 Type = "different rules"} F35 Nomen Use Statement	
5.5	CAP-subject term or classification number		F35 Nomen Use Statement R56 has related use {R56.1 has type E55 Type = "corresponding classification number"} F35 Nomen Use Statement	
5.5	CAP-identifier relationships		F35 Nomen Use Statement R56 has related	

FRAD section number	Unit of Information	Condition	Mapping	Comment
			use {R56.1 has type E55 Type = “identifier for entity”} F35 Nomen Use Statement	
3.4	Rules		subclass of E29 Design or Procedure a superclass of F43 Identifier Rule	It pertains to identifier formulation and/or recording. The restriction of Rules to CAP generation rules can be seen as implicit in the use of the rule for creating a CAP.
5.2	Rules: govern		F43 Identifier Rule R52i was the rule used in F40 Identifier Assignment R46 assigned F50 Controlled Access Point	
5.2	Rules: are applied by		F43 Identifier Rule R52 was the rule used in F40 Identifier Assignment P14 carried out by F44 Bibliographic Agency	
3.4	Agency		F44 Bibliographic Agency and subclass of F11 Corporate Body	
5.2	Agency: creates/modifies		F44 Bibliographic Agency P14i performed F40 Identifier Assignment R46 assigned F13 Identifier	An identifier is never modified. A new instance of F35 Nomen Use Statement may be created, and the “previous one” declared obsolete or deleted.
5.2	Agency: applies		F44 Bibliographic Agency P14i performed F40 Identifier Assignment R52 used rule F43 Identifier Rule	
5.3.7	Relationships among WEMI: Equivalence	Manifestation	F3 Manifestation Product Type P130i features are also found on {P130.1 kind of similarity: E55 Type = “[type of alternate format]”} F3 Manifestation Product Type F3 Manifestation Product Type P130 shows features of {P130.1 kind of similarity: E55 Type = “[type of alternate format]”} F3 Manifestation Product Type	FRBR 5.3.4: Manifestation: has an alternate FRBR 5.3.4: Manifestation: is an alternate to
5.3.7	Relationships among WEMI: Equivalence	Manifestation : [generic case]	[F3 Manifestation Product Type or F4 Manifestation Singleton] P130i features are also found on {P130.1 kind of similarity: E55 Type = “Reproduction”} [F3 Manifestation Product Type or F4 Manifestation Singleton]	FRBR 5.3.4: Manifestation: has a reproduction (a Manifestation) FRBR 5.3.4: Manifestation: is a

FRAD section number	Unit of Information	Condition	Mapping	Comment
			[F3 Manifestation Product Type or F4 Manifestation Singleton] P130 shows features of {P130.1 kind of similarity: E55 Type = "Reproduction"} [F3 Manifestation Product Type or F4 Manifestation Singleton] [F3 Manifestation Product Type or F4 Manifestation Singleton] P130 shows features of {P130.1 kind of similarity: E55 Type = "Reproduction"} F5 Item	reproduction of (a Manifestation) FRBR 5.3.5: Manifestation: is a reproduction of (an Item)
5.3.7	Relationships among WEMI: Equivalence	Item : [generic case]	F5 Item P130i features are also found on {P130.1 kind of similarity: E55 Type = "Reproduction"} [F3 Manifestation Product Type or F4 Manifestation Singleton] F5 Item P130i features are also found on {P130.1 kind of similarity: E55 Type = "Reproduction"} F5 Item F5 Item P130 shows features of {P130.1 kind of similarity: E55 Type = "Reproduction"} F5 Item	FRBR 5.3.5: Item: has a reproduction (a Manifestation) FRBR 5.3.6: Item: has a reproduction (an Item) FRBR 5.3.6: Item: is a reproduction of (an Item)
5.3.7	Relationships among WEMI: Derivative	Work	F1 Work R2 is derivative of {R2.1 has type E55 Type = "Adaptation"} F1 Work F1 Work R2i has derivative {R2.1 has type E55 Type = "Adaptation"} F1 Work	FRBR 5.3.1: Work: is an adaptation of (Work, Expression) FRBR 5.3.1: Work: has adaptation (Work)
5.3.7	Relationships among WEMI: Derivative	Work	F1 Work R2 is derivative of {R2.1 has type E55 Type = "Transformation"} F1 Work F1 Work R2i has derivative {R2.1 has type E55 Type = "Transformation"} F1 Work	FRBR 5.3.1: Work: is a transformation of (Work, Expression) FRBR 5.3.1: Work: has a transformation (Work)
5.3.7	Relationships among WEMI: Derivative	Work	F1 Work R2 is derivative of {R2.1 has type E55 Type = "Imitation"} F1 Work F1 Work R2i has derivative {R2.1 has type E55 Type = "Imitation"} F1 Work	FRBR 5.3.1: Work: is an imitation of (Work, Expression) FRBR 5.3.1: Work: has an imitation (Work)
5.3.7	Relationships among WEMI: Derivative	Expression	F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = "Adaptation"} F1 Work R3 is realised in F22 Self-Contained Expression	FRBR 5.3.2: Expression: has an adaptation

FRAD section number	Unit of Information	Condition	Mapping	Comment
			F2 Expression P16 was used in {P16.1 mode of use E55 Type = “adapted source”} F28 Expression Creation R17 created F22 Self-Contained Expression F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type E55 Type = “Adaptation”} F1 Work R3 is realised in F22 Self-Contained Expression F22 Self-Contained Expression R17i was created by F28 Expression Creation P16 used specific object {P16.1 mode of use E55 Type = “adapted source”} F2 Expression	FRBR 5.3.2: Expression: is an adaptation of
5.3.7	Relationships among WEMI: Derivative	Expression	F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = “Transformation”} F1 Work R3 is realised in F22 Self-Contained Expression F2 Expression P16 was used in {P16.1 mode of use E55 Type = “transformed source”} F28 Expression Creation R17 created F22 Self-Contained Expression F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type E55 Type = “Transformation”} F1 Work R3 is realised in F22 Self-Contained Expression F22 Self-Contained Expression R17i was created by F28 Expression Creation P16 used specific object {P16.1 mode of use E55 Type = “transformed source”} F2 Expression	FRBR 5.3.2: Expression: has a transformation FRBR 5.3.2: Expression: is a transformation of
5.3.7	Relationships among WEMI: Derivative	Expression	F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = “Imitation”} F1 Work R3 is realised in F22 Self-Contained Expression	FRBR 5.3.2: Expression: has an imitation

FRAD section number	Unit of Information	Condition	Mapping	Comment
			F2 Expression P16 was used in {P16.1 mode of use E55 Type = “imitated source”} F28 Expression Creation R17 created F22 Self-Contained Expression F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type: E55 Type = “Imitation”} F1 Work R3 is realised in F22 Self-Contained Expression F22 Self-Contained Expression R17i was created by F28 Expression Creation P16 used specific object {P16.1 mode of use E55 Type = “imitated source”} F2 Expression	FRBR 5.3.2: Expression: is an imitation of
5.3.7	Relationships among WEMI: Derivative	Expression	F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = “Abridgement”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type E55 Type = “Abridgement”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work	FRBR 5.3.2: Expression: has an abridgement FRBR 5.3.2: Expression: is an abridgement of
5.3.7	Relationships among WEMI: Derivative	Expression	F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = “Revision”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work	FRBR 5.3.2: Expression: has a revision

FRAD section number	Unit of Information	Condition	Mapping	Comment
			F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type E55 Type = “Revision”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work	FRBR 5.3.2: Expression: is a revision of
5.3.7	Relationships among WEMI: Derivative	Expression	F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = “Translation”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work F22 Self-Contained Expression (instantiated as E33 Linguistic Object) P73 has translation E33 Linguistic Object F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type E55 Type = “Translation”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work F22 Self-Contained Expression (instantiated as E33 Linguistic Object) P73i is translation of E33 Linguistic Object	FRBR 5.3.2: Expression: has a translation FRBR 5.3.2: Expression: is a translation of
5.3.7	Relationships among WEMI: Derivative	Expression	F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = “Arrangement”} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is	FRBR 5.3.2: Expression: has an arrangement

FRAD section number	Unit of Information	Condition	Mapping	Comment
			member of F15 Complex Work R10 has member F1 Work F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type E55 Type = "Arrangement"} F1 Work and F22 Self-Contained Expression R9i realises F14 Individual Work R10i is member of F15 Complex Work R10 has member F1 Work	FRBR 5.3.2: Expression: is an arrangement of
5.3.7	Relationships among WEMI: Descriptive	Work	F1 Work R2 is derivative of {R2.1 has type E55 Type = "Summary"} F1 Work F1 Work R2i has derivative {R2.1 has type E55 Type = "Summary"} F1 Work	FRBR 5.3.1: Work: is a summary of (Work) FRBR 5.3.1: Work: has a summary (Work)
5.3.7	Relationships among WEMI: Descriptive	Expression	F22 Self-Contained Expression R9i realises F14 Individual Work R2i has derivative {R2.1 has type E55 Type = "Summary"} F1 Work R3 is realised in F22 Self-Contained Expression F22 Self-Contained Expression R9i realises F14 Individual Work R2 is derivative of {R2.1 has type E55 Type = "Summary"} F1 Work R3 is realised in F22 Self-Contained Expression	FRBR 5.3.2: Expression: has a summary FRBR 5.3.2: Expression: is a summary of
5.3.7	Relationships among WEMI: Whole/Part	Work	F15 Complex Work R10 has member F1 Work F1 Work R10i is member of F15 Complex Work	FRBR 5.3.1.1: Work: has part (Work) FRBR 5.3.1.1: Work: is part of (Work)
5.3.7	Relationships among WEMI: Whole/Part	Expression	F2 Expression R15 has fragment F23 Expression Fragment F2 Expression R5 has component F22 Self-Contained Expression F23 Expression Fragment R15i is fragment of F2 Expression F22 Self-Contained Expression R5i is component of F2 Expression	FRBR 5.3.2.1: Expression: has part FRBR 5.3.2.1: Expression: is part of
5.3.7	Relationships among WEMI: Whole/Part	Manifestation	F3 Manifestation Product Type CLP46 should be composed of F3 Manifestation Product Type	FRBR 5.3.4.1: Manifestation: has part

FRAD section number	Unit of Information	Condition	Mapping	Comment
			F4 Manifestation Singleton P46 is composed of F4 Manifestation Singleton F3 Manifestation Product Type CLP46i may form part of F3 Manifestation Product Type F4 Manifestation Singleton P46i forms part of F4 Manifestation Singleton	FRBR 5.3.4.1: Manifestation: is part of
5.3.7	Relationships among WEMI: Whole/Part	Item	F5 Item P46 is composed of F5 Item F4 Manifestation Singleton P46 is composed of F4 Manifestation Singleton F5 Item P46i forms part of F5 Item F4 Manifestation Singleton P46i forms part of F4 Manifestation Singleton	FRBR 5.3.4.1: Item: has part FRBR 5.3.4.1: Item: is part of
5.3.7	Relationships among WEMI: Accompanying	Work	F1 Work R16i was initiated by F27 Work Conception P16 used specific object {P16.1 mode of use: E55 Type = “supplemented work”} F1 Work F1 Work P16i was used for {P16.1 mode of use: E55 Type = “supplemented work”} F27 Work Conception R16 initiated F1 Work	FRBR 5.3.1: Work: has a supplement (Work) FRBR 5.3.1: Work: supplements (Work)
5.3.7	Relationships among WEMI: Accompanying	Work	F15 Complex Work R10 has member F1 Work R3 is realised in F22 Self-Contained Expression P165 incorporates F22 Self-Contained Expression R3i realises F1 Work F1 Work R3 is realised in F22 Self-Contained Expression P165i is incorporated in F22 Self-Contained Expression R3i realises F1 Work R10i is member of F15 Complex Work	FRBR 5.3.1: Work: has a complement (Work) FRBR 5.3.1: Work: complements (Work)
5.3.7	Relationships among WEMI: Accompanying	Expression	F22 Self-Contained Expression R17i was created by F28 Expression Creation P16 used specific object {P16.1 mode of use E55 Type = “supplemented expression”} F2 Expression	FRBR 5.3.2: Expression: has a supplement

FRAD section number	Unit of Information	Condition	Mapping	Comment
			F2 Expression P16 was used in {P16.1 mode of use E55 Type = “supplemented expression”} F28 Expression Creation R17 created F22 Self-Contained Expression	FRBR 5.3.2: Expression: supplements
5.3.7	Relationships among WEMI: Accompanying	Expression	F22 Self-Contained Expression P165 incorporates F2 Expression F2 Expression P165 is incorporated in F22 Self-Contained Expression	FRBR 5.3.2: Expression: has a complement FRBR 5.3.2: Expression: complements
5.3.7	Relationships among WEMI: Sequential	Work	F1 Work R1i has successor F1 Work F1 Work R1 is logical successor of F1 Work	FRBR 5.3.1: Work: has a successor (Work) FRBR 5.3.1: Work: is a successor to (Work)
5.3.7	Relationships among WEMI: Sequential	Expression	F22 Self-Contained Expression R9i realises F14 Individual Work R1i has successor F1 Work F22 Self-Contained Expression R9i realises F14 Individual Work R1 is logical successor of F1 Work	FRBR 5.3.2: Expression: has a successor FRBR 5.3.2: Expression: is a successor to
5.3.7	Relationships among WEMI: Shared characteristic relationships			Not mapped. This serves to express the situation where multiple instances share the same value of some property. Regarded as irrelevant.

4. Referred to CIDOC CRM Classes and Properties

Since FRBR_{OO} refers to and reuses, wherever appropriate, large parts of ISO21127, the CIDOC Conceptual Reference Model, this section provides a comprehensive list of all constructs used from ISO21127, together with their definitions following version 6.0 maintained by CIDOC. Use in this context includes: reference as immediate superclass, superproperty or element of a path expression in a mapping statement.

Some of these constructs appear only in the mapping in section 5 (above) and not in section 4, because they are generic in nature. For instance, we regarded it as better not to overload the description of FRBR_{OO} with generic notions such as carrying out activities or using things.

4.1. List of Referred to CIDOC CRM Classes

In this section we present the classes of the CIDOC CRM Conceptual Reference Model version 6.0 referred to by FRBR_{OO} as a list. The classes that appear indirectly in the FRBR_{OO} Model, i.e. either as superclasses of classes defined in the model, or as the domain or range of referred CRM properties are marked in bold.

E1	CRM Entity
E2	Temporal Entity
E3	Condition State
E4	Period
E5	Event
E7	Activity
E11	Modification
E12	Production
E13	Attribute Assignment
E15	Identifier Assignment
E18	Physical Thing
E19	Physical Object
E21	Person
E22	Man-Made Object
E24	Physical Man-Made Thing
E26	Physical Feature
E27	Site
E28	Conceptual Object
E29	Design or Procedure
E30	Right
E31	Document
E32	Authority Document
E33	Linguistic Object
E35	Title
E36	Visual Item
E37	Mark
E39	Actor
E40	Legal Body
E41	Appellation
E42	Identifier
E44	Place Appellation
E47	Spatial Coordinates
E49	Time Appellation
E50	Date
E52	Time-Span
E53	Place

E54	Dimension
E55	Type
E56	Language
E57	Material
E59	Primitive Value
E60	Number
E61	Time Primitive
E62	String
E63	Beginning of Existence
E64	End of Existence
E65	Creation
E66	Formation
E67	Birth
E69	Death
E70	Thing
E71	Man-Made Thing
E72	Legal Object
E73	Information Object
E74	Group
E77	Persistent Item
E82	Actor Appellation
E84	Information Carrier
E89	Propositional Object
E90	Symbolic Object

4.2. List of Referred to CIDOC CRM Properties

In this section we present the properties of the CIDOC CRM 6.0 referred to by FRBR_{OO} as a list. The properties that appear indirectly in the FRBR_{OO} Model, i.e. as superproperties of properties defined in the model, are marked in bold.

Property id	Property Name	Entity – Domain	Entity – Range
P1	is identified by (identifies)	E1 CRM Entity	E41 Appellation
P2	has type (is type of)	E1 CRM Entity	E55 Type
P3	has note	E1 CRM Entity	E62 String
P4	has time-span (is time-span of)	E2 Temporal Entity	E52 Time-Span
P7	took place at (witnessed)	E4 Period	E53 Place
P9	consists of (forms part of)	E4 Period	E4 Period
P12	occurred in the presence of (was present at)	E5 Event	E77 Persistent Item
P14	carried out by (performed)	E7 Activity	E39 Actor
P15	was influenced by (influenced)	E7 Activity	E1 CRM Entity
P16	used specific object (was used for)	E7 Activity	E70 Thing
P31	has modified (was modified by)	E11 Modification	E24 Physical Man-Made Thing
P33	used specific technique (was used by)	E7 Activity	E29 Design or Procedure
P37	assigned (was assigned by)	E15 Identifier Assignment	E42 Identifier
P43	has dimension (is dimension of)	E70 Thing	E54 Dimension
P44	has condition (condition of)	E18 Physical Thing	E3 Condition State
P45	consists of (is incorporated in)	E18 Physical Thing	E57 Material
P46	is composed of (forms part of)	E18 Physical Thing	E18 Physical Thing
P49	has former or current keeper (is former or current keeper of)	E18 Physical Thing	E39 Actor
P50	has current keeper (is current keeper of)	E18 Physical Thing	E39 Actor
P51	has former or current owner (is former or current owner of)	E18 Physical Thing	E39 Actor
P57	has number of parts	E19 Physical Object	E60 Number
P59	has section (is located on or within)	E18 Physical Thing	E53 Place
P65	shows visual item (is shown by)	E24 Physical Man-Made Thing	E36 Visual Item
P67	refers to(is referred to by)	E89 Propositional Object	E1 CRM Entity
P69	has association with (is associated with)	E29 Design or Procedure	E29 Design or Procedure
P71	lists (is listed in)	E32 Authority Document	E1 CRM Entity
P72	has language (is language of)	E33 Linguistic Object	E56 Language
P74	has current or former residence (is current or former residence of)	E39 Actor	E53 Place
P75	possesses (is possessed by)	E39 Actor	E30 Right
P78	is identified by (identifies)	E52 Time-Span	E49 Time Appellation
P82	at some time within	E52 Time-Span	E61 Time Primitive
P87	is identified by (identifies)	E53 Place	E44 Place Appellation
P94	has created (was created by)	E65 Creation	E28 Conceptual Object
P95	has formed (was formed by)	E66 Formation	E74 Group
P98	brought into life (was born)	E67 Birth	E21 Person
P100	was death of (died in)	E69 Death	E21 Person
P102	has title (is title of)	E71 Man-Made Thing	E35 Title

P103	was intended for (was intention of)	E71 Man-Made Thing	E55 Type
P104	is subject to (applies to)	E72 Legal Object	E30 Right
P105	right held by (has right on)	E72 Legal Object	E39 Actor
P106	is composed of (forms part of)	E90 Symbolic Object	E90 Symbolic Object
P107	has current or former member (is current or former member of)	E74 Group	E39 Actor
P108	has produced (was produced by):	E12 Production	E24 Physical Man-Made Thing
P125	used object of type (was type of object used in)	E7 Activity	E55 Type
P127	has broader term (has narrower term)	E55 Type	E55 Type
P128	carries (is carried by)	E24 Physical Man-Made Thing	E73 Information Object
P129	is about (is subject of)	E73 Information Object	E1 CRM Entity
P130	shows features of (features are also found on)	E70 Thing	E70 Thing
P131	is identified by (identifies)	E39 Actor	E82 Actor Appellation
P138	represents (has representation)	E36 Visual Item	E1 CRM Entity
P140	assigned attribute to (was attributed by)	E13 Attribute Assignment	E1 CRM Entity
P141	assigned (was assigned by)	E13 Attribute Assignment	E1 CRM Entity
P142	used constituent (was used in)	E15 Identifier Assignment	E90 Symbolic Object
P148	has component (is component of)	E89 Propositional Object	E89 Propositional Object
P151	was formed from (participated in)	E66 Formation	E74 Group
P165	incorporates (is incorporated in)	E73 Information Object	E90 Symbolic Object

4.3. Referred to CIDOC CRM Classes

This section contains the complete definitions of the classes of the CIDOC CRM Conceptual Reference Model version 6.0 referred to by FRBR₀₀. The properties within these class definitions which are referred to in FRBR₀₀ are presented in bold face. Otherwise, we apply the same format conventions as in section 2.6.

E1 CRM Entity

Superclass of: [E2](#) Temporal Entity
[E52](#) Time-Span
[E53](#) Place
[E54](#) Dimension
[E77](#) Persistent Item

Scope note: This class comprises all things in the universe of discourse of the CIDOC Conceptual Reference Model.

It is an abstract concept providing for three general properties:

1. Identification by name or appellation, and in particular by a preferred identifier
2. Classification by type, allowing further refinement of the specific subclass an instance belongs to
3. Attachment of free text for the expression of anything not captured by formal properties

With the exception of E59 Primitive Value, all other classes within the CRM are directly or indirectly specialisations of E1 CRM Entity.

Examples:

- the earthquake in Lisbon 1755 (E5)

Properties:

P1 is identified by (identifies): [E41](#) Appellation

P2 has type (is type of): [E55](#) Type

P3 has note: [E62](#) String

(P3.1 has type: [E55](#) Type)

P48 has preferred identifier (is preferred identifier of): E42 Identifier

P137 exemplifies (is exemplified by): E55 Type

E2 Temporal Entity

Subclass of: [E1](#) CRM Entity
Superclass of: [E3](#) Condition State
[E4](#) Period

Scope note: This class comprises all phenomena, such as the instances of E4 Periods, E5 Events and states, which happen over a limited extent in time.

In some contexts, these are also called perdurants. This class is disjoint from E77 Persistent Item. This is an abstract class and has no direct instances. E2 Temporal Entity is specialized into E4 Period, which applies to a particular geographic area (defined with a greater or lesser degree of precision), and E3 Condition State, which applies to instances of E18 Physical Thing.

Examples:

- Bronze Age (E4)
- the earthquake in Lisbon 1755 (E5)
- the Peterhof Palace near Saint Petersburg being in ruins from 1944 – 1946 (E3)

Properties:

P4 has time-span (is time-span of): [E52](#) Time-Span

P114 is equal in time to: E2 Temporal Entity

P115 finishes (is finished by): E2 Temporal Entity
P116 starts (is started by): E2 Temporal Entity
P117 occurs during (includes): E2 Temporal Entity
P118 overlaps in time with (is overlapped in time by): E2 Temporal Entity
P119 meets in time with (is met in time by): E2 Temporal Entity
P120 occurs before (occurs after): E2 Temporal Entity

E3 Condition State

Subclass of: [E2](#) Temporal Entity

Scope note: This class comprises the states of objects characterised by a certain condition over a time-span.

An instance of this class describes the prevailing physical condition of any material object or feature during a specific E52 Time Span. In general, the time-span for which a certain condition can be asserted may be shorter than the real time-span, for which this condition held.

The nature of that condition can be described using *P2 has type*. For example, the E3 Condition State “condition of the SS Great Britain between 22 September 1846 and 27 August 1847” can be characterized as E55 Type “wrecked”.

Examples:

- the “Amber Room” in Tsarskoje Selo being completely reconstructed from summer 2003 until now
- the Peterhof Palace near Saint Petersburg being in ruins from 1944 – 1946
- the state of my turkey in the oven at 14:30 on 25 December, 2002 (*P2 has type: E55 Type* “still not cooked”)

Properties:

P5 consists of (forms part of): E3 Condition State

E4 Period

Subclass of: [E2](#) Temporal Entity

Superclass of: [E5](#) Event

Scope note: This class comprises sets of coherent phenomena or cultural manifestations bounded in time and space.

It is the social or physical coherence of these phenomena that identify an E4 Period and not the associated spatio-temporal bounds. These bounds are a mere approximation of the actual process of growth, spread and retreat. Consequently, different periods can overlap and coexist in time and space, such as when a nomadic culture exists in the same area as a sedentary culture.

Typically this class is used to describe prehistoric or historic periods such as the “Neolithic Period”, the “Ming Dynasty” or the “McCarthy Era”. There are however no assumptions about the scale of the associated phenomena. In particular all events are seen as synthetic processes consisting of coherent phenomena. Therefore E4 Period is a superclass of E5 Event. For example, a modern clinical E67 Birth can be seen as both an atomic E5 Event and as an E4 Period that consists of multiple activities performed by multiple instances of E39 Actor.

There are two different conceptualisations of ‘artistic style’, defined either by physical features or by historical context. For example, “Impressionism” can be viewed as a period lasting from approximately 1870 to 1905 during which paintings with particular characteristics were produced by a group of artists that included (among others) Monet, Renoir, Pissarro, Sisley and Degas. Alternatively, it can be regarded as a style applicable to all paintings sharing the characteristics of the works produced by the Impressionist painters, regardless of historical

context. The first interpretation is an E4 Period, and the second defines morphological object types that fall under E55 Type.

Another specific case of an E4 Period is the set of activities and phenomena associated with a settlement, such as the populated period of Nineveh.

Examples:

- Jurassic
- European Bronze Age
- Italian Renaissance
- Thirty Years War
- Sturm und Drang
- Cubism

Properties:

P7 took place at (witnessed): E53 Place

P8 took place on or within (witnessed): E19 Physical Object

P9 consists of (forms part of): E4 Period

P10 falls within (contains): E4 Period

P132 overlaps with: E4 Period

P133 is separated from: E4 Period

E5 Event

Subclass of: [E4](#) Period

Superclass of: [E7](#) Activity

[E63](#) Beginning of Existence

[E64](#) End of Existence

Scope note: This class comprises changes of states in cultural, social or physical systems, regardless of scale, brought about by a series or group of coherent physical, cultural, technological or legal phenomena. Such changes of state will affect instances of E77 Persistent Item or its subclasses.

The distinction between an E5 Event and an E4 Period is partly a question of the scale of observation. Viewed at a coarse level of detail, an E5 Event is an ‘instantaneous’ change of state. At a fine level, the E5 Event can be analysed into its component phenomena within a space and time frame, and as such can be seen as an E4 Period. The reverse is not necessarily the case: not all instances of E4 Period give rise to a noteworthy change of state.

Examples:

- the birth of Cleopatra (E67)
- the destruction of Herculaneum by volcanic eruption in 79 AD (E6)
- World War II (E7)
- the Battle of Stalingrad (E7)
- the Yalta Conference (E7)
- my birthday celebration 28-6-1995 (E7)
- the falling of a tile from my roof last Sunday
- the CIDOC Conference 2003 (E7)

Properties:

P11 had participant (participated in): E39 Actor

[P12](#) occurred in the presence of (was present at): [E77](#) Persistent Item

E7 Activity

Subclass of: [E5](#) Event

Superclass of: E8 Acquisition

E9 Move

E10 Transfer of Custody

[E11](#) Modification

[E13](#) Attribute Assignment

[E65](#) Creation

[E66](#) Formation

E85 Joining

E86 Leaving

E87 Curation Activity

Scope note: This class comprises actions intentionally carried out by instances of E39 Actor that result in changes of state in the cultural, social, or physical systems documented.

This notion includes complex, composite and long-lasting actions such as the building of a settlement or a war, as well as simple, short-lived actions such as the opening of a door.

Examples:

- the Battle of Stalingrad
- the Yalta Conference
- my birthday celebration 28-6-1995
- the writing of “Faust” by Goethe (E65)
- the formation of the Bauhaus 1919 (E66)
- calling the place identified by TGN ‘7017998’ ‘Quyunjig’ by the people of Iraq

Properties:

[P14](#) carried out by (performed): [E39](#) Actor
(P14.1 in the role of: [E55](#) Type)

[P15](#) was influenced by (influenced): [E1](#) CRM Entity

[P16](#) used specific object (was used for): [E70](#) Thing
(P16.1 mode of use: [E55](#) Type)

P17 was motivated by (motivated): E1 CRM Entity

P19 was intended use of (was made for): E71 Man-Made Thing
(P19.1 mode of use: E55 Type)

P20 had specific purpose (was purpose of): E5 Event

P21 had general purpose (was purpose of): E55 Type

P32 used general technique (was technique of): E55 Type

[P33](#) used specific technique (was used by): [E29](#) Design or Procedure

[P125](#) used object of type (was type of object used in): [E55](#) Type

P134 continued (was continued by): E7 Activity

E11 Modification

Subclass of: [E7](#) Activity

Superclass of: [E12](#) Production

E79 Part Addition

E80 Part Removal

Scope note: This class comprises all instances of E7 Activity that create, alter or change E24 Physical Man-Made Thing.

This class includes the production of an item from raw materials, and other so far undocumented objects, and the preventive treatment or restoration of an object for conservation.

Since the distinction between modification and production is not always clear, modification is regarded as the more generally applicable concept. This implies that some items may be consumed or destroyed in a Modification, and that others may be produced as a result of it. An event should also be documented using E81 Transformation if it results in the destruction of one or more objects and the simultaneous production of others using parts or material from the originals. In this case, the new items have separate identities.

If the instance of the E29 Design or Procedure utilized for the modification prescribes the use of specific materials, they should be documented using property *P68 foresees use of (use foreseen by)*: E57 Material of E29 Design or Procedure, rather than via *P126 employed (was employed in)*: E57 Material.

Examples:

- the construction of the SS Great Britain (E12)
- the impregnation of the Vasa warship in Stockholm for preservation after 1956
- the transformation of the Enola Gay into a museum exhibit by the National Air and Space Museum in Washington DC between 1993 and 1995 (E12, E81)
- the last renewal of the gold coating of the Toshogu shrine in Nikko, Japan

Properties:

[P31](#) has modified (was modified by): [E24](#) Physical Man-Made Thing

P126 employed (was employed in): E57 Material

E12 Production

Subclass of: [E11](#) Modification

[E63](#) Beginning of Existence

Scope note: This class comprises activities that are designed to, and succeed in, creating one or more new items.

It specializes the notion of modification into production. The decision as to whether or not an object is regarded as new is context sensitive. Normally, items are considered “new” if there is no obvious overall similarity between them and the consumed items and material used in their production. In other cases, an item is considered “new” because it becomes relevant to documentation by a modification. For example, the scribbling of a name on a potsherd may make it a voting token. The original potsherd may not be worth documenting, in contrast to the inscribed one.

This entity can be collective: the printing of a thousand books, for example, would normally be considered a single event.

An event should also be documented using E81 Transformation if it results in the destruction of one or more objects and the simultaneous production of others using parts or material from the

originals. In this case, the new items have separate identities and matter is preserved, but identity is not.

Examples:

- the construction of the SS Great Britain
- the first casting of the Little Mermaid from the harbour of Copenhagen
- Rembrandt's creating of the seventh state of his etching "Woman sitting half dressed beside a stove", 1658, identified by Bartsch Number 197 (E12,E65,E81)

Properties:

[P108](#) has produced (was produced by): [E24](#) Physical Man-Made Thing

E13 Attribute Assignment

Subclass of: [E7](#) Activity

Superclass of: E14 Condition Assessment

[E15](#) Identifier Assignment

E16 Measurement

E17 Type Assignment

Scope note: This class comprises the actions of making assertions about properties of an object or any relation between two items or concepts.

This class allows the documentation of how the respective assignment came about, and whose opinion it was. All the attributes or properties assigned in such an action can also be seen as directly attached to the respective item or concept, possibly as a collection of contradictory values. All cases of properties in this model that are also described indirectly through an action are characterised as "short cuts" of this action. This redundant modelling of two alternative views is preferred because many implementations may have good reasons to model either the action or the short cut, and the relation between both alternatives can be captured by simple rules.

In particular, the class describes the actions of people making propositions and statements during certain museum procedures, e.g. the person and date when a condition statement was made, an identifier was assigned, the museum object was measured, etc. Which kinds of such assignments and statements need to be documented explicitly in structures of a schema rather than free text, depends on if this information should be accessible by structured queries.

Examples:

- the assessment of the current ownership of Martin Doerr's silver cup in February 1997

Properties:

[P140](#) assigned attribute to (was attributed by): [E1](#) CRM Entity

[P141](#) assigned (was assigned by): [E1](#) CRM Entity

E15 Identifier Assignment

Subclass of: [E13](#) Attribute Assignment

Scope note: This class comprises activities that result in the allocation of an identifier to an instance of E1 CRM Entity. An E15 Identifier Assignment may include the creation of the identifier from multiple constituents, which themselves may be instances of E41 Appellation. The syntax and kinds of constituents to be used may be declared in a rule constituting an instance of E29 Design or Procedure.

Examples of such identifiers include Find Numbers, Inventory Numbers, uniform titles in the sense of librarianship and Digital Object Identifiers (DOI). Documenting the act of identifier assignment and deassignment is especially useful when objects change custody or the identification system of an organization is changed. In order to keep track of the identity of things in such cases, it is important to document by whom, when and for what purpose an identifier is assigned to an item.

The fact that an identifier is a preferred one for an organisation can be expressed by using the property *E1 CRM Entity. P48 has preferred identifier (is preferred identifier of): E42 Identifier*. It can better be expressed in a context independent form by assigning a suitable E55 Type, such as “preferred identifier assignment”, to the respective instance of E15 Identifier Assignment via the *P2 has type* property.

Examples:

- Replacement of the inventory number TA959a by GE34604 for a 17th century lament cloth at the Museum Benaki, Athens
- Assigning the author-uniform title heading “Goethe, Johann Wolfgang von, 1749-1832. Faust. 1. Theil.” for a work (E28)
- On June 1, 2001 assigning the personal name heading “Guillaume, de Machaut, ca. 1300-1377” (E42,E82) to Guillaume de Machaut (E21)

Properties:

[P37](#) assigned (was assigned by): [E42](#) Identifier
[P38](#) deassigned (was deassigned by): [E42](#) Identifier
[P142](#) used constituent (was used in): [E90](#) Symbolic Object

E18 Physical Thing

Subclass of: [E72](#) Legal Object

Superclass of: [E19](#) Physical Object

[E24](#) Physical Man-Made Thing

[E26](#) Physical Feature

Scope Note: This class comprises all persistent physical items with a relatively stable form, man-made or natural.

Depending on the existence of natural boundaries of such things, the CRM distinguishes the instances of E19 Physical Object from instances of E26 Physical Feature, such as holes, rivers, pieces of land etc. Most instances of E19 Physical Object can be moved (if not too heavy), whereas features are integral to the surrounding matter.

The CRM is generally not concerned with amounts of matter in fluid or gaseous states.

Examples:

- the Cullinan Diamond (E19)
- the cave “Ideon Andron” in Crete (E26)
- the Mona Lisa (E22)

Properties:

[P44](#) has condition (condition of): [E3](#) Condition State

[P45](#) consists of (is incorporated in): [E57](#) Material

[P46](#) is composed of (forms part of): [E18](#) Physical Thing

[P49](#) has former or current keeper (is former or current keeper of): [E39](#) Actor

[P50](#) has current keeper (is current keeper of): [E39](#) Actor

[P51](#) has former or current owner (is former or current owner of): [E39](#) Actor

P52 has current owner (is current owner of): E39 Actor

P53 has former or current location (is former or current location of): E53 Place

P58 has section definition (defines section): E46 Section Definition

[P59](#) has section (is located on or within): [E53](#) Place

E19 Physical Object

Subclass of: [E18](#) Physical Thing

Superclass of: E20 Biological Object

[E22](#) Man-Made Object

Scope note: This class comprises items of a material nature that are units for documentation and have physical boundaries that separate them completely in an objective way from other objects.

The class also includes all aggregates of objects made for functional purposes of whatever kind, independent of physical coherence, such as a set of chessmen. Typically, instances of E19 Physical Object can be moved (if not too heavy).

In some contexts, such objects, except for aggregates, are also called “bona fide objects” (Smith & Varzi, 2000, pp.401-420), i.e. naturally defined objects.

The decision as to what is documented as a complete item, rather than by its parts or components, may be a purely administrative decision or may be a result of the order in which the item was acquired.

Examples:

- John Smith
- Aphrodite of Milos
- the Palace of Knossos
- the Cullinan Diamond
- Apollo 13 at the time of launch

Properties:

P54 has current permanent location (is current permanent location of): E53 Place

P55 has current location (currently holds): E53 Place

P56 bears feature (is found on): E26 Physical Feature

[P57](#) has number of parts: [E60](#) Number

E21 Person

Subclass of: E20 Biological Object

[E39](#) Actor

Scope note: This class comprises real persons who live or are assumed to have lived.

Legendary figures that may have existed, such as Ulysses and King Arthur, fall into this class if the documentation refers to them as historical figures. In cases where doubt exists as to whether several persons are in fact identical, multiple instances can be created and linked to indicate their relationship. The CRM does not propose a specific form to support reasoning about possible identity.

Examples:

- Tut-Ankh-Amun
- Nelson Mandela

E22 Man-Made Object

Subclass of: [E19](#) Physical Object
[E24](#) Physical Man-Made Thing
Superclass of: [E84](#) Information Carrier

Scope note: This class comprises physical objects purposely created by human activity.

No assumptions are made as to the extent of modification required to justify regarding an object as man-made. For example, an inscribed piece of rock or a preserved butterfly are both regarded as instances of E22 Man-Made Object.

Examples:

- Mallard (the World's fastest steam engine)
- the Portland Vase
- the Coliseum

E24 Physical Man-Made Thing

Subclass of: [E18](#) Physical Thing
[E71](#) Man-Made Thing
Superclass of: [E22](#) Man-Made Object
E25 Man-Made Feature
E78 Collection

Scope Note: This class comprises all persistent physical items that are purposely created by human activity.

This class comprises man-made objects, such as a swords, and man-made features, such as rock art. No assumptions are made as to the extent of modification required to justify regarding an object as man-made. For example, a “cup and ring” carving on bedrock is regarded as instance of E24 Physical Man-Made Thing.

Examples:

- the Forth Railway Bridge (E22)
- the Channel Tunnel (E25)
- the Historical Collection of the Museum Benaki in Athens (E78)

Properties:

P62 depicts (is depicted by): E1 CRM Entity
(P62.1 mode of depiction: E55 Type)

[P65](#) shows visual item (is shown by): [E36](#) Visual Item

[P128](#) carries (is carried by): [E73](#) Information Object

E25 Man-Made Feature

Subclass of: [E24](#) Physical Man-Made Thing
[E26](#) Physical Feature

Scope Note: This class comprises physical features that are purposely created by human activity, such as scratches, artificial caves, artificial water channels, etc.

No assumptions are made as to the extent of modification required to justify regarding a feature as man-made. For example, rock art or even “cup and ring” carvings on bedrock are regarded as types of E25 Man-Made Feature.

Examples:

- the Manchester Ship Canal
- Michael Jackson’s nose following plastic surgery

E26 Physical Feature

Subclass of: [E18](#) Physical Thing

Superclass of: [E25](#) Man-Made Feature
[E27](#) Site

Scope Note: This class comprises identifiable features that are physically attached in an integral way to particular physical objects.

Instances of E26 Physical Feature share many of the attributes of instances of E19 Physical Object. They may have a one-, two- or three-dimensional geometric extent, but there are no natural borders that separate them completely in an objective way from the carrier objects. For example, a doorway is a feature but the door itself, being attached by hinges, is not.

Instances of E26 Physical Feature can be features in a narrower sense, such as scratches, holes, reliefs, surface colours, reflection zones in an opal crystal or a density change in a piece of wood. In the wider sense, they are portions of particular objects with partially imaginary borders, such as the core of the Earth, an area of property on the surface of the Earth, a landscape or the head of a contiguous marble statue. They can be measured and dated, and it is sometimes possible to state who or what is or was responsible for them. They cannot be separated from the carrier object, but a segment of the carrier object may be identified (or sometimes removed) carrying the complete feature.

This definition coincides with the definition of “fiat objects” (Smith & Varzi, 2000, pp.401-420), with the exception of aggregates of “bona fide objects”.

Examples:

- the temple in Abu Simbel before its removal, which was carved out of solid rock
- Albrecht Durer’s signature on his painting of Charles the Great
- the damage to the nose of the Great Sphinx in Giza
- Michael Jackson’s nose prior to plastic surgery

E27 Site

Subclass of: [E26](#) Physical Feature

Scope Note: This class comprises pieces of land or sea floor.

In contrast to the purely geometric notion of E53 Place, this class describes constellations of matter on the surface of the Earth or other celestial body, which can be represented by photographs, paintings and maps.

Instances of E27 Site are composed of relatively immobile material items and features in a particular configuration at a particular location.

Examples:

- the Amazon river basin
- Knossos
- the Apollo 11 landing site
- Heathrow Airport

- the submerged harbour of the Minoan settlement of Gournia, Crete

E28 Conceptual Object

Subclass of: [E71](#) Man-Made Thing

Superclass of: [E55](#) Type

[E89](#) Propositional Object

[E90](#) Symbolic Object

Scope note: This class comprises non-material products of our minds and other human produced data that have become objects of a discourse about their identity, circumstances of creation or historical implication. The production of such information may have been supported by the use of technical devices such as cameras or computers.

Characteristically, instances of this class are created, invented or thought by someone, and then may be documented or communicated between persons. Instances of E28 Conceptual Object have the ability to exist on more than one particular carrier at the same time, such as paper, electronic signals, marks, audio media, paintings, photos, human memories, etc.

They cannot be destroyed. They exist as long as they can be found on at least one carrier or in at least one human memory. Their existence ends when the last carrier and the last memory are lost.

Examples:

- Beethoven's 'Ode an die Freude' (Ode to Joy), (E73)
- the definition of "ontology" in the Oxford English Dictionary
- the knowledge about the victory at Marathon carried by the famous runner

Properties: P149 is identified by (identifies): E75 Conceptual Object Appellation

E29 Design or Procedure

Subclass of: [E73](#) Information Object

Scope note: This class comprises documented plans for the execution of actions in order to achieve a result of a specific quality, form or contents. In particular it comprises plans for deliberate human activities that may result in the modification or production of instances of E24 Physical Thing.

Instances of E29 Design or Procedure can be structured in parts and sequences or depend on others. This is modelled using *P69 is associated with*.

Designs or procedures can be seen as one of the following:

1. A schema for the activities it describes
2. A schema of the products that result from their application.
3. An independent intellectual product that may have never been applied, such as Leonardo da Vinci's famous plans for flying machines.

Because designs or procedures may never be applied or only partially executed, the CRM models a loose relationship between the plan and the respective product.

Examples:

- the ISO standardisation procedure
- the musical notation for Beethoven's "Ode to Joy"
- the architectural drawings for the Kölner Dom in Cologne, Germany
- the drawing on the folio 860 of the Codex Atlanticus from Leonardo da Vinci, 1486-1490, kept in the Biblioteca Ambrosiana in Milan

Properties:

P68 foresees use of (use foreseen by): E57 Material

[P69](#) has association with (is associated with): [E29](#) Design or Procedure

(P69.1 has type: [E55](#) Type)

E30 Right

Subclass of: [E89](#) Propositional Object

Scope Note: This class comprises legal privileges concerning material and immaterial things or their derivatives.

These include reproduction and property rights.

Examples:

- copyright held by ISO on ISO/CD 21127
- ownership of the “Mona Lisa” by the Louvre

E31 Document

Subclass of: [E73](#) Information Object

Superclass of: [E32](#) Authority Document

Scope note: This class comprises identifiable immaterial items that make propositions about reality.

These propositions may be expressed in text, graphics, images, audiograms, videograms or by other similar means. Documentation databases are regarded as a special case of E31 Document. This class should not be confused with the term “document” in Information Technology, which is compatible with E73 Information Object.

Examples:

- the Encyclopaedia Britannica (E32)
- the photo of the Allied Leaders at Yalta published by UPI, 1945
- the Doomsday Book

Properties:

P70 documents (is documented in): E1 CRM Entity

E32 Authority Document

Subclass of: [E31](#) Document

Scope note: This class comprises encyclopaedia, thesauri, authority lists and other documents that define terminology or conceptual systems for consistent use.

Examples:

- Webster’s Dictionary
- Getty Art and Architecture Thesaurus
- the CIDOC Conceptual Reference Model

Properties:

[P71](#) lists (is listed in): [E1](#) CRM Entity

E33 Linguistic Object

Subclass of: [E73](#) Information Object

Superclass of: [E34](#) Inscription

[E35](#) Title

Scope note: This class comprises identifiable expressions in natural language or languages.

Instances of E33 Linguistic Object can be expressed in many ways: e.g. as written texts, recorded speech or sign language. However, the CRM treats instances of E33 Linguistic Object independently from the medium or method by which they are expressed. Expressions in formal languages, such as computer code or mathematical formulae, are not treated as instances of E33 Linguistic Object by the CRM. These should be modelled as instances of E73 Information Object.

The text of an instance of E33 Linguistic Object can be documented in a note by P3 has note: E62 String

Examples:

- the text of the Ellesmere Chaucer manuscript
- the lyrics of the song “Blue Suede Shoes”
- the text of the Jabberwocky by Lewis Carroll
- the text of “Doktoro Jekyll kaj Sinjoro Hyde” (an Esperanto translation of Dr Jekyll and Mr Hyde)

Properties:

[P72](#) has language (is language of): [E56](#) Language

[P73](#) has translation (is translation of): E33 Linguistic Object

E35 Title

Subclass of: [E33](#) Linguistic Object

[E41](#) Appellation

Scope note: This class comprises the names assigned to works, such as texts, artworks or pieces of music.

Titles are proper noun phrases or verbal phrases, and should not be confused with generic object names such as “chair”, “painting” or “book” (the latter are common nouns that stand for instances of E55 Type). Titles may be assigned by the creator of the work itself, or by a social group.

This class also comprises the translations of titles that are used as surrogates for the original titles in different social contexts.

Examples:

- “The Merchant of Venice”
- “Mona Lisa”
- “La Pie or The Magpie”
- “Lucy in the Sky with Diamonds”

E36 Visual Item

Subclass of: [E73](#) Information Object

Superclass of: [E37](#) Mark

E38 Image

Scope Note: This class comprises the intellectual or conceptual aspects of recognisable marks and images.

This class does not intend to describe the idiosyncratic characteristics of an individual physical embodiment of a visual item, but the underlying prototype. For example, a mark such as the ICOM logo is generally considered to be the same logo when used on any number of publications. The size, orientation and colour may change, but the logo remains uniquely identifiable. The same is true of images that are reproduced many times. This means that visual items are independent of their physical support.

The class E36 Visual Item provides a means of identifying and linking together instances of E24 Physical Man-Made Thing that carry the same visual symbols, marks or images etc. The property *P62 depicts (is depicted by)* between E24 Physical Man-Made Thing and depicted subjects (E1 CRM Entity) can be regarded as a short-cut of the more fully developed path from E24 Physical Man-Made Thing through *P65 shows visual item (is shown by)*, E36 Visual Item, *P138 represents (has representation)* to E1CRM Entity, which in addition captures the optical features of the depiction.

Examples:

- the visual appearance of Monet's "La Pie" (E38)
- the Coca-Cola logo (E34)
- the Chi-Rho (E37)
- the communist red star (E37)

Properties:

**P138 represents (has representation): E1 CRM Entity
(P138.1 mode of representation: E55 Type)**

E37 Mark

Subclass of: [E36](#) Visual Item

Superclass of: E34 Inscription

Scope note: This class comprises symbols, signs, signatures or short texts applied to instances of E24 Physical Man-Made Thing by arbitrary techniques in order to indicate the creator, owner, dedications, purpose, etc.

This class specifically excludes features that have no semantic significance, such as scratches or tool marks. These should be documented as instances of E25 Man-Made Feature.

Examples:

- Minoan double axe mark
- ©
- ☺

E39 Actor

Subclass of: [E77](#) Persistent Item

Superclass of: [E21](#) Person

[E74](#) Group

Scope note: This class comprises people, either individually or in groups, who have the potential to perform intentional actions for which they can be held responsible.

The CRM does not attempt to model the inadvertent actions of such actors. Individual people should be documented as instances of E21 Person, whereas groups should be documented as instances of either E74 Group or its subclass E40 Legal Body.

Examples:

- London and Continental Railways (E40)
- the Governor of the Bank of England in 1975 (E21)
- Sir Ian McKellan (E21)

Properties:

[P74](#) has current or former residence (is current or former residence of): [E53](#) Place

[P75](#) possesses (is possessed by): [E30](#) Right

[P76](#) has contact point (provides access to): [E51](#) Contact Point

[P131](#) is identified by (identifies): [E82](#) Actor Appellation

E40 Legal Body

Subclass of: [E74](#) Group

Scope Note: This class comprises institutions or groups of people that have obtained a legal recognition as a group and can act collectively as agents.

This means that they can perform actions, own property, create or destroy things and can be held collectively responsible for their actions like individual people. The term ‘personne morale’ is often used for this in French.

Examples:

- Greenpeace
- Paveprime Ltd
- the National Museum of Denmark

E41 Appellation

Subclass of: [E90](#) Symbolic Object

Superclass of: [E35](#) Title

[E42](#) Identifier

[E44](#) Place Appellation

[E49](#) Time Appellation

[E51](#) Contact Point

[E75](#) Conceptual Object Appellation

[E82](#) Actor Appellation

Scope note: This class comprises signs, either meaningful or not, or arrangements of signs following a specific syntax, that are used or can be used to refer to and identify a specific instance of some class within a certain context.

Instances of E41 Appellation do not identify things by their meaning, even if they happen to have one, but by convention, tradition, or agreement. Instances of E41 Appellation are cultural constructs; as such, they have a context, a history, and a use in time and space by some group of users. A given instance of E41 Appellation can have alternative forms, i.e., other instances of E41 Appellation that are always regarded as equivalent independent from the thing it denotes.

Specific subclasses of E41 Appellation should be used when instances of E41 Appellation of a characteristic form are used for particular objects. Instances of E49 Time Appellation, for example, which take the form of instances of E50 Date, can be easily recognised.

E41 Appellation should not be confused with the act of naming something. *Cf.* E15 Identifier Assignment

Examples:

- “Martin”
- “the Forth Bridge”
- “the Merchant of Venice” (E35)
- “*Spigelia marilandica* (L.) L.” [not the species, just the *name*]
- “information science” [not the science itself, but the name through which we refer to it in an English-speaking context]
- “安” [Chinese “an”, meaning “peace”]

Properties:

P139 has alternative form: E41 Appellation
P139.1 has type: E55 Type

E42 Identifier

Subclass of: [E41](#) Appellation

Scope note: This class comprises strings or codes assigned to instances of E1 CRM Entity in order to identify them uniquely and permanently within the context of one or more organisations. Such codes are often known as inventory numbers, registration codes, etc. and are typically composed of alphanumeric sequences. The class E42 Identifier is not normally used for machine-generated identifiers used for automated processing unless these are also used by human agents.

Examples:

- “MM.GE.195”
- “13.45.1976”
- “OXCMS: 1997.4.1”
- ISSN “0041-5278”
- ISRC “FIFIN8900116”
- Shelf mark “Res 8 P 10”
- “Guillaume de Machaut (1300?-1377)” [a controlled personal name heading that follows the French rules]

E44 Place Appellation

Subclass of: [E41](#) Appellation

Superclass of E45 Address
E46 Section Definition
[E47](#) Spatial Coordinates
E48 Place Name

Scope Note: This class comprises any sort of identifier characteristically used to refer to an E53 Place.

Instances of E44 Place Appellation may vary in their degree of precision and their meaning may vary over time – the same instance of E44 Place Appellation may be used to refer to several places, either because of cultural shifts, or because objects used as reference points have moved around. Instances of E44 Place Appellation can be extremely varied in form: postal addresses, instances of E47 Spatial Coordinate, and parts of buildings can all be considered as instances of E44 Place Appellation.

Examples:

- “Vienna”
- “CH-1211, Genève”
- “Aquae Sulis Minerva”
- “Bath”
- “Cambridge”
- “the Other Place”
- “the City”

E47 Spatial Coordinates

Subclass of: [E44](#) Place Appellation

Scope Note: This class comprises the textual or numeric information required to locate specific instances of E53 Place within schemes of spatial identification.

Coordinates are a specific form of E44 Place Appellation, that is, a means of referring to a particular E53 Place. Coordinates are not restricted to longitude, latitude and altitude. Any regular system of reference that maps onto an E19 Physical Object can be used to generate coordinates.

Examples:

- “6°5’29”N 45°12’13”W”
- “Black queen’s bishop 4” [chess coordinate]

E49 Time Appellation

Subclass of: [E41](#) Appellation

Superclass of [E50](#) Date

Scope Note: This class comprises all forms of names or codes, such as historical periods, and dates, which are characteristically used to refer to a specific E52 Time-Span.

The instances of E49 Time Appellation may vary in their degree of precision, and they may be relative to other time frames, “Before Christ” for example. Instances of E52 Time-Span are often defined by reference to a cultural period or an event e.g. ‘the duration of the Ming Dynasty’.

Examples:

- “Meiji” [Japanese term for a specific time-span]
- “1st half of the XX century”
- “Quaternary”
- “1215 Hegira” [a date in the Islamic calendar]
- “Last century”

E50 Date

Subclass of: [E49](#) Time Appellation

Scope Note: This class comprises specific forms of E49 Time Appellation.
Dates may vary in their degree of precision.

Examples:

- “1900”
- “4-4-1959”
- “19-MAR-1922”
- “19640604”

E52 Time-Span

Subclass of: [E1](#) CRM Entity

Scope note: This class comprises abstract temporal extents, in the sense of Galilean physics, having a beginning, an end and a duration.

Time Span has no other semantic connotations. Time-Spans are used to define the temporal extent of instances of E4 Period, E5 Event and any other phenomena valid for a certain time. An E52 Time-Span may be identified by one or more instances of E49 Time Appellation.

Since our knowledge of history is imperfect, instances of E52 Time-Span can best be considered as approximations of the actual Time-Spans of temporal entities. The properties of E52 Time-Span are intended to allow these approximations to be expressed precisely. An extreme case of approximation, might, for example, define an E52 Time-Span having unknown beginning, end and duration. Used as a common E52 Time-Span for two events, it would nevertheless define them as being simultaneous, even if nothing else was known.

Automatic processing and querying of instances of E52 Time-Span is facilitated if data can be parsed into an E61 Time Primitive.

Examples:

- 1961
- From 12-17-1993 to 12-8-1996
- 14h30 – 16h22 4th July 1945
- 9.30 am 1.1.1999 to 2.00 pm 1.1.1999
- duration of the Ming Dynasty

Properties:

[P78](#) is identified by (identifies): [E49](#) Time Appellation

[P79](#) beginning is qualified by: E62 String

[P80](#) end is qualified by: E62 String

[P81](#) ongoing throughout: E61 Time Primitive

[P82](#) at some time within: [E61](#) Time Primitive

[P83](#) had at least duration (was minimum duration of): E54 Dimension

[P84](#) had at most duration (was maximum duration of): E54 Dimension

[P86](#) falls within (contains): E52 Time-Span

E53 Place

Subclass of: [E1](#) CRM Entity

Scope note: This class comprises extents in space, in particular on the surface of the earth, in the pure sense of physics: independent from temporal phenomena and matter.

The instances of E53 Place are usually determined by reference to the position of “immobile” objects such as buildings, cities, mountains, rivers, or dedicated geodetic marks. A Place can be determined by combining a frame of reference and a location with respect to this frame. It may be identified by one or more instances of E44 Place Appellation.

It is sometimes argued that instances of E53 Place are best identified by global coordinates or absolute reference systems. However, relative references are often more relevant in the context of cultural documentation and tend to be more precise. In particular, we are often interested in position in relation to large, mobile objects, such as ships. For example, the Place at which Nelson died is known with reference to a large mobile object – H.M.S Victory. A resolution of this Place in terms of absolute coordinates would require knowledge of the movements of the vessel and the precise time of death, either of which may be revised, and the result would lack historical and cultural relevance.

Any object can serve as a frame of reference for E53 Place determination. The model foresees the notion of a “section” of an E19 Physical Object as a valid E53 Place determination.

Examples:

- the extent of the UK in the year 2003
- the position of the hallmark on the inside of my wedding ring
- the place referred to in the phrase: “Fish collected at three miles north of the confluence of the Arve and the Rhone”
- here -> <-

Properties:

[P87](#) is identified by (identifies): [E44](#) Place Appellation

[P89](#) falls within (contains): E53 Place

[P121](#) overlaps with: E53 Place

[P122](#) borders with: E53 Place

E54 Dimension

Subclass of: [E1](#) CRM Entity

Scope note: This class comprises quantifiable properties that can be measured by some calibrated means and can be approximated by values, i.e. points or regions in a mathematical or conceptual space, such as natural or real numbers, RGB values etc.

An instance of E54 Dimension represents the true quantity, independent from its numerical approximation, e.g. in inches or in cm. The properties of the class E54 Dimension allow for expressing the numerical approximation of the values of an instance of E54 Dimension. If the true values belong to a non-discrete space, such as spatial distances, it is recommended to record them as approximations by intervals or regions of indeterminacy enclosing the assumed true values. For instance, a length of 5 cm may be recorded as 4.5-5.5 cm, according to the precision of the respective observation. Note, that interoperability of values described in different units depends critically on the representation as value regions.

Numerical approximations in archaic instances of E58 Measurement Unit used in historical records should be preserved. Equivalents corresponding to current knowledge should be recorded as additional instances of E54 Dimension as appropriate.

Examples:

- currency: £26.00
- length: 3.9-4.1 cm
- diameter 26 mm
- weight 150 lbs
- density: 0.85 gm/cc
- luminescence: 56 ISO lumens
- tin content: 0.46 %
- taille au garot: 5 hands
- calibrated C14 date: 2460-2720 years, etc

Properties:

P90 has value: E60 Number

P91 has unit (is unit of): E58 Measurement Unit

E55 Type

Subclass of: [E28](#) Conceptual Object

Superclass of: [E56](#) Language

[E57](#) Material

E58 Measurement Unit

Scope note: This class comprises concepts denoted by terms from thesauri and controlled vocabularies used to characterize and classify instances of CRM classes. Instances of E55 Type represent concepts in contrast to instances of E41 Appellation which are used to name instances of CRM classes.

E55 Type is the CRM's interface to domain specific ontologies and thesauri. These can be represented in the CRM as subclasses of E55 Type, forming hierarchies of terms, i.e. instances of E55 Type linked via P127 has broader term (has narrower term). Such hierarchies may be extended with additional properties.

Examples:

- weight, length, depth [types of E54]
- portrait, sketch, animation [types of E38]
- French, English, German [E56]
- excellent, good, poor [types of E3]
- Ford Model T, chop stick [types of E22]
- cave, doline, scratch [types of E26]
- poem, short story [types of E33]
- wedding, earthquake, skirmish [types of E5]

Properties:

[P127](#) has broader term (has narrower term): [E55](#) Type

P150 defines typical parts of (define typical wholes for): E55 Type

E56 Language

Subclass of: [E55](#) Type

Scope note: This class is a specialization of E55 Type and comprises the natural languages in the sense of concepts.

This type is used categorically in the model without reference to instances of it, i.e. the Model does not foresee the description of instances of instances of E56 Language, e.g.: “instances of Mandarin Chinese”.

It is recommended that internationally or nationally agreed codes and terminology are used to denote instances of E56 Language, such as those defined in ISO 639:1988.

Examples:

- el [Greek]
- en [English]
- eo [Esperanto]
- es [Spanish]
- fr [French]

E57 Material

Subclass of: [E55](#) Type

Scope note: This class is a specialization of E55 Type and comprises the concepts of materials.

Instances of E57 Material may denote properties of matter before its use, during its use, and as incorporated in an object, such as ultramarine powder, tempera paste, reinforced concrete. Discrete pieces of raw-materials kept in museums, such as bricks, sheets of fabric, pieces of metal, should be modelled individually in the same way as other objects. Discrete used or processed pieces, such as the stones from Nefer Titi’s temple, should be modelled as parts (cf. *P46 is composed of*).

This type is used categorically in the model without reference to instances of it, i.e. the Model does not foresee the description of instances of instances of E57 Material, e.g.: “instances of gold”.

It is recommended that internationally or nationally agreed codes and terminology are used.

Examples:

- brick
- gold
- aluminium
- polycarbonate
- resin

E59 Primitive Value

Superclass of: [E60](#) Number
[E61](#) Time Primitive
[E62](#) String

Scope Note: This class comprises primitive values used as documentation elements, which are not further elaborated upon within the model.

As such they are not considered as elements within our universe of discourse. No specific implementation recommendations are made. It is recommended that the primitive value system from the implementation platform be used to substitute for this class and its subclasses.

Examples:

- ABCDEFG (E62)
- 3.14 (E60)
- 0
- 1921-01-01 (E61)

E60 Number

Subclass of: [E59](#) Primitive Value

Scope Note: This class comprises any encoding of computable (algebraic) values such as integers, real numbers, complex numbers, vectors, tensors etc., including intervals of these values to express limited precision.

Numbers are fundamentally distinct from identifiers in continua, such as instances of E50 Date and E47 Spatial Coordinate, even though their encoding may be similar. Instances of E60 Number can be combined with each other in algebraic operations to yield other instances of E60 Number, e.g., $1+1=2$. Identifiers in continua may be combined with numbers expressing distances to yield new identifiers, e.g., $1924-01-31 + 2 \text{ days} = 1924-02-02$. Cf. E54 Dimension.

Examples:

- 5
- $3+2i$
- 1.5e-04
- (0.5, - 0.7,88)

E61 Time Primitive

Subclass of: [E59](#) Primitive Value

Scope Note: This class comprises instances of E59 Primitive Value for time that should be implemented with appropriate validation, precision and interval logic to express date ranges relevant to cultural documentation.

E61 Time Primitive is not further elaborated upon within the model.

Examples:

- 1994 – 1997
- 13 May 1768
- 2000/01/01 00:00:59.7
- 85th century BC

E62 String

Subclass of: [E59](#) Primitive Value

Scope Note: This class comprises the instances of E59 Primitive Values used for documentation such as free text strings, bitmaps, vector graphics, etc.

E62 String is not further elaborated upon within the model.

Examples:

- the Quick Brown Fox Jumps Over the Lazy Dog
- 6F 6E 54 79 70 31 0D 9E

E63 Beginning of Existence

Subclass of: [E5](#) Event
Superclass of: [E12](#) Production
[E65](#) Creation
[E66](#) Formation
[E67](#) Birth
E81 Transformation

Scope note: This class comprises events that bring into existence any E77 Persistent Item.

It may be used for temporal reasoning about things (intellectual products, physical items, groups of people, living beings) beginning to exist; it serves as a hook for determination of a terminus post quem and ante quem.

Examples:

- the birth of my child
- the birth of Snoopy, my dog
- the calving of the iceberg that sank the Titanic
- the construction of the Eiffel Tower

Properties:

P92 brought into existence (was brought into existence by): E77 Persistent Item

E64 End of Existence

Subclass of: [E5](#) Event
Superclass of: E6 Destruction
E68 Dissolution
[E69](#) Death
E81 Transformation

Scope note: This class comprises events that end the existence of any E77 Persistent Item.

It may be used for temporal reasoning about things (physical items, groups of people, living beings) ceasing to exist; it serves as a hook for determination of a terminus postquem and antequem. In cases where substance from a Persistent Item continues to exist in a new form, the process would be documented by E81 Transformation.

Examples:

- the death of Snoopy, my dog
- the melting of the snowman
- the burning of the Temple of Artemis in Ephesos by Herostratos in 356BC

Properties:

P93 took out of existence (was taken out of existence by): E77 Persistent Item

E65 Creation

Subclass of: [E7](#) Activity
[E63](#) Beginning of Existence

Superclass of: E83 Type Creation

Scope note: This class comprises events that result in the creation of conceptual items or immaterial products, such as legends, poems, texts, music, images, movies, laws, types etc.

Examples:

- the framing of the U.S. Constitution
- the drafting of U.N. resolution 1441

Properties:

[P94](#) has created (was created by): [E28](#) Conceptual Object

E66 Formation

Subclass of: [E7](#) Activity

[E63](#) Beginning of Existence

Scope note: This class comprises events that result in the formation of a formal or informal [E74](#) Group of people, such as a club, society, association, corporation or nation.

E66 Formation does not include the arbitrary aggregation of people who do not act as a collective.

The formation of an instance of [E74](#) Group does not mean that the group is populated with members at the time of formation. In order to express the joining of members at the time of formation, the respective activity should be simultaneously an instance of both E66 Formation and [E85](#) Joining.

Examples:

- the formation of the CIDOC CRM Special Interest Group
- the formation of the Soviet Union
- the conspiring of the murderers of Caesar

Properties:

[P95](#) has formed (was formed by): [E74](#) Group

[P151](#) was formed from (participated in): [E74](#) Group

E67 Birth

Subclass of: [E63](#) Beginning of Existence

Scope note: This class comprises the births of human beings. E67 Birth is a biological event focussing on the context of people coming into life. ([E63](#) Beginning of Existence comprises the coming into life of any living beings).

Twins, triplets etc. are brought into life by the same E67 Birth event. The introduction of the E67 Birth event as a documentation element allows the description of a range of family relationships in a simple model. Suitable extensions may describe more details and the complexity of motherhood with the intervention of modern medicine. In this model, the biological father is not seen as a necessary participant in the E67 Birth event.

Examples:

- the birth of Alexander the Great

Properties:

P96 by mother (gave birth): [E21](#) Person

P97 from father (was father for): [E21](#) Person

[P98](#) brought into life (was born): [E21](#) Person

E69 Death

Subclass of: [E64](#) End of Existence

Scope note: This class comprises the deaths of human beings.
If a person is *killed*, their death should be instantiated as E69 Death and as E7 Activity. The death or perishing of other living beings should be documented using E64 End of Existence.

Examples:

- the murder of Julius Caesar (E69, E7)
- the death of Senator Paul Wellstone

Properties:

[P100](#) was death of (died in): [E21](#) Person

E70 Thing

Subclass of: [E77](#) Persistent Item

Superclass of: [E71](#) Man-Made Thing

[E72](#) Legal Object

Scope note: This general class comprises usable discrete, identifiable, instances of E77 Persistent Item that are documented as single units.

They can be either intellectual products or physical things, and are characterized by relative stability. They may for instance either have a solid physical form, an electronic encoding, or they may be logical concept or structure.

Examples:

- my photograph collection (E78)
- the bottle of milk in my refrigerator (E22)
- the plan of the Strassburger Muenster (E29)
- the thing on the top of Otto Hahn's desk (E19)
- the form of the no-smoking sign (E36)
- the cave of Dirou, Mani, Greece (E27)

Properties:

[P43](#) has dimension (is dimension of): [E54](#) Dimension

P101 had as general use (was use of): E55 Type

**[P130](#) shows features of (features are also found on): [E70](#) Thing
(P130.1 kind of similarity: [E55](#) Type)**

E71 Man-Made Thing

Subclass of: [E70](#) Thing

Superclass of: [E24](#) Physical Man-Made Thing

[E28](#) Conceptual Object

Scope note: This class comprises discrete, identifiable man-made items that are documented as single units.

These items are either intellectual products or man-made physical things, and are characterized by relative stability. They may for instance have a solid physical form, an electronic encoding, or they may be logical concepts or structures.

Examples:

- Beethoven's 5th Symphony (E73)
- Michelangelo's David
- Einstein's Theory of General Relativity (E73)
- the taxon '*Fringilla coelebs* Linnaeus,1758' (E55)

Properties:

[P102](#) has title (is title of): [E35](#) Title
([P102.1](#) has type: [E55](#) Type)
[P103](#) was intended for (was intention of): [E55](#) Type

E72 Legal Object

Subclass of: [E70](#) Thing

Superclass of: [E18](#) Physical Thing

[E90](#) Symbolic Object

Scope note: This class comprises those material or immaterial items to which instances of [E30](#) Right, such as the right of ownership or use, can be applied.

This is true for all [E18](#) Physical Thing. In the case of instances of [E28](#) Conceptual Object, however, the identity of the [E28](#) Conceptual Object or the method of its use may be too ambiguous to reliably establish instances of [E30](#) Right, as in the case of taxa and inspirations. Ownership of corporations is currently regarded as out of scope of the CRM.

Examples:

- the Cullinan diamond ([E19](#))
- definition of the CIDOC Conceptual Reference Model Version 2.1 ([E73](#))

Properties:

[P104](#) is subject to (applies to): [E30](#) Right
[P105](#) right held by (has right on): [E39](#) Actor

E73 Information Object

Subclass of: [E89](#) Propositional Object

[E90](#) Symbolic Object

Superclass of: [E29](#) Design or Procedure

[E31](#) Document

[E33](#) Linguistic Object

[E36](#) Visual Item

Scope note: This class comprises identifiable immaterial items, such as a poems, jokes, data sets, images, texts, multimedia objects, procedural prescriptions, computer program code, algorithm or mathematical formulae, that have an objectively recognizable structure and are documented as single units.

An [E73](#) Information Object does not depend on a specific physical carrier, which can include human memory, and it can exist on one or more carriers simultaneously.

Instances of [E73](#) Information Object of a linguistic nature should be declared as instances of the [E33](#) Linguistic Object subclass. Instances of [E73](#) Information Object of a documentary nature should be declared as instances of the [E31](#) Document subclass. Conceptual items such as types and classes are not instances of [E73](#) Information Object, nor are ideas without a reproducible expression.

Examples:

- image BM000038850.JPG from the Clayton Herbarium in London
- E. A. Poe's "The Raven"

- the movie “The Seven Samurai” by Akira Kurosawa
- the Maxwell Equations

Properties: P165 incorporates (is incorporated in): E90 Symbolic Object

E74 Group

Subclass of: [E39](#) Actor

Superclass of: [E40](#) Legal Body

Scope note: This class comprises any gatherings or organizations of two or more people that act collectively or in a similar way due to any form of unifying relationship. In the wider sense this class also comprises official positions which used to be regarded in certain contexts as one actor, independent of the current holder of the office, such as the president of a country.

A gathering of people becomes an E74 Group when it exhibits organizational characteristics usually typified by a set of ideas or beliefs held in common, or actions performed together. These might be communication, creating some common 173ulgate173, a common purpose such as study, worship, business, sports, etc. Nationality can be 173ulgate173 as membership in an E74 Group (cf. HumanML markup). Married couples and other concepts of family are regarded as particular examples of E74 Group.

Examples:

- the impressionists
- the Navajo
- the Greeks
- the peace protestors in New York City on February 15 2003
- Exxon-Mobil
- King Solomon and his wives
- The President of the Swiss Confederation

Properties:

[P107](#) has current or former member (is current or former member of): [E39](#) Actor
(P107.1 kind of member: [E55](#) Type)

E77 Persistent Item

Subclass of: [E1](#) CRM Entity

Superclass of: [E39](#) Actor
[E70](#) Thing

Scope note: This class comprises items that have a persistent identity, sometimes known as “endurants” in philosophy.

They can be repeatedly recognized within the duration of their existence by identity criteria rather than by continuity or observation. Persistent Items can be either physical entities, such as people, animals or things, or conceptual entities such as ideas, concepts, products of the imagination or common names.

The criteria that determine the identity of an item are often difficult to establish –; the decision depends largely on the judgement of the observer. For example, a building is regarded as no longer existing if it is dismantled and the materials reused in a different configuration. On the other hand, human beings go through radical and profound changes during their life-span,

affecting both material composition and form, yet preserve their identity by other criteria. Similarly, inanimate objects may be subject to exchange of parts and matter. The class E77 Persistent Item does not take any position about the nature of the applicable identity criteria and if actual knowledge about identity of an instance of this class exists. There may be cases, where the identity of an E77 Persistent Item is not decidable by a certain state of knowledge.

The main classes of objects that fall outside the scope the E77 Persistent Item class are temporal objects such as periods, events and acts, and descriptive properties.

Examples:

- Leonard da Vinci
- Stonehenge
- the hole in the ozone layer
- the First Law of Thermodynamics
- the Bermuda Triangle

E82 Actor Appellation

Subclass of: [E41](#) Appellation

Scope note: This class comprises any sort of name, number, code or symbol characteristically used to identify an E39 Actor.

An E39 Actor will typically have more than one E82 Actor Appellation, and instances of E82 Actor Appellation in turn may have alternative representations. The distinction between corporate and personal names, which is particularly important in library applications, should be made by explicitly linking the E82 Actor Appellation to an instance of either E21 Person or E74 Group/E40 Legal Body. If this is not possible, the distinction can be made through the use of the *P2 has type* mechanism.

Examples:

- “John Doe”
- “Doe, J”
- “the U.S. Social Security Number 246-14-2304”
- “the Artist Formerly Known as Prince”
- “the Master of the Flemish Madonna”
- “Raphael’s Workshop”
- “the Brontë Sisters”
- “ICOM”
- “International Council of Museums”

E84 Information Carrier

Subclass of: [E22](#) Man-Made Object

Scope note: This class comprises all instances of E22 Man-Made Object that are explicitly designed to act as persistent physical carriers for instances of E73 Information Object.

This allows a relationship to be asserted between an E19 Physical Object and its immaterial information contents. An E84 Information Carrier may or may not contain information, e.g., a diskette. Note that any E18 Physical Thing may carry information, such as an E34 Inscription. However, unless it was specifically designed for this purpose, it is not an Information Carrier. Therefore the property *P128 carries (is carried by)* applies to E18 Physical Thing in general.

Examples:

- the Rosetta Stone
- my paperback copy of Crime & Punishment
- the computer disk at ICS-FORTH that stores the canonical Definition of the CIDOC CRM

Properties:

E89 Propositional Object

Subclass of: [E28](#) Conceptual Object

Superclass of: [E73](#) Information Object

[E30](#) Right

Scope note: This class comprises immaterial items, including but not limited to stories, plots, procedural prescriptions, algorithms, laws of physics or images that are, or represent in some sense, sets of propositions about real or mental things and that are documented as single units or serve as topic of discourse.

This class also comprises items that are “about” something in the sense of a subject. In the wider sense, this class includes expressions of psychological value such as non-figural art and musical themes. However, conceptual items such as types and classes are not instances of E89 Propositional Object. This should not be confused with the definition of a type, which is indeed an instance of E89 Propositional Object.

Examples:

- Maxwell’s Equations
- The ideational contents of Aristotle’s book entitled ‘Metaphysics’ as rendered in the Greek texts translated in ... Oxford edition...
- The underlying prototype of any “no-smoking” sign (E36)
- The common ideas of the plots of the movie “The Seven Samurai” by Akira Kurosawa and the movie “The Magnificent Seven” by John Sturges
- The image content of the photo of the Allied Leaders at Yalta 1945 (E38)

Properties:

[P67](#) refers to (is referred to by): [E1](#) CRM Entity
(P67.1 has type: [E55](#) Type)

[P129](#) is about (is subject of): [E1](#) CRM Entity

[P148](#) has component (is component of): [E89](#) Propositional Object

E90 Symbolic Object

Subclass of: [E28](#) Conceptual Object

[E72](#) Legal Object

Superclass of: [E73](#) Information Object

[E41](#) Appellation

Scope note: This class comprises identifiable symbols and any aggregation of symbols, such as characters, identifiers, traffic signs, emblems, texts, data sets, images, musical scores, multimedia objects, computer program code or mathematical formulae that have an objectively recognizable structure and that are documented as single units.

It includes sets of signs of any nature, which may serve to designate something, or to communicate some propositional content.

An instance of E90 Symbolic Object does not depend on a specific physical carrier, which can include human memory, and it can exist on one or more carriers simultaneously. An instance of E90 Symbolic Object may or may not have a specific meaning, for example an arbitrary character string.

In some cases, the content of an instance of E90 Symbolic Object may completely be represented by a serialized content model, such.. as the property P3 has note allows for describing this content model...P3.1 has type: E55 Type to specify the encoding..

Examples:

- ‘ecognizabl’
- The “no-smoking” sign (E36)
- ‘BM000038850.JPG’ (E75)
- image BM000038850.JPG from the Clayton Herbarium in London (E38)
- The distribution of form, tone and colour found on Leonardo da Vinci’s painting named “Mona Lisa” (E38)
- The Italian text of Dante’s “Divina Commedia” as found in the authoritative critical edition *La Commedia secondo l’antica vulgate a cura di Giorgio Petrocchi*, Milano: Mondadori, 1966-67 (= Le Opere di Dante Alighieri, Edizione Nazionale a cura della Società Dantesca Italiana, VII, 1-4) (E33)

Properties:

P106 is composed of (forms part of): E90 Symbolic Object

4.4. Referred to CIDOC CRM Properties

This section contains the complete definitions of the properties of the CIDOC CRM Conceptual Reference Model version 6.0 referred to by FRBR_{OO}. We apply the same format conventions as in section 2.7.

P1 is identified by (identifies)

Domain: [E1](#) CRM Entity

Range: [E41](#) Appellation

Superproperty of: E1 CRM Entity. P48 has preferred identifier (is preferred identifier of): E42 Identifier

[E52](#) Time-Span. [P78](#) is identified by (identifies): [E49](#) Time Appellation

[E53](#) Place. [P87](#) is identified by (identifies): [E44](#) Place Appellation

[E71](#) Man-Made Thing. [P102](#) has title (is title of): [E35](#) Title

[E39](#) Actor. [P131](#) is identified by (identifies): [E82](#) Actor Appellation

[E28](#) Conceptual Object. [P149](#) is identified by (identifies): [E75](#) Conceptual Object Appellation

Quantification: many to many (0,n:0,n)

Scope note: This property describes the naming or identification of any real world item by a name or any other identifier.

This property is intended for identifiers in general use, which form part of the world the model intends to describe, and not merely for internal database identifiers which are specific to a technical system, unless these latter also have a more general use outside the technical context. This property includes in particular identification by mathematical expressions such as coordinate systems used for the identification of instances of E53 Place. The property does not reveal anything about when, where and by whom this identifier was used. A more detailed representation can be made using the fully developed (i.e. indirect) path through E15 Identifier Assignment.

Examples:

- the capital of Italy (E53) *is identified by* “Rome” (E48)
- text 25014–32 (E33) *is identified by* “The Decline and Fall of the Roman Empire” (E35)

P2 has type (is type of)

Domain: [E1](#) CRM Entity

Range: [E55](#) Type

Superproperty of: E1 CRM Entity. P137 exemplifies (is exemplified by): E55 Type

Quantification: many to many (0,n:0,n)

Scope note: This property allows sub typing of CRM entities - a form of specialisation – through the use of a terminological hierarchy, or thesaurus.

The CRM is intended to focus on the high-level entities and relationships needed to describe data structures. Consequently, it does not specialise entities any further than is required for this immediate purpose. However, entities in the isA hierarchy of the CRM may be specialised into any number of sub entities, which can be defined in the E55 Type hierarchy. E51 Contact Point, for example, may be specialised into “e-mail address”, “telephone number”, “post office box”, “URL” etc. none of which figures explicitly in the CRM hierarchy. Sub typing obviously requires consistency between the meaning of the terms assigned and the more general intent of the CRM entity in question.

Examples:

- “enquiries@cidoc-crm.org” (E51) *has type* e-mail address (E55)

P3 has note

Domain: [E1](#) CRM Entity

Range: [E62](#) String

Superproperty of: E52 Time-Span. P79 beginning is qualified by: E62 String

E52 Time-Span. P80 end is qualified by: E62 String

Quantification: one to many (0,n;0,1)

Scope note: This property is a container for all informal descriptions about an object that have not been expressed in terms of CRM constructs.

In particular it captures the characterisation of the item itself, its internal structures, appearance etc.

Like property *P2 has type (is type of)*, this property is a consequence of the restricted focus of the CRM. The aim is not to capture, in a structured form, everything that can be said about an item; indeed, the CRM formalism is not regarded as sufficient to express everything that can be said. Good practice requires use of distinct note fields for different aspects of a characterisation. The *P3.1 has type* property of *P3 has note* allows differentiation of specific notes, e.g. “construction”, “decoration” etc.

An item may have many notes, but a note is attached to a specific item.

Examples:

- coffee mug – OXCMS:1983.1.1 (E19) *has note* chipped at edge of handle (E62) *has type* Condition (E55)

Properties: **P3.1 has type: [E55](#) Type**

P4 has time-span (is time-span of)

Domain: [E2](#) Temporal Entity

Range: [E52](#) Time-Span

Quantification: many to one, necessary, dependent (1,1;1,n)

Scope note: This property describes the temporal confinement of an instance of an E2 Temporal Entity.

The related E52 Time-Span is understood as the real Time-Span during which the phenomena were active, which make up the temporal entity instance. It does not convey any other meaning than a positioning on the “time-line” of chronology. The Time-Span in turn is approximated by a set of dates (E61 Time Primitive). A temporal entity can have in reality only one Time-Span, but there may exist alternative opinions about it, which we would express by assigning multiple Time-Spans. Related temporal entities may share a Time-Span. Time-Spans may have completely unknown dates but other descriptions by which we can infer knowledge.

Examples:

- the Yalta Conference (E7) *has time-span* Yalta Conference time-span (E52)

P7 took place at (witnessed)

Domain: [E4](#) Period

Range: [E53](#) Place

Superproperty of: E9 Move. P26 moved to (was destination of): E53 Place

E9 Move. P27 moved from (was origin of): E53 Place

Quantification: many to many, necessary (1,n:0,n)

Scope note: This property describes the spatial location of an instance of E4 Period.

The related E53 Place should be seen as an approximation of the geographical area within which the phenomena that characterise the period in question occurred. *P7 took place at (witnessed)* does not convey any meaning other than spatial positioning (generally on the surface of the earth). For example, the period “Révolution française” can be said to have taken place in “France”, the “Victorian” period, may be said to have taken place in “Britain” and its colonies, as well as other parts of Europe and north America.

A period can take place at multiple locations.

Examples:

- the period “Révolution française” (E4) *took place at* France (E53)

P9 consists of (forms part of)

Domain: [E4](#) Period

Range: [E4](#) Period

Quantification: one to many, (0,n:0,1)

Scope note: This property describes the decomposition of an instance of E4 Period into discrete, subsidiary periods.

The sub-periods into which the period is decomposed form a logical whole - although the entire picture may not be completely known - and the sub-periods are constitutive of the general period.

Examples:

- Cretan Bronze Age (E4) *consists of* Middle Minoan (E4)

P12 occurred in the presence of (was present at)

Domain: [E5](#) Event

Range: [E77](#) Persistent Item

Superproperty of: E5 Event. P11 had participant (participated in): E39 Actor

[E7](#) Activity. [P16](#) used specific object (was used for): [E70](#) Thing

E9 Move. P25 moved (moved by): E19 Physical Object

[E11](#) Modification. [P31](#) has modified (was modified by): [E24](#) Physical Man-Made Thing

E63 Beginning of Existence. P92 brought into existence (was brought into existence by): E77 Persistent Item

E64 End of Existence. P93 took out of existence (was taken out of existence by): E77 Persistent Item

Quantification: many to many, necessary (1,n:0,n)

Scope note: This property describes the active or passive presence of an E77 Persistent Item in an E5 Event without implying any specific role.

It connects the history of a thing with the E53 Place and E50 Date of an event. For example, an

object may be the desk, now in a museum on which a treaty was signed. The presence of an immaterial thing implies the presence of at least one of its carriers.

Examples:

- Deckchair 42 (E19) *was present at* The sinking of the Titanic (E5)

P14 carried out by (performed)

Domain: [E7](#) Activity

Range: [E39](#) Actor

Subproperty of: E5 Event. P11 had participant (participated in): E39 Actor

Superproperty of: E8 Acquisition. P22 transferred title to (acquired title through): E39 Actor

E8 Acquisition. P23 transferred title from (surrendered title through): E39 Actor

E10 Transfer of Custody. P28 custody surrendered by (surrendered custody through): E39 Actor

E10 Transfer of Custody. P29 custody received by (received custody through): E39 Actor

Quantification: many to many, necessary (1,n:0,n)

Scope note: This property describes the active participation of an E39 Actor in an E7 Activity.

It implies causal or legal responsibility. The *P14.1 in the role of* property of the property allows the nature of an Actor's participation to be specified.

Examples:

- the painting of the Sistine Chapel (E7) *carried out by* Michaelangelo Buonaroti (E21) *in the role of* master craftsman (E55)

Properties: P14.1 in the role of: [E55](#) Type

P15 was influenced by (influenced)

Domain: [E7](#) Activity

Range: [E1](#) CRM Entity

Superproperty of: [E7](#) Activity. [P16](#) used specific object (was used for): [E70](#) Thing

E7 Activity. P17 was motivated by (motivated): E1 CRM Entity

E7 Activity. P134 continued (was continued by): E7 Activity

E83 Type Creation. P136 was based on (supported type creation): E1 CRM Entity

Quantification: many to many (0,n:0,n)

Scope note: This is a high level property, which captures the relationship between an E7 Activity and anything that may have had some bearing upon it.

The property has more specific sub properties.

Examples:

- the designing of the Sydney Harbour Bridge (E7) *was influenced by* the Tyne bridge (E22)

P16 used specific object (was used for)

Domain: [E7](#) Activity

Range: [E70](#) Thing

Subproperty of: [E5 Event](#). [P12 occurred in the presence of \(was present at\)](#): [E77 Persistent Item](#)
[E7 Activity](#). [P15 was influenced by \(influenced\)](#): [E1 CRM Entity](#)

Superproperty of: [E7 Activity](#). [P33 used specific technique \(was used by\)](#): [E29 Design or Procedure](#)
[E15 Identifier Assignment](#). [P142 used constituent \(was used in\)](#): [E41 Appellation](#)

Quantification: many to many (0,n:0,n)

Scope note: This property describes the use of material or immaterial things in a way essential to the performance or the outcome of an E7 Activity.

This property typically applies to tools, instruments, moulds, raw materials and items embedded in a product. It implies that the presence of the object in question was a necessary condition for the action. For example, the activity of writing this text required the use of a computer. An immaterial thing can be used if at least one of its carriers is present. For example, the software tools on a computer.

Another example is the use of a particular name by a particular group of people over some span to identify a thing, such as a settlement. In this case, the physical carriers of this name are at least the people understanding its use.

Examples:

- the writing of this scope note (E7) *used specific object* Nicholas Crofts' computer (E22) *mode of use* Typing Tool; Storage Medium (E55)
- the people of Iraq calling the place identified by TGN '7017998' (E7) *used specific object* "Quyunjig" (E44) *mode of use* Current; Vernacular (E55)

Properties: P16.1 mode of use: [E55 Type](#)

P31 has modified (was modified by)

Domain: [E11 Modification](#)

Range: [E24 Physical Man-Made Thing](#)

Subproperty of: [E5 Event](#). [P12 occurred in the presence of \(was present at\)](#): [E77 Persistent Item](#)

Superproperty of: [E12 Production](#). [P108 has produced \(was produced by\)](#): [E24 Physical Man-Made Thing](#)

E79 Part Addition. P110 augmented (was augmented by): E24 Physical Man-Made Thing

E80 Part Removal. P112 diminished (was diminished by): E24 Physical Man-Made Thing

Quantification: many to many, necessary (1,n:0,n)

Scope note: This property identifies the E24 Physical Man-Made Thing modified in an E11 Modification.

If a modification is applied to a non-man-made object, it is regarded as an E22 Man-Made Object from that time onwards.

Examples:

- rebuilding of the Reichstag (E11) *has modified* the Reichstag in Berlin (E24)

P33 used specific technique (was used by)

Domain: [E7 Activity](#)

Range: [E29 Design or Procedure](#)

Subproperty of: [E7 Activity](#). [P16 used specific object \(was used for\)](#): [E70 Thing](#)

Quantification: many to many (0,n:0,n)

Scope note: This property identifies a specific E29 Design or Procedure in order to carry out an instance of

E7 Activity or parts of it.

The property differs from *P32 used general technique (was technique of)* in that the E29 Design or Procedure referred to is specific and documented rather than simply being a term in the E55 Type hierarchy. Typical examples would include intervention plans for conservation.

Typical examples would include intervention plans for conservation or the construction plans of a building.

Examples:

- Ornamentation of silver cup 232 (E11) *used specific technique* 'Instructions for golden chase work by A N Other' (E29)
- Rebuilding of Reichstag (E11) *used specific technique* Architectural plans by Foster and Partners (E29)

P37 assigned (was assigned by)

Domain: [E15](#) Identifier Assignment

Range: [E42](#) Identifier

Subproperty of: [E13](#) Attribute Assignment. [P141](#) assigned (was assigned by): [E1](#) CRM Entity

Quantification: many to many (0,n:0,n)

Scope note: This property records the identifier that was assigned to an item in an Identifier Assignment activity.

The same identifier may be assigned on more than one occasion.

An Identifier might be created prior to an assignment.

Examples:

- 01 June 1997 Identifier Assignment of the silver cup donated by Martin Doerr (E15) *assigned* "232" (E42)

P43 has dimension (is dimension of)

Domain: [E70](#) Thing

Range: [E54](#) Dimension

Quantification: one to many, dependent (0,n:1,1)

Scope note: This property records a E54 Dimension of some E70 Thing.

It is a shortcut of the more fully developed path from E70 Thing through *P39 measured (was measured by)*, E16 Measurement *P40 observed dimension (was observed in)* to E54 Dimension. It offers no information about how and when an E54 Dimension was established, nor by whom.

An instance of E54 Dimension is specific to an instance of E70 Thing.

Examples:

- silver cup 232 (E22) *has dimension* height of silver cup 232 (E54) *has unit* mm (E58), *has value* 224 (E60)

P44 has condition (is condition of)

Domain: [E18](#) Physical Thing

Range: [E3](#) Condition State

Quantification: one to many, dependent (0, n: 1,1)

Scope note: This property records an E3 Condition State for some E18 Physical Thing.

It is a shortcut of the more fully developed path from E18 Physical Thing through *P34 concerned (was assessed by)*, E14 Condition Assessment *P35 has identified (identified by)* to E3 Condition State. It offers no information about how and when the E3 Condition State was established, nor by whom.

An instance of Condition State is specific to an instance of Physical Thing.

Examples:

- silver cup 232 (E22) *has condition* oxidation traces were present in 1997 (E3) *has type* oxidation traces (E55)

P45 consists of (is incorporated in)

Domain: [E18](#) Physical Thing

Range: [E57](#) Material

Quantification: many to many, necessary (1,n:0,n)

Scope note: This property identifies the instances of E57 Materials of which an instance of E18 Physical Thing is composed.

All physical things consist of physical materials. *P45 consists of (is incorporated in)* allows the different Materials to be recorded. *P45 consists of (is incorporated in)* refers here to observed Material as opposed to the consumed raw material.

A Material, such as a theoretical alloy, may not have any physical instances.

Examples:

- silver cup 232 (E22) *consists of* silver (E57)

P46 is composed of (forms part of)

Domain: [E18](#) Physical Thing

Range: [E18](#) Physical Thing

Superproperty of: E19 Physical Object. P56 bears feature (is found on): E26 Physical Feature

Quantification: many to many (0,n:0,n)

Scope note: This property allows instances of E18 Physical Thing to be analysed into component elements.

Component elements, since they are themselves instances of E18 Physical Thing, may be further analysed into sub-components, thereby creating a hierarchy of part decomposition. An instance of E18 Physical Thing may be shared between multiple wholes, for example two buildings may share a common wall.

This property is intended to describe specific components that are individually documented, rather than general aspects. Overall descriptions of the structure of an instance of E18 Physical Thing are captured by the *P3 has note* property.

The instances of E57 Materials of which an item of E18 Physical Thing is composed should be documented using *P45 consists of (is incorporated in)*.

Examples:

- the Royal carriage (E22) *forms part of* the Royal train (E22)
- the “Hog’s Back” (E24) *forms part of* the “Fosseway” (E24)

P49 has former or current keeper (is former or current keeper of)

Domain: [E18](#) Physical Thing

Range: [E39](#) Actor

Superproperty of: [E18](#) Physical Thing. [P50](#) has current keeper (is current keeper of): [E39](#) Actor

Quantification: many to many (0,n:0,n)

Scope note: This property identifies the E39 Actor or Actors who have or have had custody of an instance of E18 Physical Thing at some time.

The distinction with *P50 has current keeper (is current keeper of)* is that *P49 has former or current keeper (is former or current keeper of)* leaves open the question as to whether the specified keepers are current.

P49 has former or current keeper (is former or current keeper of) is a shortcut for the more detailed path from E18 Physical Thing through *P30 transferred custody of (custody transferred through)*, E10 Transfer of Custody, *P28 custody surrendered by (surrendered custody through)* or *P29 custody received by (received custody through)* to E39 Actor.

Examples:

- paintings from The Iveagh Bequest (E18) *has former or current keeper* Secure Deliveries Inc. (E40)

P50 has current keeper (is current keeper of)

Domain: [E18](#) Physical Thing

Range: [E39](#) Actor

Subproperty of: [E18](#) Physical Thing. [P49](#) has former or current keeper (is former or current keeper of): [E39](#) Actor

Quantification: many to many (0,n:0,n)

Scope note: This property identifies the E39 Actor or Actors who had custody of an instance of E18 Physical Thing at the time this property was recorded.

P50 has current keeper (is current keeper of) is a shortcut for the more detailed path from E18 Physical Thing through *P30 transferred custody of (custody transferred through)*, E10 Transfer of Custody, *P29 custody received by (received custody through)* to E39 Actor.

Examples:

- paintings from The Iveagh Bequest (E18) *has current keeper* The National Gallery (E40)

P51 has former or current owner (is former or current owner of)

Domain: [E18](#) Physical Thing

Range: [E39](#) Actor

Superproperty of: E18 Physical Thing. P52 has current owner (is current owner of): E39 Actor

Quantification: many to many (0,n:0,n)

Scope note: This property identifies the E39 Actor that is or has been the legal owner (i.e. title holder) of an instance of E18 Physical Thing at some time.

The distinction with *P52 has current owner (is current owner of)* is that *P51 has former or current owner (is former or current owner of)* does not indicate whether the specified owners are current. *P51 has former or current owner (is former or current owner of)* is a shortcut for the

more detailed path from E18 Physical Thing through *P24 transferred title of (changed ownership through)*, E8 Acquisition, *P23 transferred title from (surrendered title through)*, or *P22 transferred title to (acquired title through)* to E39 Actor.

Examples:

- paintings from the Iveagh Bequest (E18) *has former or current owner* Lord Iveagh (E21)

P57 has number of parts

Domain: [E19](#) Physical Object

Range: [E60](#) Number

Quantification: many to one (0,1:0,n)

Scope note: This property documents the E60 Number of parts of which an instance of E19 Physical Object is composed.

This may be used as a method of checking inventory counts with regard to aggregate or collective objects. What constitutes a part or component depends on the context and requirements of the documentation. Normally, the parts documented in this way would not be considered as worthy of individual attention.

For a more complete description, objects may be decomposed into their components and constituents using *P46 is composed of (forms parts of)* and *P45 consists of (is incorporated in)*. This allows each element to be described individually.

Examples:

- chess set 233 (E22) *has number of parts* 33 (E60)

P59 has section (is located on or within)

Domain: [E18](#) Physical Thing

Range: [E53](#) Place

Quantification: one to many (0,n:0,1)

Scope note: This property links an area to the instance of E18 Physical Thing upon which it is found.

It is typically used when a named E46 Section Definition is not appropriate.

E18 Physical Thing may be subdivided into arbitrary regions.

P59 has section (is located on or within) is a shortcut. If the E53 Place is identified by a Section Definition, a more detailed representation can make use of the fully developed (i.e. indirect) path from E18 Physical Thing through *P58 has section definition (defines section)*, E46 Section Definition, *P87 is identified by (identifies)* to E53 Place. A Place can only be located on or within one Physical Object.

Examples:

- HMS Victory (E22) *has section* HMS Victory section B347.6 (E53)

P65 shows visual item (is shown by)

Domain: [E24](#) Physical Man-Made Thing

Range: [E36](#) Visual Item

Subproperty of: [E24](#) Physical Man-Made Thing. [P128](#) carries (is carried by): [E73](#) Information Object

Quantification: many to many (0,n:0,n)

Scope note: This property documents an E36 Visual Item shown by an instance of E24 Physical Man-Made Thing.

This property is similar to *P62 depicts (is depicted by)* in that it associates an item of E24 Physical Man-Made Thing with a visual representation. However, *P65 shows visual item (is shown by)* differs from the *P62 depicts (is depicted by)* property in that it makes no claims about what the E36 Visual Item is deemed to represent. E36 Visual Item identifies a recognisable image or visual symbol, regardless of what this image may or may not represent.

For example, all recent British coins bear a portrait of Queen Elizabeth II, a fact that is correctly documented using *P62 depicts (is depicted by)*. Different portraits have been used at different periods, however. *P65 shows visual item (is shown by)* can be used to refer to a particular portrait.

P65 shows visual item (is shown by) may also be used for Visual Items such as signs, marks and symbols, for example the 'Maltese Cross' or the 'copyright symbol' that have no particular representational content.

This property is part of the fully developed path from E24 Physical Man-Made Thing through *P65 shows visual item (is shown by)*, E36 Visual Item, *P138 represents (has representation)* to E1 CRM Entity which is shortcut by, *P62 depicts (is depicted by)*.

Examples:

- My T-Shirt (E22) *shows visual item* Mona Lisa (E38)

P67 refers to (is referred to by)

Domain: [E89](#) Propositional Object

Range: [E1](#) CRM Entity

Superproperty of: [E31](#) Document. [P70](#) documents (is documented in): [E1](#) CRM Entity
[E32](#) Authority Document. [P71](#) lists (is listed in): [E1](#) CRM Entity
[E89](#) Propositional Object. [P129](#) is about (is subject of): [E1](#) CRM Entity
[E36](#) Visual Item. [P138](#) represents (has representation): [E1](#) CRM Entity
[E29](#) Design or Procedure. [P68](#) foresees use of (use foreseen by): [E57](#) Material

Quantification: many to many (0,n:0,n)

Scope note: This property documents that an E89 Propositional Object makes a statement about an instance of E1 CRM Entity. *P67 refers to (is referred to by)* has the *P67.1 has type* link to an instance of E55 Type. This is intended to allow a more detailed description of the type of reference. This differs from *P129 is about (is subject of)*, which describes the primary subject or subjects of the E89 Propositional Object.

Examples:

the eBay auction listing of 4 July 2002 (E73) *refers to* silver cup 232 (E22) *has type* item for sale (E55)

Properties: *P67.1* has type: [E55](#) Type

P69 has association with (is associated with)

Domain: [E29](#) Design or Procedure

Range: [E29](#) Design or Procedure

Quantification: many to many (0,n:0,n)

Scope note: This symmetric property describes the association of an E29 Design or Procedure with other Designs or Procedures.

Any instance of E29 Design or Procedure may be associated with other designs or procedures.

The *P69.1 has type* property of *P69 is associated with* allows the nature of the association to be specified; examples of types of association between instances of E29 Design or Procedure include: whole-part, sequence, prerequisite, etc

Examples:

- procedure for glass blowing (E29) *has association with* procedure for glass heating (E29)

Properties: P69.1 has type: [E55](#) Type

P71 lists (is listed in)

Domain: [E32](#) Authority Document

Range: [E1](#) CRM Entity

Subproperty of: [E89](#) Propositional Object. [P67](#) refers to (is referred to by): [E1](#) CRM Entity

Quantification: many to many (0,n:0,n)

Scope note: This property documents a source E32 Authority Document for an instance of an E1 CRM Entity.

Examples:

- the Art & Architecture Thesaurus (E32) *lists* alcazars (E55)

P72 has language (is language of)

Domain: [E33](#) Linguistic Object

Range: [E56](#) Language

Quantification: many to many, necessary (0,n:0,n)

Scope note: This property describes the E56 Language of an E33 Linguistic Object.

Linguistic Objects are composed in one or more human Languages. This property allows these languages to be documented.

Examples:

- the American Declaration of Independence (E33) *has language* 18th Century English (E56)

P74 has current or former residence (is current or former residence of)

Domain: [E39](#) Actor

Range: [E53](#) Place

Quantification: many to many (0,n:0,n)

Scope note: This property describes the current or former E53 Place of residence of an E39 Actor.

The residence may be either the Place where the Actor resides, or a legally registered address of any kind.

Examples:

- Queen Elizabeth II (E39) *has current or former residence* Buckingham Palace (E53)

P75 possesses (is possessed by)

Domain: [E39](#) Actor

Range: [E30](#) Right

Quantification: many to many (0,n:0,n)

Scope note: This property identifies former or current instances of E30 Rights held by an E39 Actor.

Examples:

- Michael Jackson (E21) *possesses* Intellectual property rights on the Beatles' back catalogue (E30)

P78 is identified by (identifies)

Domain: [E52](#) Time-Span

Range: [E49](#) Time Appellation

Subproperty of: [E1](#) CRM Entity. [P1](#) is identified by (identifies): [E41](#) Appellation

Quantification: many to many (0,n:0,n)

Scope note: This property identifies an E52 Time-Span using an E49Time Appellation.

Examples:

- the time span 1926 to 1988 (E52) *is identified by* "Showa" (Japanese time appellation) (E49)

P82 at some time within

Domain: [E52](#) Time-Span

Range: [E61](#) Time Primitive

Quantification: many to one, necessary (1,1:0,n)

Scope note: This property describes the maximum period of time within which an E52 Time-Span falls.

Since Time-Spans may not have precisely known temporal extents, the CRM supports statements about the minimum and maximum temporal extents of Time-Spans. This property allows a Time-Span's maximum temporal extent (i.e. its outer boundary) to be assigned an E61 Time Primitive value. Time Primitives are treated by the CRM as application or system specific date intervals, and are not further analysed.

Examples:

- the time-span of the development of the CIDOC CRM (E52) *at some time within* 1992-infinity (E61)

P87 is identified by (identifies)

Domain: [E53](#) Place

Range: [E44](#) Place Appellation

Subproperty of: [E1](#) CRM Entity. [P1](#) is identified by (identifies): [E41](#) Appellation

Quantification: many to many (0,n:0,n)

Scope note: This property identifies an E53 Place using an E44 Place Appellation.

Examples of Place Appellations used to identify Places include instances of E48 Place Name, addresses, E47 Spatial Coordinates etc.

Examples:

- the location of the Duke of Wellington's House (E53) *is identified by* "No 1 London" (E45)

P94 has created (was created by)

Domain: [E65](#) Creation

Range: [E28](#) Conceptual Object

Subproperty of: E63 Beginning of Existence. P92 brought into existence (was brought into existence by): E77 Persistent Item

Superproperty of: E83 Type Creation. P135 created type (was created by): E55 Type

Quantification: one to many, necessary, dependent (1,n:1,1)

Scope note: This property allows a conceptual E65 Creation to be linked to the E28 Conceptual Object created by it.

It represents the act of conceiving the intellectual content of the E28 Conceptual Object. It does not represent the act of creating the first physical carrier of the E28 Conceptual Object. As an example, this is the composition of a poem, not its commitment to paper.

Examples:

- the composition of “The Four Friends” by A. A. Milne (E65) *has created* “The Four Friends” by A. A. Milne (E28)

P95 has formed (was formed by)

Domain: [E66](#) Formation

Range: [E74](#) Group

Subproperty of: E63 Beginning of Existence. P92 brought into existence (was brought into existence by): E77 Persistent Item

Quantification: one to many, necessary, dependent (1,n:1,1)

Scope note: This property links the founding or E66 Formation for an E74 Group with the Group itself.

Examples:

- the formation of the CIDOC CRM SIG at the August 2000 CIDOC Board meeting (E66) *has formed* the CIDOC CRM Special Interest Group (E74)

P98 brought into life (was born)

Domain: [E67](#) Birth

Range: [E21](#) Person

Subproperty of: E63 Beginning of Existence. P92 brought into existence (was brought into existence by): E77 Persistent Item

Quantification: one to many, dependent (0,n:1,1)

Scope note: This property links an E67 Birth event to an E21 Person in the role of offspring.

Twins, triplets etc. are brought into life by the same Birth event. This is not intended for use with general Natural History material, only people. There is no explicit method for modelling conception and gestation except by using extensions.

Examples:

- the Birth of Queen Elizabeth II (E67) *brought into life* Queen Elizabeth II (E21)

P100 was death of (died in)

Domain: [E69](#) Death

Range: [E21](#) Person

Subproperty of: E64 End of Existence. P93 took out of existence (was taken out of existence by): E77 Persistent Item

Quantification: one to many, necessary (1,n:0,n)

Scope note: This property links an E69 Death event to the E21 Person that died.

A Death event may involve multiple people, for example in the case of a battle or disaster.

This is not intended for use with general Natural History material, only people.

Examples:

- Mozart's death (E69) *was death of* Mozart (E21)

P102 has title (is title of)

Domain: [E71](#) Man-Made Thing

Range: [E35](#) Title

Subproperty of: [E1](#) CRM Entity. [P1](#) is identified by (identifies): [E41](#) Appellation

Quantification: many to many (0,n:0,n)

Scope note: This property describes the E35 Title applied to an instance of E71 Man-Made Thing. The E55 Type of Title is assigned in a sub property.

The *P102.1 has type* property of the *P102 has title (is title of)* property enables the relationship between the Title and the thing to be further clarified, for example, if the Title was a given Title, a supplied Title etc.

It allows any man-made material or immaterial thing to be given a Title. It is possible to imagine a Title being created without a specific object in mind.

Examples:

- the first book of the Old Testament (E33) *has title* "Genesis" (E35)
has type translated (E55)

Properties: P102.1 has type: [E55](#) Type

P103 was intended for (was intention of)

Domain: [E71](#) Man-Made Thing

Range: [E55](#) Type

Quantification: many to many (0,n:0,n)

Scope note: This property links an instance of E71 Man-Made Thing to an E55 Type of usage.

It creates a property between specific man-made things, both physical and immaterial, to Types of intended methods and techniques of use. Note: A link between specific man-made things and a specific use activity should be expressed using *P19 was intended use of (was made for)*.

Examples:

- this plate (E22) *was intended for* being destroyed at wedding reception (E55)

P104 is subject to (applies to)

Domain: [E72](#) Legal Object

Range: [E30](#) Right

Quantification: many to many (0,n:0,n)

Scope note: This property links a particular E72 Legal Object to the instances of E30 Right to which it is subject.

The Right is held by an E39 Actor as described by *P75 possesses (is possessed by)*.

Examples:

- Beatles back catalogue (E72) *is subject to* reproduction right on Beatles back catalogue (E30)

P105 right held by (has right on)

Domain: [E72](#) Legal Object

Range: [E39](#) Actor

Superproperty of: E18 Physical Thing. P52 has current owner (is current owner of): E39 Actor

Quantification: many to many (0,n:0,n)

Scope note: This property identifies the E39 Actor who holds the instances of E30 Right to an E72 Legal Object.

It is a superproperty of *P52 has current owner (is current owner of)* because ownership is a right that is held on the owned object.

P105 right held by (has right on) is a shortcut of the fully developed path from E72 Legal Object through *P104 is subject to (applies to)*, E30 Right, *P75 possesses (is possessed by)* to E39 Actor.

Examples:

- Beatles back catalogue (E73) *right held by* Michael Jackson (E21)

P106 is composed of (forms part of)

Domain: [E90](#) Symbolic Object

Range: [E90](#) Symbolic Object

Quantification: many to many (0,n:0,n)

Scope note: This property associates an instance of E90 Symbolic Object with a part of it that is by itself an instance of E90 Symbolic Object, such as fragments of texts or clippings from an image.

Examples:

- This Scope note (E33) *P106 is composed of* ‘fragments of texts’ (E33)
- ‘recognizable’ (E90) *P106 is composed of* ‘recognizable’ (E90)

P107 has current or former member (is current or former member of)

Domain: [E74](#) Group

Range: [E39](#) Actor

Quantification: many to many (0,n:0,n)

Scope note: This property relates an E39 Actor to the E74 Group of which that E39 Actor is a member.

Groups, Legal Bodies and Persons, may all be members of Groups. A Group necessarily consists of more than one member.

This property is a shortcut of the more fully developed path from E74 Group through P144 joined with (gained member by), E85 Joining, P143 joined (was joined by) to E39 Actor. The property P107.1 *kind of member* can be used to specify the type of membership or the role the member has in the group.

Examples:

- Moholy Nagy (E21) *is current or former member of* Bauhaus (E74)
- National Museum of Science and Industry (E40) *has current or former member* The National Railway Museum (E40)
- The married couple Queen Elisabeth and Prince Phillip (E74) *has current or former member* Prince Phillip (E21) with P107.1 *kind of member* husband (E55 Type)

Properties: P107.1 *kind of member*: [E55](#) Type

P108 has produced (was produced by)

Domain: [E12](#) Production

Range: [E24](#) Physical Man-Made Thing

Subproperty of: [E11](#) Modification. [P31](#) has modified (was modified by): [E24](#) Physical Man-Made Thing

E63 Beginning of Existence. P92 brought into existence (was brought into existence by): E77 Persistent Item

Quantification: one to many, necessary, dependent (1,n:1,1)

Scope note: This property identifies the E24 Physical Man-Made Thing that came into existence as a result of an E12 Production.

The identity of an instance of E24 Physical Man-Made Thing is not defined by its matter, but by its existence as a subject of documentation. An E12 Production can result in the creation of multiple instances of E24 Physical Man-Made Thing.

Examples:

- The building of Rome (E12) *has produced* The Colosseum (E22)

P125 used object of type (was type of object used in)

Domain: [E7](#) Activity

Range: [E55](#) Type

Subproperty of:

Superproperty of: E7 Activity. P32 used general technique (was technique of): E55 Type

Quantification: many to many (0,n:0,n)

Scope note: This property defines the kind of objects used in an E7 Activity, when the specific instance is either unknown or not of interest, such as use of "a hammer".

Examples:

- at the Battle of Agincourt (E7), the English archers *used object of type* long bow (E55)

P127 has broader term (has narrower term)

Domain: [E55](#) Type

Range: [E55](#) Type

Quantification: many to many (0,n:0,n)

Scope note: This property identifies a super-Type to which an E55 Type is related.

It allows Types to be organised into hierarchies. This is the sense of "broader term generic (BTG)" as defined in ISO 2788.

Examples:

- *dime (E55) has broader term coin (E55)*

P128 carries (is carried by)

Domain: [E24](#) Physical Man-Made Thing

Range: [E90](#) Symbolic Object

Subproperty of: **E70 Thing. P130 shows features of (features are also found on): E70 Thing**

Superproperty of: [E24](#) Physical Man-Made Thing. [P65](#) shows visual item (is shown by): [E36](#) Visual Item

Quantification: many to many (0,n:0,n)

Scope note: This property identifies an E90 Symbolic Object carried by an instance of E24 Physical Man-Made Thing.

In general this would be an E84 Information Carrier. *P65 shows visual item (is shown by)* is a specialisation of *P128 carries (is carried by)* which should be used for carrying visual items.

Examples:

- Matthew's paperback copy of Reach for the Sky (E84) *carries* the text of Reach for the Sky (E73)

P129 is about (is subject of)

Domain: [E89](#) Propositional Object

Range: [E1](#) CRM Entity

Subproperty: [E89](#) Propositional Object. [P67](#) refers to (is referred to by): [E1](#) CRM Entity

Quantification: many to many (0,n:0,n)

Scope note: This property documents that an E89 Propositional Object has as subject an instance of E1 CRM Entity.

This differs from *P67 refers to (is referred to by)*, which refers to an E1 CRM Entity, in that it describes the primary subject or subjects of an E89 Propositional Object.

Examples:

- The text entitled 'Reach for the sky' (E33) *is about* Douglas Bader (E21)

P130 shows features of (features are also found on)

Domain: [E70](#) Thing

Range: [E70](#) Thing

Superproperty: E33 Linguistic Object. P73 has translation (is translation of): E33 Linguistic Object

Quantification: many to many (0,n:0,n)

Scope note: This property generalises the notions of "copy of" and "similar to" into a dynamic, asymmetric relationship, where the domain expresses the derivative, if such a direction can be established.

Otherwise, the relationship is symmetric. It is a short-cut of *P15 was influenced by (influenced)* in a creation or production, if such a reason for the similarity can be verified. Moreover it expresses similarity in cases that can be stated between two objects only, without historical knowledge about its reasons.

Examples:

- the Parthenon Frieze on the Acropolis in Athens (E22) *shows features of* the Original Parthenon Frieze in the British museum (E22). *Kind of similarity:* Copy (E55)

Properties: P130.1 kind of similarity: [E55](#) Type

P131 is identified by (identifies)

Domain: [E39](#) Actor

Range: [E82](#) Actor Appellation

Subproperty: [E1](#) CRM Entity. [P1](#) is identified by (identifies): [E41](#) Appellation

Quantification: many to many (0,n:0,n)

Scope note: This property identifies a name used specifically to identify an E39 Actor.

This property is a specialisation of *P1 is identified by (identifies)* is identified by.

Examples:

- Tyler Withersopp IV (E39) *is identified by* US social security number “619-17-4204” (E82)

P138 represents (has representation)

Domain: [E36](#) Visual Item

Range: [E1](#) CRM Entity

Subproperty: [E89](#) Propositional Object. [P67](#) refers to (is referred to by): [E1](#) CRM Entity

Quantification: many to many (0,n:0,n)

Scope note: This property establishes the relationship between an E36 Visual Item and the entity that it visually represents.

Any entity may be represented visually. This property is part of the fully developed path from E24 Physical Man-Made Thing through *P65 shows visual item (is shown by)*, E36 Visual Item, *P138 represents (has representation)* to E1 CRM Entity, which is shortcut by *P62 depicts (is depicted by)*. P138.1 mode of representation allows the nature of the representation to be refined.

Examples:

- the design on the reverse of a Swiss coin (E36) *represents* Helvetia (E28) *mode of representation* Profile (E55)

Properties: P138.1 mode of representation: [E55](#) Type

P140 assigned attribute to (was attributed by)

Domain: [E13](#) Attribute Assignment

Range: [E1](#) CRM Entity

Superproperty of: [E14](#) Condition Assessment. P34 concerned (was assessed by): [E18](#) Physical Thing

[E16](#) Measurement. P39 measured (was measured by): [E70](#) Thing

[E17](#) Type Assignment. P41 classified (was classified by): [E1](#) CRM Entity

Quantification: many to many (0,n:0,n)

Scope note: This property indicates the item to which an attribute or relation is assigned.

Examples:

- February 1997 Current Ownership Assessment of Martin Doerr's silver cup ([E13](#)) *assigned attribute to* Martin Doerr's silver cup ([E19](#))
- 01 June 1997 Identifier Assignment of the silver cup donated by Martin Doerr ([E15](#)) *assigned attribute to* silver cup 232 ([E19](#))

P141 assigned (was assigned by)

Domain: [E13](#) Attribute Assignment

Range: [E1](#) CRM Entity

Superproperty of: [E14](#) Condition Assessment. P35 has identified (identified by): [E3](#) Condition State

[E15](#) Identifier Assignment. [P37](#) assigned (was assigned by): [E42](#) Identifier

[E15](#) Identifier Assignment. P38 deassigned (was deassigned by): [E42](#) Identifier

[E16](#) Measurement. P40 observed dimension (was observed in): [E54](#) Dimension

[E17](#) Type Assignment. P42 assigned (was assigned by): [E55](#) Type

Quantification: many to many (0,n:0,n)

Scope note: This property indicates the attribute that was assigned or the item that was related to the item denoted by a property P140 assigned attribute to in an Attribute assignment action.

Examples:

- February 1997 Current Ownership Assessment of Martin Doerr's silver cup ([E13](#)) *assigned* Martin Doerr ([E21](#))
- 01 June 1997 Identifier Assignment of the silver cup donated by Martin Doerr ([E15](#)) *assigned* object identifier "232" ([E42](#))

P142 used constituent (was used in)

Domain: [E15](#) Identifier Assignment

Range: [E90](#) Symbolic Object

Subproperty of: [E7](#) Activity. [P16](#) used specific object (was used for): [E70](#) Thing

Quantification: (0,n:0,n)

Scope note: This property associates the event of assigning an instance of [E42](#) Identifier with the instances of [E90](#) Symbolic Object that were used as constituents of the identifier.

Examples:

- On June 1, 2001 assigning the personal name identifier “Guillaume, de Machaut, ca. 1300-1377” (E15) *used constituent* “ca. 1300-1377” (E49)
- Assigning a uniform title to the anonymous textual work known as ‘The Adoration of the Shepherds’(E15) *used constituent* ‘Coventry’ (E48)
- Assigning a uniform title to Pina Bausch’s choreographic work entitled ‘Rite of spring’ (E15) *used constituent* ‘(Choreographic Work: Bausch)’(E90)
- Assigning a uniform title to the motion picture directed in 1933 by Merian C. Cooper and Ernest B. Schoedsack and entitled ‘King Kong’ (E15) *used constituent* ‘1933’ (E50)
- Assigning the corporate name identifier ‘Univerza v Ljubljani. Oddelek za bibliotekarstvo’ to The Department for library science of the University of Ljubljana (E15) *used constituent* ‘Univerza v Ljubljani’ (E42)

P148 has component (is component of)

Domain: [E89](#) Propositional Object

Range: [E89](#) Propositional Object

Superproperty of:

Quantification: (0,n;0,n)

Scope note: This property associates an instance of E89 Propositional Object with a structural part of it that is by itself an instance of E89 Propositional Object.

Examples:

- Dante’s “Divine Comedy” (E89) *has component* Dante’s “Hell” (E89)

P151 was formed from (participated in)

Domain: [E66](#) Formation

Range: [E74](#) Group

Subproperty of: E5 Event. P11 had participant (participated in): E39 Actor

Quantification: (0,n;0:n)

Scope note: This property associates an instance of E66 Formation with an instance of E74 Group from which the new group was formed preserving a sense of continuity such as in mission, membership or tradition.

Examples:

- The formation of the House of Bourbon-Conti in 1581 (E66) *was formed from* House of Condé (E74)

P165 incorporates (is incorporated in)

Domain: [E73](#) Information Object

Range: [E90](#) Symbolic Object

Subproperty of: [E90](#) Symbolic Object. [P106](#) is composed of (forms part of): [E90](#) Symbolic Object

Quantification: (0,n;0,n)

Scope note: This property associates an instance of E73 Information Object with an instance of E90 Symbolic Object (or any of its subclasses) that was included in it.

This property makes it possible to recognise the autonomous status of the incorporated signs, which were created in a distinct context, and can be incorporated in many distinct self-contained expressions, and to highlight the difference between structural and accidental whole-part relationships between conceptual entities.

It accounts for many cultural facts that are quite frequent and significant: the inclusion of a poem in an anthology, the re-use of an operatic aria in a new opera, the use of a reproduction of a painting for a book cover or a CD booklet, the integration of textual quotations, the presence of lyrics in a song that sets those lyrics to music, the presence of the text of a play in a movie based on that play, etc.

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6. Amendments

6.1. Amendments to Version 1.0.1

In the 21th Joint meeting of the CIDOC CRM SIG and ISO/TC46/SC4/WG9 and 15th FRBR – CIDOC CRM Harmonization meeting, 27th – 30th of January 2010, National Gallery, Helsinki, the following has been decided:

F2 Expression

The forth example has been changed for better understanding

FROM

Christian Morgenstern's 'Fisches Nachtgesang' [a poem consisting simply of “-” and “~” signs, arranged in a determined combination] (F22)

TO

The signs which make up Christian Morgenstern's 'Fisches Nachtgesang' Christian Morgenstern's 'Fisches Nachtgesang' [a poem consisting simply of “-” and “~” signs, arranged in a determined combination] (F22)

F10 Person

The following examples have been added to F10.

Examples: Margaret Atwood
 Hans Christian Andersen
 Queen Victoria

F21 Recording Work

There was a problem with F21 declared as a subclass of F16 Container Work because individual instances of F21 may or may not be related to expressions of other works. The SIG decided that F21 isn't subclass of F16. F21 is subclass of F1. In the case where the recorded perdurant expresses some Work, the respective instance of F21 is also an F16 Container Work.

The following updates have been made:

- a. Figure 6, 7, 10 has been updated
- b. Page 9: the second bullet has been changed from
 - Works that consist in establishing all the features of recordings of sounds and/or images (either natural or involving human activity). This is modelled as: F21 Recording Work *is a* F16 Container Work, F21 Recording Work *R13 is realised in (realises)* F26 Recording, and F26 Recording *is a* F22 Self-Contained Expression.
- To:
 - Works that consist in establishing all the features of recordings of sounds and/or images (either natural or involving human activity). This is modelled as: F21 Recording Work *is a* F1 Work, F21 Recording Work *R13 is realised in (realises)* F26 Recording, and F26 Recording *is a* F22 Self-Contained Expression.
- c. Page 27: 2.5.1, the table was updated
- d. Page 29: 2.5.2, the table was updated
- e. Page 36: F1 Work, the superclass of part has been updated
- f. Page 46,47: F21 Recording Work, the subclass of part has been updated. The scope note has been updated from :

This class comprises works that conceptualise the capturing of features of perdurants. The characteristics of the manifestation of a recording work are those of the product of the capture process. The characteristics of any other works recorded are distinct from those of the recording work itself.

To:

This class comprises works that conceptualise the capturing of features of perdurants. The characteristics of the manifestation of a recording work are those of the product of the capture

process. The characteristics of any other works recorded are distinct from those of the recording work itself. In the case where the recorded perdurant expresses some Work, the respective instance of F21 is also an F16 Container Work

F23 Expression Fragment

The following examples (third and the fourth example) have been deleted because they are controversial.

The notes G-G-G-Eflat (opening of the 1st movement of Ludwig van Beethoven's 5th symphony) performed by an orchestra, recorded, and broadcast by the BBC during World War II (the rhythm of this musical fragment corresponds to the Morse code for the initial V for Victory)

The graphic content of a digitised enlarged detail of Mona Lisa's left eye

F24 Publication Expression

The scope note of this entity has been reworded for better understanding

FROM

Scope note: This class comprises the complete layout and content provided by a publisher (in the broadest sense of the term) in a given publication and not just what was added by the publisher to the authors' expressions. It comprises the expressions of the authors' Works that constitute the *raison d'être* for the publication. Frequently, it also comprises illustrations selected by the publisher from different artists.

TO

Scope note: "This class comprises complete sets of signs present in publications, reflecting publishers' final decisions as to both content and layout of the publications."

F29 Recording Event

In the scope note of F26 Recording was not clear if the actual recordings are not just the result of instances of F29 Recording Event, but also of the whole process of post-production. Or is the notion of post-production implicitly included in the notion of F29 Recording Event? The SIG decided to add a sentence in the scope note of F29. The scope note of F29 Recording Event has been changed from :

This class comprises activities that intend to convey (and preserve) the content of events in a recording, such as a live recording of a performance, a documentary, or other capture of a perdurant. Such activities may follow the directions of a recording plan.

To:

This class comprises activities that intend to convey (and preserve) the content of events in a recording, such as a live recording of a performance, a documentary, or other capture of a perdurant. Such activities may follow the directions of a recording plan. They may include postproduction

R1 is logical successor of (has successor)

The example 1 is actually inaccurate, as the action of "H.—the story of Heathcliff" does not take place after the end of "Wuthering Heights", but right in the middle of it. The SIG decided to accept the proposal of PLB for replacing this example with the following.

FROM

The novel entitled 'H.--: the story of Heathcliff's journey back to Wuthering Heights', authored by the person named 'Lin Haire-Sargeant' (F1) *R1 is logical successor of* The novel entitled 'Wuthering Heights', authored by the person named 'Emily Brontë' (F1)

TO

Albrecht Dürer's woodcut from 'The Large Woodcut Passion' entitled 'The Agony in the Garden' (F1, conceived ca 1496-98) *R1 is logical successor of* Albrecht Dürer's woodcut from 'The Large Woodcut

Passion' entitled 'The Last Supper' (F1, dated 1510)

R4 carriers provided by (comprises carriers of)

This property is declared incorrectly as a subproperty of *R41 has representative manifestation product type (is representative manifestation product type for)*. It is superproperty of R41. Also the path of which R4 is a shortcut has been declared inversed and the corresponding superproperty of CRM has added. The following changes have been made.

FROM

Domain: [F2](#) Expression
Range: [F3](#) Manifestation Product Type
Superproperty of:
Subproperty of: [F2](#) Expression.[R41](#) has representative manifestation product type (is representative manifestation product type for): [F3](#) Manifestation Product Type
Quantification: (1:n,0,n)

Scope note: This property associates a publication, i.e. an instance of F3 Manifestation Product Type, with an instance of F2 Expression, which all exemplars of that publication should carry, as long as they are recognised as complete exemplars of that publication. Typically, this property is observed on one exemplar of a publication, and extrapolated to all other exemplars of the same publication. This property is a shortcut of: F3 Manifestation Product Type *CLR6 should carry* F24 Publication Expression *R14 incorporates* F2 Expression

TO:

Domain: [F2](#) Expression
Range: [F3](#) Manifestation Product Type
Superproperty of: [F2](#) Expression.[R41](#) has representative manifestation product type (is representative manifestation product type for): [F3](#) Manifestation Product Type
Subproperty of: E73 Information Object.P128B is carried by: E24 Physical Man-Made Thing.P2 has type:E55 Type
Quantification: (1:n,0,n)
Scope note: This property associates a publication, i.e. an instance of F3 Manifestation Product Type, with an instance of F2 Expression, which all exemplars of that publication should carry, as long as they are recognised as complete exemplars of that publication. Typically, this property is observed on one exemplar of a publication, and extrapolated to all other exemplars of the same publication. This property is a shortcut of: F2 Expression *R14B is incorporated in* F24 Publication Expression *CLR6B should be carried by* F3 Manifestation Product Type.

R8 consists of (forms part of)

The CRM property added to the superproperty list of R8. The change is.

FROM

Domain: [F13](#) Identifier
Range: [F12](#) Name
Superproperty of:
Subproperty of:

TO

Domain: [F13](#) Identifier
Range: [F12](#) Name
Superproperty of:
Subproperty of: E90 Symbolic Object.P106 is composed of (forms part of): E90 Symbolic Object

R10 has member (is member of)

The CRM property added to the superproperty list of R10. The change is.

FROM

Domain: [F15](#) Complex Work
Range: [F1](#) Work
Superproperty of:
Subproperty of:
Quantification: (2:n,0:n)

TO

Domain: [F15](#) Complex Work
Range: [F1](#) Work
Superproperty of:
Subproperty of: E89 Propositional Object.P148 has component (is component of): E89 Propositional Object
Quantification: (2:n,0:n)

R11 has issuing rule (is issuing rule of)

The CRM property added to the superproperty list of R11. The change is.

FROM

Domain: [F18](#) Serial Work
Range: [E29](#) Design or Procedure
Superproperty of:
Subproperty of:

TO

Domain: [F18](#) Serial Work
Range: [E29](#) Design or Procedure
Superproperty of:
Subproperty of: E70 Thing.P16B was used for:E7 Activity. P33 used specific technique: E29 Design or Procedure

R20 recorded (was recorded through)

The CRM superproperty has been added.

Subproperty of: E7 Activity.P15 was influenced by (influenced) E5 Event.P9B forms part of: E5 Event.P9 consists of: E5 Event

R26 produced things of type (was produced by)

The CRM superproperty has been added.

Subproperty of: E12 Production.P108 produced: E24 Physical Man-MadeThing. P2 has type: E55 Type

R41 has representative manifestation product type (is representative manifestation product type for)

The superproperty / subproperty part has been updated, following the changes to R4.

FROM

Domain: [F2](#) Expression
Range: [F3](#) Manifestation Product Type

Superproperty of: [F2](#) Expression. [R4](#) carriers provided by (comprises carriers of): [F3](#) Manifestation Product Type

Subproperty of:

TO

Domain: [F2](#) Expression

Range: [F3](#) Manifestation Product Type

Superproperty of:

Subproperty of: [F2](#) Expression. [R4](#) carriers provided by (comprises carriers of): [F3](#) Manifestation Product Type

Proofreading

Page 9: The list of the meetings was updated.

p. 30: F17 Aggregation Work was not in the right place in the class hierarchy, it had to swap lines with F21 Recording Work.

Page 36: property “R2.1 has type: E55 Type” of R2 property has been added.

p. 38: a portion of text that was repeated twice is deleted.

p. 49: the inverted commas around the text of the scope note of F24 Publication Expression are removed.

Page 57: R7, the label of the inverse property in the example has been deleted because we do not repeat the label for the reverse property in examples

General Notice 1. Updates to the following CRM referred Entities and properties has been made. These are

<i>Page no.</i>	<i>Entity / Property code</i>	<i>Page no.</i>	<i>Entity / Property code</i>
89	E1	111	P4
91	E4	112	P14
93	E12	114	P44
96	E29	117	P65
97	E35	124	P148
98	E41		
99	E44		
99	E47		
100	E49		
100	E50		
102	E55		
105	E70		
107	E82		

6.2. Amendments to Version 1.0.2

F40 Identifier Assignment

In the 17th FRBR-CIDOC CRM meeting joined with the 23rd meeting of the CIDOC CRM SIG and ISO/TC46/SC4/WG9, it is decided to change the scope note of F40 for being compatible with Controlled Access Points from FRAD. So the scope note has been changed from:

Scope note: This class comprises activities that result in the allocation of an identifier to any Entity. An Identifier Assignment may include the creation of the identifier from multiple constituents. The syntax and kinds of constituents to be used may be declared in a rule. It also includes the assignment of uniform titles.

Examples: Assigning the author-title heading ‘Goethe, Johann Wolfgang von, 1749-1832. Faust. 1. Theil.’ as a uniform title for a work

Assigning the title heading ‘Bible. English. American Standard’ as a uniform title for an

expression

To:

Scope note: This class comprises activities that result in the allocation of an identifier to any E1 CRM Entity. An F40 Identifier Assignment may include the creation of the identifier from multiple constituents. The syntax of the identifier and the kinds of constituents to be used in constructing it may be declared in a rule. The construction of controlled access points for the names of persons, families and corporate bodies following specific cataloguing rules is a typical library application. F40 Identifier Assignment also includes the assignment of uniform titles as controlled access points for works or expressions

Examples: Assigning the name heading 'William, Prince, Duke of Cambridge, 1982-' as a controlled access point for a personal name using the *Anglo-American Cataloguing Rules*, 2nd edition

Assigning the name heading 'Library and Archives Canada' as an authorised controlled access point for a corporate body name using the *Anglo-American Cataloguing Rules*, 2nd edition

Assigning the name heading 'Bibliothèque et Archives Canada' as an authorised controlled access point for a corporate body name using the *Règles de catalogage anglo-américaines*, 2e édition

Assigning the author-title heading 'Goethe, Johann Wolfgang von, 1749-1832. Faust. 1. Theil.' as a uniform title for a work

Assigning the title heading 'Bible. English. American Standard' as a uniform title for an expression

F43 Identifier Rule

In the 18th FRBR - CIDOC CRM Harmonization meeting joined with the 24th meeting of the CIDOC CRM SIG and ISO/TC46/SC4/WG9, it is decided that F43 Design or Procedure is a subclass of F2 Expression. This was one of the outcomes of the discussion about FRAD Rules. This decision produced the following changes in the document:

Page 129: The F43 Identifier Rule has been added to the superproperty section of F2

Page 131: The F2 is declared to Subproperty section of F43 Identifier Rule in the table of FRBROO Property Hierarchy aligned with (part of) CIDOC CRM Property Hierarchy

R14 incorporates (is incorporated in)

In the 18th FRBR - CIDOC CRM Harmonization meeting joined with the 24th meeting of the CIDOC CRM SIG and ISO/TC46/SC4/WG9, it is decided that R14 is also subproperty of *P106 is composed of (forms part of)*. This decision produced the following changes in the document:

Page 59: The P106 has been added to the Subproperty section of the definition of R14

Page 34: The R14 is declared to Subproperty of P106 in the table of FRBROO Property Hierarchy aligned with (part of) CIDOC CRM Property Hierarchy

This is related to the **issue 192**

R15 has fragment (is fragment of)

In the 18th FRBR - CIDOC CRM Harmonization meeting joined with the 24th meeting of the CIDOC CRM SIG and ISO/TC46/SC4/WG9, it is decided that R15 is not subproperty of *P148 has component (is component of)*.

R40 has representative expression (is representative expression for)

For consistency purposes the frbr-crm group decided by email to make the following updates to the range, subproperty and scope note of R40.

From:

Domain: F1 Work

Range: F22 Self-contained Expression

Superproperty of:

Subproperty of: F1 Work.R3 is realised in (realises): F22 Self-contained Expression

Quantification: (0:n,0:n)

Scope note: This property identifies an instance of F2 Expression that has been chosen as the most characteristic expression of the instance of F15 Complex Work of which it is an expression.

Typically, any expression that is not regarded as “representative” for the work it expresses, would require a uniform title, with qualifiers specifying the differences between that expression and a representative expression, although this may not always be done in practice. The title of a Work may not be one taken from a representative expression.

A given work can have more than one representative expression, provided the differences between these expressions are not deemed “substantial.” If the anticipated needs of users are not considered to call for bibliographic distinctions between variant expressions of a work, then even expressions that differ significantly from each other can be regarded as equally representative for the work. (See FRBR: Final Report, p. 19-20).

A given expression can be deemed representative for a work with regard to some of its aspects (e.g., the text contained in an edition the title proper of which reads “The tragicall historie of HAMLET Prince of DenmarkeA, and the language of that text), and not representative for it with regard to some other aspects (e.g., the title proper “The tragicall historie of HAMLET Prince of DenmarkeA itself, which, being different from the title that is regarded as “representative” for ShakespeareAs work, will require the use of a uniform title).

R40 has representative expression is a shortcut of the more developed path F1 Work R50B was assigned by F42 Representative Expression Assignment R51 assigned F2 Expression. "

To:

Domain: F1 Work

Range:F2 Expression

Superproperty of:

Subproperty of: E70 Thing. P130 shows features of (features are also found on): E70 Thing

Quantification: (0:n,0:n)

Scope note: This property identifies an instance of F2 Expression that has been chosen as the most characteristic expression of the instance of F1 Work of which it is an expression.

Typically, any expression that is not regarded as “representative” for the work it expresses, would require a uniform title, with qualifiers specifying the differences between that expression and a representative expression, although this may not always be done in practice. The title of a Work may not be one taken from a representative expression.

A given work can have more than one representative expression, provided the differences between these expressions are not deemed “substantial.” If the anticipated needs of users are not considered to call for bibliographic distinctions between variant expressions of a work, then even expressions that differ significantly from each other can be regarded as equally representative for the work. (See FRBR: Final Report, p. 19-20).

A given expression can be deemed representative for a work with regard to some of its aspects (e.g., the text contained in an edition the title proper of which reads “The tragicall historie of HAMLET Prince of DenmarkeA, and the language of that text), and not representative for it with regard to some other aspects (e.g., the title proper “The tragicall historie of HAMLET Prince of DenmarkeA itself, which, being different from the title that is regarded as “representative” for ShakespeareAs work, will require the use of a uniform title).

R40 has representative expression is a shortcut of the more developed path F1 Work R50B was assigned by F42 Representative Expression Assignment R51 assigned F2 Expression.

This change produced the following updates:

Page 127: Figure 12 has been updated

Page 133: update property hierarchy table

Page 134: update the definition of R40

R47 used constituent (was used in)

In the 18th FRBR-CIDOC CRM meeting joined with the 24rd meeting of the CIDOC CRM SIG and ISO/TC46/SC4/WG9, it is decided to change the range of R47 in order to cover situations where “whole and part” are applied in subject Authority Systems and the forms of names are not only represented in natural language but also are represented in artificial languages as codes, formulas, graphs etc as it is stated in FRISAD.

Thus the range of R47 was changed from F12 Name to E90 Symbolic Object. This change produces the following updates:

Page 126: Figure 11 has been updated

Page 130: update the range of R47 from F12 Name to E90 Symbolic Object in the property definition section of F40

Page 133: update the entry of R47 in Property Hierarchy Table

Proofreading

Page 9: The version number of CIDOC CRM from 5.0.1 became 5.0.4

Page 10: The word “draft” is deleted from the first sentence in the introduction.

Page 17: In the figure 4 the name of the property P128 has been corrected.

Page 19: In c. the code number of the relation R5 has been corrected from R15 to R5 and the figure 6 is updated.

Page 28: The class hierarchy of F16 Container work and F21 Recording Work has been corrected

Page 29: The FRBR_{OO} Class Hierarchy aligned with (part of) CIDOC CRM Class Hierarchy is updated

Page 32, 35 : In the line of R15, the domain and the range of the property has been corrected by interchanging them. This produced the following changes:

Page 19: In item b, the domain and the range of the property R15 has been corrected

Page 48: the declared property R15 in the definition of F23 has been deleted

Page 38: the R15 has been added to the properties of F2

Page 60: The domain and the range of the property R15 has been interchanged. In the section of the examples the direction of the property is corrected.

Page 127: Figure 12 has been updated

Page 34: The Property Hierarchy aligned with (part of) CIDOC CRM Property Hierarchy updated

Page 38: The underline has been removed from the text “The signs which make up” in the fourth example of F2

Page 42: the text “that was used to create a bibliographic record for that publication” from the second example of F9 has been dropped.

Page 126: The notation of R52 property has been corrected

Page 128: The class hierarchy of the FRBR_{OO} Identifier Creation Model Aligned with CIDOC CRM has been updated. The entries of F42, F43 have been corrected

Page 133: Minor corrections to the property hierarchy table

Page 144: Updates to the referred CIDOC CRM Entities

6.3. Amendments to Version 2.0 (18th FRBR – CIDOC CRM)

R14 incorporates (is incorporated in)

In the 18th FRBR - CIDOC CRM Harmonization meeting joined with the 24th meeting of the CIDOC CRM SIG and ISO/TC46/SC4/WG9, it is decided that R14 is also subproperty of *P106 is composed of (forms part of)*. This decision produced the following changes in the document:

Page 64: The P106 was been added to the Subproperty section of the definition of R14

Page 35: The R14 is declared to Subproperty of P106 in the table of FRBR_{OO} Property Hierarchy aligned with (part of) CIDOC CRM Property Hierarchy

This is related to the **issue 192**

R15 has fragment (is fragment of)

In the 18th FRBR - CIDOC CRM Harmonization meeting joined with the 24th meeting of the CIDOC CRM SIG and ISO/TC46/SC4/WG9, it is decided that R15 is not subproperty of *P148 has component (is component of)*.

The class F12 Name

During the harmonization process of FRSAD with CIDOC CRM and considering that FRSAD Nomen is more general than FRBR Name. and it is decided to change the name of the class from Name to Nomen and to be subclass of E41. This decision created the following changes:

Page 28: In the FRBR_{OO} Class Hierarchy table, name change

Page 30,31: In the FRBR_{OO} Class Hierarchy aligned with CIDOC CRM table, name-super class change

Page 32,44,61,62, name change

Page 78,79,80,86,90: In the List of mappings of FRBRER, name change

Page 145: in Chapter 7.3, name change

Page 154: In Property Hierarchy, name change

Page 160: In the example of 147, name change

F11 is subclass of E40

Proofreading

Page 10: The word “draft” is deleted from the first sentence in the introduction.

Page 19: In c. the code number of the relation R5 has been corrected from R15 to R5 and the figure 6 is updated.

Page 28: The class hierarchy of F16 Container work and F21 Recording Work has been corrected

Page 32, 35 : In the line of R15, the domain and the range of the property has been corrected by interchanging them. This produced the following changes:

Page 19: In item b, the domain and the range of the property R15 has been corrected

Page 49: the declared property R15 in the definition of F23 has been deleted

Page 38: the R15 has been added to the properties of F2

Page 64: The domain and the range of the property R15 has been corrected by interchanging them

Page 38: The underline has been removed from the text “The signs which make up” in the fourth example of F2

Page 42: the text “that was used to create a bibliographic record for that publication” from the second example of F9 has been dropped.

6.4 Amendments to Version 2.0 (20th FRBR – CIDOC CRM)

In the 20th FRBR - CIDOC CRM Harmonization meeting joined with the 26th meeting of the CIDOC CRM SIG and ISO/TC46/SC4/WG9, the following changes have been.

1 Introduction

The text is changed:

FROM:

This document is the definition of **FRBR**¹⁰ object-oriented version, harmonised with CIDOC CRM, hereafter referred to as **FRBR_{oo}**, a formal ontology intended to capture and represent the underlying semantics of bibliographic information and to facilitate the integration, mediation, and interchange of bibliographic and museum information. Such a common view is necessary to provide interoperable information systems for those users interested in accessing common or related content. Beyond that, it results in a formalisation which is more suited for the implementation of FRBR concepts with object-oriented tools, and which facilitates the testing and adoption of FRBR concepts in implementations with different functional specifications and different environments. It applies empirical analysis and ontological structure to the entities and processes associated with works, to their properties, and to the relationships among them. Thereby it reveals a web of interrelationships, which is also applicable to information objects in non-bibliographic arenas¹¹, and is useful to justify the need of information elements in different environments.

The FRBR model was originally designed as an entity-relationship model by a study group appointed by the International Federation of Library Associations and Institutions (IFLA) during the period 1991-1997, and was published in 1998. The original entity-relationship definition of FRBR is referred to hereafter as FRBR_{ER}.

Quite independently, the CIDOC CRM¹² model was being developed from 1996 under the auspices of the ICOM-CIDOC (International Council for Museums – International Committee on Documentation) Documentation Standards Working Group. The definition of the CIDOC CRM model has now become ISO standard 21127.

The idea that both the library and museum communities might benefit from harmonising the two models was first expressed in 2000, on the occasion of ELAG's (European Library Automation Group) 24th Library Systems Seminar in Paris, with Nicholas Crofts and Dan Matei drafting on the spot a preliminary object-oriented representation of the FRBR model entities roughly mapped to CIDOC CRM classes. This idea grew up in the following years and eventually led to the formation in 2003 of the International Working Group on FRBR/CIDOC CRM Harmonisation, that brings together representatives from both communities with the common goals of: a)

¹⁰ “FRBR” stands for: “Functional Requirements for Bibliographic Records,” after the name of the IFLA Study Group that developed the model. However, current use and understanding of the FRBR model go well beyond that, and the term “FRBR” has now turned to a noun in its own right, used without particular intention to refer to “functionalities,” nor to “requirements,” but rather to the *semantics* of bibliographic records. The *Final Report on Functional Requirements for Bibliographic Records* published in 1998 contained both a study on functional requirements for bibliographic records, and a description of the model known today as “FRBR.”

¹¹ Coleman, Anita S. 2002. Scientific models as works. *Cataloging & classification quarterly* 33n3/4: 129-59.

¹² “CIDOC CRM” stands for “Comité international de documentation [= International Committee on Documentation] Conceptual Reference Model,” which, when isolated from any context, is not particularly meaningful (CIDOC is affiliated to ICOM, the International Council of Museums). Just like FRBR, the acronym, rather meaningless by itself, has now turned to a noun in its own right.

Expressing the IFLA FRBR model with the concepts, tools, mechanisms, and notation conventions provided by the CIDOC CRM, and: b) Aligning (possibly even merging) the two object-oriented models thus obtained.

The International Working Group on FRBR/CIDOC CRM Harmonisation, chaired by Martin Doerr (ICS FORTH, Greece) and Patrick Le Bœuf (BnF, France), is affiliated at the same time to the IFLA FRBR Review Group and the CIDOC CRM Special Interest Group (CRM-SIG). The present definition of FRBR_{OO} was developed through email exchange among members of the Working Group, and more importantly during the following series of meetings:

- Meeting #1: 2003, Nov. 12-14, Paris;
- Meeting #2: 2004, March 22-25, Heraklion, Greece;
- Meeting #3: 2005, February 14-16, London;
- Meeting #4: 2005, July 4-6, Heraklion, Greece;
- Meeting #5: 2005, November 16-18, Nuremberg, Germany;
- Meeting #6: 2006, March 27-29, London;
- Meeting #7: 2006, June 26-29, Trondheim, Norway;
- Meeting #8: 2006, October 25-27, Heraklion, Greece;
- Meeting #9: 2007, March 14-16, Paris;
- Meeting #10: 2007, July 9-10, Edinburgh, Scotland;
- Meeting #11: 2007, December 4-7, Nuremberg, Germany;
- Meeting #12: 2008, May 12-15, Heraklion, Greece;
- Meeting #13: 2008, November 5-7, London;
- Meeting #14: 2009, May 20-22, London;
- Meeting #15: 2010, January 27, Helsinki;
- Meeting #16: 2010, December 20-22, Nuremberg;
- Meeting #17: 2011, May 17-20, Heraklion, Greece;
- Meeting #18: 2011, November 14-17, Amsterdam;
- Meeting #19: 2011, April 30-May 2, Heraklion, Greece.

More information on the activities of the Group can be found on http://archive.ifla.org/VII/s13/wgfrbr/FRBR-CRMDialogue_wg.htm and on http://cidoc.ics.forth.gr/frbr_inro.html.

We express our gratitude to the European funded Project DELOS NoE for providing financial help for several of these meetings.

TO:

This document is the definition of the object-oriented version of the **FRBR**¹³ family of conceptual models¹⁴, harmonised with CIDOC CRM, hereafter referred to as **FRBR_{OO}**, a formal ontology that captures and represents the underlying semantics of bibliographic information and therefore facilitates the integration, mediation, and interchange of bibliographic and museum information. Such a common view is necessary for the development of interoperable information systems serving users interested in accessing common or related content. Beyond that, it results in a formalisation which is more suited for the implementation of concepts from the FRBR family of conceptual models with object-oriented tools, and which facilitates the testing and adoption of these concepts in implementations with different functional specifications and beyond the library domain. It applies empirical analysis and ontological structure to the entities and processes associated with the bibliographic universe, to their properties, and to the relationships among them. It thereby reveals a web of interrelationships, which are also applicable to information objects in non-bibliographic arenas¹⁵.

The FRBR model was designed as an entity-relationship model by a study group appointed by the International Federation of Library Associations and Institutions (IFLA) during the period 1991-1997, it was approved by the IFLA Cataloguing Section in 1997, and was published in 1998. The original entity-relationship definition of FRBR is referred to hereafter as FRBR_{ER}.

¹³ “FRBR” stands for: “Functional Requirements for Bibliographic Records,” after the name of the IFLA Study Group that developed the model. However, current use and understanding of the FRBR model go well beyond that, and the term “FRBR” has now turned to a noun in its own right, used without particular intention to refer to “functionalities,” nor to “requirements,” but rather to the *semantics* of bibliographic records. The *Final Report on Functional Requirements for Bibliographic Records* published in 1998 contained both a study on functional requirements for bibliographic records, and a description of the model known today as “FRBR.”

¹⁴ In addition to FRBR itself, the FRBR family of conceptual models includes the *Functional Requirements for Authority Data* (FRAD), published in 2009, and the *Functional Requirements for Subject Authority Data* (FRSAD), published in 2011.

¹⁵ Coleman, Anita S. 2002. Scientific models as works. *Cataloging & classification quarterly* 33n3/4: 129-59.

Quite independently, the CIDOC CRM¹⁶ model was being developed, beginning in 1996, under the auspices of the ICOM-CIDOC (International Council for Museums – International Committee on Documentation) Documentation Standards Working Group. The definition of the CIDOC CRM model was adopted as ISO standard 21127.¹⁷

The idea that both the library and museum communities might benefit from harmonising their two models was first expressed in 2000, on the occasion of ELAG's (European Library Automation Group) 24th Library Systems Seminar in Paris. This idea led to the formation, in 2003, of the International Working Group on FRBR/CIDOC CRM Harmonisation, that brings together representatives from both communities with the common goals of: a) Expressing the IFLA FRBR model with the concepts, tools, mechanisms, and notation conventions provided by the CIDOC CRM, and: b) Aligning (possibly even merging) the two object-oriented models thus obtained.

The International Working Group on FRBR/CIDOC CRM Harmonisation, chaired by Martin Doerr (ICS FORTH, Greece) and Patrick Le Bœuf (BnF, France) [and Pat Riva?], is affiliated at the same time to the IFLA FRBR Review Group and the CIDOC CRM Special Interest Group (CRM-SIG). The present definition of FRBR_{OO} was developed through email exchange among members of the Working Group, and more importantly during a series of meetings.

Version 1.0 of FRBR_{OO} was finally approved and issued in January 2010; it covered the entities and concepts from FRBR and included an appendix on identifier creation. The focus of later meetings has been to extend the model to fully encompass the published versions of the FRAD and FRSAD models. Version 2.0 is the result of this expansion.

More information on the activities of the Group, minutes of the meetings and all previous versions can be found on http://archive.ifla.org/VII/s13/wgfrbr/FRBR-CRMdialogue_wg.htm and on http://cidoc.ics.forth.gr/frbr_inro.html.

1.1 Purposes

The text is changed:

FROM:

This model attempts to **represent FRBR** by modelling in a sufficiently consistent way the conceptualisation of the reality behind library practice, as it is apparent from or implicit in FRBR. It is important to keep in mind that the aim is not to transform the IFLA FRBR model into something totally different or better, nor of course to reject it or replace it – but to express the conceptualisation of FRBR with the object-oriented methodology instead of the entity-relationship methodology, as an alternative. Nor is it the intention to force museums' concerns and viewpoints into the bibliographic universe, or libraries' concerns and viewpoints into the museum universe. Rather, the point is to identify the common ground in the universe both sides share and to ensure mutual benefit by pursuing the following objectives.

TO:

This model represents FRBR, FRAD and FRSAD through modelling the conceptualisation of the reality behind library practice, as it is apparent from or implicit in the FRBR family of models. It is important to keep in mind that the aim is not to transform the IFLA models into something conceptually different, but to express the conceptualisation of the FRBR family within the object-oriented methodology instead of the entity-relationship methodology. Furthermore, the intention is to identify the common ground that memory institutions share and to exploit it by pursuing the following objectives

1.1.2 A verification of FRBR's internal consistency

The text is changed:

FROM:

¹⁶ “CIDOC CRM” stands for “Comité international de documentation [= International Committee on Documentation] Conceptual Reference Model,” which, when isolated from any context, is not particularly meaningful (CIDOC is affiliated to ICOM, the International Council of Museums). Just like FRBR, the acronym, rather meaningless in itself, has now turned into a noun in its own right.

¹⁷ *Information and documentation – a reference ontology for the interchange of cultural heritage information*. ISO 21127:2006. Geneva: ISO, 2006.

Expressing the FRBR models in a different formalism than the one in which it was originally developed provides a means to evaluate the model in terms of its internal consistency. It is also a good opportunity to correct some semantic inconsistencies or inaccuracies in the formulation of FRBR that may be regarded as negligible when FRBR_{ER} is only used in a library catalogue context, but that prove to be quite crucial from the moment one strives to design an overall model for the integration of cultural heritage related information.

TO:

Expressing the FRBR family in a different formalism than the one in which it was originally developed provides a means to evaluate the models in terms of their internal consistency. It is also a good opportunity to make adjustments to avoid some semantic inconsistencies and imprecisions in the formulation of the FRBR family, which prove to be crucial in the design of an overall model for the integration of cultural heritage related information. Additionally, the clarifications helped in the further development of the FRBR model itself, such as the interpretation of aggregates and aggregating work and understanding the dual nature of Manifestation.

1.1.3 An enablement of information interoperability and integration

The last sentence is changed:

FROM

Besides, CIDOC CRM is explicitly compatible in formalism with the World Wide Web Consortium's Resource Description Framework (RDF), which can only be beneficial for FRBR

TO

Besides, CIDOC CRM is explicitly compatible in formalism with the World Wide Web Consortium's Resource Description Framework (RDF), which is also beneficial for the IFLA models

1.1.5 An extension of the scope of FRBR and the CIDOC CRM

The text is changed in the last sentence:

FROM:

...Consequently, it also extends the scope of FRBR to cultural materials, since FRBR inherits all concepts of the CIDOC CRM, and opens the way for FRBR to benefit from further extensions of the scope of CIDOC CRM, such as the scientific heritage of observations and experiments.

TO:

...Consequently, it also extends the scope of the FRBR family of conceptual models to cultural materials, since FRBR_{OO} inherits all concepts of the CIDOC CRM, and opens the way for the IFLA models to benefit from further extensions of the scope of CIDOC CRM, such as the scientific heritage of observations and experiments.

1.1.7 Understanding the attributes and relationships

The text is changed

FROM:

... FRBR_{ER}. During its meetings, the International Working Group on FRBR/CIDOC CRM Harmonisation strove to extract their semantics as accurately as possible, to express them as "properties" in the sense of CIDOC CRM, and to compare them with possibly existing CIDOC CRM properties. Entities, or classes in the terminology adopted by the CIDOC CRM, play a nearly secondary role as the maximal sets of things for which a property is applicable.

TO:

... FRBR family. During its meetings, the International Working Group on FRBR/CIDOC CRM Harmonisation strove to extract the semantics as accurately as possible, to express them as "properties" in the sense of CIDOC CRM, and to relate them to CIDOC CRM properties where possible. Entities, or classes in the terminology adopted by the CIDOC CRM, play a nearly secondary role as the maximal sets of things for which a property is applicable.

1.1.8 Transforming attributes into properties

The text is changed:

FROM:

The CIDOC CRM model declares no “attributes” at all (except implicitly in its “scope notes” for classes), but regards any information element as a “property” (or “relationship”) between two classes. The semantics extracted from FRBR_{ER} attributes are therefore rendered in FRBR_{OO} as properties, according to the same principles as the CIDOC CRM model.

TO:

The CIDOC CRM model declares no “attributes” at all, but regards any information element as a “property” (or “relationship”) between two classes. The semantics extracted from FRBR_{ER}, FRAD and FRSAD attributes are therefore rendered in FRBR_{OO} as properties, according to the same principles as the CIDOC CRM model.

1.1.10 By-product 2: Adding a bibliographic flavour to CIDOC CRM

The text is changed

FROM:

The second by-product was that the analysis provided for bibliographic processes in FRBR_{OO} paved the way to the introduction of refinements into CIDOC CRM, so that the museum community’s model could give a better account for mass production phenomena (such as the printing of engravings, for instance), or the relation between creating immaterial content and physical carrier. Further, it introduces a basic model of intellectual conception and derivation applicable to all art forms, which the museum community has been hesitating so far to formally analyse. (MD)

TO:

The second by-product was that the analysis provided for bibliographic processes in FRBR_{OO} and for the processes of naming entities in FRAD and FRSAD, paved the way for the introduction of refinements in the CIDOC CRM. This enabled the museum community’s model to give a better account of mass production phenomena (such as the printing of engravings), the relation between creating immaterial content and physical carriers and the practices of identifying or naming things. Further, it introduces a basic model of intellectual conception and derivation applicable to all art forms, which the museum community had been hesitating to formally analyse.

1.2.1 Introduction of temporal entities, events and time processes

The text in the first two paragraphs is changed

FROM:

... Since FRBR_{OO} borrows structures from the CIDOC CRM to express the concepts declared in FRBR_{ER}, ‘temporal entities’ had inevitably to be introduced into FRBR_{OO}. Besides, some FRBR commentators had already made the point that time issues are insufficiently addressed in FRBR_{ER}¹⁸; the task of harmonising FRBR with the CIDOC CRM was an opportunity to fix that. Temporal entities were introduced into FRBR_{OO} by declaring some of the classes of FRBR_{OO} as subclasses of the following classes from CIDOC CRM: E65 Creation, E12 Production, and E13 Attribute Assignment.

Figure 1 show how the classes F27 Work Conception and F28 Expression Creation serve to link an E39 Actor, a E52 Time and a E53 Place to the F1 Work, F2 Expression and F4 Manifestation Singleton that are created by

¹⁸ HEANEY, Michael. *Time is of the essence*: some thoughts occasioned by the papers contributed to the International Conference on the Principles and Future Development of AACR [on line]. Oxford: Bodleian Library, 1997 [cited 19 January 2008]. Available from World Wide Web: <<http://www.bodley.ox.ac.uk/users/mh/time978a.htm>>.

LAGOZE, Carl. Business unusual: how “event-awareness” may breathe life into the catalog?. In: *Conference on bibliographic control in the new millennium* [on line]. Washington: Library of Congress, October 19, 2000 [cited 19 January 2008]. Available from Internet: <http://lcweb.loc.gov/catdir/bibcontrol/lagoze_paper.html>.

FITCH, Kent. *ALEG Data Model. Inventory* [on line]. [Brisbane]: AustLit Gateway, revised 27 July 2000 [cited 26 March 2004]. Available from World Wide Web: <<http://www.austlit.edu.au:7777/DataModel/inventory.html>>.

DOERR, Martin; HUNTER, Jane; LAGOZE, Carl. Towards a core ontology for information integration. In: *Journal of Digital Information* [on line]. 2003-04-09, Vol. 4, No. 1 [cited 19 January 2008]. Available from World Wide Web: <<http://journals.tdl.org/jodi/article/view/92/91>>.

those processes. In the lower part of the figure the work elaboration process is shown along a time axis. First, the activity F27 Work Conception produces an idea, then the F28 Expression Creation activity produces simultaneously an F2 Expression and its first manifestation (in the form of a F4 Manifestation Singleton), which together realise a work (F1).

TO:

... Since FRBR_{OO} borrows structures from the CIDOC CRM to express the concepts declared in FRBR_{ER}, “temporal entities” had inevitably to be introduced into FRBR_{OO}. Besides, some FRBR commentators had already made the point that time issues are insufficiently addressed in FRBR_{ER};¹⁹ the task of harmonising FRBR with the CIDOC CRM was an opportunity to fix that. Temporal entities were introduced into FRBR_{OO} by declaring some of the classes of FRBR_{OO} as subclasses of the following classes from CIDOC CRM: E65 Creation, E12 Production, E7 Activity, and E13 Attribute Assignment.

Figure 1 shows how the classes F27 Work Conception and F28 Expression Creation serve to link an E39 Actor, an E52 Time-Span and an E53 Place to the F1 Work, F2 Expression and F4 Manifestation Singleton that are created by those processes. In the lower part of the figure the work elaboration process is shown along a time axis. First, the activity F27 Work Conception produces an idea, then the F28 Expression Creation activity produces simultaneously an F2 Expression and its first manifestation (in the form of a F4 Manifestation Singleton), which together realise a work (F1).

1.2.2. Refinement of Group 1 Entities

In the second paragraph, in the middle and in the last paragraphs, the text is changed

FROM:

. Just like any product of the human mind, a Work necessarily begins to exist in the material world at a given point in time (even if it is a recollection of a Platonic form); this is the reason why FRBR_{OO} introduces the notion of F27 Work Conception. It makes the meaning of the FRBR_{ER} attribute ‘4.2.3 date of Work’ explicit.

.....
While it can be said that a typical bibliographic record created by a national bibliographic agency describes, in FRBR_{ER}, an instance of the Manifestation entity, this is no longer true in FRBR_{OO}. Actually, what a typical bibliographic record covers depends on the nature of the thing described, and, to a lesser degree, on the cataloguing policy that was followed when creating it.

.....
In the case of serials, since the scope note for F18 Serial Work indicates that “there is in general no single expression or manifestation representing a complete serial work, unless the serial work is ended,” what the bibliographic record describes is actually an instance of the F18 Serial Work itself. Information elements that, in the FRBR_{ER} conceptualisation, were attached to the Expression and Manifestation entities, are actually part of the instance of E29 Design or Procedure that serves as an issuing rule for the serial work. It belongs to the very definition of the instance of F18 Serial Work that it consists of issues published by a given publisher and containing texts in a given language. However, those information elements may change over time while the serial work retains its identity; in that case, the instance of F18 Serial Work has several distinct issuing rules over time. This is what is meant when a single bibliographic record shows that at a given date, the publisher and/or place of publication have changed.

TO:

. Just like any product of the human mind, a Work necessarily begins to exist in the material world at a given point in time; this is the reason why FRBR_{OO} introduces the notion of F27 Work Conception. It makes the meaning of the FRBR_{ER} attribute ‘4.2.3 date of Work’ explicit.

¹⁹ HEANEY, Michael. *Time is of the essence*: some thoughts occasioned by the papers contributed to the International Conference on the Principles and Future Development of AACR [on line]. Oxford: Bodleian Library, 1997 [cited 19 January 2008]. Available from World Wide Web: <<http://www.bodley.ox.ac.uk/users/mh/time978a.htm>>.

LAGOZE, Carl. Business unusual: how “event-awareness” may breathe life into the catalog?. In: *Conference on bibliographic control in the new millennium* [on line]. Washington: Library of Congress, October 19, 2000 [cited 19 January 2008]. Available from Internet: <http://lcweb.loc.gov/catdir/bibcontrol/lagoze_paper.html>.

FITCH, Kent. *ALEG Data Model. Inventory* [on line]. [Brisbane]: AustLit Gateway, revised 27 July 2000 [cited 26 March 2004]. Available from World Wide Web: <<http://www.austlit.edu.au:7777/DataModel/inventory.html>>.

DOERR, Martin; HUNTER, Jane; LAGOZE, Carl. Towards a core ontology for information integration. In: *Journal of Digital Information* [on line]. 2003-04-09, Vol. 4, No. 1 [cited 19 January 2008]. Available from World Wide Web: <<http://journals.tdl.org/jodi/article/view/92/91>>.

.....

While it can be said that the attributes in FRBR_{ER} still reflect to a certain degree traditional cataloguing policy, this is no longer true in FRBR_{OO}. Actually, what a bibliographic record should cover, following the intentions of FRBR, depends on the nature of the thing described, and, to a lesser degree, on the cataloguing policy that was followed when creating it. Interpreting FRBR, FRBR_{OO} strictly associates attributes (or “properties”) with the entity of the bibliographic discourse they actually belong to. Only this form allows for the explanation and reconciliation of the various application dependent simplifications a particular implementation might choose. Some prominent cases are:

In the case of serials, since the scope note for F18 Serial Work indicates that “there is in general no single Expression or Manifestation representing a complete serial work, unless the serial work has ended,” what the bibliographic record describes is actually an instance of the F18 Serial Work itself. Information elements that, in the FRBR_{ER} conceptualisation, were directly attached to the Expression and Manifestation entities, are in FRBR_{OO} seen as being in reality part of the issuing rule for the serial work (represented as an instance of E29 Design or Procedure). It is at the very core of the definition of F18 Serial Work that it plans that issues are published by a particular publisher and contain texts in a particular form. However, those information elements may change over time while the serial work retains its identity; in that case, the instance of F18 Serial Work has several distinct issuing rules over time, a case not modelled in FRBR_{ER}. This is what is meant when a single bibliographic record shows that at a given date, the publisher and/or place of publication have changed.

1.3. Differences in the FRBR Family between FRAD/FRSAD and FRBROO

The following text has been added to this section.

The FRAD model puts its emphasis on the separation between the bibliographic entities themselves (person, family, corporate body, work), their names as found in the non-bibliographic universe and the controlled access points constructed to represent the instances of the entities in bibliographic contexts through the use of rules as applied by bibliographic agencies.

In FRBROO the mechanism that underlies these relationships is made explicit by the introduction of classes and properties that capture the link between instances of persons, etc. and the names they use when performing different activities over time. Any activity in which one can see the use of a name in a given context is an F52 Name Use Activity. In the library domain, the F35 Nomen Use Statement records an appellation used within a KOS, including specific identifiers that provide controlled access points, which are generally recorded in authority records. Name authority files, subject headings files, classification systems are all typical library examples that are modelled as F34 KOS.

Taking as an example a Library of Congress name authority record for a corporate body (selected fields shown below), the information recorded in MARC21 field 670 is modelled by means of an F52 Name Use Activity. It indicates that in the book whose title is *Lo Scavo di S. Giovanni di Ruoti ed il periodo tradonantico in Basilicata* [This example is problematic – see ‘Issues’] published in 1983, the name of the centre is given in Italian on the title page (Centro accademico canadese in Italia) and in English on page 6 (Canadian Academic Centre in Italy). Thus this one book exemplifies two distinct F52 Name Use Activities which associate (R63 named) two different E41 Appellations (R64 used name) with the same instance of F11 Corporate body by means of two F35 Nomen Use Statements. These statements, created using the appropriate cataloguing rules, record the assignment of the identifiers Canadian Academic Centre in Italy (as the preferred form, MARC 21 field 110) and Centro accademico canadese in Italia (as a variant form, MARC 21 field 410) as controlled identifiers for this centre. These controlled access points (F35 Nomen Use Statements within the authority record) are R32 warranted by the F52 Name Use Activity which took place in the 1983 book.

```
010    __|a n 85118480
110     2_|a Canadian Academic Centre in Italy
410     2_|a Centro accademico canadese in Italia
670    __|a Lo Scavo di S. Giovanni di Ruoti ed il eriod tradonantico in Basilicata, 1983: |b t.p. (Centro accademico canadese in Italia) p. 6 (Canadian Academic Centre in Italy)
```

The FRBROO model clarifies that multiple bibliographic identities (such as pseudonyms coexisting with real names in different publications) are to be understood as instances of multiple name use by a single F10 Person in different contexts. This permits a new interpretation of the FRAD Person (An individual or a persona or identity established or adopted by an individual or group) as different F52 Name Use Activities of the same person taking place in specific contexts (R61). This insight is simpler than defining personas as classes within the model, as well as being considerably more flexible. The same modelling technique subsumes the apparently different situations of multiple pseudonyms used in different contexts thereby creating multiple bibliographic identities and variant forms of names used simultaneously. The cardinality of the relationships between instances of F10 Person and the names that persons use can be one-to-one, one-to-many, or many-to-many (allowing for joint pseudonyms).

Similarly, in FRSAD the basic model indicates that thema is distinct from the nomens used to represent it. In FRBROO this is modelled in the same way: F52 Name Use Activity links the statements found in reference sources that attest to the use of a particular name for a particular concept (each of these statements being an instance of F35 Nomen Use Statement in the context of a particular KOS).

In a typical Library of Congress Subject Headings (LCSH) authority record (selected fields shown below), the Thema, as represented by the record number (sh 8507420 in MARC 21 field 010), is associated with the nomen Lamniformes (an F35 Nomen Use Statement). An instance of F52 Name Use Activity is recorded on page 51 of the book *Fishes of the world* by J.S. Nelson published in 1994 that confirms the use of this term for this thema.

A broader term relationship is stated within LCSH (an instance of a F34 KOS) between this thema and two other thema, this is encoded in the MARC 21 550 fields (code value g in subfield w indicates broader term). This same authority record shows the use of MARC 21 field 053 to encode the assignment of the nomen QL638.94.L36 to this thema, this time within the Library of Congress Classification (LCC).

```
010      __ |a sh 85074230
053      _0 |a QL638.94.L36 |c Zoology
150      __ |a Lamniformes
550      __ |w g |a Chondrichthyes
550      __ |w g |a Sharks
670      __ |a Nelson, J.S. Fishes of the world, 1994: |b p. 51 (Order Lamniformes (mackerel sharks). Seven families with 10
genera and 16 species)
```

2.1 Graphic Overview of the Object-Oriented Definition of FRBR

This section has been separated in subsections following the figures. Also the Appendix 5 has been moved to 2.1.6 subsection of 2.1. In the first subsection, in the first paragraph as well as in the second part of the n^{th} the following changes have been made:

FROM :

Figure 6 shows the relations that exist between “works” and “expressions” and the subclasses of both concepts, independently from any dynamic aspects involving the activities of creation and modification. It shows an analysis of the original FRBR_{ER} concepts Work and Expression into the more detailed ones that appear only indirectly in FRBR_{ER} via attributes that are specific to these detailed concepts rather than to Work and Expression in general. The reader may find the actual relation of these concepts to the FRBR_{ER} attributes in section 5.3 below.

In detail:

.....

n. The notion of “work” is actually a vague one, which covers three more specific notions:

-
- The concept of re-using some already existing material or of using some event (either natural or involving human activity) in order to produce some new creation. This is modelled as: F16 Container Work *is a* F1 Work, F1 Work *R3 is realised in (realises)* F22 Self-Contained Expression, and (unless a natural event is being used) F22 Self-Contained Expression *R14 incorporates (is incorporated in)* F2 Expression.

.....
 Figure 10 lingers on the way FRBR_{OO} models live performing arts. In contrast to the other figures, it makes use of a concrete example. It demonstrate how successive intellectual processes incorporate Expressions from previous ones, add new elements of different natures, and thereby “add value” to the previous steps. In this sense, the performance adds movement and sound to a text, the recording adds points of view or ways of seeing. In a detailed example:

.....

TO:

2.1.1. Static View of the Work and Expression Classes

Figure 6 shows the relations that exist between works and expressions and the subclasses of both concepts, independently from any dynamic aspects involving the activities of creation and modification. It shows an analysis of the original FRBR_{ER} concepts Work and Expression into the more detailed ones that appear only indirectly in FRBR_{ER} via attributes that are specific to these detailed concepts rather than to Work and Expression in general. The reader may find the actual relation of these concepts to the FRBR_{ER} attributes in section 3.3 below.

.....
 Comments on Figure 6:

.....
 n. The notion of “work” is actually a vague one, which covers three more specific notions:

-
- The concept of re-using some already existing material in order to produce some new creation. This is modelled as: F16 Container Work *is a* F1 Work, F1 Work *R3 is realised in (realises)* F22 Self-Contained Expression, and F22 Self-Contained Expression *R14 incorporates (is incorporated in)* F2 Expression.

...

2.1.2 Dynamic View of the Work and Expression Classes

....

2.1.3 Dynamic View of the Manifestation and Item Classes

Comments on Figure 8:

.....

2.1.4 Static View of the Manifestation and Item Classes

Comments on Figure 9:

.....

2.1.5. Performing Arts as an Example for the Incorporation of Expressions in Expressions of Other Works

Figure 10 illustrates the way FRBR_{OO} models live performing arts. In contrast to the other figures, it makes use of a concrete example. It demonstrates how successive intellectual processes incorporate Expressions from previous ones, add new elements of different natures, and thereby “add value” to the previous steps. In this sense, the performance adds movement and sound to a text, the recording adds points of view or ways of seeing.

.....

Figure 13

Comments on Figure 10:

Shakespeare’s *Henry IV* is a play in 2 parts: each part is a self-contained play in its own right, but form nevertheless parts of a larger overall F15 Complex Work.

2.1.6. Creation and Assignment of Controlled Access Points

During the cataloguing process, one important phase is the creation of what are termed controlled access points (formerly known as “headings”). Controlled access points enable a given instance of a given bibliographic entity to be consistently referred to in a given bibliographic database. Controlled access points are, in general, composed of parts, which consist of signs, and some of which are appellations in their own right. They are created to identify persons, corporate bodies, geographic areas, works, etc.

Specific sections of cataloguing rules specify the steps that have to be followed when creating a controlled access point and how to ensure its uniqueness. The steps include the choice of the preferred form of the name, the choice of qualifiers, their form, punctuation and order. During an F40 Identifier Assignment the cataloguer composes the identifier and makes informed decisions. That complex process is modelled using three distinct classes: F12 Nomen, F13 Identifier, and F50 Controlled Access Point. F13 Identifier corresponds to both standardised strings such as uniform titles (which are instances of F50 Controlled Access Point), and the notion of numeric identifiers such as international standard numbers defined in ISO standards (such as ISBN, ISSN, ISRC, etc.). Any qualifier used in cataloguing practice to disambiguate access points is an instance of E90 Symbolic Object (or, in many cases, of its subclass E41 Appellation; e.g., dates are appellations of instances of E52 Time-Span).

Figure 11 is a representation of such an activity.

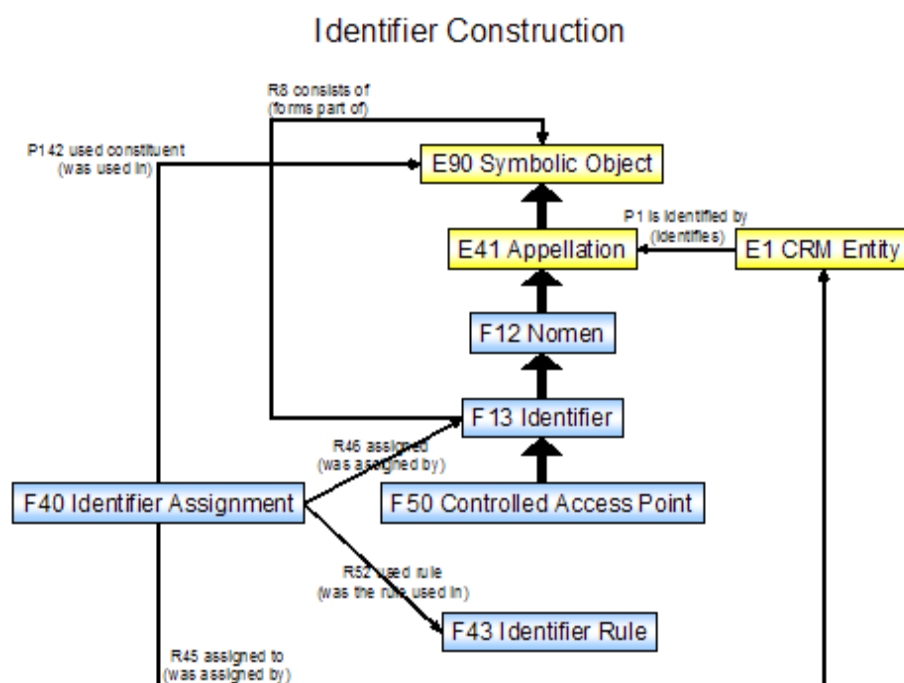


Figure 14

The model described in Figure 11 is relevant beyond library practice. It allows for the implementation of the reasoning processes involved in analysing the information encoded in the parts of an identifier. Therefore the CIDOC CRM has incorporated this model.

The process of creating controlled access points that are as specific, accurate, and “unique” as possible is particularly interesting when applied to instances of Work and Expression, as they lack a material nature. Figure 12 shows how the instance of an F2 Expression common to some instances of F3 Manifestation Product Type or F4 Manifestation Singleton would be identified with the one manifestation of a particular expression that has been selected to be “representative” for this expression. This model does not correspond to any explicit library information. Rather, it describes an implicit process behind the appellation creation process: The very fact of composing an identifier for an instance of F2 Expression using the identifier of one particular manifestation makes the latter “representative” for the corresponding expression. Note that the “representative” manifestation is not necessarily the historically first or earliest one. It might be the most popular one, or the first one detected. It might be a fragment of a whole detected later. As this introduces a kind of arbitrariness, the authors found this model interesting when discussing the respective practices. Similarly, an instance of F1 Work can only be identified if a corresponding instance of F2 Expression is selected as “representative”.

How to identify Work?

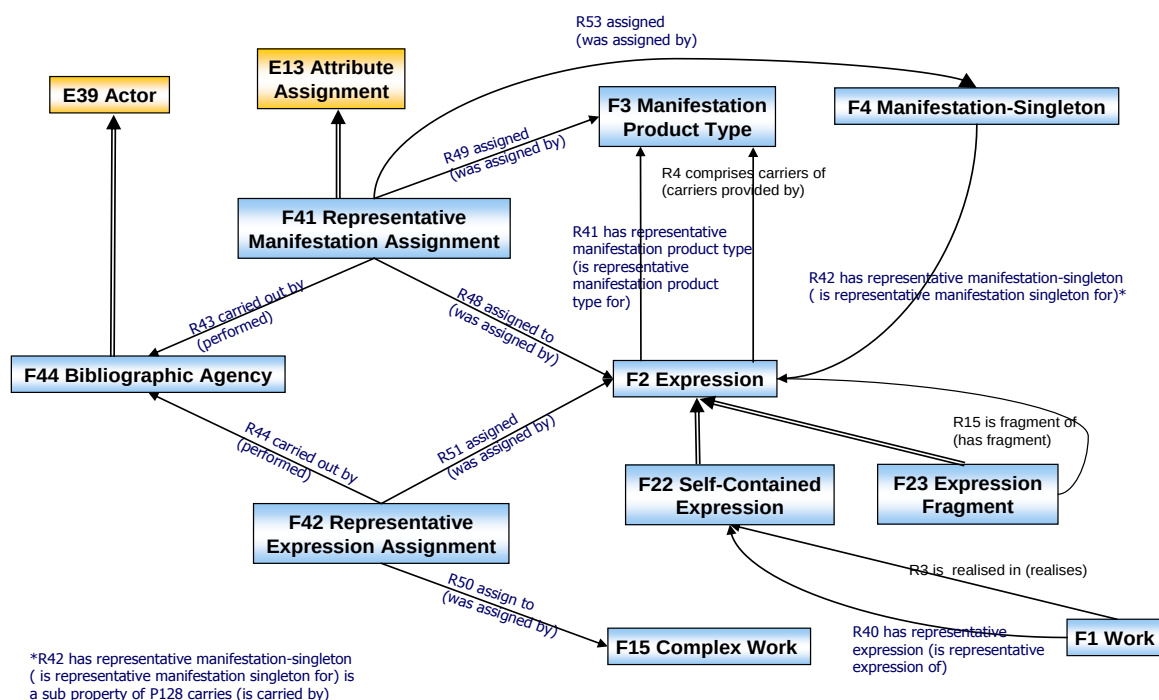


Figure 15

2.4. Presentation Conventions

Between the first and the second paragraph a paragraph is inserted. The following text is changed
FROM:

All instances of E41 Appellation are presented within single quotation marks, whether they are used for themselves or just to refer to the things they name. Any punctuation mark that follows an instance of E41 Appellation is placed outside the single quotation marks, as it does not belong to the appellation itself.

.....

TO:

All instances of E41 Appellation are presented within single quotation marks, whether they are used for themselves or just to refer to the things they name. Any punctuation mark that follows an instance of E41 Appellation is placed outside the single quotation marks, as it does not belong to the appellation itself.

Furthermore, all references to instances of E90 Symbolic Object in the form of a content model are presented within single quotation marks, such as 'abc'. By content model we mean the symbol sequence the symbolic object consists of.

.....

Proofreading

First page: Pat Riva is added to the editors list. Also the contributors list was re arranged.

Page 11: minor grammatical changes are made to the section **1.1.4 An opportunity for mutual enrichment for FRBR and CIDOC CRM**

FROM:

The CIDOC CRM model is influenced by the process of FRBR's re-formulation as well. Modelling bibliographic information highlights some issues that may have been overlooked during the development of CIDOC CRM, and the way such issues were addressed in FRBR₀₀ resulted in some cases in making changes in the CIDOC CRM model. These changes are so significant that an anticipated revision of the

ISO standard 21127 was required.

TO:

The CIDOC CRM model is influenced by the process of FRBR's re-formulation as well. Modelling bibliographic information highlights some issues that may have been overlooked during the development of CIDOC CRM, and the way such issues are addressed in FRBR_{OO} resulted in some cases in making changes in the CIDOC CRM model. These changes are so significant that a revision of the ISO standard 21127 was required.

Page 12: editorial updates are made to **1.1.6 Sources**

FROM

.....:

IFLA Study Group on the functional requirements for bibliographic records. *Functional requirements for bibliographic records: final report* [printed text]. Munich, Germany: K. G. Saur, 1998. Also available online from World Wide Web: <<http://www.ifla.org/en/publications/functional-requirements-for-bibliographic-records>>.

.....:

ICOM/CIDOC Documentation Standards Group; & CIDOC CRM Special Interest Group. *Definition of the CIDOC Conceptual Reference Model: version 5.0.4*, November 2011 [electronic resource]. [Heraklion, Greece]: [ICS-FORTH], 2011. Available online at: <http://www.cidoc-crm.org/docs/cidoc_crm_version_5.0.4.pdf>, or: <http://www.cidoc-crm.org/docs/cidoc_crm_version_5.0.4.doc>.

The FRBR model was complemented over years with two additional models: FRAD ("Functional Requirements for Authority Data"), and FRSAR ("Functional Requirements for Subject Authority Data"). The documents that contain a definition for those two models were also used during the process of elaborating the definition of the FRBR_{OO} model:

IFLA Working Group on Functional Requirements and Numbering of Authority Records (FRANAR). *Functional requirements for authority data: a conceptual model*. Munich, Germany: K. G. Saur, 2009.

IFLA Working Group on Functional Requirements for Subject Authority Records (FRSAR). *Functional requirements for subject authority data (FRSAD): a conceptual model*. Berlin: De Gruyter Saur, 2011. Also available online: <<http://www.ifla.org/files/classification-and-indexing/functional-requirements-for-subject-authority-data/frsad-final-report.pdf>>.

TO:

.....:

IFLA Study Group on the functional requirements for bibliographic records. *Functional requirements for bibliographic records: final report*. Munich, Germany: K. G. Saur, 1998. Also available online from World Wide Web: <<http://www.ifla.org/en/publications/functional-requirements-for-bibliographic-records>>.

.....:

ICOM/CIDOC Documentation Standards Group; CIDOC CRM Special Interest Group. *Definition of the CIDOC Conceptual Reference Model: version 5.1*, May 2012. [Heraklion, Greece]: [ICS-FORTH], 2012. Available online at: <http://cidoc.ics.forth.gr/docs/cidoc_crm_version_5.1.doc>, or: <http://www.cidoc-crm.org/docs/cidoc_crm_version_5.1.pdf>.

In the preparation of version 2.0 of FRBR_{OO} the final approved statements of the FRAD and FRSAD models were used.

IFLA Working Group on Functional Requirements and Numbering of

Authority Records (FRANAR); Glenn E. Patton, ed., *Functional requirements for authority data: a conceptual model*. München: K.G. Saur, 2009.

IFLA Working Group on the Functional Requirements for Subject Authority Records (FRSAR); Marcia Lei Zeng, Maja Žumer and Athena Salaba, ed., *Functional requirements for subject authority data (FRSAD): a conceptual model*. Berlin: De Gruyter Saur, 2011. Also available online from World Wide Web: <<http://www.ifla.org/node/1297>>.

Page 29: Minor editorial changes are made.

Page 31-40: the listed tables are updated

Page 41: The following notation is added:

- The line “Equal to:” declares the CIDOC CRM class that covers the same concept as the FRBR_{OO} class;

6.5 Amendments to Version 2.0 (22nd FRBR – CIDOC CRM)

In the introduction Pat Riva has been added to the international group.

The International Working Group on FRBR/CIDOC CRM Harmonisation, chaired by Martin Doerr (ICS FORTH, Greece), Patrick Le Bœuf (BnF, France), and Pat Riva (BAnQ, Canada),

6.6 Amendments to Version 2.0 (23rd FRBR – CIDOC CRM)

Modifications were made in the following portions of text:

1.1.6. Sources: URLs were updated.

1.2.1. Introduction of temporal entities..., footnote 7: URLs were checked, and updated where needed.

1.2.2. Refinement of Group 1 entities: Paragraph devoted to F27 Work Conception rephrased.

1.2.3. Analysis of creation and production processes: the notion of F53 Material Copy was introduced. **Figure 5 is still to be updated.**

2.1.2. Dynamic view of the Work and Expression classes: text devoted to F27 work Conception rephrased.

2.1.7. Photographs and animated images: this Section was introduced.

2.3. Property quantifiers: quantification conventions were harmonized with CIDOC CRM.

2.5.1., 2.5.2. FRBRoo class hierarchy: F53 and F54 were introduced (+ E25 Man-Made Feature, as superclass of F53).

2.5.3., 2.5.4. FRBRoo property hierarchy: R65 was introduced.

2.6.:

F5 Item: “Subclass of” and “Properties” modified.

F21 Recording Work: “Scope note” and “Examples” modified.

F26 Recording: “Examples” modified.

F27 Work Conception: “Scope note” and “Examples” modified.

F28 Expression Creation: “Scope note” and “Examples” modified.

F29 Recording Event: “Examples” modified, R65 property added.

F32 Carrier Production Event: “Scope note” modified, range of R28 preproperty modified.

F52 Name Use Activity: “Scope note” and “Examples” modified.

F53 Material Copy introduced.

F54 Utilized Information Carrier introduced.

2.7.:

Introduction: bullet 7 modified, bullet 12 introduced.

All properties: quantification conventions harmonized with CIDOC CRM.

R3: “Scope note” and “Examples” modified, R3.1 introduced.

R6: “Domain” modified.

R13: further “Examples” added.

R14: “Range”, “Scope note” and “Examples” modified.

R16: “Scope note” and “Examples” modified.

R21: further “Examples” added.

R22: further “Examples” added.

R28: “Range” and “Scope note” modified.

R34: “Subproperty of” modified.

R56: “Examples” modified.

R57: “Examples” **is still to be modified.**

R59: “Subproperty of” modified.

R60: “Subproperty of” **still needs “theoretical investigation”.**

R61: “Subproperty of” **still to be drafted.**

R62: “Subproperty of” modified (**proposal to be checked.**)

R65 introduced.

4.3.:

E25 Man-Made Feature introduced (as superclass of F53 Material Copy).

+ Minor editorial amendments (typos, spelling...).

6.7 Amendments to Version 2.1 (25th FRBR – CIDOC CRM)

Throughout the document:

All references to CIDOC CRM version 5.1 replaced with references to CIDOC CRM version 6.0.

All occurrences of “R14 incorporates...” replaced with “P165 incorporates...”

All reverse properties denoted by an “i” following the property’s code, instead of a “B”.

1. Introduction:

P. 10, 5th paragraph: date of formation of the WG added.

P. 10, footnote 5: a statement about the publication of revised ISO standard in 2014 added.

P. 12, 1.1.6. Sources: URLs of CIDOC CRM version 5.1 replaced with URLs of CIDOC CRM version 6.0.

P. 19, 5th paragraph: One sentence added: “In other terms, the thema (an instance of E55 Type) identified by the nomen “Lamniformes” *P127 has broader term* the thema (a distinct instance of E55 Type) identified by the nomen “Sharks”.”

2. Description of the Model:

P. 19, last paragraph: a statement about the modelling of photographs and animated images added.

P.21, bracket “g”: one footnote added: “Alternatively, users who would be eager to avoid the confusion between structural parts and successive members of complex works can use the CIDOC CRM property *P148 has component (is component of)*, instead of *R10 has member*, to record the relationship between an instance of F15 Complex Work and its structural parts.”

P24, last line: “writer” generalized to “creator”.

P25, 1st paragraph: parenthesis deleted: “(Not all manuscripts, however, are necessarily produced by an instance of F28 Expression Creation: a perfect copy of a brief text, made by a highly trained scribe from an original, and checked several times with the original to contain no alteration of the text, could be regarded as just the result of an instance of E12 Production; but as a rule, no two mediaeval manuscripts carry exactly the same text).”

P. 29, 1st paragraph: statement about codes for inverted properties added (Rni instead of RnB).

P. 38: R14 deleted.

P. 40: R66 added.

P. 41: R66 added.

P. 42: R66 added and R14 deleted.

P. 43: R14 deleted.

P. 46, F1 Work, Scope note rephrased.

P. 47, F2 Expression, Examples: some Examples instantiated as both expressions and linguistic objects.

P. 54, F15 Complex Work, Scope note rephrased.

P. 58, F22 Self-Contained Expression, Properties: R14 deleted.

P. 66, F40 Identifier Assignment, Scope Note rephrased.

P. 78-79, *R14 incorporates (is incorporated in)* deleted.

P. 101, *R65 recorded aspects of*: second “Shortcut of” deleted, and “Subproperty of shortcut of: [F29](#) Recording Event. [R20](#) recorded: [E5](#) Event. [P12](#) occurred in the presence of: [E18](#) Physical Thing” added.

P. 101, *R66 included performed version of* added.

P. 200: *P165 incorporates (is incorporated in)* added.